

mielke-geometroynamics — formula report

1794 inline formulas (section omitted; all rendered), 1149 display equations, 5 tables, 18 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_313 EQ0966 (14.2.7) [conf 0.024]		■ 0.024 ● W +1	$\begin{aligned} d\,\hat{s}^2 &= \frac{1}{4+K} \operatorname{Tr} \left(\hat{\vartheta} \otimes_s \hat{\vartheta}^T \right) = \hat{g}_{AB} \hat{\vartheta}^A \otimes_s \hat{\vartheta}^B; \quad A, B, = 0, \dots, 3 + K \\ &= \frac{1}{4+K} \operatorname{Tr} \left[\begin{array}{c c} \vartheta \otimes \vartheta + \hat{\ell}^2 \omega \otimes \omega & \hat{\ell} \omega \otimes \overset{\circ}{\vartheta} \\ \hline \hat{\ell} \vartheta \otimes \omega & \begin{array}{c} \ddots \\ \vartheta \otimes \end{array} \end{array} \right]. \end{aligned}$	$d\hat{s}^2 = \frac{1}{4+K} \operatorname{Tr}(\hat{\vartheta} \otimes_s \hat{\vartheta}^T) = \hat{g}_{AB} \hat{\vartheta}^A \otimes_s \hat{\vartheta}^B; \quad A, B, = 0, \dots, 3 + K$ $= \frac{1}{4+K} \operatorname{Tr} \left[\frac{\vartheta \otimes \vartheta + \hat{\ell}^2 \omega \otimes \omega}{\hat{\ell} \overset{\circ}{\vartheta} \otimes \omega} \middle \frac{\hat{\ell} \omega \otimes \overset{\circ}{\vartheta}}{\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta}} \right]. \quad (1.$	
mielke-geometroynamics_134 EQ0403 (6.4.23) [conf 0.028]		■ 0.028 ● C +4	$\Psi_{\parallel}(\vartheta) = \exp \left(- \int_{M_3} \overset{(\pm)}{C}_{\text{TT}} \right) = \exp \left(\frac{-1}{2\ell^2} \int_{M_3} \left[\overset{*}{+}\vartheta_B \wedge \overset{*}{T}^B \right] \right)$	$\Psi_{\parallel}(\vartheta) = \exp \left(- \int_{M_3} \overset{(\pm)}{C}_{\text{TT}} \right) = \exp \left(\frac{-1}{2\ell^2} \int_{M_3} \left[\overset{*}{+}\vartheta_B \wedge \overset{*}{T}^B \right] \right)$	$\Psi_{\parallel}(\vartheta) = \exp \left(- \int_{M_3} \overset{(\pm)}{C}_{\text{TT}} \right) = \exp \left(\frac{-1}{2\ell^2} \int_{M_3} \left[\overset{*}{+}\vartheta_B \wedge \overset{*}{T}^B \right] \right)$
mielke-geometroynamics_275 EQ0850 (12.3.3) [conf 0.032]		■ 0.032 ● C +3	$\begin{aligned} \not{D}^2 &= -\frac{1}{2} \gamma^\alpha \gamma^\beta \left(\left\{ D_\alpha^{\{\}}, D_\beta^{\{\}} \right\} + \left[D_\alpha^{\{\}}, D_\beta^{\{\}} \right] \right) - 2im \not{D}^{\{\}} \\ &\cong -\square - \frac{1}{8} \sigma^{\alpha\beta} R_{\alpha\beta\mu\nu} \sigma^{\mu\nu} \\ &= -\frac{i}{4} \gamma_5 \left(\not{D}^{\{\}} \not{\not{a}} \right) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \mathscr{A}_\alpha D_\beta^{\{\}} - \frac{1}{16} \mathscr{A}_\alpha \mathscr{A}^\alpha - m^2 \end{aligned}$	$\mathcal{D}^2 = -\frac{1}{2} \gamma^\alpha \gamma^\beta \left(\left\{ D_\alpha^{\{\}}, D_\beta^{\{\}} \right\} + \left[D_\alpha^{\{\}}, D_\beta^{\{\}} \right] \right) - 2im \mathcal{D}^{\{\}}$ $= -\frac{i}{4} \gamma_5 \left(\mathcal{D}^{\{\}} \not{\not{a}} \right) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \mathscr{A}_\alpha D_\beta^{\{\}} + m^2 - \frac{1}{2} m \gamma_5 \not{\not{a}} - \frac{1}{16} \not{\not{a}} \not{\not{a}}$ $\cong -\square - \frac{1}{8} \sigma^{\alpha\beta} R_{\alpha\beta\mu\nu} \sigma^{\mu\nu}$ $= -\frac{i}{4} \gamma_5 \left(\mathcal{D}^{\{\}} \not{b} \right) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \mathscr{A}_\alpha D_\beta^{\{\}} - \frac{1}{16} \mathscr{A}_\alpha \mathscr{A}^\alpha - m^2$	$\mathcal{D}^2 = -\frac{1}{2} \gamma^\alpha \gamma^\beta \left(\{ D_\alpha^{\{\}}, D_\beta^{\{\}} \} + [D_\alpha^{\{\}}, D_\beta^{\{\}}] \right) - 2im \mathcal{D}^{\{\}}$ $= -\frac{i}{4} \gamma_5 (\mathcal{D}^{\{\}} \not{\not{a}}) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \mathscr{A}_\alpha D_\beta^{\{\}} + m^2 - \frac{1}{2} m \gamma_5 \not{\not{a}} - \frac{1}{16} \not{\not{a}} \not{\not{a}}$ $\cong -\square - \frac{1}{8} \sigma^{\alpha\beta} R_{\alpha\beta\mu\nu} \sigma^{\mu\nu}$ $= -\frac{i}{4} \gamma_5 (\mathcal{D}^{\{\}} \not{b}) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \mathscr{A}_\alpha D_\beta^{\{\}} - \frac{1}{16} \mathscr{A}_\alpha \mathscr{A}^\alpha - m^2$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_134 EQ0402 (6.4.22) [conf 0.042]		■ 0.042 ● C +5	$\begin{array}{ll} \text{q: } & \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta), \\ \text{\scriptsize $\pm$ p: } & \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{*}_{\alpha}} \Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta),$ $\text{\scriptsize $\pm$ p: } \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{*}_{\alpha}} \Psi_{\parallel}(\vartheta),$	$q: \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta),$ $\text{\scriptsize $\pm$ p: } \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{*}_{\alpha}} \Psi_{\parallel}(\vartheta),$
mielke-geometrodynamics_101 EQ0293 (4.5.4) [conf 0.052]		■ 0.052 ● W +0	$\xi \downarrow L \equiv (\xi \downarrow \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \downarrow D\Psi) \wedge \frac{\partial L}{\partial D\Psi}.$	$\xi \downarrow L \equiv (\xi \downarrow \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \downarrow D\Psi) \wedge \frac{\partial L}{\partial D\Psi}.$	
mielke-geometrodynamics_047 EQ0064 (2.8.6) [conf 0.054]		■ 0.054 ● N -2	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \text{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \text{Tr}(\Omega \wedge \Omega) - \text{Tr}(\Omega) \wedge \text{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}^2(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \text{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \text{Tr}(\Omega \wedge \Omega) - \text{Tr}(\Omega) \wedge \text{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}^2(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	
mielke-geometrodynamics_133 EQ0400 (6.4.19) [conf 0.060]		■ 0.060 ● W +1	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \underline{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \underline{D} \stackrel{(\pm)}{*} \underline{A}_{\alpha} = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \underline{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \underline{D} \stackrel{(\pm)}{*} \underline{A}_{\alpha} = 0,$	
mielke-geometrodynamics_229 EQ0718 (10.9.4) [conf 0.066]		■ 0.066 ● N +0	$\Sigma_{\alpha} := e_{\alpha} \downarrow L_{\Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L_{\Psi}}{\partial \Psi} - (e_{\alpha} \downarrow \bar{\Psi}) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi} - (e_{\alpha} \downarrow D\bar{\Psi}) \wedge \frac{\partial L_{\Psi}}{\partial D\bar{\Psi}};$	$\Sigma_{\alpha} := e_{\alpha} \downarrow L_{\Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L_{\Psi}}{\partial \Psi} - (e_{\alpha} \downarrow \bar{\Psi}) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi} - (e_{\alpha} \downarrow D\bar{\Psi}) \wedge \frac{\partial L_{\Psi}}{\partial D\bar{\Psi}};$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_131 E00391 (6.4.6) [conf 0.066]	131	■ 0.066 ● W +0	$\begin{aligned} \Sigma_{\alpha} &:= \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ &\quad - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}. \end{aligned}$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ \quad - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ \quad - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$
mielke-geometrodynamics_168 E00520 (7.7.48) [conf 0.078]	168	■ 0.078 ● C +3	$\begin{aligned} 2 \ell^2 d C_{\mathrm{T}}^* &:= 2 d \left(\vartheta^{\alpha} \wedge T_{\alpha} \right) \\ &= R^{\{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} \\ &\quad - T^{\alpha} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta} \rfloor T^{\beta}) - \frac{1}{2} e_{\alpha} \rfloor (T^{\beta} \wedge \vartheta_{\beta}) \right] \\ &= R^{\{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} - T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right), \end{aligned}$	$2\ell^2 d C_{\mathrm{T}}^* := 2d(\vartheta^{\alpha} \wedge T_{\alpha}) \\ = R^{\{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} \\ - T^{\alpha} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta} \rfloor T^{\beta}) - \frac{1}{2} e_{\alpha} \rfloor (T^{\beta} \wedge \vartheta_{\beta}) \right] \\ = R^{\{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} - T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right),$	$2\ell^2 d C_{\mathrm{T}}^* := 2d(\vartheta^{\alpha} \wedge T_{\alpha}) \quad (7.7.48) \\ = R^{\{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} \\ - T^{\alpha} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta} \rfloor T^{\beta}) - \frac{1}{2} e_{\alpha} \rfloor (T^{\beta} \wedge \vartheta_{\beta}) \right] \\ = R^{\{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} - T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right),$
mielke-geometrodynamics_129 E00382 (6.2.33) [conf 0.087]	129	■ 0.087 ● W +1	$\begin{aligned} \stackrel{\circ}{\Gamma}_{\alpha}^{\beta} &:= \Gamma_{\alpha}^{\beta} + i A_{\alpha}^{\beta}, \quad A_{\alpha}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta}, \\ \stackrel{\circ}{\Gamma}_{\alpha}^{\beta} &:= \Gamma_{\alpha}^{\beta} + i A_{\alpha}^{\beta}, \quad A_{\alpha}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta}, \end{aligned}$	$\stackrel{(\pm)}{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + i A_{\alpha}^{\beta}, \quad A_{\alpha}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta},$	$\stackrel{(\pm)}{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + i A_{\alpha}^{\beta}, \quad A_{\alpha}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta},$
mielke-geometrodynamics_135 E00405 (6.4.25) [conf 0.098]	135	■ 0.098 ● C +8	$\begin{aligned} \underline{D}^* \left(\frac{\delta}{\delta \underline{\vartheta}^B} \right) &= \underline{D}^{(\pm)} \underline{T}^B \equiv \underline{R}_C^{(\pm)B} \underline{\vartheta}^C = 0, \\ \underline{D}^* \left(\frac{\delta}{\delta \underline{\vartheta}^B} \right) &= \underline{D}^{(\pm)} \underline{T}^B \equiv \underline{R}_C^{(\pm)B} \underline{\vartheta}^C = 0, \end{aligned}$	$-\ell^2 \underline{D}^* \left(\frac{\delta}{\delta \underline{\vartheta}^B} \right) = \underline{D}^{(\pm)} \underline{T}^B \equiv \underline{R}_C^{(\pm)B} \underline{\vartheta}^C = 0,$	$-\ell^2 \underline{D}^* \left(\frac{\delta}{\delta \underline{\vartheta}^B} \right) = \underline{D}^{(\pm)} \underline{T}^B \equiv \underline{R}_C^{(\pm)B} \underline{\vartheta}^C = 0,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_168 EQ0529 (7.7.49) [conf 0.099]	168	■ 0.099 ● C +6	$\begin{aligned} L_{\text{Euler}} &:= (-1)^{\text{sig}+1} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ &\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \\ &\equiv -L_{\text{SKY}} \\ &\quad - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \right) \end{aligned}$	$\begin{aligned} L_{\text{Euler}} &:= (-1)^{\text{sig}+1} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ &\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \\ &\equiv -L_{\text{SKY}} \\ &\quad - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \right) \end{aligned}$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}$ $= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right)$ $\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta}$ $\equiv -L_{\text{SKY}}$ $- 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^* \text{Ric}_{\beta}^{\beta} \right)$
mielke-geometrodynamics_315 EQ0974 (14.2.15) [conf 0.099]	315	■ 0.099 ● N +0	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\pi\overset{\circ}{\Theta} = 0} \right],$
mielke-geometrodynamics_180 EQ0553 (8.4.7)	180	■ 0.100 ● W +2	$\begin{aligned} L_{\text{SKY}}^{(*)} &= \frac{1}{2} \text{Tr} \left(\overset{\pm}{\Omega}^g \wedge {}^* \overset{\pm}{\Omega}^g \right) \\ &= -\frac{1}{2} R^{\alpha\beta} \wedge {}^* R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= -\frac{1}{4} \left(R_{\alpha\beta} \pm {}^* R_{\alpha\beta}^{(*)} \right) \wedge {}^* \left(R^{\alpha\beta} \pm {}^* R^{\alpha\beta(*)} \right). \end{aligned}$	$\begin{aligned} L_{\text{SKY}}^{(*)} &= \frac{1}{2} \text{Tr} \left(\overset{\pm}{\Omega}^g \wedge {}^* \overset{\pm}{\Omega}^g \right) \\ &= -\frac{1}{2} R^{\alpha\beta} \wedge {}^* R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= -\frac{1}{4} \left(R_{\alpha\beta} \pm {}^* R_{\alpha\beta}^{(*)} \right) \wedge {}^* \left(R^{\alpha\beta} \pm {}^* R^{\alpha\beta(*)} \right). \end{aligned}$	$L_{\text{SKY}}^{(*)} = \frac{1}{2} \text{Tr} \left(\overset{\pm}{\Omega}^g \wedge {}^* \overset{\pm}{\Omega}^g \right)$ $= -\frac{1}{2} R^{\alpha\beta} \wedge {}^* R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}$ $= -\frac{1}{4} \left(R_{\alpha\beta} \pm {}^* R_{\alpha\beta}^{(*)} \right) \wedge {}^* \left(R^{\alpha\beta} \pm {}^* R^{\alpha\beta(*)} \right).$
Conf. MathPix confidence: ■ ■ ■ — ■ ■ ■ — no value Residual render vs scan (inkdrill): ● ● ● ● ● ● C W S N K not measured component weak stable noise clean					

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mielke-geometrodynamics_104 EQ0311 (4.6.3)	104	■ 0.111 ● C +6	$\begin{aligned} & \widetilde{\leftarrow}_{\widetilde{y}} \widetilde{D} \widetilde{\xi}^{\alpha} = \widetilde{y} \left(\widetilde{D} \widetilde{D} \widetilde{\xi}^{\alpha} \right) + \widetilde{D} (\widetilde{y} \widetilde{D} \widetilde{\xi}^{\alpha}) \\ & = y J \left(D D \xi^{\alpha} + R^{(T) \alpha} \right) = 0, \end{aligned}$	$\widetilde{\mathbf{L}}_{\widetilde{y}} \widetilde{D} \widetilde{\xi}^{\alpha} = \widetilde{y} \left(\widetilde{D} \widetilde{D} \widetilde{\xi}^{\alpha} \right) + \widetilde{D} (\widetilde{y} \widetilde{D} \widetilde{\xi}^{\alpha}) = y \left(D D \xi^{\alpha} + R^{(T) \alpha} \right) = 0,$	
mielke-geometrodynamics_128 EQ0378 (6.2.29)	128	■ 0.136 ● C +6	$D \, D \, \vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \, D \, D \, \Pi_{\alpha} = - \, R_{\alpha}^{\beta} \wedge \Pi_{\beta} = \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \lrcorner \Pi^{\beta} \right) \wedge \Sigma_{\beta} \cong 0.$	$D D \vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \, D \, D \, \Pi_{\alpha} = - \, R_{\alpha}^{\beta} \wedge \Pi_{\beta} = \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \lrcorner \Pi^{\beta} \right) \wedge \Sigma_{\beta} \cong 0.$	
mielke-geometrodynamics_346 EQ1059 (15.2.16)	346	■ 0.147 ● C +12	$\begin{aligned} \psi = & \left[\begin{array}{ccc c} \mathcal{P}_a & \mathcal{P}_b & \mathcal{P}_c & \nu_e \\ \eta_a & \eta_b & \eta_c & e^- \\ \lambda_a & \lambda_b & \lambda_c & \mu^- \\ \hline c_a & c_b & c_c & \nu_{\mu} \end{array} \right] \begin{array}{c} \text{up} \\ \text{down} \\ \text{strange} \\ \text{charm} \end{array} \begin{array}{c} \uparrow \\ \text{flavors} \\ \downarrow \end{array} \\ & \text{red} \quad \text{yellow} \quad \text{blue} \quad \text{lilac} \end{aligned}$	$\psi = \left[\begin{array}{ccc c} \mathcal{P}_a & \mathcal{P}_b & \mathcal{P}_c & \nu_e \\ \eta_a & \eta_b & \eta_c & e^- \\ \lambda_a & \lambda_b & \lambda_c & \mu^- \\ \hline c_a & c_b & c_c & \nu_{\mu} \end{array} \right] \begin{array}{c} \text{up} \\ \text{down} \\ \text{strange} \\ \text{charm} \end{array} \begin{array}{c} \uparrow \\ \text{flavors} \\ \downarrow \end{array}$	
mielke-geometrodynamics_314 EQ0972 (14.2.13)	314	■ 0.153 ● K +0	$\stackrel{\circ}{D} \alpha := \stackrel{\circ}{d} \alpha - \omega^g \wedge \alpha - \alpha \wedge \stackrel{\circ}{\omega}$	$\stackrel{\circ}{D} \alpha := \stackrel{\circ}{d} \alpha - \omega^g \wedge \alpha - \alpha \wedge \stackrel{\circ}{\omega}$	
mielke-geometrodynamics_095 EQ0264 (4.3.29)	095	■ 0.154 ● W +2	$\Gamma_{ij}^{\prime} = \left\{ \begin{array}{c} \kappa \\ ij \end{array} \right\} - K_{ij}^{\cdot \kappa},$	$\Gamma_{ij}^{\cdot \kappa} = \left\{ \begin{array}{c} \kappa \\ ij \end{array} \right\} - K_{ij}^{\cdot \kappa},$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E00292 (4.5.3)	100	■0.165 ●W +2	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \lrcorner \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \lrcorner D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \lrcorner \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \lrcorner D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \lrcorner \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} \end{aligned}$	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \lrcorner \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \lrcorner D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \lrcorner \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \lrcorner D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \lrcorner \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} \end{aligned}$	
mielke-geometrodynamics_E00734 (11.1.1)	237	■0.220 ●W +1	$\pi_1 \left(S^0(s, n-s), e \right) = \begin{cases} \mathbb{Z}_2 & n = 2k \\ \mathbb{Z}_2 \oplus \mathbb{Z}_2 & n = 4k \\ \mathbb{Z}_4 & n = 2(2k+1) \end{cases}$	$\pi_1 \left(SO_o(s, n-s), e \right) = \begin{cases} \mathbb{Z}_2 & n = 2k \\ \mathbb{Z}_2 \oplus \mathbb{Z}_2 & n = 4k \\ \mathbb{Z}_4 & n = 2(2k+1) \end{cases}$	$\pi_1(SO_o(s, n-s), e) = \begin{cases} \mathbb{Z}_2 & n = 2k \\ \mathbb{Z}_2 \oplus \mathbb{Z}_2 & n = 4k \\ \mathbb{Z}_4 & n = 2(2k+1). \end{cases}$
mielke-geometrodynamics_E00961 (14.2.1)	312	■0.220 ●C +3	$\hat{\vartheta} = \hat{\vartheta}(m, g) = \underbrace{\left[\begin{array}{c c} \vartheta & \hat{\ell}\omega \\ \hline 0 & \vartheta \end{array} \right]}_4 \}^4 K$	$\hat{\vartheta} = \hat{\vartheta}(m, g) = \underbrace{\left[\begin{array}{c c} \vartheta & \hat{\ell}\omega \\ \hline 0 & \vartheta \end{array} \right]}_4 \}^4 K$	$\hat{\vartheta} = \hat{\vartheta}(m, g) = \underbrace{\left[\begin{array}{c c} \vartheta & \hat{\ell}\omega \\ \hline 0 & \vartheta \end{array} \right]}_4 \}^4 K$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_085 E00211 (4.2.3)		■ 0.223 ● C +4	$\begin{aligned} & * \widetilde{\omega} & \& \\ & = \stackrel{*}{\chi} \left(d \widetilde{\omega} - \frac{1}{2} [\widetilde{\omega}, \widetilde{\omega}] \right) \\ & = d \omega^L - \omega^L \wedge \omega^L + d \omega^T - \omega^T \wedge \omega^L - \omega^L \wedge \omega^T + \frac{1}{2} [\omega^T, \omega^T] \\ & = \Omega^L + \Omega^T \end{aligned}$	$\begin{aligned} * \widetilde{\Omega} &= \stackrel{*}{\chi} \left(d \widetilde{\omega} - \frac{1}{2} [\widetilde{\omega}, \widetilde{\omega}] \right) \\ &= d \omega^L - \omega^L \wedge \omega^L + d \omega^T - \omega^T \wedge \omega^L - \omega^L \wedge \omega^T + \frac{1}{2} [\omega^T, \omega^T] \\ &= \Omega^L + \Omega^T \end{aligned}$	$\begin{aligned} \stackrel{*}{\chi} \widetilde{\Omega} &= \stackrel{*}{\chi} (d \widetilde{\omega} - \frac{1}{2} [\widetilde{\omega}, \widetilde{\omega}]) \\ &= d \omega^L - \omega^L \wedge \omega^L + d \omega^T - \omega^T \wedge \omega^L - \omega^L \wedge \omega^T + \frac{1}{2} [\omega^T, \omega^T] \\ &= \Omega^L + \Omega^T \end{aligned}$
mielke-geometrodynamics_095 E00266 (4.3.31)		■ 0.227 ● W -2	$R_{\{i\ j\}c}^{\cdot d} = R_{ijc}^{\{! \dots d\}} - 2 D_{[i} K_{j]c}^{\cdot d} + 2 K_{[i c}^{\cdot h} K_{j]h}^{\cdot d}.$	$R_{ijc}^{::d} = R_{ijc}^{\{! \dots d\}} - 2 D_{[i} K_{j]c}^{\cdot d} + 2 K_{[i c}^{\cdot h} K_{j]h}^{\cdot d}.$	$R_{ijc}^{..d} = R_{ijc}^{\{! \dots d\}} - 2 D_{[i} K_{j]c}^{\cdot d} + 2 K_{[i c}^{\cdot h} K_{j]h}^{\cdot d}.$
mielke-geometrodynamics_099 E00286 (4.4.15)		■ 0.229 ● K +0	$D \Pi_{\square}^T - E_{\square} = \Sigma_{[l}$	$D \Pi_{\square}^T - E_{\square} = \Sigma_{[l}$	$D \Pi_{\square}^T - E_{\square} = \Sigma_{[l}$
mielke-geometrodynamics_114 E00337 (5.3.14)		■ 0.239 ● C +4	$\begin{aligned} & R^{\{\alpha\beta\}} \wedge \eta_{\alpha\beta} \equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} - K^{\alpha\mu} \wedge K_{\mu}^{\beta} \wedge \eta_{\alpha\beta} + K^{\alpha\beta} \wedge T^{\gamma} \wedge \eta_{\alpha\beta\gamma} \\ & \quad + d(K^{\alpha\beta} \wedge \eta_{\alpha\beta}) \\ & = R^{\alpha\beta} \wedge \eta_{\alpha\beta} + T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right) + 2dC_{TT}^* \end{aligned}$	$\begin{aligned} R^{\{\alpha\beta\}} \wedge \eta_{\alpha\beta} &\equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} - K^{\alpha\mu} \wedge K_{\mu}^{\beta} \wedge \eta_{\alpha\beta} + K^{\alpha\beta} \wedge T^{\gamma} \wedge \eta_{\alpha\beta\gamma} \\ &\quad + d(K^{\alpha\beta} \wedge \eta_{\alpha\beta}) \\ &= R^{\alpha\beta} \wedge \eta_{\alpha\beta} + T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right) + 2dC_{TT}^* \end{aligned}$	$\begin{aligned} R^{\{ \alpha \beta \}} \wedge \eta_{\alpha \beta} &\equiv R^{\alpha \beta} \wedge \eta_{\alpha \beta} - K^{\alpha \mu} \wedge K_{\mu}^{\beta} \wedge \eta_{\alpha \beta} + K^{\alpha \beta} \wedge T^{\gamma} \wedge \eta_{\alpha \beta \gamma} \\ &\quad + d(K^{\alpha \beta} \wedge \eta_{\alpha \beta}) \\ &= R^{\alpha \beta} \wedge \eta_{\alpha \beta} + T^{\alpha} \wedge * \Big(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \Big) + 2dC_{TT}^* \end{aligned} \tag{5.3}$
mielke-geometrodynamics_271 E00835 (12.2.3)		■ 0.247 ● C -3	$\begin{aligned} & L_{\mathrm{D}} & = L \left(\gamma, \psi, D^{(f)} \psi \right) - \frac{i}{2} \bar{\psi} \left(* \gamma \wedge K - K \wedge * \gamma \right) \psi + A \wedge j \\ & = L \left(\gamma, \psi, D^{(\dagger)} \psi \right) + \frac{1}{4} \mathscr{A} \wedge j_5 + A \wedge j \\ & = L \left(\gamma, \psi, D^{(i)} \psi \right) - T^{\alpha} \wedge \mu_{\alpha} + A \wedge j. \end{aligned}$	$\begin{aligned} L_D &= L(\gamma, \psi, D^{(f)} \psi) - \frac{i}{2} \bar{\psi} (* \gamma \wedge K - K \wedge * \gamma) \psi + A \wedge j \\ &= L(\gamma, \psi, D^{(\dagger)} \psi) + \frac{1}{4} \mathscr{A} \wedge j_5 + A \wedge j \\ &= L(\gamma, \psi, D^{(i)} \psi) - T^{\alpha} \wedge \mu_{\alpha} + A \wedge j. \end{aligned}$	$\begin{aligned} L_D &= L(\gamma, \psi, D^{\dagger} \psi) - \frac{i}{2} \bar{\psi} (* \gamma \wedge K - K \wedge * \gamma) \psi + A \wedge j \\ &= L(\gamma, \psi, D^{\dagger} \psi) + \frac{1}{4} \mathscr{A} \wedge j_5 + A \wedge j \\ &= L(\gamma, \psi, D^{\dagger} \psi) - T^{\alpha} \wedge \mu_{\alpha} + A \wedge j. \end{aligned}$
mielke-geometrodynamics_248 E00787 (11.3.6)		■ 0.258 ● C +3	$i \gamma \wedge * D^{[1} \psi - 6 \ell^{*2} \gamma^{n+1} \gamma \wedge * (\bar{\psi} \gamma^{n+1} \gamma \psi) \psi - m \psi \eta = 0$	$i \gamma \wedge * D^{\{1} \psi - 6 \ell^{*2} \gamma^{n+1} \gamma \wedge * (\bar{\psi} \gamma^{n+1} \gamma \psi) \psi - m \psi \eta = 0 \Big ,$	$i \gamma \wedge * D^{\{1} \psi - 6 \ell^{*2} \gamma^{n+1} \gamma \wedge * (\bar{\psi} \gamma^{n+1} \gamma \psi) \psi - m \psi \eta = 0 \Big ,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_037 EQ1078 (A.3)	365	■ 0.258 ● W +1	$\begin{aligned} & \left(\sigma^a \right)^T := \left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right) \equiv \mathbb{1}, \quad \sigma^1 := \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right), \\ & \sigma^2 := \left(\begin{array}{ll} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 := \left(\begin{array}{ll} 1 & 0 \\ 0 & -1 \end{array} \right). \end{aligned}$	$(\sigma^a)^T := \left(\begin{array}{ll} \sigma^0, \sigma^k & k = 1, 2, 3 \\ \sigma^0 := \left(\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right) \equiv \mathbb{1}, & \sigma^1 := \left(\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array} \right) \\ \sigma^2 := \left(\begin{array}{ll} 0 & -i \\ i & 0 \end{array} \right), & \sigma^3 := \left(\begin{array}{ll} 1 & 0 \\ 0 & -1 \end{array} \right). \end{array} \right) ..$	$(\sigma^a)^T := (\sigma^0, \sigma^k) \quad k = 1, 2, 3.$ $\sigma^0 := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \equiv \mathbb{1}, \quad \sigma^1 := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$ $\sigma^2 := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$
mielke-geometrodynamics_037 EQ0018 (2.3.4)	037	■ 0.261 ● N +0	$\delta : \left\{ \begin{array}{ccc} G \times P & \rightarrow & P \\ w & \psi & \psi \\ g_p \cdot & p_o = p & \end{array} \right e \cdot p_o = p_o \in P \right\}.$	$\delta : \left\{ \begin{array}{ccc} G \times P & \rightarrow & P \\ \psi & \psi & \psi \\ g_p \cdot & p_o = p & \end{array} \right e \cdot p_o = p_o \in P \right\}.$	$\delta : \left\{ \begin{array}{ccc} G \times P & \rightarrow & P \\ \psi & \psi & \psi \\ g_p \cdot & p_o = p & \end{array} \right e \cdot p_o = p_o \in P \right\}.$
mielke-geometrodynamics_177 EQ0544 (8.3.5)	177	■ 0.261 ● S +1	$\exp(\zeta) := \mathbf{1} + \zeta + \frac{1}{2!} \zeta^2 + \frac{1}{3!} \zeta^3 + \dots$	$\exp(\zeta) := \mathbf{1} + \zeta + \frac{1}{2!} \zeta^2 + \frac{1}{3!} \zeta^3 + \dots$	$\exp(\zeta) := \mathbf{1} + \zeta + \frac{1}{2!} \zeta^2 + \frac{1}{3!} \zeta^3 + \dots$
mielke-geometrodynamics_199 EQ0605 (9.2.4)	199	■ 0.263 ● N +0	$\chi(M) := \sum_{p=0}^n (-1)^p b_p = \int_M \pi(\gamma_2^g)$	$\chi(M) := \sum_{p=0}^n (-1)^p b_p = \int_M \pi(\gamma_2^g)$	$\chi(M) := \sum_{p=0}^n (-1)^p b_p = \int_M \pi(\gamma_2^g)$
mielke-geometrodynamics_211 EQ0627 (10.1.15)	211	■ 0.271 ● N +0	$E_\alpha := \frac{\partial L}{\partial \vartheta^\alpha} = e_\alpha \lrcorner L + (e_\alpha \lrcorner T^\beta) \wedge H_\beta + 2(-1)^s (e_\alpha \lrcorner R^{*\beta}) \wedge H_\beta^*,$	$E_\alpha := \frac{\partial L}{\partial \vartheta^\alpha} = e_\alpha \lrcorner L + (e_\alpha \lrcorner T^\beta) \wedge H_\beta + 2(-1)^s (e_\alpha \lrcorner R^{*\beta}) \wedge H_\beta^*,$	$E_\alpha := \frac{\partial L}{\partial \vartheta^\alpha} = e_\alpha \lrcorner L + (e_\alpha \lrcorner T^\beta) \wedge H_\beta + 2(-1)^s (e_\alpha \lrcorner R^{*\beta}) \wedge H_\beta^*,$
mielke-geometrodynamics_316 EQ0979 (14.2.20)	316	■ 0.272 ● N +0	$\hat{\ell} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* / \sqrt{\alpha_g} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* \frac{\sqrt{\hbar c}}{g}$	$\hat{\ell} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* / \sqrt{\alpha_g} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* \frac{\sqrt{\hbar c}}{g}$	$\hat{\ell} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* / \sqrt{\alpha_g} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* \frac{\sqrt{\hbar c}}{g}$
mielke-geometrodynamics_216 EQ0654 (10.3.11)	216	■ 0.274 ● N +0	$E_\alpha = e_\alpha \lrcorner L + [(e_\alpha \lrcorner T^\beta) + (-1)^s 2\delta \ell (e_\alpha \lrcorner R^{*\beta})] \wedge H_\beta + (-1)^s \frac{2\gamma}{\ell} (e_\alpha \lrcorner R^{*\beta}) \wedge \vartheta_\beta,$	$E_\alpha = e_\alpha \lrcorner L + [(e_\alpha \lrcorner T^\beta) + (-1)^s 2\delta \ell (e_\alpha \lrcorner R^{*\beta})] \wedge H_\beta + (-1)^s \frac{2\gamma}{\ell} (e_\alpha \lrcorner R^{*\beta}) \wedge \vartheta_\beta,$	$E_\alpha = e_\alpha \lrcorner L + [(e_\alpha \lrcorner T^\beta) + (-1)^s 2\delta \ell (e_\alpha \lrcorner R^{*\beta})] \wedge H_\beta + (-1)^s \frac{2\gamma}{\ell} (e_\alpha \lrcorner R^{*\beta}) \wedge \vartheta_\beta,$
mielke-geometrodynamics_095 EQ0260 (4.3.33)	095	■ 0.276 ● W -2	$R_{i[j\kappa l]}^{(l)} \equiv 0$	$R_{i[j\kappa l]}^{(l)} \equiv 0$	$R_{i[j\kappa l]}^{(l)} \equiv 0$
mielke-geometrodynamics_213 EQ0639 (10.2.10)	213	■ 0.278 ● N -1	$R_\alpha^{i*} = -\Lambda_{\text{eff}} \eta_\alpha,$	$R_\alpha^{i*} = -\Lambda_{\text{eff}} \eta_\alpha,$	$R_\alpha^{(l)*} = -\Lambda_{\text{eff}} \eta_\alpha,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_009527 (7.7.47)	168	0.282 N -1	$\begin{aligned} L_{\mathrm{Pontr}} &:= dC_{\mathrm{RR}} = \frac{1}{2} R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha} = -\frac{1}{2} R_{\alpha\beta}^{\{\}} \wedge R^{\{\}\alpha\beta} \\ &\quad - \frac{1}{12} d \left[{}^* \mathcal{A} \wedge R^{\{\}} - \frac{1}{3} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{9} {}^* \mathcal{A} \wedge {}^* (\mathcal{A} \wedge {}^* \mathcal{A}) \right]. \end{aligned}$	$L_{\mathrm{Pontr}} := dC_{\mathrm{RR}} = \frac{1}{2} R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha} = -\frac{1}{2} R_{\alpha\beta}^{\{\}} \wedge R^{\{\j\alpha\beta} - \frac{1}{12} d \left[{}^* \mathcal{A} \wedge R^{\{\}} - \frac{1}{3} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{9} {}^* \mathcal{A} \wedge {}^* (\mathcal{A} \wedge {}^* \mathcal{A}) \right].$	$L_{\mathrm{Pontr}} := dC_{\mathrm{RR}} = \frac{1}{2} R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha} = -\frac{1}{2} R_{\alpha\beta}^{\{\}} \wedge R^{\{\}\alpha\beta} - \frac{1}{12} d \left[{}^* \mathcal{A} \wedge R^{\{\}} - \frac{1}{3} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{9} {}^* \mathcal{A} \wedge {}^* (\mathcal{A} \wedge {}^* \mathcal{A}) \right].$
mielke-geometrodynamics_009942 (14.1.23)	306	0.287 W +0	$\xi \wedge {}^* \xi = \pm \frac{\Omega \wedge {}^* \Omega \operatorname{Tr}^*(\Omega \wedge {}^* \Omega) - (-1)^{2z} \Omega \wedge \Omega \operatorname{Tr}^*(\Omega \wedge \Omega)}{\sqrt{(\operatorname{Tr}^*(\Omega \wedge {}^* \Omega))^2 + (-1)^{2z} (\operatorname{Tr}^*(\Omega \wedge \Omega))^2}}.$	$\xi \wedge {}^* \xi = \pm \frac{\Omega \wedge {}^* \Omega \operatorname{Tr}^*(\Omega \wedge {}^* \Omega) - (-1)^{2z} \Omega \wedge \Omega \operatorname{Tr}^*(\Omega \wedge \Omega)}{\sqrt{(\operatorname{Tr}^*(\Omega \wedge {}^* \Omega))^2 + (-1)^{2z} (\operatorname{Tr}^*(\Omega \wedge \Omega))^2}}.$	$\xi \wedge {}^* \xi = \pm \frac{\Omega \wedge {}^* \Omega \operatorname{Tr}^*(\Omega \wedge {}^* \Omega) - (-1)^{2z} \Omega \wedge \Omega \operatorname{Tr}^*(\Omega \wedge \Omega)}{\sqrt{(\operatorname{Tr}^*(\Omega \wedge {}^* \Omega))^2 + (-1)^{2z} (\operatorname{Tr}^*(\Omega \wedge \Omega))^2}}.$
mielke-geometrodynamics_00440 (6.8.11)	143	0.288 N +0	$\left. D \Sigma_{\alpha} \right \cong (e_{\alpha} \rfloor T^{\gamma}) \wedge \Sigma_{\gamma} + (e_{\alpha} \rfloor R^{\gamma\delta}) \wedge \tau_{\gamma\delta}$	$D\Sigma_{\alpha} \cong (e_{\alpha} \rfloor T^{\gamma}) \wedge \Sigma_{\gamma} + (e_{\alpha} \rfloor R^{\gamma\delta}) \wedge \tau_{\gamma\delta}$	$D\Sigma_{\alpha} \cong (e_{\alpha} \rfloor T^{\gamma}) \wedge \Sigma_{\gamma} + (e_{\alpha} \rfloor R^{\gamma\delta}) \wedge \tau_{\gamma\delta}$
mielke-geometrodynamics_00315 (4.6.7)	105	0.292 N +1	$\Delta\Phi(\widetilde{\Gamma},\gamma)\simeq\frac{i}{\hbar}\int_S\left(R^{(T)\alpha}P_{\alpha}+R_{\alpha}^{(L)\beta}L^{\alpha}_{\beta}\right)$	$\Delta\Phi(\widetilde{\Gamma},\gamma)\simeq\frac{i}{\hbar}\int_S\left(R^{(T)\alpha}P_{\alpha}+R_{\alpha}^{(L)\beta}L^{\alpha}_{\beta}\right)$	$\Delta\Phi(\widetilde{\Gamma},\gamma)\simeq\frac{i}{\hbar}\int_S(R^{(T)\alpha}P_{\alpha}+R_{\alpha}^{(L)\beta}L^{\alpha}_{\beta}).$
mielke-geometrodynamics_009965 (14.2.5)	313	0.299 N +0	$\begin{aligned} &\stackrel{\circ}{\Theta}:=\dot{\stackrel{\circ}{\vartheta}}-[\stackrel{\circ}{\omega},\stackrel{\circ}{\vartheta}]= (1-2b)\stackrel{\circ}{\vartheta}\wedge\stackrel{\circ}{\vartheta}, \\ &\stackrel{\circ}{\Omega}:=\dot{\stackrel{\circ}{\omega}}-\stackrel{\circ}{\omega}\wedge\stackrel{\circ}{\omega}=b(1-b)\stackrel{\circ}{\vartheta}\wedge\stackrel{\circ}{\vartheta}. \end{aligned}$	$\begin{aligned} \stackrel{\circ}{\Theta} &:= \dot{\stackrel{\circ}{\vartheta}} - [\stackrel{\circ}{\omega}, \stackrel{\circ}{\vartheta}] = (1 - 2b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}, \\ \stackrel{\circ}{\Omega} &:= \dot{\stackrel{\circ}{\omega}} - \stackrel{\circ}{\omega} \wedge \stackrel{\circ}{\omega} = b(1 - b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}. \end{aligned}$	$\begin{aligned} \stackrel{\circ}{\Theta} &:= \dot{\stackrel{\circ}{\vartheta}} - [\stackrel{\circ}{\omega}, \stackrel{\circ}{\vartheta}] = (1 - 2b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}, \\ \stackrel{\circ}{\Omega} &:= \dot{\stackrel{\circ}{\omega}} - \stackrel{\circ}{\omega} \wedge \stackrel{\circ}{\omega} = b(1 - b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}. \end{aligned}$
mielke-geometrodynamics_00226 (4.2.19)	087	0.299 C +4	$\begin{aligned} A^{-1}\omega^{\mathrm{T}} &= A^{-1}\widetilde{\vartheta} - G^{-1}DG^{A^{-1}}\widetilde{\varphi} \\ &= G^{-1}\widetilde{\vartheta} - G^{-1}D\widetilde{\varphi}, \end{aligned}$	$\begin{aligned} A^{-1}\omega^{\mathrm{T}} &= A^{-1}\widetilde{\vartheta} - G^{-1}DG^{A^{-1}}\widetilde{\varphi} \\ &= G^{-1}\widetilde{\vartheta} - G^{-1}D\widetilde{\varphi}, \end{aligned}$	$\begin{aligned} A^{-1}\omega^{\mathrm{T}} &= A^{-1}\widetilde{\vartheta} - G^{-1}DG^{A^{-1}}\widetilde{\varphi} \\ &= G^{-1}\widetilde{\vartheta} - G^{-1}D\widetilde{\varphi}, \end{aligned}$
mielke-geometrodynamics_00407 (6.4.27)	135	0.315 N +0	$\Psi_{\Lambda}(A)=\exp\left(\frac{3}{\Lambda}\int_{M_3}{}^{(\pm)}\overline{C}_{\mathrm{RR}}(\,)\right),$	$\Psi_{\Lambda}(A)=\exp\left(\frac{3}{\Lambda}\int_{M_3}{}^{(\pm)}\overline{C}_{\mathrm{RR}}^{\{\}}(\,)\right),$	$\Psi_{\Lambda}(A)=\exp\left(\frac{3}{\Lambda}\int_{M_3}{}^{(\pm)}\overline{C}_{\mathrm{RR}}^{\{\}}(\,)\right),$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured

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mielke-geometrodynamics_082 EQ0196 (4.1.8)		■ 0.322 ● N +1	$\begin{aligned} & \left[P_{\alpha}, P_{\beta} = 0 \right] \quad (P_{\alpha} := \partial / \partial x^{\alpha}), \\ & [P_{\alpha}, L_{\gamma \delta}] = g_{\alpha \delta} P_{\gamma} - g_{\alpha \gamma} P_{\delta}, \\ & [L_{\alpha \beta}, L_{\gamma \delta}] = c_{\alpha \beta \gamma \delta}^{\varepsilon} L_{\varepsilon \gamma}, \\ & c_{\alpha \beta \delta \gamma}^{\varepsilon \zeta} := \delta_{(\alpha}^{\varepsilon} \delta_{\delta}^{\zeta]} \eta_{\beta \gamma)}. \end{aligned}$	$\begin{aligned} & [P_{\alpha}, P_{\beta} = 0] \quad (P_{\alpha} := \partial / \partial x^{\alpha}), \\ & [P_{\alpha}, L_{\gamma \delta}] = g_{\alpha \delta} P_{\gamma} - g_{\alpha \gamma} P_{\delta}, \\ & [L_{\alpha \beta}, L_{\gamma \delta}] = c_{\alpha \beta \gamma \delta}^{\varepsilon} L_{\varepsilon \gamma}, \\ & c_{\alpha \beta \delta \gamma}^{\varepsilon \zeta} := \delta_{(\alpha}^{\varepsilon} \delta_{\delta}^{\zeta]} \eta_{\beta \gamma)}. \end{aligned}$	$[P_{\alpha}, P_{\beta} = 0] \quad (P_{\alpha} := \partial / \partial x^{\alpha}),$ $[P_{\alpha}, L_{\gamma \delta}] = g_{\alpha \delta} P_{\gamma} - g_{\alpha \gamma} P_{\delta},$ $[L_{\alpha \beta}, L_{\gamma \delta}] = c_{\alpha \beta \gamma \delta}^{\varepsilon} L_{\varepsilon \gamma},$ $c_{\alpha \beta \delta \gamma}^{\varepsilon \zeta} := \delta_{(\alpha}^{\varepsilon} \delta_{\delta}^{\zeta]} \eta_{\beta \gamma)}.$
mielke-geometrodynamics_196 EQ0597 (9.1.15)		■ 0.323 ● W +2	$\left[\mathrm{T}_{\alpha \beta} \right]_{\gamma}^{\delta} = \left[\begin{array}{cccc} f & -h & \cdot & \cdot \\ \cdot & \cdot & -k & \cdot \\ \cdot & \cdot & \cdot & -k \\ \cdot & \cdot & \cdot & -g \\ \cdot & \cdot & \cdot & -g \\ 0 & 1 & 2 & 3 \end{array} \right] \begin{array}{l} 01 \\ 02 \\ 03 \\ 23 \\ 31 \\ 12 \end{array}$	$\left[\mathrm{T}_{\alpha \beta} \right]_{\gamma}^{\delta} = \left[\begin{array}{cccc} f & -h & \cdot & \cdot \\ \cdot & \cdot & -k & \cdot \\ \cdot & \cdot & \cdot & -k \\ \cdot & \cdot & \cdot & -g \\ \cdot & \cdot & \cdot & -g \\ 0 & 1 & 2 & 3 \end{array} \right] \begin{array}{l} 01 \\ 02 \\ 03 \\ 23 \\ 31 \\ 12 \end{array}$	$[T_{\alpha \beta}{}^{\gamma}] = \begin{bmatrix} f & -h & \cdot & \cdot \\ \cdot & \cdot & -k & \cdot \\ \cdot & \cdot & \cdot & -k \\ \cdot & \cdot & \cdot & -g \\ \cdot & \cdot & \cdot & -g \\ 0 & 1 & 2 & 3 \end{bmatrix} \begin{array}{l} 01 \\ 02 \\ 03 \\ 23 \\ 31 \\ 12 \end{array}$
mielke-geometrodynamics_255 EQ0803 (11.4.15)		■ 0.324 ● W +0	$\psi = \frac{4}{\ell} \left(\frac{2\pi m}{3} \right)^{1/2} e^{-\mu/2 - \nu/4 - i\omega m t} \left[\begin{array}{cc} iH(\rho) & x_{\kappa}^m \\ F(\rho) & \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } x_{\kappa}^m \end{array} \right].$	$\psi = \frac{4}{\ell} \left(\frac{2\pi m}{3} \right)^{1/2} e^{-\mu/2 - \nu/4 - i\omega m t} \left[\begin{array}{cc} iH(\rho) & x_{\kappa}^m \\ F(\rho) & \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } x_{\kappa}^m \end{array} \right].$	$\psi = \frac{4}{\ell} \left(\frac{2\pi m}{3} \right)^{1/2} e^{-\mu/2 - \nu/4 - i\omega m t} \left[\begin{array}{cc} iH(\rho) & x_{\kappa}^m \\ F(\rho) & \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } x_{\kappa}^m \end{array} \right].$
mielke-geometrodynamics_365 EQ1082 (A.7)		■ 0.326 ● W +1	$\begin{aligned} & \gamma^0 := \left(\begin{array}{cc} \sigma^0 & 0 \\ 0 & \sigma^0 \end{array} \right), \quad \gamma^k := \left(\begin{array}{cc} 0 & \sigma^k \\ -\sigma^k & 0 \end{array} \right); \\ & \gamma^5 := i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \left(\begin{array}{cc} 0 & \sigma^0 \\ \sigma^0 & 0 \end{array} \right). \end{aligned}$	$\gamma^0 := \left(\begin{array}{cc} \sigma^0 & 0 \\ 0 & \sigma^0 \end{array} \right), \quad \gamma^k := \left(\begin{array}{cc} 0 & \sigma^k \\ -\sigma^k & 0 \end{array} \right);$ $\gamma^5 := i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \left(\begin{array}{cc} 0 & \sigma^0 \\ \sigma^0 & 0 \end{array} \right).$	$\gamma^0 := \left(\begin{array}{cc} \sigma^0 & 0 \\ 0 & \sigma^0 \end{array} \right), \quad \gamma^k := \left(\begin{array}{cc} 0 & \sigma^k \\ -\sigma^k & 0 \end{array} \right);$ $\gamma^5 := i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \left(\begin{array}{cc} 0 & \sigma^0 \\ \sigma^0 & 0 \end{array} \right).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_00978 (14.2.19)	316	■0.331 ●W -1	\begin{aligned} \pi\,\hat L_{ }&= -\frac{\hat\ell^2}{\ell^{*2}}\operatorname{Tr}(\Omega'\wedge{}^*\Omega'\wedge\underbrace{\vartheta\wedge\dots\wedge\vartheta}_{K\text{ terms}})\\ &= -\frac{1}{\alpha_g}\operatorname{Tr}(\Omega'\wedge{}^*\Omega')\sqrt{\det\overset{\circ}{g}} = L'_{\mathrm YM}d_\mu(G)\end{aligned}	$\pi\hat L_{ }=-\frac{\hat{\ell}^2}{\ell^{*2}}\mathrm{Tr}(\Omega'\wedge{}^*\Omega'\wedge\underbrace{\vartheta\wedge\cdots\wedge\vartheta}_{{\rm K~terms}})$ $=-\frac{1}{\alpha_g}\mathrm{Tr}(\Omega'\wedge{}^*\Omega')\sqrt{\det\overset{\circ}{g}}=L'_{\mathrm YM}d_\mu(G)$	$\pi\hat L_{ }= -\frac{\hat{\ell}^2}{\ell^{*2}}\mathrm{Tr}(\mathcal{O}'\wedge{}^*\mathcal{O}'\wedge\underbrace{\vartheta\wedge\cdots\wedge\vartheta}_{\mathrm{K~terms}})$ $= -\frac{1}{\alpha_g}\mathrm{Tr}(\mathcal{O}'\wedge{}^*\mathcal{O}')\sqrt{\det\overset{\circ}{g}}=L'_{\mathrm YM}d_\mu(G)$
mielke-geometrodynamics_00579 (8.6.17)	186	■0.351 ●K +0	\left.\left.\left.\left.2\right.\left(e^\beta\right)\right.\right.\right.DH_{\alpha\beta}-\frac{1}{4}\vartheta_\alpha\wedge e^\gamma\Big]e^\delta\lrcorner DH_{\gamma\delta}\Big)-E_\alpha=\Sigma_\alpha-D\mu_\alpha,	$2D\left(e^\beta\rfloor DH_{\alpha\beta}-\frac{1}{4}\vartheta_\alpha\wedge e^\gamma\rfloor e^\delta\lrcorner DH_{\gamma\delta}\right)-E_\alpha=\Sigma_\alpha-D\mu_\alpha,$	$2D\left(e^\beta\rfloor DH_{\alpha\beta}-\frac{1}{4}\vartheta_\alpha\wedge e^\gamma\rfloor e^\delta\rfloor DH_{\gamma\delta}\right)-E_\alpha=\Sigma_\alpha-D\mu_\alpha,$
mielke-geometrodynamics_00956 (14.1.38)	309	■0.353 ●W -1	\begin{aligned} d\,\delta&\stackrel{!}{=}(-1)^{-z}*{\Sigma_{\mathrm YM}}(\Omega)\wedge D{\Sigma_{\mathrm YM}}(\Omega)/(*{\Sigma_{\mathrm YM}}\wedge\Sigma_{\mathrm YM})\\ &\stackrel{!}{=}(-1)^{-z}*(\vartheta\wedge{}^*\Omega^g)\wedge D(\vartheta\wedge{}^*\Omega^g)/*(\vartheta\wedge{}^*\Omega^g\wedge{}^*(\vartheta\wedge{}^*\Omega^g)).\end{aligned}	$d\delta=(-1)^{-z}*{\Sigma_{\mathrm YM}}(\Omega)\wedge D{\Sigma_{\mathrm YM}}(\Omega)/(*{\Sigma_{\mathrm YM}}\wedge\Sigma_{\mathrm YM})$ $=(-1)^{-z}*(\vartheta\wedge{}^*\Omega^g)\wedge D(\vartheta\wedge{}^*\Omega^g)/*(\vartheta\wedge{}^*\Omega^g\wedge{}^*(\vartheta\wedge{}^*\Omega^g))\,.$	$d\delta=(-1)^{-z}*{\Sigma_{\mathrm YM}}(\Omega)\wedge D{\Sigma_{\mathrm YM}}(\Omega)/(*{\Sigma_{\mathrm YM}}\wedge\Sigma_{\mathrm YM})$ $=(-1)^{-z}*(\vartheta\wedge{}^*\Omega^g)\wedge D(\vartheta\wedge{}^*\Omega^g)/*(\vartheta\wedge{}^*\Omega^g\wedge{}^*(\vartheta\wedge{}^*\Omega^g))\,.$
mielke-geometrodynamics_00555 (8.4.9)	180	■0.371 ●N +0	\left.\left.E_\alpha\right.=\frac{1}{2}\operatorname{Tr}\left[\left(\stackrel{\pm}{\Omega}\wedge e_\alpha\right)^*\stackrel{\pm}{\Omega}-e_\alpha\right]^\pm\Omega^g\wedge{}^*\stackrel{\pm}{\Omega}\right)=0,	$E_\alpha=\frac{1}{2}\mathrm{Tr}\left(\overset{\pm}{\Omega}\wedge e_\alpha\right]^*\overset{\pm}{\Omega}-e_\alpha\Big]^\pm\Omega^g\wedge{}^*\overset{\pm}{\Omega}\Big)=0,$	$E_\alpha=\frac{1}{2}\mathrm{Tr}\left(\overset{\pm}{\Omega}^g\wedge e_\alpha\rfloor{}^*\overset{\pm}{\Omega}^g-e_\alpha\rfloor\overset{\pm}{\Omega}^g\wedge{}^*\overset{\pm}{\Omega}^g\right)=0,$
mielke-geometrodynamics_00562 (8.5.5)	182	■0.377 ●C +9	\begin{aligned} \widetilde{L}_{\mathrm SKY}^{(*)}&\stackrel{!}{=}\frac{1}{2}\operatorname{Tr}\left[\left(\otimes^g-i\frac{\theta_{\mathrm T}^*}{4\ell^2}\sigma\right)\wedge{}^*\left(\overset{t}{\Omega}-i\frac{\theta_{\mathrm T}^*}{4\ell^2}\sigma\right)\right]\\ &=-\frac{\theta_{\mathrm T}^*}{2\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\theta_{\mathrm T}^*\Lambda_{\mathrm eff}\,\eta-\frac{1}{2}R^{\alpha\beta}\wedge{}^*R_{\alpha\beta}\mp\frac{(-1)^{\mathrm sig}}{2}R^{\alpha\beta}\wedge R_{\alpha\beta}^{(*)}\end{aligned}	$\tilde{L}_{\mathrm SKY}^{(*)}=\frac{1}{2}\mathrm{Tr}\left[\left(\otimes^g-i\frac{\theta_{\mathrm T}^*}{4\ell^2}\sigma\right)\wedge{}^*\left(\overset{t}{\Omega}-i\frac{\theta_{\mathrm T}^*}{4\ell^2}\sigma\right)\right]$ $=-\frac{\theta_{\mathrm T}^*}{2\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\theta_{\mathrm T}^*\Lambda_{\mathrm eff}\,\eta-\frac{1}{2}R^{\alpha\beta}\wedge{}^*R_{\alpha\beta}\mp\frac{(-1)^{\mathrm sig}}{2}R^{\alpha\beta}\wedge R_{\alpha\beta}^{(*)}$	$\tilde{L}_{\mathrm SKY}^{(*)}=\frac{1}{2}\mathrm{Tr}\left[\left(\overset{\pm}{\Omega}^g-i\frac{\theta_{\mathrm T}^*}{4\ell^2}\sigma\right)\wedge{}^*\left(\overset{\pm}{\Omega}^g-i\frac{\theta_{\mathrm T}^*}{4\ell^2}\sigma\right)\right]\tag{8.5}$ $=-\frac{\theta_{\mathrm T}^*}{2\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\theta_{\mathrm T}^*\Lambda_{\mathrm eff}\,\eta-\frac{1}{2}R^{\alpha\beta}\wedge{}^*R_{\alpha\beta}\mp\frac{(-1)^{\mathrm sig}}{2}R^{\alpha\beta}\wedge R_{\alpha\beta}^{(*)}\,.$
Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

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mielke-geometrodynamics_132 EQ0393 (6.4.8)		■ 0.382 ● W +1	$\begin{aligned} n^{\alpha} \Sigma_{\alpha} &= n \downarrow L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \downarrow \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ &\cong n \downarrow L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}]. \end{aligned}$	$n^{\alpha} \Sigma_{\alpha} = n \downarrow L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \downarrow \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}]$ $\cong n \downarrow L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$	$n^{\alpha} \Sigma_{\alpha} = n \downarrow L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \downarrow \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}]$ $\cong n \downarrow L - (\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$
mielke-geometrodynamics_094 EQ0261 (4.3.26)		■ 0.394 ● N +0	$K_{\{. \ \alpha \beta\}^c} = -\frac{1}{2} \left(T_{\alpha \beta}^c - T_{\alpha, \beta}^{\cdot c} + T_{\beta, \alpha}^{\cdot c} \right) = -K_{\cdot \beta \alpha}^c.$	$K_{\cdot \alpha \beta}^c = -\frac{1}{2} \left(T_{\cdot \alpha \beta}^c - T_{\alpha, \beta}^{\cdot c} + T_{\beta, \alpha}^{\cdot c} \right) = -K_{\cdot \beta \alpha}^c.$	$K_{\cdot \alpha \beta}^c = -\frac{1}{2} \left(T_{\cdot \alpha \beta}^c - T_{\alpha, \beta}^{\cdot c} + T_{\beta, \alpha}^{\cdot c} \right) = -K_{\cdot \beta \alpha}^c.$
mielke-geometrodynamics_128 EQ0377 (6.2.28)		■ 0.396 ● W +0	$\stackrel{\scriptscriptstyle{(\pm)}}{L}_{\parallel} = -\frac{1}{2} \left[\vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{D}^{\scriptscriptstyle{(\pm)}}_{\alpha} + d \left(\vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_{\alpha} \right) \right] = -\frac{1}{2} \left(\stackrel{\scriptscriptstyle{(\pm)}}{D} \vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_{\alpha}, \right.$	$\stackrel{\scriptscriptstyle{(\pm)}}{L}_{\parallel} = -\frac{1}{2} \left[\vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{D}^{\scriptscriptstyle{(\pm)}}_{\alpha} + d \left(\vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_{\alpha} \right) \right] = -\frac{1}{2} \left(\stackrel{\scriptscriptstyle{(\pm)}}{D} \vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_{\alpha}, \right.$	$\stackrel{\scriptscriptstyle{(\pm)}}{L}_{\parallel} = -\frac{1}{2} \left[\vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{D}^{\scriptscriptstyle{(\pm)}} \Pi_{\alpha} + d(\vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_{\alpha}) \right] = -\frac{1}{2} \stackrel{\scriptscriptstyle{(\pm)}}{D} \vartheta^{\alpha} \wedge \stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_{\alpha},$
mielke-geometrodynamics_094 EQ0262 (4.3.27)		■ 0.399 ● C +3	$\begin{aligned} \Omega^g &= \Omega^{\mathfrak{I}} - dK + K \wedge \omega^{\mathfrak{I}} + \omega^{\mathfrak{I}} \wedge K - K \wedge K \\ &= \Omega^{\mathfrak{I}} - D^{\mathfrak{I}} K - K \wedge K = \Omega^{\mathfrak{I}} - DK + K \wedge K \end{aligned}$	$\Omega^g = \Omega^{\mathfrak{I}} - dK + K \wedge \omega^{\mathfrak{I}} + \omega^{\mathfrak{I}} \wedge K - K \wedge K$ $= \Omega^{\mathfrak{I}} - D^{\mathfrak{I}} K - K \wedge K = \Omega^{\mathfrak{I}} - DK + K \wedge K$	$\Omega^g = \Omega^{\mathfrak{I}} - dK + K \wedge \omega^{\mathfrak{I}} + \omega^{\mathfrak{I}} \wedge K - K \wedge K$ $= \Omega^{\mathfrak{I}} - D^{\mathfrak{I}} K - K \wedge K = \Omega^{\mathfrak{I}} - DK + K \wedge K$
mielke-geometrodynamics_184 EQ0570 (8.6.8)		■ 0.403 ● K +0	$\left. E_{\alpha} := \partial L / \partial \vartheta^{\alpha} = e_{\alpha} \downarrow L + (e_{\alpha} \downarrow T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \downarrow R^{\beta \gamma}) \wedge H_{\beta \gamma} \right.$	$E_{\alpha} := \partial L / \partial \vartheta^{\alpha} = e_{\alpha} \downarrow L + (e_{\alpha} \downarrow T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \downarrow R^{\beta \gamma}) \wedge H_{\beta \gamma}$	$E_{\alpha} := \partial L / \partial \vartheta^{\alpha} = e_{\alpha} \downarrow L + (e_{\alpha} \downarrow T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \downarrow R^{\beta \gamma}) \wedge H_{\beta \gamma}$
mielke-geometrodynamics_103 EQ0306 (4.5.16)		■ 0.406 ● N +0	$\left. \widetilde{\varepsilon}_{\mathrm{MA}} := \varepsilon_{\mathrm{RC}} + \xi \lrcorner L \cong - \left(\mathcal{L}_{\xi} \Gamma_{\beta}^{\gamma} \right) \wedge H^{\beta}{}_{\gamma} + dH_{\infty} \right.$	$\widetilde{\varepsilon}_{\mathrm{MA}} := \varepsilon_{\mathrm{RC}} + \xi \lrcorner L \cong - \left(\mathcal{L}_{\xi} \Gamma_{\beta}^{\gamma} \right) \wedge H^{\beta}{}_{\gamma} + dH_{\infty}$	$\widetilde{\varepsilon}_{\mathrm{MA}} := \varepsilon_{\mathrm{RC}} + \xi \downarrow L \cong - \left(\mathcal{L}_{\xi} \Gamma_{\beta}^{\gamma} \right) \wedge H^{\beta}{}_{\gamma} + dH_{\infty}$
mielke-geometrodynamics_114 EQ0336 (5.3.13)		■ 0.407 ● N -1	$R^{\alpha \beta} = R^{\{j \alpha \beta\} - D^{\mathfrak{I}} K^{\alpha \beta} - K^{\alpha}{}_{\mu} \wedge K^{\mu \beta}$	$R^{\alpha \beta} = R^{\{j \alpha \beta\} - D^{\mathfrak{I}} K^{\alpha \beta} - K^{\alpha}{}_{\mu} \wedge K^{\mu \beta}$	$R^{\alpha \beta} = R^{\mathfrak{I} \alpha \beta} - D^{\mathfrak{I}} K^{\alpha \beta} - K^{\alpha}{}_{\mu} \wedge K^{\mu \beta}$
mielke-geometrodynamics_375 EQ1136 (C.4)		■ 0.409 ● N +0	$c_{\{i \ j\}^k} = -c_{\{j \ i\}^k}$	$c_{ij}^k = -c_{ji}^k$	$c_{ij}{}^k = -c_{ji}{}^k$
mielke-geometrodynamics_279 EQ0861 (12.4.8)		■ 0.420 ● N -1	$\begin{aligned} dC_{\mathrm{RR}} &= -\mathrm{Tr}(\Omega \wedge \Omega) \\ &= \frac{1}{2} R_{\alpha \beta}^{\mathfrak{I}} \wedge R^{\{\beta \alpha \beta\}} \\ &+ \frac{1}{12} d \left[{}^* \mathcal{A} \wedge R^{\mathfrak{I}} - \frac{1}{3} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{9} {}^* \mathcal{A} \wedge {}^* (\mathcal{A} \wedge {}^* \mathcal{A}) \right]. \end{aligned}$	$dC_{\mathrm{RR}} = -\mathrm{Tr}(\Omega \wedge \Omega)$ $= \frac{1}{2} R_{\alpha \beta}^{\mathfrak{I}} \wedge R^{\{\beta \alpha \beta\}}$ $+ \frac{1}{12} d \left[{}^* \mathcal{A} \wedge R^{\mathfrak{I}} - \frac{1}{3} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{9} {}^* \mathcal{A} \wedge {}^* (\mathcal{A} \wedge {}^* \mathcal{A}) \right].$	$dC_{\mathrm{RR}} = -\mathrm{Tr}(\Omega \wedge \Omega)$ $= \frac{1}{2} R_{\alpha \beta}^{\mathfrak{I}} \wedge R^{\mathfrak{I} \alpha \beta}$ $+ \frac{1}{12} d \left[{}^* \mathcal{A} \wedge R^{\mathfrak{I}} - \frac{1}{3} \mathcal{A} \wedge d\mathcal{A} + \frac{1}{9} {}^* \mathcal{A} \wedge {}^* (\mathcal{A} \wedge {}^* \mathcal{A}) \right].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_151 EQ0451 (7.2.4)		■ 0.424 ● N +0	$\stackrel{+}{\Omega} := \frac{1}{2} \left(\Omega^g \pm {}^* \Omega^{g(*)} \right),$	$\stackrel{+g}{\Omega} := \frac{1}{2} \left(\Omega^g \pm {}^* \Omega^{g(*)} \right),$	$\stackrel{\pm}{\Omega}^g := \frac{1}{2} \left(\Omega^g \pm {}^* \Omega^{g(*)} \right),$
mielke-geometrodynamics_127 EQ0372 (6.2.23)		■ 0.437 ● C -3	$\left. \left(\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha \mp \frac{i}{4} \ell^2 e_\alpha \right) \right] \left(\stackrel{(+)}{\Pi}^\beta \wedge \stackrel{(-)}{\Pi}_\beta \right) = \Sigma_\alpha,$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha \mp \frac{i}{4} \ell^2 e_\alpha \Big] \left(\stackrel{(+)}{\Pi}^\beta \wedge \stackrel{(-)}{\Pi}_\beta \right) = \Sigma_\alpha,$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha \mp \frac{i}{4} \ell^2 e_\alpha \Big] \left(\stackrel{(+)}{\Pi}^\beta \wedge \stackrel{(-)}{\Pi}_\beta \right) = \Sigma_\alpha,$
mielke-geometrodynamics_322 EQ997 (14.3.10)		■ 0.442 ● N -1	$D^{\text{int}} = \stackrel{\circ}{\gamma} \wedge {}^* \stackrel{\circ}{D} = \stackrel{\circ}{\gamma} \wedge {}^* (\stackrel{\circ}{d} + i \stackrel{\circ}{\tilde{\omega}})$	$D^{(\text{int})} = \stackrel{\circ}{\gamma} \wedge {}^* \stackrel{\circ}{D} = \stackrel{\circ}{\gamma} \wedge {}^* (\stackrel{\circ}{d} + i \stackrel{\circ}{\tilde{\omega}})$	$\not{D}^{(int)} = \stackrel{\circ}{\gamma} \wedge {}^* \stackrel{\circ}{D} = \stackrel{\circ}{\gamma} \wedge {}^* (\stackrel{\circ}{d} + i \stackrel{\circ}{\tilde{\omega}})$
mielke-geometrodynamics_156 EQ0474 (7.4.2)		■ 0.444 ● N -1	$R_{\alpha\beta} \equiv C_{\alpha\beta} + \frac{1}{2} Ri^\tau_{\alpha\beta} - \frac{R}{12} \vartheta_\alpha \wedge \vartheta_\beta,$	$R_{\alpha\beta} \equiv C_{\alpha\beta} + \frac{1}{2} Ri^\tau_{\alpha\beta} - \frac{R}{12} \vartheta_\alpha \wedge \vartheta_\beta,$	$R_{\alpha\beta} \equiv C_{\alpha\beta} + \frac{1}{2} Rj\check{c}_{\alpha\beta} - \frac{R}{12} \vartheta_\alpha \wedge \vartheta_\beta,$
mielke-geometrodynamics_221 EQ0681 (10.3.38)		■ 0.452 ● C +4	$\begin{gathered} \chi = \theta_{\mathrm{L}} \varepsilon - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}, \quad \Lambda = \frac{(-1)^s}{4\ell^2} (\theta_{\mathrm{L}} \varepsilon^3 - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}^3), \quad \theta_{\mathrm{T}} = \frac{(-1)^s}{2} (\theta_{\mathrm{L}} \varepsilon^2 - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}^2), \\ \Delta\theta_{\mathrm{L}} = \theta_{\mathrm{L}} - \tilde{\theta}_{\mathrm{L}}, \quad \Delta\theta_{\mathrm{TL}} = \frac{1}{2} (\theta_{\mathrm{L}} \varepsilon - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}). \end{gathered}$	$\chi = \theta_{\mathrm{L}} \varepsilon - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}, \quad \Lambda = \frac{(-1)^s}{4\ell^2} (\theta_{\mathrm{L}} \varepsilon^3 - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}^3), \quad \theta_{\mathrm{T}} = \frac{(-1)^s}{2} (\theta_{\mathrm{L}} \varepsilon^2 - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}^2), \\ \Delta\theta_{\mathrm{L}} = \theta_{\mathrm{L}} - \tilde{\theta}_{\mathrm{L}}, \quad \Delta\theta_{\mathrm{TL}} = \frac{1}{2} (\theta_{\mathrm{L}} \varepsilon - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}).$	$\chi = \theta_{\mathrm{L}} \varepsilon - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}, \quad \Lambda = \frac{(-1)^s}{4\ell^2} (\theta_{\mathrm{L}} \varepsilon^3 - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}^3), \quad \theta_{\mathrm{T}} = \frac{(-1)^s}{2} (\theta_{\mathrm{L}} \varepsilon^2 - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}^2), \\ \Delta\theta_{\mathrm{L}} = \theta_{\mathrm{L}} - \tilde{\theta}_{\mathrm{L}}, \quad \Delta\theta_{\mathrm{TL}} = \frac{1}{2} (\theta_{\mathrm{L}} \varepsilon - \tilde{\theta}_{\mathrm{L}} \tilde{\varepsilon}).$
mielke-geometrodynamics_273 EQ0845 (12.2.13)		■ 0.457 ● N +0	$C_{\mathrm{TT}} \cong \frac{1}{4} j_5, \quad \mathcal{A} = 2\ell^2 {}^* C_{\mathrm{TT}} = (\ell^2/2) \bar{\psi} \gamma_5 \gamma \psi.$	$C_{\mathrm{TT}} \cong \frac{1}{4} j_5, \quad \mathcal{A} = 2\ell^2 {}^* C_{\mathrm{TT}} = (\ell^2/2) \bar{\psi} \gamma_5 \gamma \psi.$	$C_{\mathrm{TT}} \cong \frac{1}{4} j_5, \quad \mathcal{A} = 2\ell^2 {}^* C_{\mathrm{TT}} = (\ell^2/2) \bar{\psi} \gamma_5 \gamma \psi.$
mielke-geometrodynamics_128 EQ0375 (6.2.26)		■ 0.466 ● S +2	$\stackrel{(\pm)}{L}_\parallel = \mp \frac{i}{4} \ell^2 \stackrel{(\pm)}{\Pi}^\alpha \wedge \stackrel{(\pm)}{\Pi}_\alpha, \quad \overline{\stackrel{(+)}{L}_\parallel} = \stackrel{(-)}{L}_\parallel,$	$\stackrel{(\pm)}{L}_\parallel = \mp \frac{i}{4} \ell^2 \stackrel{(\pm)}{\Pi}^\alpha \wedge \stackrel{(\pm)}{\Pi}_\alpha, \quad \overline{\stackrel{(+)}{L}_\parallel} = \stackrel{(-)}{L}_\parallel,$	$\stackrel{(\pm)}{L}_\parallel = \mp \frac{i}{4} \ell^2 \stackrel{(\pm)}{\Pi}^\alpha \wedge \stackrel{(\pm)}{\Pi}_\alpha, \quad \overline{\stackrel{(+)}{L}_\parallel} = \stackrel{(-)}{L}_\parallel,$
mielke-geometrodynamics_087 EQ0222 (4.2.15)		■ 0.466 ● N +0	$\widetilde{\varphi}: A(M) \rightarrow C^\infty(\overset{\circ}{V})$	$\widetilde{\varphi}: A(M) \rightarrow C^\infty(\overset{\circ}{V})$	$\widetilde{\varphi}: A(M) \rightarrow C^\infty(\overset{\circ}{V})$
mielke-geometrodynamics_199 EQ0604 (9.2.3)		■ 0.474 ● W +1	$\begin{aligned} L_{\text{Euler}} &= {}^* (\gamma_2^g) = \frac{1}{(4\pi)^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right) \\ &= \frac{(-1)^{s/4}}{32\pi^2} R^{\alpha\beta\gamma\delta} {}^* R_{\alpha\beta\gamma\delta} \sqrt{ g } d^4 x. \end{aligned}$	$\begin{aligned} L_{\text{Euler}} &= {}^* (\gamma_2^g) = \frac{1}{(4\pi)^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right) \\ &= \frac{(-1)^{s/4}}{32\pi^2} R^{\alpha\beta\gamma\delta} {}^* R_{\alpha\beta\gamma\delta} \sqrt{ g } d^4 x. \end{aligned}$	$\begin{aligned} L_{\text{Euler}} &= \pi^*(\gamma_2^g) = \frac{1}{(4\pi)^2} \text{Tr}(\Omega^g \wedge \Omega^{g(*)}) \\ &= \frac{(-1)^{s/4}}{32\pi^2} R^{\alpha\beta\gamma\delta} {}^* R_{\alpha\beta\gamma\delta} \sqrt{ g } d^4 x. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00401 (6.4.20)	133	■ 0.482 ● W +1	\mathscr{H}_{\{\}}=\stackrel{(±)}{\underbrace{\hspace{-1pt}\! \! (\pm)}}{\rm A}_{{\hat{0}}}^{\hspace{-1pt}\! \! (\pm)}=0,	$\mathcal{H}_{ } = \underline{D} - \overset{(\pm)}{\underset{\circ}{\mathcal{A}}_0} = 0,$	$\mathcal{H}_{ } = \overset{(\pm)}{\underline{D}} - \overset{(\pm)}{\underset{\circ}{\mathcal{A}}_0} = 0,$
mielke-geometrodynamics_E00750 (11.1.17)	241	■ 0.484 ● U +23	D_{\left.\right \overparen{S_0(1,3)}}^{\mathrm{F}}\wedge\mathrm{F}=\underbrace{D^{\frac{1}{2}}\times\cdots\times D^{\frac{1}{2}}}_{(n-4)/4}\times\mathrm{d}x^1\wedge\cdots\wedge\mathrm{d}x^{n-4}	(not rendered)	$D_{ \overline{S_0(1,3)}}^{\mathrm{F}} = \underbrace{D^{\frac{1}{2}} \times \cdots \times D^{\frac{1}{2}}}_{(n-4)/4}.$
mielke-geometrodynamics_E00686 (10.3.44)	222	■ 0.497 ● N +0	\left.\left.\left.C_l=E_l^\alpha*\left[D^{\{\}}\left[2\left(e^\beta\lrcorner D^{\{\}}H_{[\alpha\beta]}\right)+\frac{1}{2}\vartheta_\alpha\wedge\left(e^\beta\lrcorner e^\gamma\lrcorner D^{\{\}}H_{[\beta\gamma]}\right)\right]\right]\right.\right.\right.	$C_l = E_l^\alpha * \left\{ D^{\{\}} \left[2 \left(e^\beta \lrcorner D^{\{\}} H_{[\alpha\beta]} \right) + \frac{1}{2} \vartheta_\alpha \wedge \left(e^\beta \lrcorner e^\gamma \lrcorner D^{\{\}} H_{[\beta\gamma]} \right) \right] \right\}$	$C_l = E_l^\alpha * \left\{ D^{\{\}} \left[2 \left(e^\beta \lrcorner D^{\{\}} H_{[\alpha\beta]} \right) + \frac{1}{2} \vartheta_\alpha \wedge \left(e^\beta \lrcorner e^\gamma \lrcorner D^{\{\}} H_{[\beta\gamma]} \right) \right] \right\}$
mielke-geometrodynamics_E00515 (7.7.33)	166	■ 0.497 ● N +0	\left.\left.T:=\frac{1}{4}\operatorname{Tr}(\check{\gamma}\rfloor\Theta)=e_\alpha\right\rfloor T^\alpha,\quad\mathscr{A}:=\frac{1}{4}*{\operatorname{Tr}(\gamma\wedge\Theta)}=*({\vartheta}_\alpha\wedge T^\alpha).	$T := \frac{1}{4} \operatorname{Tr}(\tilde{\gamma} \rfloor \Theta) = e_\alpha \rfloor T^\alpha, \quad \mathscr{A} := \frac{1}{4} * \operatorname{Tr}(\gamma \wedge \Theta) = * (\vartheta_\alpha \wedge T^\alpha).$	$T := \frac{1}{4} \operatorname{Tr}(\check{\gamma} \rfloor \Theta) = e_\alpha \rfloor T^\alpha, \quad \mathscr{A} := \frac{1}{4} * \operatorname{Tr}(\gamma \wedge \Theta) = * (\vartheta_\alpha \wedge T^\alpha).$
mielke-geometrodynamics_E00799 (11.4.11)	254	■ 0.505 ● K +0	d s^2=\frac{1}{\lambda^2}(R_\circ^2 d\lambda^2-\text{o}_{ij}dy^i\otimes_s dy^j)	$ds^2 = \frac{1}{\lambda^2} \left(R_\circ^2 d\lambda^2 - \text{o}_{ij} dy^i \otimes_s dy^j \right).$	$ds^2 = \frac{1}{\lambda^2} \left(R_\circ^2 d\lambda^2 - \text{o}_{ij} dy^i \otimes_s dy^j \right).$
mielke-geometrodynamics_E00960 (14.1.42)	309	■ 0.508 ● W -1	\xi_{\mu\nu}\xi_{\rho\sigma}=-\frac{1}{2}\bar{R}_{\mu\nu\rho\sigma}-\frac{1}{2}\frac{\bar{R}_{\mu\nu\gamma\delta}\bar{R}_{\rho\sigma}{}^{\gamma\delta}}{R_{\alpha\beta}{}^{\{\}}R^{\{\}}{}^{\alpha\beta}}	$\xi_{\mu\nu}\xi_{\rho\sigma} = -\frac{1}{2}\bar{R}_{\mu\nu\rho\sigma} - \frac{1}{2}\frac{\bar{R}_{\mu\nu\gamma\delta}\bar{R}_{\rho\sigma}{}^{\gamma\delta}}{R_{\alpha\beta}{}^{\{\}}R^{\{\}}{}^{\alpha\beta}}.$	$\xi_{\mu\nu}\xi_{\rho\sigma} = -\frac{1}{2}\bar{R}_{\mu\nu\rho\sigma} - \frac{1}{2}\frac{\bar{R}_{\mu\nu\gamma\delta}\bar{R}_{\rho\sigma}{}^{\gamma\delta}}{R_{\alpha\beta}{}^{\{\}}R^{\{\}}{}^{\alpha\beta}}.$
mielke-geometrodynamics_E01131 (B.43)	373	■ 0.509 ● W +2	\langle , \rangle_M : \begin{cases} \bigwedge^p T^*(M) \times C_p(M, \mathbb{R}) \rightarrow \mathbb{R} \\ \langle \alpha^{(p)}, c_p \rangle_M \rightarrow \int_{c_p} \alpha^{(p)} \end{cases}	$\langle , \rangle_M : \begin{cases} \bigwedge^p T^*(M) \times C_p(M, \mathbb{R}) \rightarrow \mathbb{R} \\ \langle \alpha^{(p)}, c_p \rangle_M \rightarrow \int_{c_p} \alpha^{(p)} \end{cases}$	$\langle , \rangle_M : \begin{cases} \bigwedge^p T^*(M) \times C_p(M, \mathbb{R}) \rightarrow \mathbb{R} \\ \langle \alpha^{(p)}, c_p \rangle_M \rightarrow \int_{c_p} \alpha^{(p)} \end{cases}$
mielke-geometrodynamics_E00294 (4.5.5)	101	■ 0.510 ● N +0	\Sigma_\alpha=e_\alpha\lrcorner L-(e_\alpha\lrcorner D\Psi)\wedge\frac{\partial L}{\partial D\Psi}-(e_\alpha\lrcorner\Psi)\wedge\frac{\partial L}{\partial\Psi}.	$\Sigma_\alpha = e_\alpha \lrcorner L - (e_\alpha \lrcorner D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_\alpha \lrcorner \Psi) \wedge \frac{\partial L}{\partial \Psi}.$	$\Sigma_\alpha = e_\alpha \lrcorner L - (e_\alpha \lrcorner D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_\alpha \lrcorner \Psi) \wedge \frac{\partial L}{\partial \Psi}.$
mielke-geometrodynamics_E00118 (3.3.19)	059	■ 0.513 ● N -1	L_{\mathrm{Pontr}}:=-\frac{1}{2} F \wedge F = \mathbf{E} \cdot \mathbf{B} \eta.	$L_{\mathrm{Pontr}} := -\frac{1}{2} F \wedge F = \mathbf{E} \cdot \mathbf{B} \eta.$	$L_{\mathrm{Pontr}} := -\frac{1}{2} F \wedge F = \mathbf{E} \cdot \mathbf{B} \eta.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_169 EQ0533 (7.7.53)	169	■ 0.513 ● C +5	$\begin{aligned} G \ \& \ :=\frac{1}{2} \ R^{\beta\gamma} \ \wedge \ \wedge \eta_{\alpha\beta\gamma} \gamma^{\alpha} \\\wedge \eta_{\alpha\beta\gamma} \gamma^{\alpha} \ \wedge \ \wedge \\\left.\left$		

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_082 EQ0198 (4.1.11)		■ 0.533 ● W +0	$\begin{aligned} e_{\alpha}(m) &\rightarrow e_{\alpha}'(m) = g_{\alpha}^{\beta} e_{\beta}(m), \\ g &= [g_{\alpha}^{\beta}] \in GL(n, \mathbb{R}). \end{aligned}$	$e_{\alpha}(m) \rightarrow e_{\alpha}'(m) = g_{\alpha}^{\beta} e_{\beta}(m),$ $g = [g_{\alpha}^{\beta}] \in GL(n, \mathbb{R}).$	$e_{\alpha}(m) \rightarrow e_{\alpha}'(m) = g_{\alpha}^{\beta} e_{\beta}(m),$ $g = [g_{\alpha}^{\beta}] \in GL(n, \mathbb{R}).$
mielke-geometrodynamics_196 EQ0599 (9.1.17)		■ 0.534 ● W +2	$\begin{aligned} & f = -\frac{1}{r} \left(1 + \frac{\kappa}{2\ell^{*2}} r^2 \right) e^{-\nu/2}, \quad g = \frac{1}{r} e^{\nu/2} \\ & h = -\frac{1}{(+)\ell^{*}} \frac{\sqrt{-\kappa}}{3}, \quad k = +\frac{1}{(-)2\ell^{*}} \sqrt{-3\kappa}. \end{aligned}$	$f = -\frac{1}{r} \left(1 + \frac{\kappa}{2\ell^{*2}} r^2 \right) e^{-\nu/2}, g = \frac{1}{r} e^{\nu/2}$ $h = -\frac{1}{(+)\ell^{*}} \frac{\sqrt{-\kappa}}{3}, \quad k = +\frac{1}{(-)2\ell^{*}} \sqrt{-3\kappa}$	$f = -\frac{1}{r} \left(1 + \frac{\kappa}{2\ell^{*2}} r^2 \right) e^{-\nu/2}, g = \frac{1}{r} e^{\nu/2}$ $h = -\frac{1}{(+)\ell^{*}} \frac{\sqrt{-\kappa}}{3}, \quad k = +\frac{1}{(-)2\ell^{*}} \sqrt{-3\kappa}.$
mielke-geometrodynamics_241 EQ0752 (11.1.19)		■ 0.535 ● K +0	$\Psi = \Psi \otimes \tilde{b} \in C^{\infty} \left(V^{\frac{1}{2}} \right).$	$\Psi = \Psi \otimes \tilde{b} \in C^{\infty} \left(V^{\frac{1}{2}} \right).$	$\Psi = \psi \otimes \tilde{b} \in C^{\infty}(V^{\frac{1}{2}}).$
mielke-geometrodynamics_344 EQ1047 (15.2.4)		■ 0.536 ● W +0	$\begin{aligned} \bar{E}_i &= \frac{1}{2} \left\{ E_{i\alpha}^{(f)j} \gamma^{\alpha} + i^{*} E_{i\alpha}^{(f)j} \gamma^{\alpha} \gamma^5 \right\} \lambda_j^{(f)} \\ &\oplus \frac{1}{2} \left\{ E_{i\alpha}^{(c)j} \gamma^{\alpha} + i^{*} E_{i\alpha}^{(c)j} \gamma^{\alpha} \gamma^5 \right\} \lambda_j^{(c)} \end{aligned}$	$\bar{E}_i = \frac{1}{2} \left\{ E_{i\alpha}^{(f)j} \gamma^{\alpha} + i^{*} E_{i\alpha}^{(f)j} \gamma^{\alpha} \gamma^5 \right\} \lambda_j^{(f)}$ $\oplus \frac{1}{2} \left\{ E_{i\alpha}^{(c)j} \gamma^{\alpha} + i^{*} E_{i\alpha}^{(c)j} \gamma^{\alpha} \gamma^5 \right\} \lambda_j^{(c)}$	$\bar{E}_i = \frac{1}{2} \left\{ E_{i\alpha}^{(f)j} \gamma^{\alpha} + i^{*} E_{i\alpha}^{(f)j} \gamma^{\alpha} \gamma^5 \right\} \lambda_j^{(f)}$ $\oplus \frac{1}{2} \left\{ E_{i\alpha}^{(c)j} \gamma^{\alpha} + i^{*} E_{i\alpha}^{(c)j} \gamma^{\alpha} \gamma^5 \right\} \lambda_j^{(c)}$
mielke-geometrodynamics_317 EQ0982 (14.2.23)		■ 0.537 ● N +0	$\int_G \pi \hat{L}_{\mathrm{E}} = \frac{1}{\ell^{*2}} L_{\mathrm{W}} + L'_{\mathrm{YM}} + \frac{1}{2\ell^{*2}} \vartheta \wedge \vartheta \wedge \vartheta \wedge \vartheta \int_G \overset{\circ}{R} d\mu(G).$	$\int_G \pi \hat{L}_{\mathrm{E}} = \frac{1}{\ell^{*2}} L_{\mathrm{W}} + L'_{\mathrm{YM}} + \frac{1}{2\ell^{*2}} \vartheta \wedge \vartheta \wedge \vartheta \wedge \vartheta \int_G \overset{\circ}{R} d\mu(G).$	$\int_G \pi \hat{L}_{\mathrm{E}} = \frac{1}{\ell^{*2}} L_{\mathrm{W}} + L'_{\mathrm{YM}} + \frac{1}{2\ell^{*2}} \vartheta \wedge \vartheta \wedge \vartheta \wedge \vartheta \int_G \overset{\circ}{R} d\mu(G).$
mielke-geometrodynamics_375 EQ1134 (C.2)		■ 0.539 ● K +0	$\exp : \mathfrak{g} \rightarrow G \ni g = \exp \left(\imath \xi^j I_j \right)$	$\exp : \mathfrak{g} \rightarrow G \ni g = \exp \left(\imath \xi^j I_j \right)$	$\exp : \mathfrak{g} \rightarrow G \ni g = \exp(\imath \xi^j I_j)$
mielke-geometrodynamics_323 EQ1002 (14.3.15)		■ 0.541 ● N +0	$(i \not{D})^2 = (-1)^{3n-1+s} \operatorname{Tr} \{ \gamma \wedge \gamma \wedge^{*} (D D) - \gamma \wedge (D \gamma) \wedge^{*} D \}$	$(i \not{D})^2 = (-1)^{3n-1+s} \operatorname{Tr} \{ \gamma \wedge \gamma \wedge^{*} (D D) - \gamma \wedge (D \gamma) \wedge^{*} D \}$	$(i \not{D})^2 = (-1)^{3n-1+s} \operatorname{Tr} \{ \gamma \wedge \gamma \wedge^{*} (D D) - \gamma \wedge (D \gamma) \wedge^{*} D \}$
mielke-geometrodynamics_365 EQ1085 (A.11)		■ 0.541 ● K +0	$\begin{aligned} \mathbb{C}(0,1) &= \mathbb{C} \quad (\text{field of complex numbers}), \\ \mathbb{C}(1,0) &= \mathbb{R} \oplus \mathbb{R}, \end{aligned}$	$\mathbb{C}(0,1) = \mathbb{C} \quad (\text{field of complex numbers}),$ $\mathbb{C}(1,0) = \mathbb{R} \oplus \mathbb{R},$	$\mathbb{C}(0,1) = \mathbb{C} \quad (\text{field of complex numbers}),$ $\mathbb{C}(1,0) = \mathbb{R} \oplus \mathbb{R},$
mielke-geometrodynamics_033 EQ0004 (2.1.3)		■ 0.543 ● N +0	$e_{\alpha}(m) = e_{,\alpha}^i(m) \partial_i.$	$e_{\alpha}(m) = e_{,\alpha}^i(m) \partial_i.$	$e_{\alpha}(m) = e_{,\alpha}^i(m) \partial_i.$
mielke-geometrodynamics_181 EQ0558 (8.5.1)		■ 0.544 ● N +0	$\Delta L_{\text{FP}} = s \operatorname{Tr} \left\{ \frac{\theta_{\text{T}}^{*i}}{4\ell^2} \bar{\chi} \wedge^{*} \sigma - \bar{c} d^{*} \gamma \right\}$	$\Delta L_{\text{FP}} = s \operatorname{Tr} \left\{ \frac{\theta_{\text{T}}^{*i}}{4\ell^2} \bar{\chi} \wedge^{*} \sigma - \bar{c} d^{*} \gamma \right\}$	$\Delta L_{\text{FP}} = s \operatorname{Tr} \left\{ \frac{\theta_{\text{T}}^{*i}}{4\ell^2} \bar{\chi} \wedge^{*} \sigma - \bar{c} d^{*} \gamma \right\}$
mielke-geometrodynamics_102 EQ0304 (4.5.14)		■ 0.545 ● N +1	$\begin{gathered} D \Sigma_{\alpha} \simeq 0 \quad \forall D \left(\tau_{\alpha \beta} + x_{[\alpha} \wedge \Sigma_{\beta]} \right) \simeq 0. \end{gathered}$	$D \Sigma_{\alpha} \simeq 0$ $D \left(\tau_{\alpha \beta} + x_{[\alpha} \wedge \Sigma_{\beta]} \right) \simeq 0.$	$D \Sigma_{\alpha} \simeq 0,$ $D(\tau_{\alpha \beta} + x_{[\alpha} \wedge \Sigma_{\beta]}) \simeq 0.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_00889 (13.4.8)	286	0.576 ● N +0	$\begin{aligned} & \Gamma^{\{T\}} \rightarrow \Gamma^{\{T\}'} = \Lambda^{-1}(x) \left[\Gamma^{\{T\}} + D\tau(x) \right], \\ & \Gamma_*^{\{T\}} \xrightarrow{A^{-1}(x)} \Gamma_*^{\{T\}'} = \left[\Gamma_*^{\{T\}} + D\tau_*(x) \right] \Lambda(x); \end{aligned}$	$\Gamma^{\{T\}} \xrightarrow{A^{-1}(x)} \Gamma^{\{T\}'} = \Lambda^{-1}(x) \left[\Gamma^{\{T\}} + D\tau(x) \right],$ $\Gamma_*^{\{T\}} \xrightarrow{A^{-1}(x)} \Gamma_*^{\{T\}'} = \left[\Gamma_*^{\{T\}} + D\tau_*(x) \right] \Lambda(x);$	$\Gamma^{\{T\}} \xrightarrow{A^{-1}(x)} \Gamma^{\{T\}'} = \Lambda^{-1}(x) \left[\Gamma^{\{T\}} + D\tau(x) \right],$ $\Gamma_*^{\{T\}} \xrightarrow{A^{-1}(x)} \Gamma_*^{\{T\}'} = \left[\Gamma_*^{\{T\}} + D\tau_*(x) \right] \Lambda(x);$
mielke-geometrodynamics_00116 (3.3.17)	059	0.585 ● N +0	$dH = j \left[\begin{array}{l} \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{c^2 \partial t} = \mu \mathbf{J} \\ \nabla \cdot \mathbf{E} = \rho / \varepsilon. \end{array} \right]$	$dH = j \left\{ \begin{array}{l} \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{c^2 \partial t} = \mu \mathbf{J} \\ \nabla \cdot \mathbf{E} = \rho / \varepsilon. \end{array} \right.$	$dH = j \quad \left\{ \begin{array}{l} \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{c^2 \partial t} = \mu \mathbf{J} \\ \nabla \cdot \mathbf{E} = \rho / \varepsilon. \end{array} \right.$
mielke-geometrodynamics_00381 (6.2.32)	129	0.588 ● W +1	$\stackrel{(\pm)}{D} \Pi_{\alpha} := D \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \downarrow \stackrel{(\mp)}{\Pi} \right) \wedge \stackrel{(\pm)}{\Pi}_{\beta},$	$\stackrel{(\pm)}{D} \Pi_{\alpha} := D \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \downarrow \stackrel{(\mp)}{\Pi} \right) \wedge \stackrel{(\pm)}{\Pi}_{\beta},$	$\stackrel{(\pm)}{D} \Pi_{\alpha} := D \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \downarrow \stackrel{(\mp)}{\Pi} \right) \wedge \stackrel{(\pm)}{\Pi}_{\beta},$
mielke-geometrodynamics_00856 (2.7.6)	045	0.592 ● N +0	$\phi = \varphi \otimes b = G^{-1} \varphi \otimes^G b \in C^{\infty}(V^{\rho}).$	$\phi = \varphi \otimes b = G^{-1} \varphi \otimes^G b \in C^{\infty}(V^{\rho}).$	$\phi = \varphi \otimes b = G^{-1} \varphi \otimes^G b \in C^{\infty}(V^{\rho}).$
mielke-geometrodynamics_00187 (3.7.17)	074	0.593 ● N +0	$\begin{aligned} & \left(D^{\{*\}} \Omega \right) \wedge \vec{D}^{\{*\}} \Omega = \left(D^{\{*\}} \Omega + \left[\Omega, D^{\{*\}} \Omega \right] \right) \wedge \vec{D}^{\{*\}} \Omega = \\ & \left(D^{\{*\}} \Omega + \left[\Omega, D^{\{*\}} \Omega \right] \right) \wedge \vec{D}^{\{*\}} \Omega = 0. \end{aligned}$	$(D^* \Omega) = \vec{D}^* \vec{\Omega} + [\omega_0, \Omega_0]$ $= \vec{D}^* \vec{\Omega} + [\omega_0, \Theta^{\perp}] = 0.$	$(D^* \Omega) = \vec{D}^* \vec{\Omega} + [\omega_0, \Omega_0]$ $= \vec{D}^* \vec{\Omega} + [\omega_0, \Theta^{\perp}] = 0.$
mielke-geometrodynamics_00581 (8.6.19)	186	0.593 ● K +0	$\widehat{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha},$	$\widehat{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha},$	$\widehat{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha},$
mielke-geometrodynamics_00948 (14.1.29)	307	0.593 ● N +0	$\begin{aligned} & D^{\{*\}} \Omega \wedge \left(D^{\{*\}} \Omega + \left[\Omega, D^{\{*\}} \Omega \right] \right) \cos \delta - \\ & (-1)^z (D\xi + (-1)^{z*} \xi \wedge D\delta) \sin \delta = \tau_e \end{aligned}$	$D^* \Omega = (D^* \xi - (-1)^z \xi \wedge D\delta) \cos \delta$ $- (-1)^z (D\xi + (-1)^{z*} \xi \wedge D\delta) \sin \delta = \tau_e$	$D^* \Omega = (D^* \xi - (-1)^z \xi \wedge D\delta) \cos \delta$ $- (-1)^z (D\xi + (-1)^{z*} \xi \wedge D\delta) \sin \delta = \tau_e$
mielke-geometrodynamics_00357 (6.2.8)	125	0.600 ● N +0	$L_{\mathrm{HE}} = -\frac{1}{2\kappa} R_{\alpha\beta}^{\{\}} \wedge^* (\vartheta^{\alpha} \wedge \vartheta^{\beta})$	$L_{\mathrm{HE}} = -\frac{1}{2\kappa} R_{\alpha\beta}^{\{\}} \wedge^* (\vartheta^{\alpha} \wedge \vartheta^{\beta})$	$L_{\mathrm{HE}} = -\frac{1}{2\kappa} R_{\alpha\beta}^{\{\}} \wedge^* (\vartheta^{\alpha} \wedge \vartheta^{\beta})$
mielke-geometrodynamics_00784 (10.6.3)	225	0.601 ● N +0	$\square F - m_{\mathrm{photon}}^2 F = 0,$	$\square F - m_{\mathrm{photon}}^2 F = 0,$	$\square F - m_{\mathrm{photon}}^2 F = 0,$
mielke-geometrodynamics_00927 (14.1.8)	303	0.604 ● W -1	$\begin{aligned} & \Sigma_{\mathrm{YM}} = \frac{(-1)^s}{2\alpha_g} \mathrm{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) +^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\} \\ & = \frac{(-1)^s}{4\alpha_g} \mathrm{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) -^* \Omega \wedge^* (\vartheta \wedge \Omega) \right\}. \end{aligned}$	$\Sigma_{\mathrm{YM}} = \frac{(-1)^s}{2\alpha_g} \mathrm{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) +^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\}$ $= \frac{(-1)^s}{4\alpha_g} \mathrm{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) -^* \Omega \wedge^* (\vartheta \wedge \Omega) \right\}.$	$\Sigma_{\mathrm{YM}} = \frac{(-1)^s}{2\alpha_g} \mathrm{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) +^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\}$ $= \frac{(-1)^s}{4\alpha_g} \mathrm{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) -^* \Omega \wedge^* (\vartheta \wedge \Omega) \right\}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00115 (3.3.16)	059	■ 0.630 ● N +1	$\mathbf{d} F \equiv 0 \quad \Leftrightarrow \left\{ \begin{array}{l} \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \equiv 0 \\ \nabla \cdot \mathbf{B} \equiv 0 \end{array} \right.$	$dF \equiv 0 \quad \left\{ \begin{array}{l} \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \equiv 0 \\ \nabla \cdot \mathbf{B} \equiv 0 \end{array} \right.$	$dF \equiv 0 \quad \left\{ \begin{array}{l} \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \equiv 0 \\ \nabla \cdot \mathbf{B} \equiv 0. \end{array} \right.$
mielke-geometrodynamics_E00171 (3.6.6)	070	■ 0.630 ● N +0	$i^* \gamma \wedge D_{\pm} \psi = 0 \quad \text{and} \quad F^{\pm} = \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi = m \mathcal{P}_{\pm},$	$i^* \gamma \wedge D_{\pm} \psi = 0 \quad \text{and} \quad F^{\pm} = \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi = m \mathcal{P}_{\pm},$	$i^* \gamma \wedge D_{\pm} \psi = 0 \quad \text{and} \quad F^{\pm} = \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi = m \mathcal{P}_{\pm},$
mielke-geometrodynamics_E00189 (3.7.19)	074	■ 0.631 ● N +0	$\{ \}^* \Omega = (\stackrel{+}{-}) \Omega \text{ . }$	$^* \Omega = (\stackrel{+}{-}) \Omega.$	$^* \Omega = (\stackrel{+}{-}) \Omega.$
mielke-geometrodynamics_E00788 (11.3.7)	249	■ 0.635 ● W +0	$\left\{ i \gamma^{\kappa} \nabla_{\kappa}^{()} - \frac{6}{n^2} \ell^{*2} \gamma^{n+1} \gamma^{\mu} \bar{\psi} \gamma^{n+1} \gamma_{\mu} \psi - m \right\} \psi = 0.$	$\left\{ i \gamma^{\kappa} \nabla_{\kappa}^{()} - \frac{6}{n^2} \ell^{*2} \gamma^{n+1} \gamma^{\mu} \bar{\psi} \gamma^{n+1} \gamma_{\mu} \psi - m \right\} \psi = 0.$	$\left\{ i \gamma^{\kappa} \nabla_{\kappa}^{()} - \frac{6}{n^2} \ell^{*2} \gamma^{n+1} \gamma^{\mu} \bar{\psi} \gamma^{n+1} \gamma_{\mu} \psi - m \right\} \psi = 0.$
mielke-geometrodynamics_E00164 (3.5.12)	068	■ 0.636 ● N +0	$L_{\mathrm{YM}} + L_{\infty} = (\stackrel{+}{-}) \frac{1}{2\alpha_g} \mathrm{Tr} \left(^* \Omega (\stackrel{-}{+}) \Omega \right)^2 \geq 0$	$L_{\mathrm{YM}} + L_{\infty} = (\stackrel{+}{-}) \frac{1}{2\alpha_g} \mathrm{Tr} \left(^* \Omega (\stackrel{-}{+}) \Omega \right)^2 \geq 0$	$L_{\mathrm{YM}} + L_{\infty} = (\stackrel{+}{-}) \frac{1}{2\alpha_g} \mathrm{Tr} (^* \Omega (\stackrel{-}{+}) \Omega)^2 \geq 0$
mielke-geometrodynamics_E00549 (8.4.3)	179	■ 0.637 ● W +2	$\begin{aligned} L_{\mathrm{FP}} & \& := -s \operatorname{Tr} \left[\bar{\chi} \wedge ^* \Omega^g + \bar{\Phi} D^* \Psi + \frac{1}{2} \rho \bar{\chi} \wedge ^* \beta + \bar{C} d^* \Gamma + \frac{1}{2} \bar{C} \wedge ^* \mathbb{B} \right] \\ & = \frac{1}{2} s \left[\bar{\chi}^{\alpha\beta} \wedge ^* R_{\alpha\beta}^{(\pm)} + \bar{\Phi}_{\alpha\beta} D^* \Psi^{\alpha\beta} + \frac{1}{2} \rho \bar{\chi}^{\alpha\beta} \wedge ^* b_{\alpha\beta} + \tilde{\Psi}_{\mathrm{F}} \right] \end{aligned}$	$L_{\mathrm{FP}} := -s \operatorname{Tr} \left\{ \bar{\chi} \wedge ^* \Omega^g + \bar{\Phi} D^* \Psi + \frac{1}{2} \rho \bar{\chi} \wedge ^* \beta + \bar{C} d^* \Gamma + \frac{1}{2} \bar{C} \wedge ^* \mathbb{B} \right\} \\ = \frac{1}{2} s \left[\bar{\chi}^{\alpha\beta} \wedge ^* R_{\alpha\beta}^{(\pm)} + \bar{\Phi}_{\alpha\beta} D^* \Psi^{\alpha\beta} + \frac{1}{2} \rho \bar{\chi}^{\alpha\beta} \wedge ^* b_{\alpha\beta} + \tilde{\Psi}_{\mathrm{F}} \right]$	$L_{\mathrm{FP}} := -s \operatorname{Tr} \left\{ \bar{\chi} \wedge ^* \bar{\Omega}^g + \bar{\Phi} D^* \Psi + \frac{1}{2} \rho \bar{\chi} \wedge ^* \beta + \bar{C} d^* \Gamma + \frac{1}{2} \bar{C} \wedge ^* \mathbb{B} \right\} \\ = \frac{1}{2} s \left[\bar{\chi}^{\alpha\beta} \wedge ^* R_{\alpha\beta}^{(\pm)} + \bar{\Phi}_{\alpha\beta} D^* \Psi^{\alpha\beta} + \frac{1}{2} \rho \bar{\chi}^{\alpha\beta} \wedge ^* b_{\alpha\beta} + \tilde{\Psi}_{\mathrm{F}} \right]. \quad (8)$
mielke-geometrodynamics_E01058 (15.2.15)	346	■ 0.638 ● K +0	$\bar{D}^{\{\}} := d + i \bar{\omega}^{\{\}}$	$\bar{D}^{\{\}} := d + i \bar{\omega}^{\{\}}$	$\bar{D}^{\{\}} := d + i \bar{\omega}^{\{\}}$
mielke-geometrodynamics_E01008 (14.4.3)	325	■ 0.640 ● N +1	$\begin{aligned} d s^2 & \& = -\frac{\Delta}{\rho^2} \left[c d t - J_3 \frac{\hbar}{M} \sin^2 \vartheta d \phi \right]^2 + \frac{\rho^2}{\Delta} d r + \rho^2 d \vartheta^2 \\ & + \frac{\sin^2 \vartheta}{\rho^2} \left\{ \left[r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right] d \phi - J_3 \frac{\hbar}{M} d t \right\}^2 \end{aligned}$	$d s^2 = -\frac{\Delta}{\rho^2} \left[c d t - J_3 \frac{\hbar}{M} \sin^2 \vartheta d \phi \right]^2 + \frac{\rho^2}{\Delta} d r + \rho^2 d \vartheta^2 \\ + \frac{\sin^2 \vartheta}{\rho^2} \left\{ \left[r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right] d \phi - J_3 \frac{\hbar}{M} d t \right\}^2$	$d s^2 = -\frac{\Delta}{\rho^2} \left[c d t - J_3 \frac{\hbar}{M} \sin^2 \vartheta d \phi \right]^2 + \frac{\rho^2}{\Delta} d r + \rho^2 d \vartheta^2 \\ + \frac{\sin^2 \vartheta}{\rho^2} \left\{ \left[r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right] d \phi - J_3 \frac{\hbar}{M} d t \right\}^2 .$
mielke-geometrodynamics_E00255 (4.3.20)	093	■ 0.642 ● N +0	$\stackrel{*}{\sigma} \Omega^g = \frac{1}{4} R_{\alpha\beta}^{..cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta.$	$\stackrel{*}{\sigma} \Omega^g = \frac{1}{4} R_{\alpha\beta}^{..cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta.$	$\stackrel{*}{\sigma} \Omega^g = \frac{1}{4} R_{\alpha\beta}^{..cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta.$
mielke-geometrodynamics_E01015 (14.4.10)	327	■ 0.646 ● N +1	$A = \{ \}^* \omega = A^{(\circ)} \lambda_3 .$	$A = ^* \omega = A^{(\circ)} \lambda_3.$	$A = \stackrel{*}{\sigma} \omega = A^{(\circ)} \lambda_3.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00119 (3.3.20)	059	0.646 N +0	$\begin{aligned} \Sigma_{\alpha} &:= e_{\alpha} \rfloor L - (e_{\alpha} \rfloor dA) \wedge \frac{\partial L}{\partial(dA)} \\ &= \frac{1}{2} [(e_{\alpha} \rfloor F) \wedge H - F \wedge (e_{\alpha} \rfloor H)] \end{aligned}$	$\Sigma_{\alpha} := e_{\alpha} \rfloor L - (e_{\alpha} \rfloor dA) \wedge \frac{\partial L}{\partial(dA)} = \frac{1}{2} [(e_{\alpha} \rfloor F) \wedge H - F \wedge (e_{\alpha} \rfloor H)]$	$\Sigma_{\alpha} := e_{\alpha} \rfloor L - (e_{\alpha} \rfloor dA) \wedge \frac{\partial L}{\partial(dA)} = \frac{1}{2} [(e_{\alpha} \rfloor F) \wedge H - F \wedge (e_{\alpha} \rfloor H)]$
mielke-geometrodynamics_E00120 (3.3.21)	060	0.651 K +0	$d \Sigma_{\alpha} \simeq (e_{\alpha} \rfloor F) \wedge j$	$d \Sigma_{\alpha} \simeq (e_{\alpha} \rfloor F) \wedge j$	$d \Sigma_{\alpha} \simeq (e_{\alpha} \rfloor F) \wedge j$
mielke-geometrodynamics_E00497 (7.7.15)	163	0.653 C -4	$ds^2 = o_{\alpha\beta} \vartheta^{\alpha} \otimes \vartheta^{\beta} = g_{ij} dx^i \otimes dx^j, \quad o_{\alpha\beta} := \text{diag}\{1, \underbrace{-1, \dots, -1}_{\text{sig}}\},$	$ds^2 = o_{\alpha\beta} \vartheta^{\alpha} \otimes \vartheta^{\beta} = g_{ij} dx^i \otimes dx^j, \quad o_{\alpha\beta} := \text{diag}\{1, \underbrace{-1, \dots, -1}_{\text{sig}}\},$	$ds^2 = o_{\alpha\beta} \vartheta^{\alpha} \otimes \vartheta^{\beta} = g_{ij} dx^i \otimes dx^j, \quad o_{\alpha\beta} := \text{diag}\{1, \underbrace{-1, \dots, -1}_{\text{sig}}\},$
mielke-geometrodynamics_E00800 (11.4.12)	254	0.658 N +0	$\left\{ i \gamma^k \partial_k - \frac{3i}{2R_{\text{o}} \lambda} \gamma^0 - \frac{m}{\lambda} \right\} \psi = 0,$	$\left\{ i \gamma^k \partial_k - \frac{3i}{2R_{\text{o}} \lambda} \gamma^0 - \frac{m}{\lambda} \right\} \psi = 0,$	$\left\{ i \gamma^k \partial_k - \frac{3i}{2R_{\text{o}} \lambda} \gamma^0 - \frac{m}{\lambda} \right\} \psi = 0,$
mielke-geometrodynamics_E00316 (4.6.8)	105	0.664 N +0	$\Phi(\widetilde{\Gamma}, \gamma) = (2\pi \hbar \mathcal{M}_n G / \hbar \ell c^2) = 2\pi n$	$\Phi(\widetilde{\Gamma}, \gamma) = (2\pi \hbar \mathcal{M}_n G / \hbar \ell c^2) = 2\pi n$	$\Phi(\widetilde{\Gamma}, \gamma) = (2\pi \hbar \mathcal{M}_n G / \hbar \ell c^2) = 2\pi n,$
mielke-geometrodynamics_E00335 (5.3.12)	113	0.665 W +0	$\begin{aligned} \widetilde{T}_{\mu\nu} &= T_{\mu\nu} + \ell^{*2} \left\{ -4 \tau_{\mu \cdot [\lambda]}^{\cdot \kappa} \tau_{\nu \cdot [\kappa]}^{\cdot \lambda} - 2 \tau_{\mu}^{\cdot \kappa \lambda} \tau_{\nu \kappa \lambda} + \tau_{\cdot, \mu}^{\kappa \lambda} \tau_{\kappa \lambda \nu} \right. \\ &\quad \left. + \frac{1}{2} g_{\mu\nu} \left(4 \tau_{\delta \cdot [\lambda]}^{\cdot \kappa} \tau_{\cdot, \kappa]}^{\delta \lambda} + \tau^{\delta \kappa \lambda} \tau_{\delta \kappa \lambda} \right) \right\}. \end{aligned}$	$\widetilde{T}_{\mu\nu} = T_{\mu\nu} + \ell^{*2} \left\{ -4 \tau_{\mu \cdot [\lambda]}^{\cdot \kappa} \tau_{\nu \cdot [\kappa]}^{\cdot \lambda} - 2 \tau_{\mu}^{\cdot \kappa \lambda} \tau_{\nu \kappa \lambda} + \tau_{\cdot, \mu}^{\kappa \lambda} \tau_{\kappa \lambda \nu} + \frac{1}{2} g_{\mu\nu} \left(4 \tau_{\delta \cdot [\lambda]}^{\cdot \kappa} \tau_{\cdot, \kappa]}^{\delta \lambda} + \tau^{\delta \kappa \lambda} \tau_{\delta \kappa \lambda} \right) \right\}.$	$\widetilde{T}_{\mu\nu} = T_{\mu\nu} + \ell^{*2} \left\{ -4 \tau_{\mu \cdot [\lambda]}^{\cdot \kappa} \tau_{\nu \cdot [\kappa]}^{\cdot \lambda} - 2 \tau_{\mu}^{\cdot \kappa \lambda} \tau_{\nu \kappa \lambda} + \tau_{\cdot, \mu}^{\kappa \lambda} \tau_{\kappa \lambda \nu} + \frac{1}{2} g_{\mu\nu} (4 \tau_{\delta \cdot [\lambda]}^{\cdot \kappa} \tau_{\cdot, \kappa]}^{\delta \lambda} + \tau^{\delta \kappa \lambda} \tau_{\delta \kappa \lambda}) \right\}.$
mielke-geometrodynamics_E01055 (15.2.12)	345	0.665 N +0	$\bar{L}_{\mathbf{w}.}(f) = \frac{i}{8} \text{Tr} \left(\bar{\gamma} \wedge \bar{\gamma} \wedge {}^* \bar{\Omega} \right).$	$\bar{L}_{\mathbf{w}.}(f) = \frac{i}{8} \text{Tr} \left(\bar{\gamma} \wedge \bar{\gamma} \wedge {}^* \bar{\Omega} \right).$	$\bar{L}_{\mathbf{w}.}(f) = \frac{i}{8} \text{Tr} \left(\bar{\gamma} \wedge \bar{\gamma} \wedge {}^* \bar{\Omega} \right).$
mielke-geometrodynamics_E01145 (C.13)	377	0.667 N +0	$\stackrel{\circ}{g}_{lm} := c_{lk}{}^i c_{mi}{}^k$	$\stackrel{\circ}{g}_{lm} := c_{lk}{}^i c_{mi}{}^k$	$\stackrel{\circ}{g}_{lm} := c_{lk}{}^i c_{mi}{}^k$
mielke-geometrodynamics_E00450 (7.2.3)	151	0.676 N +0	$\Omega^g = \stackrel{+}{\Omega}^g + \bar{\Omega}^g$	$\Omega^g = \stackrel{+}{\Omega}^g + \bar{\Omega}^g$	$\Omega^g = \stackrel{+}{\Omega}^g + \bar{\Omega}^g$
mielke-geometrodynamics_E00143 (3.4.14)	064	0.679 N +0	$\text{div } \iota := (-1)^{1+s} d^* \iota = 0$	$\text{div } \iota := (-1)^{1+s} d^* \iota = 0$	$\text{div } \iota := (-1)^{1+s} d^* \iota = 0$
mielke-geometrodynamics_E00472 (7.3.11)	155	0.682 U +23	$\stackrel{+}{\Omega}^{\{\}} = 0 \iff \mathcal{R}_{ij}^{\{\}} = 0 \iff R_{ij}^{\{\}} = \Lambda' g_{ij}.$	$(not\ rendered)$	$\stackrel{+}{\Omega}^{\{\}} = 0 \iff \mathcal{R}_{ij}^{\{\}} = 0 \iff R_{ij}^{\{\}} = \Lambda' g_{ij}.$
Conf. MathPix confidence: ≥ 0.9 $0.5\text{--}0.9$ < 0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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mielke-geometrodynamics_256 E08812 (11.4.24)	256	0.686 C +8	$\begin{aligned} & \partial_{\rho^*} H + \frac{1+\kappa}{\rho} e^{(\nu-\mu)/2} H = \frac{1}{\beta} \left[\omega + e^{\nu/2} - \varepsilon e^{-\mu} \left(H^2 - F^2 - \frac{2}{3} a H^2 \right) \right] F, \\ & \partial_{\rho^*} F + \frac{1-\kappa}{\rho} e^{(\nu-\mu)/2} F = \frac{1}{\beta} \left[-\omega + e^{\nu/2} - \varepsilon e^{-\mu} \left(H^2 - F^2 + \frac{2}{3} a F^2 \right) \right] H. \end{aligned}$	$\partial_{\rho^*} H + \frac{1+\kappa}{\rho} e^{(\nu-\mu)/2} H = \frac{1}{\beta} \left[\omega + e^{\nu/2} - \varepsilon e^{-\mu} \left(H^2 - F^2 - \frac{2}{3} a H^2 \right) \right] F,$ $\partial_{\rho^*} F + \frac{1-\kappa}{\rho} e^{(\nu-\mu)/2} F = \frac{1}{\beta} \left[-\omega + e^{\nu/2} - \varepsilon e^{-\mu} \left(H^2 - F^2 + \frac{2}{3} a F^2 \right) \right] H.$	$\partial_{\rho^*} H + \frac{1+\kappa}{\rho} e^{(\nu-\mu)/2} H = \frac{1}{\beta} \left[\omega + e^{\nu/2} - \varepsilon e^{-\mu} \left(H^2 - F^2 - \frac{2}{3} a H^2 \right) \right] F,$ $\partial_{\rho^*} F + \frac{1-\kappa}{\rho} e^{(\nu-\mu)/2} F = \frac{1}{\beta} \left[-\omega + e^{\nu/2} - \varepsilon e^{-\mu} \left(H^2 - F^2 + \frac{2}{3} a F^2 \right) \right] H.$
mielke-geometrodynamics_111 E09321 (5.2.5)	111	0.686 W +0	$\tau_{\alpha\beta} := \frac{\partial L_D}{\partial \Gamma^{\alpha\beta}} = \frac{1}{8} \bar{\Psi} (*\gamma\sigma_{\alpha\beta} + \sigma_{\alpha\beta} *\gamma) \Psi = \frac{1}{4} \eta_{\alpha\beta\gamma\delta} \bar{\Psi} \gamma^\delta \gamma_5 \Psi \eta^\gamma = \tau_{\alpha\beta\gamma} \eta^\gamma$	$\tau_{\alpha\beta} := \frac{\partial L_D}{\partial \Gamma^{\alpha\beta}} = \frac{1}{8} \bar{\Psi} (*\gamma\sigma_{\alpha\beta} + \sigma_{\alpha\beta} *\gamma) \Psi = \frac{1}{4} \eta_{\alpha\beta\gamma\delta} \bar{\Psi} \gamma^\delta \gamma_5 \Psi \eta^\gamma = \tau_{\alpha\beta\gamma} \eta^\gamma$	$\tau_{\alpha\beta} := \frac{\partial L_D}{\partial \Gamma^{\alpha\beta}} = \frac{1}{8} \bar{\Psi} (*\gamma\sigma_{\alpha\beta} + \sigma_{\alpha\beta} *\gamma) \Psi = \frac{1}{4} \eta_{\alpha\beta\gamma\delta} \bar{\Psi} \gamma^\delta \gamma_5 \Psi \eta^\gamma = \tau_{\alpha\beta\gamma} \eta^\gamma$
mielke-geometrodynamics_131 E09389 (6.4.4)	131	0.688 W +2	$\mathscr{H} = \mathscr{K} - \dot{\Psi} \wedge \left(\frac{\partial \mathscr{K}}{\partial \dot{\Psi}} \right),$	$\mathscr{H} = \mathscr{K} - \dot{\Psi} \wedge \left(\frac{\partial \mathscr{K}}{\partial \dot{\Psi}} \right),$	$\mathscr{H} = \mathscr{K} - \dot{\Psi} \wedge \left(\frac{\partial \mathscr{K}}{\partial \dot{\Psi}} \right),$
mielke-geometrodynamics_307 E09949 (14.1.30)	307	0.688 N +0	$\begin{aligned} D\Omega &= (D\xi + (-1)^z * \xi \wedge D\delta) \cos \delta \\ &+ (-1)^z (D^* \xi - (-1)^z \xi \wedge D\delta) \sin \delta = 0. \end{aligned}$	$D\Omega = (D\xi + (-1)^z * \xi \wedge D\delta) \cos \delta + (-1)^z (D^* \xi - (-1)^z \xi \wedge D\delta) \sin \delta = 0.$	$D\Omega = (D\xi + (-1)^z * \xi \wedge D\delta) \cos \delta + (-1)^z (D^* \xi - (-1)^z \xi \wedge D\delta) \sin \delta = 0.$
mielke-geometrodynamics_224 E08696 (10.5.7)	224	0.694 N +0	$\left. D^* \gamma \right _{\alpha} = \left. * \gamma \right _{\alpha} \wedge T, \quad T := e_{\alpha} \rfloor T^{\alpha},$	$D^* \gamma = * \gamma \wedge T, \quad T := e_{\alpha} \rfloor T^{\alpha},$	$D^* \gamma = * \gamma \wedge T, \quad T := e_{\alpha} \rfloor T^{\alpha},$
mielke-geometrodynamics_175 E09534 (8.2.1)	175	0.699 C +5	$\begin{aligned} s\Gamma &= \Psi - DC \\ sC &= \Phi - \frac{1}{2}[C, C] \\ s\Omega^g &= -D\Psi - [C, \Omega^g] \\ s\Psi &= -D\Phi - [C, \Psi] \\ s\Phi &= -[C, \Phi] \end{aligned}$	$\begin{aligned} s\Gamma &= \Psi - DC \\ sC &= \Phi - \frac{1}{2}[C, C] \\ s\Omega^g &= -D\Psi - [C, \Omega^g] \\ s\Psi &= -D\Phi - [C, \Psi] \\ s\Phi &= -[C, \Phi] \end{aligned}$	$\begin{aligned} s\Gamma &= \Psi - DC \\ sC &= \Phi - \frac{1}{2}[C, C] \\ s\Omega^g &= -D\Psi - [C, \Omega^g] \\ s\Psi &= -D\Phi - [C, \Psi] \\ s\Phi &= -[C, \Phi] \end{aligned}$
mielke-geometrodynamics_112 E09329 (5.3.6)	112	0.699 N +0	$D\Theta \equiv [\Omega^g, \gamma], \quad D\Omega^g \equiv 0,$	$D\Theta \equiv [\Omega^g, \gamma], \quad D\Omega^g \equiv 0,$	$D\Theta \equiv [\Omega^g, \gamma], \quad D\Omega^g \equiv 0,$
mielke-geometrodynamics_155 E09471 (7.3.10)	155	0.706 N +0	$\bar{\Omega}^{\{\}} = 0 \iff \Omega_C^{\{\}} = 0 \text{ and } R^{\{\}} = 0.$	$\bar{\Omega}^{\{\}} = 0 \iff \Omega_C^{\{\}} = 0 \text{ and } R^{\{\}} = 0.$	$\bar{\Omega}^{\{\}} = 0 \iff \Omega_C^{\{\}} = 0 \text{ and } R^{\{\}} = 0.$
mielke-geometrodynamics_240 E09747 (11.1.14)	240	0.706 N +0	$M = \mathbb{R} \times S^3 \#^v W^3.$	$M = \mathbb{R} \times S^3 \#^v W^3.$	$M = \mathbb{R} \times S^3 \#^v W^3.$
mielke-geometrodynamics_105 E08314 (4.6.6)	105	0.710 N +0	$\Phi(\tilde{\Gamma}, \gamma) = P \exp \left[(i/\hbar) \oint \left(\Gamma^{(T)\alpha} P_{\alpha} + \Gamma_{\alpha}^{(L)\beta} L^{\alpha}_{\beta} \right) \right]$	$\Phi(\tilde{\Gamma}, \gamma) = P \exp \left[(i/\hbar) \oint \left(\Gamma^{(T)\alpha} P_{\alpha} + \Gamma_{\alpha}^{(L)\beta} L^{\alpha}_{\beta} \right) \right]$	$\Phi(\tilde{\Gamma}, \gamma) = P \exp \left[(i/\hbar) \oint \left(\Gamma^{(T)\alpha} P_{\alpha} + \Gamma_{\alpha}^{(L)\beta} L^{\alpha}_{\beta} \right) \right]$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_058 E00108 (3.3.9)		0.711 N -1	$L_{\mathrm{Pontr}}=-\frac{1}{2} F \wedge F=-\frac{1}{2} d$ C,	$L_{\mathrm{Pontr}}=-\frac{1}{2} F \wedge F=-\frac{1}{2} dC,$	$L_{\mathrm{Pontr}}=-\frac{1}{2} F \wedge F=-\frac{1}{2} dC,$
mielke-geometrodynamics_177 E00545 (8.3.6)		0.711 N +0	$d \rightarrow \exp(\zeta \rfloor) d \exp(-\zeta \rfloor)$.	$d \rightarrow \exp(\zeta \rfloor) d \exp(-\zeta \rfloor).$	$d \rightarrow \exp(\zeta \rfloor) d \exp(-\zeta \rfloor).$
mielke-geometrodynamics_192 E00583 (9.1.1)		0.712 N +0	$d s^{\{2\}}=-e^{\{v\}} d t^{\{2\}}+e^{\{\lambda\}} d r^{\{2\}}+r^{\{2\}} d$ $\Omega^{\{n-2\}}$	$ds^2=-e^v dt^2+e^\lambda dr^2+r^2 d\Omega^{n-2}$	$ds^2=-e^v dt^2+e^\lambda dr^2+r^2 d\Omega^{n-2}$
mielke-geometrodynamics_094 E00260 (4.3.25)		0.718 N +0	$\stackrel{*}{K}=\frac{1}{2} K_{\alpha}^{\beta c} L_{\beta c} \otimes \vartheta^{\alpha}.$ $L_{\beta c} \otimes \vartheta^{\alpha}.$	$\sigma^* K=\frac{1}{2} K_{\alpha}^{\beta c} L_{\beta c} \otimes \vartheta^{\alpha}.$	$\sigma^* K=\frac{1}{2} K_{\alpha}^{\beta c} L_{\beta c} \otimes \vartheta^{\alpha}.$
mielke-geometrodynamics_285 E00878 (13.3.7)		0.722 N +0	$\mathrm{SL}(n+1,\mathbb{R}) \approx \mathrm{A}_*(n,\mathbb{R}) := \mathbb{R}^n \otimes \mathrm{GL}(n,\mathbb{R}) \otimes \mathbb{R}_*^n.$ $\mathbb{R}^n \otimes \mathrm{GL}(n,\mathbb{R}) \otimes \mathbb{R}_*^n.$	$\mathrm{SL}(n+1,\mathbb{R}) \approx \mathrm{A}_*(n,\mathbb{R}) := \mathbb{R}^n \otimes \mathrm{GL}(n,\mathbb{R}) \otimes \mathbb{R}_*^n.$	$\mathrm{SL}(n+1,\mathbb{R}) \approx \mathrm{A}_*(n,\mathbb{R}) := \mathbb{R}^n \otimes \mathrm{GL}(n,\mathbb{R}) \otimes \mathbb{R}_*^n.$
mielke-geometrodynamics_042 E00041 (2.6.4)		0.725 N +0	$\stackrel{*}{\Omega}=\frac{1}{2} F_{\alpha} \beta^j L_j \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$ $\vartheta^{\alpha} \wedge \vartheta^{\beta}$	$\sigma^* \Omega=\frac{1}{2} F_{\alpha} \beta^j L_j \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$	$\sigma^* \Omega=\frac{1}{2} F_{\alpha} \beta^j L_j \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$
mielke-geometrodynamics_185 E00574 (8.6.12)		0.725 W -2	$L_{\mathrm{eff}}:=L_{\mathrm{qPG}}(**)-dC_{\theta}=-\frac{\theta_{\mathrm{T}}^*}{2\ell^2} R^{\{\} \alpha \beta} \wedge \eta_{\alpha \beta}+\frac{\theta_{\mathrm{T}}^* \Lambda_{\mathrm{eff}}}{\ell^2} \eta.$ $\Lambda_{\mathrm{eff}}:=L_{\mathrm{qPG}}(**)-dC_{\theta}=-\frac{\theta_{\mathrm{T}}^*}{2\ell^2} R^{\{\} \alpha \beta} \wedge \eta_{\alpha \beta}+\frac{\theta_{\mathrm{T}}^* \Lambda_{\mathrm{eff}}}{\ell^2} \eta.$	$L_{\mathrm{eff}}:=L_{\mathrm{qPG}}(**)-dC_{\theta}=-\frac{\theta_{\mathrm{T}}^*}{2\ell^2} R^{\{\} \alpha \beta} \wedge \eta_{\alpha \beta}+\frac{\theta_{\mathrm{T}}^* \Lambda_{\mathrm{eff}}}{\ell^2} \eta.$	$L_{\mathrm{eff}}:=L_{\mathrm{qPG}}(**)-dC_{\theta}=-\frac{\theta_{\mathrm{T}}^*}{2\ell^2} R^{\{\} \alpha \beta} \wedge \eta_{\alpha \beta}+\frac{\theta_{\mathrm{T}}^* \Lambda_{\mathrm{eff}}}{\ell^2} \eta.$
mielke-geometrodynamics_126 E00366 (6.2.13)		0.725 N -1	$\stackrel{(\pm)}{L}_{\mathrm{EC}}:=L_{\mathrm{EC}} \pm idC_{\mathrm{TT}}=-\frac{1}{2\ell^2} R^{\{\} \alpha \beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}+\frac{\Lambda}{\ell^2} \eta.$ $L_{\mathrm{EC}} \pm idC_{\mathrm{TT}}=-\frac{1}{2\ell^2} R^{\{\} \alpha \beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}+\frac{\Lambda}{\ell^2} \eta.$	$\stackrel{(\pm)}{L}_{\mathrm{EC}}:=L_{\mathrm{EC}} \pm idC_{\mathrm{TT}}=-\frac{1}{2\ell^2} R^{\{\} \alpha \beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}+\frac{\Lambda}{\ell^2} \eta.$	$\stackrel{(\pm)}{L}_{\mathrm{EC}}:=L_{\mathrm{EC}} \pm idC_{\mathrm{TT}}=-\frac{1}{2\ell^2} R^{\{\} \alpha \beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}+\frac{\Lambda}{\ell^2} \eta.$
mielke-geometrodynamics_193 E00585 (9.1.3)		0.725 S +0	$d s^{\{2\}}=\frac{n-2}{2 \Lambda_{\mathrm{eff}}} t^2 \left(-d t^2+d r^2\right)+\frac{(n-2)(n-3)}{2 \Lambda_{\mathrm{eff}}} d \Omega^{n-2}$ $\Lambda_{\mathrm{eff}}:=\frac{n-2}{2 \Lambda_{\mathrm{eff}}} t^2 \left(-d t^2+d r^2\right)+\frac{(n-2)(n-3)}{2 \Lambda_{\mathrm{eff}}} d \Omega^{n-2}$	$ds^2=\frac{n-2}{2\Lambda_{\mathrm{eff}}} t^2 \left(-dt^2+dr^2\right)+\frac{(n-2)(n-3)}{2\Lambda_{\mathrm{eff}}} d\Omega^{n-2}$	$ds^2=\frac{n-2}{2\Lambda_{\mathrm{eff}}} t^2 \left(-dt^2+dr^2\right)+\frac{(n-2)(n-3)}{2\Lambda_{\mathrm{eff}}} d\Omega^{n-2}$
mielke-geometrodynamics_038 E00021 (2.4.3)		0.726 N -2	$\phi^{(p)}:=\varphi^p \otimes b \in C^{\infty}\left(\wedge^p T_{\mathbb{C}}^*(M) \otimes V^p\right), \quad b \in C^{\infty}\left(V^p\right).$ $\varphi^p \otimes b \in C^{\infty}\left(\wedge^p T_{\mathbb{C}}^*(M) \otimes V^p\right), \quad b \in C^{\infty}\left(V^p\right).$	$\phi^{(p)}:=\varphi^p \otimes b \in C^{\infty}\left(\wedge^p T_{\mathbb{C}}^*(M) \otimes V^p\right), \quad b \in C^{\infty}\left(V^p\right).$	$\phi^{(p)}:=\varphi^p \otimes b \in C^{\infty}\left(\wedge^p T_{\mathbb{C}}^*(M) \otimes V^p\right), \quad b \in C^{\infty}\left(V^p\right).$
mielke-geometrodynamics_085 E00209 (4.2.1)		0.728 N +0	$\stackrel{*}{\chi} \widetilde{\omega}=\omega^{\mathrm{L}} \otimes \omega^{\mathrm{T}},$ $\omega^{\mathrm{L}} \otimes \omega^{\mathrm{T}},$	$\chi^* \widetilde{\omega}=\omega^{\mathrm{L}} \otimes \omega^{\mathrm{T}},$	$\chi^* \widetilde{\omega}=\omega^{\mathrm{L}} \otimes \omega^{\mathrm{T}},$
mielke-geometrodynamics_104 E00309 (4.6.1)		0.728 N +0	$\widetilde{\Gamma}:=\Gamma^{(T) \alpha} P_{\alpha}+\Gamma_{\alpha}^{(L) \beta} L^{\alpha}{}_{\beta}$ $\Gamma^{(T) \alpha} P_{\alpha}+\Gamma_{\alpha}^{(L) \beta} L^{\alpha}{}_{\beta}$	$\widetilde{\Gamma}:=\Gamma^{(T) \alpha} P_{\alpha}+\Gamma_{\alpha}^{(L) \beta} L^{\alpha}{}_{\beta}$	$\widetilde{\Gamma}:=\Gamma^{(T) \alpha} P_{\alpha}+\Gamma_{\alpha}^{(L) \beta} L^{\alpha}{}_{\beta}$
mielke-geometrodynamics_034 E00009 (2.2.4)		0.729 N +1	$E_i^{\alpha} e^i{}_{\beta}=\delta^{\alpha}{}_{\beta}.$ $\delta^{\alpha}{}_{\beta}.$	$E_i^{\alpha} e^i{}_{\beta}=\delta^{\alpha}{}_{\beta}.$	$E_i^{\alpha} e^i{}_{\beta}=\delta^{\alpha}{}_{\beta}.$
mielke-geometrodynamics_130 E00384 (6.3.2)		0.730 C +3	$H_{\alpha}^{\parallel}=\frac{1}{\ell^2}\left[\vartheta^{\beta} \wedge *(d \vartheta_{\beta} \wedge \vartheta_{\alpha})-\frac{1}{2} \vartheta_{\alpha} \wedge *(d \vartheta^{\beta} \wedge \vartheta_{\beta})\right]=\frac{1}{2 \ell^2} \Gamma^{\{\} \beta \gamma} \wedge \eta_{\alpha \beta \gamma}$ $\vartheta^{\beta} \wedge *(d \vartheta_{\beta} \wedge \vartheta_{\alpha})-\frac{1}{2} \vartheta_{\alpha} \wedge *(d \vartheta^{\beta} \wedge \vartheta_{\beta})\right]=\frac{1}{2 \ell^2} \Gamma^{\{\} \beta \gamma} \wedge \eta_{\alpha \beta \gamma}$	$H_{\alpha}^{\parallel}=\frac{1}{\ell^2}\left[\vartheta^{\beta} \wedge *(d \vartheta_{\beta} \wedge \vartheta_{\alpha})-\frac{1}{2} \vartheta_{\alpha} \wedge *(d \vartheta^{\beta} \wedge \vartheta_{\beta})\right]=\frac{1}{2 \ell^2} \Gamma^{\{\} \beta \gamma} \wedge \eta_{\alpha \beta \gamma}$	$H_{\alpha}^{\parallel}=\frac{1}{\ell^2}\left[\vartheta^{\beta} \wedge *(d \vartheta_{\beta} \wedge \vartheta_{\alpha})-\frac{1}{2} \vartheta_{\alpha} \wedge *(d \vartheta^{\beta} \wedge \vartheta_{\beta})\right]=\frac{1}{2 \ell^2} \Gamma^{\{\} \beta \gamma} \wedge \eta_{\alpha \beta \gamma}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E08687 (10.3.45)	222	■ 0.753 ● N +0	$\left.\left.-E_{\alpha}\right\rfloor L(\star)-(-1)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\right\rfloor R^{\{\star\beta}\right)\wedge\vartheta_{\beta}=\widetilde{\ell}\widetilde{\Sigma}_{\alpha}$	$-E_{\alpha}=-e_{\alpha}\rfloor L(\star)-(-1)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\rfloor R^{\{\star\beta}\right)\wedge\vartheta_{\beta}=\widetilde{\ell}\widetilde{\Sigma}_{\alpha}$	$-E_{\alpha}=-e_{\alpha}\rfloor L(\star)-(-1)^s\frac{2\gamma}{\ell}(e_{\alpha}\rfloor R^{\{\star\beta}\wedge\vartheta_{\beta}=\ell\widetilde{\Sigma}_{\alpha}$
mielke-geometrodynamics_E08686 (9.2.5)	199	■ 0.753 ● N +1	$\chi(M)=\int_M\left\{\frac{1}{(4\pi)^2}\int_{\partial M}\left\{\alpha\wedge\Omega^{g(\star)}+\frac{2}{3}\alpha\wedge(\alpha\wedge\alpha)^{(\star)}\right\}\right\}.$	$\chi(M)=\int_M^*(\gamma_2^g)-\frac{1}{(4\pi)^2}\int_{\partial M}\left\{\alpha\wedge\Omega^{g(\star)}+\frac{2}{3}\alpha\wedge(\alpha\wedge\alpha)^{(\star)}\right\}.$	$\chi(M)=\int_M^*(\gamma_2^g)-\frac{1}{(4\pi)^2}\int_{\partial M}\left\{\alpha\wedge\Omega^{g(\star)}+\frac{2}{3}\alpha\wedge(\alpha\wedge\alpha)^{(\star)}\right\}.$
mielke-geometrodynamics_E08992 (14.3.5)	320	■ 0.754 ● N +0	$\stackrel{\circ}{\gamma}^{K+1}\chi_{\rho F}^{\pm}(\xi)=\pm\chi_{\rho F}^{\pm}(\xi)$	$\stackrel{\circ}{\gamma}^{K+1}\chi_{\rho F}^{\pm}(\xi)=\pm\chi_{\rho F}^{\pm}(\xi)$	$\stackrel{\circ}{\gamma}^{K+1}\chi_{\rho F}^{\pm}(\xi)=\pm\chi_{\rho F}^{\pm}(\xi)$
mielke-geometrodynamics_E08249 (4.3.14)	092	■ 0.755 ● N +0	$D_kg_{ij}=E_{,k}^{\alpha}\nabla_{\alpha}g_{ij}=0$	$D_kg_{ij}=E_{,k}^{\alpha}\nabla_{\alpha}g_{ij}=0$	$D_kg_{ij}=E_{,k}^{\alpha}\nabla_{\alpha}g_{ij}=0$
mielke-geometrodynamics_E08399 (6.4.17)	133	■ 0.755 ● W +0	$\begin{aligned} & \left\{ \left[D_{\alpha}, D_{\beta} \right] = -T_{\alpha\beta}{}^{\gamma}(x) D_{\gamma} + R_{\alpha\beta\gamma}{}^{\delta}(x) L^{\gamma}{}_{\delta}, \right. \\ & \left[L^{\alpha}{}_{\beta}, D_{\gamma} \right] = \delta_{\gamma}^{\alpha} D_{\beta}, \\ & \left. \left[L^{\alpha}{}_{\beta}, L^{\gamma}{}_{\delta} \right] = \delta_{\delta}^{\alpha} L^{\gamma}{}_{\beta} - \delta_{\beta}^{\gamma} L^{\alpha}{}_{\delta}, \right\} \end{aligned}$	$\begin{aligned} & \left[D_{\alpha}, D_{\beta} \right] = -T_{\alpha\beta}{}^{\gamma}(x) D_{\gamma} + R_{\alpha\beta\gamma}{}^{\delta}(x) L^{\gamma}{}_{\delta}, \\ & \left[L^{\alpha}{}_{\beta}, D_{\gamma} \right] = \delta_{\gamma}^{\alpha} D_{\beta}, \\ & \left[L^{\alpha}{}_{\beta}, L^{\gamma}{}_{\delta} \right] = \delta_{\delta}^{\alpha} L^{\gamma}{}_{\beta} - \delta_{\beta}^{\gamma} L^{\alpha}{}_{\delta}, \end{aligned}$	$\begin{aligned} & \left[D_{\alpha}, D_{\beta} \right] = -T_{\alpha\beta}{}^{\gamma}(x) D_{\gamma} + R_{\alpha\beta\gamma}{}^{\delta}(x) L^{\gamma}{}_{\delta}, \\ & \left[L^{\alpha}{}_{\beta}, D_{\gamma} \right] = \delta_{\gamma}^{\alpha} D_{\beta}, \\ & \left[L^{\alpha}{}_{\beta}, L^{\gamma}{}_{\delta} \right] = \delta_{\delta}^{\alpha} L^{\gamma}{}_{\beta} - \delta_{\beta}^{\gamma} L^{\alpha}{}_{\delta}, \end{aligned}$
mielke-geometrodynamics_E08957 (14.1.39)	309	■ 0.756 ● N +0	$\partial_{\kappa}\delta=\sqrt{ g }\varepsilon_{\kappa\lambda\rho\sigma}R^{\{\sigma}{}_{,\nu}\nabla^{\rho}R^{\}\nu\lambda}/\left(R^{\{\}_{\alpha\beta}}R^{\{\alpha\beta}\right).$	$\partial_{\kappa}\delta=\sqrt{ g }\varepsilon_{\kappa\lambda\rho\sigma}R^{\{\sigma}{}_{,\nu}\nabla^{\rho}R^{\}\nu\lambda}/\left(R^{\{\}_{\alpha\beta}}R^{\{\alpha\beta}\right).$	$\partial_{\kappa}\delta=\sqrt{ g }\varepsilon_{\kappa\lambda\rho\sigma}R^{\{\}_{\sigma}{}_{,\nu}\nabla^{\rho}R^{\{\nu\lambda}}/(R^{\{\}_{\alpha\beta}}R^{\{\alpha\beta}).$
mielke-geometrodynamics_E08970 (14.2.11)	314	■ 0.758 ● C +6	$\hat{\Omega}=\hat{d}\hat{\omega}-\hat{\omega}-\hat{\omega}\wedge\hat{\omega}=\left[\frac{\Omega^g+\alpha\wedge\alpha^T}{\circ}\middle \frac{-D\alpha}{\stackrel{\circ}{\Omega}+\alpha^T\wedge\alpha}\right].$	$\hat{\Omega}=\hat{d}\hat{\omega}-\hat{\omega}-\hat{\omega}\wedge\hat{\omega}=\left[\frac{\Omega^g+\alpha\wedge\alpha^T}{\circ}\middle \frac{-D\alpha}{\stackrel{\circ}{\Omega}+\alpha^T\wedge\alpha}\right].$	$\hat{\Omega}=\hat{d}\hat{\omega}-\hat{\omega}-\hat{\omega}\wedge\hat{\omega}=\left[\frac{\Omega^g+\alpha\wedge\alpha^T}{(\stackrel{\circ}{D}\alpha)^T}\middle \frac{-D\alpha}{\stackrel{\circ}{\Omega}+\alpha^T\wedge\alpha}\right].$
mielke-geometrodynamics_E01016 (14.4.11)	327	■ 0.758 ● N +0	$ds^2=-\left(1-\left(\frac{r_{\circ}}{r}\right)^4\right)\frac{\Delta}{r^2}dt^2+\frac{r^2}{\Delta}dr^2+r^2d\Omega^2.$	$ds^2=-\left(1-\left(\frac{r_{\circ}}{r}\right)^4\right)\frac{\Delta}{r^2}dt^2+\frac{r^2}{\Delta}dr^2+r^2d\Omega^2.$	$ds^2=-\left(1-\left(\frac{r_{\circ}}{r}\right)^4\right)\frac{\Delta}{r^2}dt^2+\frac{r^2}{\Delta}dr^2+r^2d\Omega^2.$
mielke-geometrodynamics_E08310 (4.6.2)	104	■ 0.760 ● W +2	$\begin{aligned} & \widetilde{D}\widetilde{\xi}^{\alpha}=d\xi^{\alpha}+\Gamma_{\beta}^{(L)\alpha}\wedge\xi^{\beta}+\Gamma^{(T)\alpha} \\ & =D\xi^{\alpha}+\Gamma^{(T)\alpha}=0 \end{aligned}$	$\begin{aligned} & \widetilde{D}\widetilde{\xi}^{\alpha}=d\xi^{\alpha}+\Gamma_{\beta}^{(L)\alpha}\wedge\xi^{\beta}+\Gamma^{(T)\alpha} \\ & =D\xi^{\alpha}+\Gamma^{(T)\alpha}=0 \end{aligned}$	$\begin{aligned} & \widetilde{D}\widetilde{\xi}^{\alpha}=d\xi^{\alpha}+\Gamma_{\beta}^{(L)\alpha}\wedge\xi^{\beta}+\Gamma^{(T)\alpha} \\ & =D\xi^{\alpha}+\Gamma^{(T)\alpha}=0. \end{aligned}$
mielke-geometrodynamics_E08832 (12.1.4)	270	■ 0.763 ● N +0	$\left\langle d\widetilde{j}_5\right\rangle =0,$	$\left\langle d\widetilde{j}_5\right\rangle =0,$	$\langle d\widetilde{j}_5\rangle=0,$
mielke-geometrodynamics_E08231 (4.2.24)	088	■ 0.766 ● N +2	$T_{\alpha\beta}^{\cdot\cdot c}=2D_{[\alpha}E_{\cdot \beta]}^c=2\left(\partial_{[\alpha}E_{\cdot \beta]}^c+\Gamma_{[\alpha d}^{\cdot c}E_{\cdot \beta]}^d\right).$	$T_{\alpha\beta}^{\cdot\cdot c}=2D_{[\alpha}E_{\cdot \beta]}^c=2\left(\partial_{[\alpha}E_{\cdot \beta]}^c+\Gamma_{[\alpha d}^{\cdot c}E_{\cdot \beta]}^d\right).$	$T_{\alpha\beta}^{\cdot\cdot c}=2D_{[\alpha}E_{\cdot \beta]}^c=2\left(\partial_{[\alpha}E_{\cdot \beta]}^c+\Gamma_{[\alpha d}^{\cdot c}E_{\cdot \beta]}^d\right).$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_01079 (A.4)	365	0.766 N +0	$\left[\sigma^i, \sigma^j\right]=\varepsilon^i{}_j \sigma^k$	$\left[\sigma^i, \sigma^j\right]=\varepsilon^{ij}{}_k \sigma^k$	$\left[\sigma^i, \sigma^j\right]=\varepsilon^{ij}{}_k \sigma^k$
mielke-geometrodynamics_009276 (4.4.4)	097	0.767 W +0	$\begin{aligned} & \delta L_g = \delta \vartheta \wedge E - \delta \vartheta \wedge D \Pi^T + d \left(\delta \vartheta \wedge \Pi^T \right) \\ & - \left[\delta \omega^g, \vartheta \right] \wedge \left[\Pi^T_{\square}, \vartheta \right] - \delta \omega^g \wedge D \Pi^L + d \left(\delta \omega^g \wedge \Pi^L \right), \end{aligned}$	$\delta L_g = \delta \vartheta \wedge E - \delta \vartheta \wedge D \Pi^T + d \left(\delta \vartheta \wedge \Pi^T \right) - \left[\delta \omega^g, \vartheta \right] \wedge \left[\Pi^T_{\square}, \vartheta \right] - \delta \omega^g \wedge D \Pi^L + d \left(\delta \omega^g \wedge \Pi^L \right),$	$\delta L_g = \delta \vartheta \wedge E - \delta \vartheta \wedge D \Pi^T + d(\delta \vartheta \wedge \Pi^T) - [\delta \omega^g, \vartheta] \wedge [\Pi^T_{\square}, \vartheta] - \delta \omega^g \wedge D \Pi^L + d(\delta \omega^g \wedge \Pi^L),$
mielke-geometrodynamics_008613 (10.1.1)	209	0.769 N +0	$\vartheta^{\alpha}=E_i^{\alpha}dx^i \quad \text{and} \quad \Gamma_{\alpha}^{\star}:=\frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}.$	$\vartheta^{\alpha}=E_i^{\alpha}dx^i \quad \text{and} \quad \Gamma_{\alpha}^{\star}:=\frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}.$	$\vartheta^{\alpha}=E_i^{\alpha}dx^i \quad \text{and} \quad \Gamma_{\alpha}^{\star}:=\frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}.$
mielke-geometrodynamics_01066 (16.2.5)	355	0.770 C -3	$\rho^*=\int_o^{\rho}\frac{dx}{1\pm x^2}=\left\{\begin{array}{l}\operatorname{artan}\rho,\kappa>0\\\operatorname{Arth}\rho,\kappa<0\end{array}\right.$	$\rho^*=\int_o^{\rho}\frac{dx}{1\pm x^2}=\left\{\begin{array}{l}\operatorname{artan}\rho,\kappa>0\\\operatorname{Arth}\rho,\kappa<0\end{array}\right.$	$\rho^*=\int_o^{\rho}\frac{dx}{1\pm x^2}=\left\{\begin{array}{l}\operatorname{artan}\rho,\kappa>0\\\operatorname{Arth}\rho,\kappa<0\end{array}\right).$
mielke-geometrodynamics_01030 (15.1.6)	337	0.771 W -1	$\begin{aligned} & \frac{1}{2} \vartheta^g \wedge^* \Omega^g + \Lambda^* \vartheta_g = \ell^{*2} \left\{ \Sigma_{\mathrm{m}}(\mathrm{leptons}) + \Sigma^{\mathrm{g}} \right\}, \\ & \frac{1}{2} \vartheta^f \wedge^* \Omega^f = \ell^2 \left\{ \Sigma_{\mathrm{m}}(\mathrm{hadrons}) + \Sigma^{\mathrm{f}} \right\}. \end{aligned}$	$\frac{1}{2} \vartheta^g \wedge^* \Omega^g + \Lambda^* \vartheta_g = \ell^{*2} \left\{ \Sigma_{\mathrm{m}}(\mathrm{leptons}) + \Sigma^{\mathrm{g}} \right\},$ $\frac{1}{2} \vartheta^f \wedge^* \Omega^f = \ell^2 \left\{ \Sigma_{\mathrm{m}}(\mathrm{hadrons}) + \Sigma^{\mathrm{f}} \right\}.$	$\frac{1}{2} \vartheta^g \wedge^* \Omega^g + \Lambda^* \vartheta_g = \ell^{*2} \left\{ \Sigma_{\mathrm{m}}(\mathrm{leptons}) + \Sigma^{\mathrm{g}} \right\},$ $\frac{1}{2} \vartheta^f \wedge^* \Omega^f = \ell^2 \left\{ \Sigma_{\mathrm{m}}(\mathrm{hadrons}) + \Sigma^{\mathrm{f}} \right\}.$
mielke-geometrodynamics_00112 (3.3.13)	059	0.772 W -2	$\begin{aligned} j:= & \mathscr{J} \wedge dt + \hat{\rho} \\ = & -(J_x dy \wedge dz + J_y dz \wedge dx + J_z dx \wedge dy) \wedge dt \\ & + \rho dx \wedge dy \wedge dz. \end{aligned}$	$j:= -\mathscr{J} \wedge dt + \hat{\rho} = -(J_x dy \wedge dz + J_y dz \wedge dx + J_z dx \wedge dy) \wedge dt + \rho dx \wedge dy \wedge dz$	$j:= -\mathscr{J} \wedge dt + \hat{\rho} = -(J_x dy \wedge dz + J_y dz \wedge dx + J_z dx \wedge dy) \wedge dt + \rho dx \wedge dy \wedge dz.$
mielke-geometrodynamics_00130 (3.4.1)	062	0.780 N -1	$\mathscr{F}:=\frac{1}{\alpha_g}\operatorname{Tr}\left\{^*\left(\Omega\wedge^*\Omega\right)\right\}=\frac{1}{4\alpha_g}F_{\alpha\beta}{}^jF^{\alpha\beta}{}_j$	$\mathscr{F}:=\frac{1}{\alpha_g}\operatorname{Tr}\left\{^*(\Omega\wedge^*\Omega)\right\}=\frac{1}{4\alpha_g}F_{\alpha\beta}{}^jF^{\alpha\beta}{}_j$	$\mathscr{F}:=\frac{1}{\alpha_g}\operatorname{Tr}\left\{^*(\Omega\wedge^*\Omega)\right\}=\frac{1}{4\alpha_g}F_{\alpha\beta}{}^jF^{\alpha\beta}{}_j$
mielke-geometrodynamics_00470 (7.3.9)	155	0.782 N +0	$\{ \}^{\ast} \backslash \Omega ^{g(\ast)}=\zeta \backslash \Omega ^g,$	$^*\Omega^{g(*)}=\zeta\Omega^g,$	$^*\Omega^{g(*)}=\zeta\Omega^g,$
mielke-geometrodynamics_00846 (2.6.9)	043	0.783 N +0	$\begin{aligned} \Delta b &= \int_{\partial U} \omega \otimes b = \int_U d(\omega \otimes b)_{\mathrm{hor}} = \int_U D(\omega \otimes b) \\ &= \int_U \Omega \otimes b \simeq \Omega \otimes b \end{aligned}$	$\Delta b = \int_{\partial U} \omega \otimes b = \int_U d(\omega \otimes b)_{\mathrm{hor}} = \int_U D(\omega \otimes b) = \int_U \Omega \otimes b \simeq \Omega \otimes b$	$\Delta b = \int_{\partial U} \omega \otimes b = \int_U d(\omega \otimes b)_{\mathrm{hor}} = \int_U D(\omega \otimes b) = \int_U \Omega \otimes b \simeq \Omega \otimes b$
mielke-geometrodynamics_01049 (15.2.6)	344	0.785 N +1	$\{ \}^{\ast} \backslash \bar{\omega} \backslash \Omega = \bar{\omega} \backslash \Omega _{i}^{\ast} \backslash \bar{L} _{j} \otimes dx^i = \bar{\omega} _i dx^i$	$^*\bar{\omega}=\bar{\omega}_i{}^j\bar{L}_j\otimes dx^i=\bar{\omega}_idx^i$	$^*\bar{\omega}=\bar{\omega}_i{}^j\bar{L}_j\otimes dx^i=\bar{\omega}_idx^i$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_063 EQ0133 (3.4.4)		■ 0.785 ● N -2	$L_{\omega} = L(\mathscr{F}, {}^*\mathscr{F})\eta.$		$L_{\omega} = L(\mathscr{F}, {}^*\mathscr{F})\eta.$
mielke-geometrodynamics_104 EQ0312 (4.6.4)		■ 0.786 ● W +0	$\begin{aligned} \Delta \xi^\alpha &= - \oint_C y \rfloor (DD\xi^\alpha) = \oint_C y \rfloor R^{(T)\alpha} = \int_S R^{(T)\alpha} \\ &\simeq \frac{1}{2} (T_{ij}^\alpha - R_{ij\beta}^\alpha \xi^\beta) \int_S dy^i \wedge dy^j \end{aligned}$	$\Delta \xi^\alpha = - \oint_C y \rfloor (DD\xi^\alpha) = \oint_C y \rfloor R^{(T)\alpha} = \int_S R^{(T)\alpha} \\ \simeq \frac{1}{2} (T_{ij}^\alpha - R_{ij\beta}^\alpha \xi^\beta) \int_S dy^i \wedge dy^j$	$\Delta \xi^\alpha = - \oint_C y \rfloor (DD\xi^\alpha) = \oint_C y \rfloor R^{(T)\alpha} = \int_S R^{(T)\alpha} \\ \simeq \frac{1}{2} (T_{ij}^\alpha - R_{ij\beta}^\alpha \xi^\beta) \int_S dy^i \wedge dy^j .$
mielke-geometrodynamics_087 EQ0224 (4.2.17)		■ 0.787 ● N +0	$\widetilde{\varphi} \rightarrow A^{-1} \widetilde{\varphi} = G^{-1} (\widetilde{\varphi} - T).$		$\widetilde{\varphi} \rightarrow A^{-1} \widetilde{\varphi} = G^{-1} (\widetilde{\varphi} - T).$
mielke-geometrodynamics_306 EQ0941 (14.1.22)		■ 0.788 ● N +0	$\tan 2\delta = (-1)^z \frac{\mathrm{Tr}^*(\Omega \wedge \Omega)}{\mathrm{Tr}^*(\Omega \wedge {}^*\Omega)}.$		$\tan 2\delta = (-1)^z \frac{\mathrm{Tr}^*(\Omega \wedge \Omega)}{\mathrm{Tr}^*(\Omega \wedge {}^*\Omega)}.$
mielke-geometrodynamics_242 EQ0760 (11.1.27)		■ 0.791 ● N +0	$\left(\gamma^{n+1}\right)^2=\mp \mathbb{1}.$	$\left(\gamma^{n+1}\right)^2=\mp \mathbb{1}.$	$(\gamma^{n+1})^2=\mp \mathbb{1}.$
mielke-geometrodynamics_230 EQ0725 (10.10.5)		■ 0.793 ● N +0	$\delta_{\mathrm{susy}} L \cong \bar{\sigma} \left(i \gamma^\alpha \Psi \wedge \frac{\delta L}{\delta \vartheta^\alpha} - 2 D \frac{\delta L}{\delta \bar{\Psi}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) + 2 d \left(\bar{\sigma} \wedge \frac{\delta L}{\delta \bar{\Psi}} \right).$		$\delta_{\mathrm{susy}} L \cong \bar{\sigma} \left(i \gamma^\alpha \Psi \wedge \frac{\delta L}{\delta \vartheta^\alpha} - 2 D \frac{\delta L}{\delta \bar{\Psi}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) + 2 d \left(\bar{\sigma} \wedge \frac{\delta L}{\delta \bar{\Psi}} \right).$
mielke-geometrodynamics_256 EQ0810 (11.4.22)		■ 0.793 ● N +0	$\frac{3}{8} \ell^2 \bar{\psi} \psi = 4 \pi m e^{-\mu - \nu / 2} \left(H^2 - F^2 \right) \left Y_{ \kappa - 1}^{ \kappa - 1} (\vartheta, \phi) \right ^2$		$\frac{3}{8} \ell^2 \bar{\psi} \psi = 4 \pi m e^{-\mu - \nu / 2} (H^2 - F^2) \left Y_{ \kappa - 1}^{ \kappa - 1} (\vartheta, \phi) \right ^2$
mielke-geometrodynamics_338 EQ1032 (15.1.9)		■ 0.793 ● N +0	$\frac{1}{2} * (\vartheta^g \wedge {}^* \Omega^g) + \Lambda \vartheta^g = 0,$		$\frac{1}{2} * (\vartheta^g \wedge {}^* \Omega^g) + \Lambda \vartheta^g = 0,$
mielke-geometrodynamics_103 EQ0305 (4.5.15)		■ 0.793 ● U +12	$\mathrm{varepsilon}_{\mathrm{RC}} := \xi^\alpha \Sigma_\alpha + (e_\beta \rfloor \widehat{D} \xi^\gamma) \tau^\beta{}_\gamma,$	$(not\ rendered)$	$\varepsilon_{\mathrm{RC}} := \xi^\alpha \Sigma_\alpha + (e_\beta \rfloor \widehat{D} \xi^\gamma) \tau^\beta{}_\gamma,$
mielke-geometrodynamics_274 EQ0848 (12.3.1)		■ 0.794 ● C +5	$\langle dj_5 \rangle = 2im \langle \bar{\psi} \gamma_5 \psi \rangle \eta - \frac{1}{4\pi^2} \mathrm{Tr}(G \wedge G) - \frac{1}{96\pi^2} \left[2R_{\alpha\beta}^{\{\} } \wedge R^{\{\beta\alpha\}} + d\mathscr{A} \wedge d\mathscr{A} \right]$	$\langle dj_5 \rangle = 2im \langle \bar{\psi} \gamma_5 \psi \rangle \eta - \frac{1}{4\pi^2} \mathrm{Tr}(G \wedge G) - \frac{1}{96\pi^2} \left[2R_{\alpha\beta}^{\{\} } \wedge R^{\{\beta\alpha\}} + d\mathscr{A} \wedge d\mathscr{A} \right]$	$\langle dj_5 \rangle = 2im \langle \bar{\psi} \gamma_5 \psi \rangle \eta - \frac{1}{4\pi^2} Tr(G \wedge G) - \frac{1}{96\pi^2} \left[2R_{\alpha\beta}^{\{\} } \wedge R^{\{\}\alpha\beta} + d\mathscr{A} \wedge d\mathscr{A} \right]$
mielke-geometrodynamics_110 EQ0319 (5.2.3)		■ 0.795 ● N +1	$L_{\mathrm{D}} = L\left(\gamma, \psi, D^{\{\} }\psi\right) - \frac{i}{2} \bar{\psi} \left({}^*\gamma \wedge K - K \wedge {}^*\gamma\right) \psi \\ = L\left(\gamma, \psi, D^{\mathfrak{l} }\psi\right) + \frac{1}{4} \mathscr{A} \wedge j_5.$		$L_{\mathrm{D}} = L(\gamma, \psi, D^{\{\} }\psi) - \frac{i}{2} \bar{\psi} \left({}^*\gamma \wedge K - K \wedge {}^*\gamma\right) \psi \\ = L(\gamma, \psi, D^{\{\} }\psi) + \frac{1}{4} \mathscr{A} \wedge j_5.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_202 EQ0610 (9.3.1)		■ 0.806 ● N +0	$\begin{aligned} d\,C_{\mathrm{RR}^{\{\star\}}} &:= \\ &\frac{1}{2}d\left(\Gamma_{\alpha\beta}\wedge R^{\alpha\beta(\star)}-\frac{1}{3}\Gamma_{\alpha}{}^{\beta(\star)}\wedge\Gamma_{\beta}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\alpha}\right) \\ &\equiv -L_{\mathrm{SKY}}-2\operatorname{Ric}_{\alpha\beta}\wedge{}^*\operatorname{Ric}^{\alpha\beta}+\frac{1}{2}\operatorname{Ric}_{\alpha}{}^{\alpha}\wedge{}^*\operatorname{Ric}_{\beta}{}^{\beta} \end{aligned}$	$dC_{\mathrm{RR}^{(*)}}:=\frac{1}{2}d\left(\Gamma_{\alpha\beta}\wedge R^{\alpha\beta(*)}-\frac{1}{3}\Gamma_{\alpha}{}^{\beta(*)}\wedge\Gamma_{\beta}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\alpha}\right)$ $\equiv -L_{\mathrm{SKY}}-2\operatorname{Ric}_{\alpha\beta}\wedge{}^*\operatorname{Ric}^{\alpha\beta}+\frac{1}{2}\operatorname{Ric}_{\alpha}{}^{\alpha}\wedge{}^*\operatorname{Ric}_{\beta}{}^{\beta}$	$dC_{\mathrm{RR}^{(*)}}:=\frac{1}{2}d\left(\Gamma_{\alpha\beta}\wedge R^{\alpha\beta(*)}-\frac{1}{3}\Gamma_{\alpha}{}^{\beta(*)}\wedge\Gamma_{\beta}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\alpha}\right)$ $\equiv -L_{\mathrm{SKY}}-2\operatorname{Ric}_{\alpha\beta}\wedge{}^*\operatorname{Ric}^{\alpha\beta}+\frac{1}{2}\operatorname{Ric}_{\alpha}{}^{\alpha}\wedge{}^*\operatorname{Ric}_{\beta}{}^{\beta}$
mielke-geometrodynamics_244 EQ0767 (11.1.34)		■ 0.807 ● N +0	$\psi\longrightarrow S^{-1}\psi=S^{-1}\psi$	$\psi\longrightarrow S^{-1}\psi=S^{-1}\psi$	$\psi\longrightarrow S^{-1}\psi=S^{-1}\psi$
mielke-geometrodynamics_095 EQ0267 (4.3.32)		■ 0.808 ● W +2	$\begin{aligned} &\&R_{\{i\,j\}\,\kappa l}^{\{\, \, \}}=R_{i\,j}(\, \, \kappa l)^{\{\, \, \}}=0\,\,\&\&R_{i\,j\,\kappa l}^{\{\, \, \}}=R_{\kappa l i j}^{\}\, . \end{aligned}$	$R_{(ij)\kappa l}^{\{\, \, \}}=R_{ij(\kappa l)}^{\{\, \, \}}=0$ $R_{ij\kappa l}^{\}\,=R_{\kappa l i j}^{\}\,.$	$R_{(ij)\kappa l}^{\{\, \, \}}=R_{ij(\kappa l)}^{\{\, \, \}}=0$ $R_{ij\kappa l}^{\{\, \, \}}=R_{\kappa l i j}^{\{\, \, \}}.$
mielke-geometrodynamics_101 EQ0298 (4.5.8)		■ 0.811 ● N +1	$\begin{aligned} \delta L &= \delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\alpha}} \\ &+ \delta\Gamma_{\alpha}{}^{\beta}\wedge I^{\alpha}{}_{\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}+\delta\Psi\wedge\frac{\delta L}{\delta\Psi}+D\left(\delta\Psi\wedge\frac{\partial L}{\partial D\Psi}\right) . \end{aligned}$	$\delta L = \delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\alpha}}$ $+ \delta\Gamma_{\alpha}{}^{\beta}\wedge I^{\alpha}{}_{\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}+\delta\Psi\wedge\frac{\delta L}{\delta\Psi}+D\left(\delta\Psi\wedge\frac{\partial L}{\partial D\Psi}\right)$	$\delta L = \delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\alpha}}$ $+ \delta\Gamma_{\alpha}{}^{\beta}\wedge I^{\alpha}{}_{\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}+\delta\Psi\wedge\frac{\delta L}{\delta\Psi}+D\left(\delta\Psi\wedge\frac{\partial L}{\partial D\Psi}\right).$
mielke-geometrodynamics_295 EQ0919 (13.8.5)		■ 0.811 ● N +0	$D\,T^{\{\alpha\}}\equiv R_{\{\beta\}\{\, \, \}}^{\{\, \, \}}\wedge\vartheta^{\beta}$	$DT^{\alpha}\equiv R_{\beta}{}^{\alpha}\wedge\vartheta^{\beta}$	$DT^{\alpha}\equiv R_{\beta}{}^{\alpha}\wedge\vartheta^{\beta}$
mielke-geometrodynamics_254 EQ0802 (11.4.14)		■ 0.812 ● W +0	$\begin{aligned} &\&\left[i\,\gamma^0\partial_0-e^{(\nu-\mu)/2}\left[i\vec{\gamma}\cdot\vec{\partial}+i\gamma^1\partial_r\left(\frac{\mu}{2}+\frac{\nu}{4}\right)\right]\right. \\ &\left.+\frac{3\varepsilon}{8}\ell^2e^{\nu/2}\left(\bar{\psi}\psi-\mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5\right)-e^{\nu/2}m\right]\psi=0 \end{aligned}$	$\left\{i\gamma^0\partial_0-e^{(\nu-\mu)/2}\left[i\vec{\gamma}\cdot\vec{\partial}+i\gamma^1\partial_r\left(\frac{\mu}{2}+\frac{\nu}{4}\right)\right]\right. \\ \left.+\frac{3\varepsilon}{8}\ell^2e^{\nu/2}\left(\bar{\psi}\psi-\mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5\right)-e^{\nu/2}m\right\}\psi=0$	$\left\{i\gamma^0\partial_0-e^{(\nu-\mu)/2}\left[i\vec{\gamma}\cdot\vec{\partial}+i\gamma^1\partial_r\left(\frac{\mu}{2}+\frac{\nu}{4}\right)\right]\right. \\ \left.+\frac{3\varepsilon}{8}\ell^2e^{\nu/2}(\bar{\psi}\psi-\mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5)-e^{\nu/2}m\right\}\psi=0$
mielke-geometrodynamics_255 EQ0807 (11.4.19)		■ 0.812 ● N +0	$\vec{\sigma}\cdot\vec{L}\chi_{\kappa}^m=-(\kappa+1)\chi_{\kappa}^m.$	$\vec{\sigma}\cdot\vec{L}\chi_{\kappa}^m=-(\kappa+1)\chi_{\kappa}^m.$	$\vec{\sigma}\cdot\vec{L}\chi_{\kappa}^m=-(\kappa+1)\chi_{\kappa}^m.$
mielke-geometrodynamics_132 EQ0397 (6.4.12)		■ 0.813 ● N +0	$\mathscr{H}\cong n^{\alpha}\mathscr{G}_{\alpha}+\Gamma_{\perp}{}^{\alpha\beta}\mathscr{G}_{\alpha\beta},$	$\mathscr{H}\cong n^{\alpha}\mathscr{G}_{\alpha}+\Gamma_{\perp}{}^{\alpha\beta}\mathscr{G}_{\alpha\beta},$	$\mathscr{H}\cong n^{\alpha}\mathscr{G}_{\alpha}+\Gamma_{\perp}{}^{\alpha\beta}\mathscr{G}_{\alpha\beta},$
mielke-geometrodynamics_227 EQ0709 (10.7.5)		■ 0.813 ● N +0	$R_{\{\beta\}\{\, \, \}}^{\{\, \, \}}\wedge\vartheta^{\beta}=\frac{\ell^2}{2}\varepsilon(x^{\alpha}Y\wedge X\wedge Y-y^{\alpha}X\wedge X\wedge Y)=0.$	$R_{\beta}{}^{\alpha}\wedge\vartheta^{\beta}=\frac{\ell^2}{2}\varepsilon(x^{\alpha}Y\wedge X\wedge Y-y^{\alpha}X\wedge X\wedge Y)=0.$	$R_{\beta}{}^{\alpha}\wedge\vartheta^{\beta}=\frac{\ell^2}{2}\varepsilon(x^{\alpha}Y\wedge X\wedge Y-y^{\alpha}X\wedge X\wedge Y)=0.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_085 EQ0210 (4.2.2)		0.814 N +1	$\stackrel{*}{\approx}\chi=\left[\begin{array}{c c}\omega^{\mathrm{L}}&\omega^{\mathrm{T}}\\ \hline 0&0\end{array}\right].$		$\overset{*}{\chi}\omega=\left[\begin{array}{c c}\omega^{\mathrm{L}}&\omega^{\mathrm{T}}\\ \hline 0&0\end{array}\right].$
mielke-geometrodynamics_244 EQ0769 (11.1.36)		0.817 N +0	$\widetilde{\omega}\longrightarrow{}^S\widetilde{\omega}=S\widetilde{\omega}S^{-1}+(dS)S^{-1}$		$\widetilde{\omega}\longrightarrow{}^S\widetilde{\omega}=S\widetilde{\omega}S^{-1}+(dS)S^{-1}$
mielke-geometrodynamics_133 EQ0398 (6.4.14)		0.820 W +0	$\begin{aligned} &\left(\mathscr{G}_\alpha(t,\mathbf{x}),\mathscr{G}_\beta(t,\mathbf{y})\right)=\left(-T_{\alpha\beta}{}^\gamma\mathscr{G}_\gamma+R_{\alpha\beta\gamma}{}^\delta\mathscr{G}^\gamma{}_\delta\right)\cdot\delta(\mathbf{x}-\mathbf{y}),\\ &\left(\mathscr{G}^\alpha{}_\beta(t,\mathbf{x}),\mathscr{G}_\gamma(t,\mathbf{y})\right)=\delta^\gamma{}_\beta\mathscr{G}_\beta\delta(\mathbf{x}-\mathbf{y}),\\ &\left(\mathscr{G}^\alpha{}_\beta(t,\mathbf{x}),\mathscr{G}^\gamma{}_\delta(t,\mathbf{y})\right)=\left(\delta^\alpha{}_\delta\mathscr{G}^\gamma{}_\beta-\delta^\gamma{}_\beta\mathscr{G}^\alpha{}_\delta\right)\cdot\delta(\mathbf{x}-\mathbf{y}). \end{aligned}$		$\begin{aligned} &\left(\mathscr{G}_\alpha(t,\mathbf{x}),\mathscr{G}_\beta(t,\mathbf{y})\right)=\left(-T_{\alpha\beta}{}^\gamma\mathscr{G}_\gamma+R_{\alpha\beta\gamma}{}^\delta\mathscr{G}^\gamma{}_\delta\right)\cdot\delta(\mathbf{x}-\mathbf{y}),\\ &\left(\mathscr{G}^\alpha{}_\beta(t,\mathbf{x}),\mathscr{G}_\gamma(t,\mathbf{y})\right)=\delta^\gamma{}_\beta\mathscr{G}_\beta\delta(\mathbf{x}-\mathbf{y}),\\ &\left(\mathscr{G}^\alpha{}_\beta(t,\mathbf{x}),\mathscr{G}^\gamma{}_\delta(t,\mathbf{y})\right)=\left(\delta^\alpha{}_\delta\mathscr{G}^\gamma{}_\beta-\delta^\gamma{}_\beta\mathscr{G}^\alpha{}_\delta\right)\cdot\delta(\mathbf{x}-\mathbf{y}). \end{aligned}$
mielke-geometrodynamics_045 EQ0059 (2.7.9)		0.820 K +0	$\bar{\varphi}\rightarrow{}^{G^{-1}}\bar{\varphi}:=\bar{\varphi}G\quad\in C^\infty\left(V^{\bar{s}}\right).$		$\bar{\varphi}\rightarrow{}^{G^{-1}}\bar{\varphi}:=\bar{\varphi}G\quad\in C^\infty\left(V^{\bar{s}}\right).$
mielke-geometrodynamics_320 EQ0993 (14.3.6)		0.822 W +0	$\int_G\pi\hat{L}_{\mathrm{D.}}=\frac{i}{2}\left(\bar{\psi}\gamma\wedge{}^*D^{g+\omega}\psi+\left(\overline{D^{g+\omega}\psi}\right)\wedge{}^*\gamma\psi\right)+\frac{\hat{\ell}}{2}\bar{\psi}\gamma\wedge\gamma\wedge{}^*\Omega\psi-\widehat{m}\bar{\psi}\psi\eta,$		$\int_G\pi\hat{L}_{\mathrm{D.}}=\frac{i}{2}\left(\bar{\psi}\gamma\wedge{}^*D^{g+\omega}\psi+\left(\overline{D^{g+\omega}\psi}\right)\wedge{}^*\gamma\psi\right)+\frac{\hat{\ell}}{2}\bar{\psi}\gamma\wedge\gamma\wedge{}^*\Omega\psi-\widehat{m}\bar{\psi}\psi\eta,$
mielke-geometrodynamics_273 EQ0844 (12.2.12)		0.831 N +0	$\mathrm{L}=\frac{\mathrm{i}}{2\ell^2}\mathrm{Tr}\left(\Omega\wedge{}^*\sigma\right)+\mathrm{L}_{\mathrm{D}}=\frac{1}{2\ell^2}\mathbf{R}^{\alpha\beta}\wedge\eta_{\alpha\beta}+\mathrm{L}_{\mathrm{D}},$		$\mathrm{L}=\frac{\mathrm{i}}{2\ell^2}\mathrm{Tr}\left(\Omega\wedge{}^*\sigma\right)+\mathrm{L}_{\mathrm{D}}=\frac{1}{2\ell^2}\mathbf{R}^{\alpha\beta}\wedge\eta_{\alpha\beta}+\mathrm{L}_{\mathrm{D}},$
mielke-geometrodynamics_086 EQ0218 (4.2.10)		0.836 N +0	$\begin{aligned} &\omega^{\mathrm{L}}\rightarrow{}^{A^{-1}}\omega^{\mathrm{L}}=G^{-1}\omega^{\mathrm{L}}G-G^{-1}dG,\\ &\omega^{\mathrm{T}}\rightarrow{}^{A^{-1}}\omega^{\mathrm{T}}=G^{-1}\omega^{\mathrm{T}}-G^{-1}DT. \end{aligned}$		$\begin{aligned} &\omega^{\mathrm{L}}\rightarrow{}^{A^{-1}}\omega^{\mathrm{L}}=G^{-1}\omega^{\mathrm{L}}G-G^{-1}dG,\\ &\omega^{\mathrm{T}}\rightarrow{}^{A^{-1}}\omega^{\mathrm{T}}=G^{-1}\omega^{\mathrm{T}}-G^{-1}DT. \end{aligned}$
mielke-geometrodynamics_090 EQ0240 (4.3.5)		0.836 S -1	$F(m,f\vartheta)= f F(m,\vartheta),\\ F(m,\vartheta)>0\quad\text{if}\quad\vartheta\neq0.$		$F(m,f\vartheta)= f F(m,\vartheta),\\ F(m,\vartheta)>0\quad\text{if}\quad\vartheta\neq0.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_008445 (6.9.1)	144	0.837 N +0	$ds^2 = -\left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right)^{-1} dr^2 + r^2 d\Omega^2,$	$ds^2 = -\left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2}\right)^{-1} dr^2 + r^2 d\Omega^2,$	
mielke-geometrodynamics_008227 (4.2.20)	088	0.839 N +1	$\widetilde{\vartheta} \rightarrow^{A^{-1}} \widetilde{\vartheta} = G^{-1} \widetilde{\vartheta}.$	$\widetilde{\vartheta} \rightarrow^{A^{-1}} \widetilde{\vartheta} = G^{-1} \widetilde{\vartheta}.$	
mielke-geometrodynamics_00773 (11.2.2)	245	0.840 N +0	$i\gamma \wedge {}^* D\psi - \frac{i}{2} (D \wedge {}^* \gamma) \psi - m\psi \eta = 0$	$i\gamma \wedge {}^* D\psi - \frac{i}{2} (D \wedge {}^* \gamma) \psi - m\psi \eta = 0$	$i\gamma \wedge {}^* D\psi - \frac{i}{2} (D \wedge {}^* \gamma) \psi - m\psi \eta = 0$
mielke-geometrodynamics_008811 (11.4.23)	256	0.840 N +0	$\frac{3}{8} \ell^2 \bar{\psi} \gamma^5 \psi = 4\pi m e^{-\mu - v/2} \frac{2}{3} H F \left Y_{ \kappa -1}^{ \kappa -1}(\vartheta, \phi) \right ^2$	$\frac{3}{8} \ell^2 \bar{\psi} \gamma^5 \psi = 4\pi m e^{-\mu - v/2} \frac{2}{3} H F \left Y_{ \kappa -1}^{ \kappa -1}(\vartheta, \phi) \right ^2$	$\frac{3}{8} \ell^2 \bar{\psi} \gamma^5 \psi = 4\pi m e^{-\mu - v/2} \frac{2}{3} H F \left Y_{ \kappa -1}^{ \kappa -1}(\vartheta, \phi) \right ^2$
mielke-geometrodynamics_00584 (9.1.2)	192	0.843 N +0	$\begin{aligned} e^{\{\nu\}} &= e^{\{-\lambda\}} \ \& \ = 1 - \left(\frac{2}{r} \right)^{n-3} - \frac{2\Lambda_{\text{eff.}}}{(n-1)(n-2)} r^2 \\ \mu &= \frac{M}{m^{*2}}; \quad d\Omega^2 := d\vartheta^2 + \sin^2 \vartheta d\phi^2 \end{aligned}$	$e^\nu = e^{-\lambda} = 1 - \left(\frac{2\mu}{r} \right)^{n-3} - \frac{2\Lambda_{\text{eff.}}}{(n-1)(n-2)} r^2$ $\mu = \frac{M}{m^{*2}}; \quad d\Omega^2 := d\vartheta^2 + \sin^2 \vartheta d\phi^2$	$e^\nu = e^{-\lambda} = 1 - \left(\frac{2\mu}{r} \right)^{n-3} - \frac{2\Lambda_{\text{eff.}}}{(n-1)(n-2)} r^2$ $\mu = \frac{M}{m^{*2}}; \quad d\Omega^2 := d\vartheta^2 + \sin^2 \vartheta d\phi^2$
mielke-geometrodynamics_00349 (5.4.7)	117	0.845 N +0	$\left\langle dj_5 \right\rangle = 2im \left\langle \bar{\psi} \gamma_5 \psi \right\rangle \eta - \frac{1}{96\pi^2} \left[2R_{\alpha\beta}^{\{\}} \wedge R^{\{\beta\alpha\beta} + d\mathscr{A} \wedge d\mathscr{A} \right]$	$\langle dj_5 \rangle = 2im \langle \bar{\psi} \gamma_5 \psi \rangle \eta - \frac{1}{96\pi^2} \left[2R_{\alpha\beta}^{\{\}} \wedge R^{\{\beta\alpha\beta} + d\mathscr{A} \wedge d\mathscr{A} \right]$	$\langle dj_5 \rangle = 2im \langle \bar{\psi} \gamma_5 \psi \rangle \eta - \frac{1}{96\pi^2} \left[2R_{\alpha\beta}^{\{\}} \wedge R^{\{\beta\alpha\beta} + d\mathscr{A} \wedge d\mathscr{A} \right]$
mielke-geometrodynamics_00792 (11.4.4)	253	0.845 N +0	$ds^2 = g_{kl} dx^k \otimes_s dx^l = e^\nu dt^2 - e^\mu \left(dr^2 + r^2 d\vartheta^2 + r^2 \sin^2 \vartheta d\phi^2 \right).$	$ds^2 = g_{kl} dx^k \otimes_s dx^l = e^\nu dt^2 - e^\mu \left(dr^2 + r^2 d\vartheta^2 + r^2 \sin^2 \vartheta d\phi^2 \right).$	$ds^2 = g_{kl} dx^k \otimes_s dx^l = e^\nu dt^2 - e^\mu (dr^2 + r^2 d\vartheta^2 + r^2 \sin^2 \vartheta d\phi^2).$
mielke-geometrodynamics_00742 (11.1.9)	238	0.846 W +1	$\Lambda : \left\{ \begin{array}{ll} \widetilde{SO}_{\circ}(s,n-s) & \longrightarrow SO_{\circ}(s,n-s), \\ \widetilde{\Psi} & \longrightarrow g^{\alpha}_{\beta} = -\frac{i}{N} \text{Tr} \left(\gamma^{\alpha} \text{Si} \gamma_{\beta} S^{-1} \right), \end{array} \right.$	$\Lambda : \left\{ \begin{array}{ll} \widetilde{SO}_{\circ}(s,n-s) & \longrightarrow SO_{\circ}(s,n-s), \\ \widetilde{\Psi} & \longrightarrow g^{\alpha}_{\beta} = -\frac{i}{N} \text{Tr} (\gamma^{\alpha} Si \gamma_{\beta} S^{-1}), \end{array} \right.$	$\Lambda : \left\{ \begin{array}{ll} \widetilde{SO}_{\circ}(s,n-s) & \longrightarrow SO_{\circ}(s,n-s), \\ \widetilde{\Psi} & \longrightarrow g^{\alpha}_{\beta} = -\frac{i}{N} \text{Tr} (\gamma^{\alpha} Si \gamma_{\beta} S^{-1}), \end{array} \right.$
mielke-geometrodynamics_008897 (13.5.2)	288	0.846 W +1	$\hat{R} := R_A{}^B E^A{}_B := \left[d\Gamma_A{}^B - \Gamma_A{}^C \wedge \Gamma_C{}^B \right] E^A{}_B = \left[R_{\alpha}{}^{\beta} - \frac{\Lambda}{3} \theta_{\alpha} \wedge \vartheta^{\beta} \right] L^{\alpha}{}_{\beta} + T^{\beta} P_{\beta} + D\theta_{\alpha} P^{\alpha}_{*}$	$\hat{R} := R_A{}^B \not{E}^A{}_B := \left[d\Gamma_A{}^B - \Gamma_A{}^C \wedge \Gamma_C{}^B \right] \not{E}^A{}_B = \left[R_{\alpha}{}^{\beta} - \frac{\Lambda}{3} \theta_{\alpha} \wedge \vartheta^{\beta} \right] L^{\alpha}{}_{\beta} + T^{\beta} P_{\beta} + D\theta_{\alpha} P^{\alpha}_{*},$	$\hat{R} := R_A{}^B \not{E}^A{}_B := \left[d\Gamma_A{}^B - \Gamma_A{}^C \wedge \Gamma_C{}^B \right] \not{E}^A{}_B = \left[R_{\alpha}{}^{\beta} - \frac{\Lambda}{3} \theta_{\alpha} \wedge \vartheta^{\beta} \right] L^{\alpha}{}_{\beta} + T^{\beta} P_{\beta} + D\theta_{\alpha} P^{\alpha}_{*},$

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mielke-geometrodynamics_090296 (4.5.7)	101	0.847 N +0	$F_{\alpha}=\left(e_{\alpha}\downharpoonleft D\,\Psi\right)\frac{\delta L}{\delta\Psi}+(-1)^p(e_{\alpha}\downharpoonleft\Psi)\wedge D\frac{\delta L}{\delta\Psi}$	$F_{\alpha}=(e_{\alpha}\downharpoonleft D\Psi)\frac{\delta L}{\delta\Psi}+(-1)^p(e_{\alpha}\downharpoonleft\Psi)\wedge D\frac{\delta L}{\delta\Psi}$	$F_{\alpha}=(e_{\alpha}\downharpoonleft D\Psi)\frac{\delta L}{\delta\Psi}+(-1)^p(e_{\alpha}\downharpoonleft\Psi)\wedge D\frac{\delta L}{\delta\Psi}$
mielke-geometrodynamics_090241 (4.3.6)	091	0.848 N +0	$g_{\{\alpha\,\beta\}}(m,\,\vartheta):=\frac{1}{2}\,\frac{\partial^2}{\partial\vartheta^{\alpha}\otimes_s\partial\vartheta^{\beta}}F^{\{2\}}(m,\,\vartheta)\{\partial\vartheta^{\alpha}\otimes_s\partial\vartheta^{\beta}\}\,.$	$g_{\alpha\beta}(m,\vartheta):=\frac{1}{2}\frac{\partial^2F^2(m,\vartheta)}{\partial\vartheta^{\alpha}\otimes_s\partial\vartheta^{\beta}}.$	$g_{\alpha\beta}(m,\vartheta):=\frac{1}{2}\frac{\partial^2F^2(m,\vartheta)}{\partial\vartheta^{\alpha}\otimes_s\partial\vartheta^{\beta}}.$
mielke-geometrodynamics_090478 (7.4.6)	157	0.849 N -1	$\operatorname{Ric}_{\{\alpha\,\beta\}}:=\operatorname{Ric}_{\{\alpha\,\beta\}}-\frac{1}{4}\operatorname{Rg}_{\{\alpha\,\beta\}}=0\,.$	$\operatorname{Ric}_{\alpha\beta}:=\operatorname{Ric}_{\alpha\beta}-\frac{1}{4}\operatorname{Rg}_{\alpha\beta}=0.$	$R\hat{\jmath}\mathcal{Z}_{\alpha\beta}:=\operatorname{Ric}_{\alpha\beta}-\frac{1}{4}Rg_{\alpha\beta}=0.$
mielke-geometrodynamics_090295 (4.5.6)	101	0.850 N +0	$\begin{aligned} D\,\Sigma_{\{\alpha\}}\,\&\,\left.\left.\left.\left(e_{\{\alpha\}}\right)\right\rfloor T^{\{\beta\}}\right)\right\rfloor\wedge\Sigma_{\{\beta\}}+\left.\left.\left.\left(e_{\{\alpha\}}\right)\right\rfloor R^{\{\beta\gamma\}}\right)\right\rfloor\wedge\tau_{\beta\gamma}+F_{\alpha}\\ \&\,\left.\left.\left.\left(e_{\{\alpha\}}\right)\right\rfloor T^{\{\beta\}}\right)\right\rfloor\wedge\Sigma_{\beta}+\left.\left.\left.\left(e_{\{\alpha\}}\right)\right\rfloor R^{\{\beta\gamma\}}\right)\right\rfloor\wedge\tau_{\beta\gamma}. \end{aligned}$	$D\Sigma_{\alpha}=(e_{\alpha}\downharpoonleft T^{\beta})\wedge\Sigma_{\beta}+(e_{\alpha}\downharpoonleft R^{\beta\gamma})\wedge\tau_{\beta\gamma}+F_{\alpha}$ $\simeq(e_{\alpha}\downharpoonleft T^{\beta})\wedge\Sigma_{\beta}+(e_{\alpha}\downharpoonleft R^{\beta\gamma})\wedge\tau_{\beta\gamma}.$	$D\Sigma_{\alpha}=(e_{\alpha}\downharpoonleft T^{\beta})\wedge\Sigma_{\beta}+(e_{\alpha}\downharpoonleft R^{\beta\gamma})\wedge\tau_{\beta\gamma}+F_{\alpha}$ $\simeq(e_{\alpha}\downharpoonleft T^{\beta})\wedge\Sigma_{\beta}+(e_{\alpha}\downharpoonleft R^{\beta\gamma})\wedge\tau_{\beta\gamma}.$
mielke-geometrodynamics_090842 (12.2.10)	272	0.852 N +0	$\mu:=e_{\{\alpha\}}\downharpoonleft\mu^{\{\alpha\}}=\frac{3}{4}\ast j_5$	$\mu:=e_{\alpha}\downharpoonleft\mu^{\alpha}=\frac{3}{4}\ast j_5$	$\mu:=e_{\alpha}\downharpoonleft\mu^{\alpha}=\frac{3}{4}\ast j_5$
mielke-geometrodynamics_090506 (7.7.24)	164	0.852 K +0	$\gamma_5:=(i\,/\,4!)^{\ast}(\gamma\wedge\gamma\wedge\gamma\wedge\gamma)$	$\gamma_5:=(i/4!)^*(\gamma\wedge\gamma\wedge\gamma\wedge\gamma)$	$\gamma_5:=(i/4!)^*(\gamma\wedge\gamma\wedge\gamma\wedge\gamma)$
mielke-geometrodynamics_091014 (14.4.9)	326	0.856 N +0	$M^{\{2\}}=\left(M_{\mathrm{ir}}+\frac{\alpha_eM^{*2}}{4M_{\mathrm{ir}}}\operatorname{Tr}\bar{Q}^2\right)^2+\frac{1}{4}J(J+1)\frac{M^{*4}}{M_{\mathrm{ir}}^2},$	$M^2=\left(M_{\mathrm{ir}}+\frac{\alpha_eM^{*2}}{4M_{\mathrm{ir}}}\operatorname{Tr}\bar{Q}^2\right)^2+\frac{1}{4}J(J+1)\frac{M^{*4}}{M_{\mathrm{ir}}^2},$	$M^2=\left(M_{\mathrm{ir}}+\frac{\alpha_eM^{*2}}{4M_{\mathrm{ir}}}\operatorname{Tr}\bar{Q}^2\right)^2+\frac{1}{4}J(J+1)\frac{M^{*4}}{M_{\mathrm{ir}}^2},$
mielke-geometrodynamics_090753 (11.1.20)	241	0.859 U +17	$\hat{\Psi}_{\mid\overparen{S\,0}\circ(1,3)}=\left[\begin{array}{l}\Psi_1\\\Psi_2\\\vdots\\\Psi_{\lfloor\frac{n}{4}\rfloor}\end{array}\right].$	<i>(not rendered)</i>	$\hat{\Psi}_{\mid\overparen{S\,0}\circ(1,3)}=\left[\begin{array}{l}\Psi_1\\\Psi_2\\\vdots\\\Psi_{\lfloor\frac{n}{4}\rfloor}\end{array}\right].$
mielke-geometrodynamics_090783 (11.3.2)	247	0.862 N +0	$\begin{aligned} L_{\mathrm{D}}\,\&\,=\frac{i}{2}\left(\overline{\psi}\,\gamma\wedge^{\ast}D^{\{\}}\psi+\overline{\left(D^{\{\}}\psi\right)}\wedge^{\ast}\gamma\psi\right)-m\overline{\psi}\psi\eta\\ &+\frac{1}{2}\overline{\psi}\left[\widetilde{K},\ast\gamma\right]\psi=:L_{\mathrm{D}}^{\{\}}+L_{\mathrm{NL}}. \end{aligned}$	$L_{\mathrm{D}}=\frac{i}{2}\left(\bar{\psi}\gamma\wedge^{\ast}D^{\{\}}\psi+\overline{\left(D^{\{\}}\psi\right)}\wedge^{\ast}\gamma\psi\right)-m\bar{\psi}\psi\eta$ $+\frac{1}{2}\bar{\psi}\left[\widetilde{K},\ast\gamma\right]\psi=:L_{\mathrm{D}}^{\{\}}+L_{\mathrm{NL}}.$	$L_{\mathrm{D}}=\frac{i}{2}\left(\bar{\psi}\gamma\wedge^{\ast}D^{\{\}}\psi+\overline{\left(D^{\{\}}\psi\right)}\wedge^{\ast}\gamma\psi\right)-m\bar{\psi}\psi\eta$ $+\frac{1}{2}\bar{\psi}[\widetilde{K},\ast\gamma]\psi=:L_{\mathrm{D}}^{\{\}}+L_{\mathrm{NL}}.$

Conf. MathPix confidence:

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Residual render vs scan (inkdrill):

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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_01100 (B.12)	369	■ 0.864 ● N +0	$\begin{aligned} \{ \}^{\ast} \alpha^{\{p\}} &:= \frac{1}{(n-p)! \cdot (p!)} \sqrt{ \det g_{ij} } \varepsilon_{\alpha_1 \cdots \alpha_p \beta_1 \cdots \beta_{n-p}} \\ &\quad \times A^{\alpha_1 \cdots \alpha_p} \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_{n-p}}. \end{aligned}$	${}^* \alpha^{(p)} := \frac{1}{(n-p)! \cdot (p!)} \sqrt{ \det g_{ij} } \varepsilon_{\alpha_1 \cdots \alpha_p \beta_1 \cdots \beta_{n-p}} \times A^{\alpha_1 \cdots \alpha_p} \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_{n-p}}.$	${}^* \alpha^{(p)} := \frac{1}{(n-p)! \cdot (p!)} \sqrt{ \det g_{ij} } \varepsilon_{\alpha_1 \cdots \alpha_p \beta_1 \cdots \beta_{n-p}} \times A^{\alpha_1 \cdots \alpha_p} \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_{n-p}}.$
mielke-geometrodynamics_00184 (3.7.14)	074	■ 0.864 ● C +16	$\vec{\omega} = d \vec{\omega} - (\overrightarrow{\omega} \wedge)$	$\vec{\Omega} = d\vec{\omega} - (\overrightarrow{\omega} \overrightarrow{\wedge})$	$\vec{\Omega} = d\vec{\omega} - (\overrightarrow{\omega} \wedge \vec{\omega}) = d\vec{\omega} - \vec{\omega} \wedge \vec{\omega}.$
mielke-geometrodynamics_00165 (3.5.13)	069	■ 0.865 ● N +0	$\{ \}^{\ast} \Omega = (\stackrel{+}{-}) \Omega \; ;$	${}^* \Omega = (\overset{+}{-}) \Omega ;$	${}^* \Omega = (\overset{+}{-}) \Omega ;$
mielke-geometrodynamics_00737 (11.1.4)	238	■ 0.867 ● N +0	$x^{\alpha} \rightarrow x^{\prime \alpha} = g_{\beta}^{\alpha} \quad x^{\beta}, [g_{\beta}^{\alpha}] \in SO(s, n-s), \quad x^{\alpha} \in \mathbf{E}^n.$	$x^{\alpha} \rightarrow x^{\prime \alpha} = g_{\beta}^{\alpha} \quad x^{\beta}, [g_{\beta}^{\alpha}] \in SO(s, n-s), \quad x^{\alpha} \in \mathbf{E}^n.$	$x^{\alpha} \rightarrow x^{\prime \alpha} = g_{\beta}^{\alpha} \quad x^{\beta}, [g_{\beta}^{\alpha}] \in SO(s, n-s), \quad x^{\alpha} \in \mathbf{E}^n.$
mielke-geometrodynamics_00853 (12.3.6)	276	■ 0.868 ● C +3	$\left\langle \vartheta^{\alpha} \wedge \sigma_{\alpha} \right\rangle = -\frac{1}{3\pi^2} \left[\text{Tr} (G \wedge {}^* G) + \frac{1}{24} \left(2R^{\alpha\beta\{\}} \wedge R_{\alpha\beta}^{\{\}j(\star)} + d\mathscr{A} \wedge {}^* d\mathscr{A} \right) \right],$	$\langle \vartheta^{\alpha} \wedge \sigma_{\alpha} \rangle = -\frac{1}{3\pi^2} \left[\text{Tr} (G \wedge {}^* G) + \frac{1}{24} \left(2R^{\alpha\beta\{\}} \wedge R_{\alpha\beta}^{\{\}j(\star)} + d\mathscr{A} \wedge {}^* d\mathscr{A} \right) \right],$	$\langle \vartheta^{\alpha} \wedge \sigma_{\alpha} \rangle = -\frac{1}{3\pi^2} \left[\text{Tr} (G \wedge {}^* G) + \frac{1}{24} \left(2R^{\alpha\beta\{\}} \wedge R_{\alpha\beta}^{\{\}j(\star)} + d\mathscr{A} \wedge {}^* d\mathscr{A} \right) \right],$
mielke-geometrodynamics_00774 (11.2.3)	245	■ 0.868 ● N +0	$i(\overline{D\psi}) \wedge {}^* \gamma + \frac{i}{2} \bar{\psi} D \wedge {}^* \gamma + m \bar{\psi} \eta = 0,$	$i(\overline{D\psi}) \wedge {}^* \gamma + \frac{i}{2} \bar{\psi} D \wedge {}^* \gamma + m \bar{\psi} \eta = 0,$	$i(\overline{D\psi}) \wedge {}^* \gamma + \frac{i}{2} \bar{\psi} D \wedge {}^* \gamma + m \bar{\psi} \eta = 0,$
mielke-geometrodynamics_00548 (8.4.2)	178	■ 0.871 ● N -1	$s_{L_{\text{Pontr}}} = -2 d \operatorname{Tr} \left\{ \Psi \wedge \Omega^g \right\} = -d \left(\Psi_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} \right).$	$s_{L_{\text{Pontr}}} = -2d \operatorname{Tr} \{ \Psi \wedge \Omega^g \} = -d \left(\Psi_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} \right).$	$s_{L_{\text{Pontr}}} = -2d \operatorname{Tr} \{ \Psi \wedge \Omega^g \} = -d (\Psi_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha}).$
mielke-geometrodynamics_00244 (4.3.9)	091	■ 0.873 ● N +1	$g_{\{i\}j} e_{\alpha}^i \wedge e_{\beta}^j = g_{\alpha\beta}.$	$g_{ij} e_{\alpha}^i e_{\beta}^j = g_{\alpha\beta}.$	$g_{ij} e_{\alpha}^i e_{\beta}^j = g_{\alpha\beta}.$
mielke-geometrodynamics_00890 (13.4.9)	287	■ 0.876 ● N +0	$\hat{\mathbf{x}} = \binom{\xi}{1} = \binom{\xi^{\alpha} P_{\alpha}}{1}, \quad \hat{\xi}_{*} = (\xi_{*}, 1) = (\xi_{\beta}^{*} P_{*}^{\beta}, 1)$	$\hat{\xi} = \binom{\xi}{1} = \binom{\xi^{\alpha} P_{\alpha}}{1}, \quad \hat{\xi}_{*} = (\xi_{*}, 1) = (\xi_{\beta}^{*} P_{*}^{\beta}, 1)$	$\hat{\xi} = \binom{\xi}{1} = \binom{\xi^{\alpha} P_{\alpha}}{1}, \quad \hat{\xi}_{*} = (\xi_{*}, 1) = (\xi_{\beta}^{*} P_{*}^{\beta}, 1)$
mielke-geometrodynamics_01007 (14.4.2)	325	■ 0.877 ● N +1	$\begin{aligned} A^{\circ} &= \sigma^{\circ} \omega^{\circ} = -\frac{r^2}{\rho^2} \left[dt - J_3 \frac{\hbar}{Mc} \sin^2 \vartheta d\phi \right] Q^{\circ} \cos \delta \\ &\quad - \frac{\cos \vartheta}{\rho^2} \left[\left(r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right) d\phi - J_3 \frac{\hbar}{M} dt \right] P^{\circ} \sin \delta \end{aligned}$	$A^{(\circ)} = \sigma^{\circ} \omega^{(\circ)} = -\frac{r^2}{\rho^2} \left[dt - J_3 \frac{\hbar}{Mc} \sin^2 \vartheta d\phi \right] Q^{(\circ)} \cos \delta - \frac{\cos \vartheta}{\rho^2} \left[\left(r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right) d\phi - J_3 \frac{\hbar}{M} dt \right] P^{(\circ)} \sin \delta$	$A^{(\circ)} = \sigma^{\circ} \omega^{(\circ)} = -\frac{r^2}{\rho^2} \left[dt - J_3 \frac{\hbar}{Mc} \sin^2 \vartheta d\phi \right] Q^{(\circ)} \cos \delta - \frac{\cos \vartheta}{\rho^2} \left[\left(r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right) d\phi - J_3 \frac{\hbar}{M} dt \right] P^{(\circ)} \sin \delta.$
mielke-geometrodynamics_01062 (16.2.1)	355	■ 0.877 ● N +0	$\left[\square - (\mathrm{mc}/\hbar)^2 \right] \varphi = 0$	$[\square - (\mathrm{mc}/\hbar)^2] \varphi = 0$	$[\square - (mc/\hbar)^2] \varphi = 0$
mielke-geometrodynamics_00851 (2.7.1)	044	■ 0.881 ● N -2	$G(p) : P \rightarrow P, \quad G(p) \in \mathscr{G}_p$	$G(p) : P \rightarrow P, \quad G(p) \in \mathscr{G}_p$	$G(p) : P \rightarrow P, \quad G(p) \in \mathscr{G}_p$
mielke-geometrodynamics_00178 (3.7.8)	072	■ 0.881 ● K +0	$\left\langle O \right ^{\lambda} \varphi \left O \right\rangle = \mu / \sqrt{\lambda}$	$\left\langle O \right ^{\lambda} \varphi \left O \right\rangle = \mu / \sqrt{\lambda}$	$\langle O ^{\lambda} \varphi O \rangle = \mu / \sqrt{\lambda}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_040 EQ0628 (2.5.1)		0.885 N +0	$V_{\{p\}^{\wedge\ast}}=T_{\{p\}^{\wedge\ast}}\left(F_{\{m\}}(M)\right),$	$V_p^{\ast}=T_p^{\ast}\left(F_m(M)\right),$	$V_p^{\ast}=T_p^{\ast}(F_m(M)),$
mielke-geometrodynamics_295 EQ0920 (13.8.6)		0.888 N -2	$\begin{aligned} d\,C_{\{\mathrm{RR}\}(\star)}\,\&:=\frac{1}{2}d\left(\Gamma_{\alpha\beta}\wedge R^{\alpha\beta(\ast)}-\frac{1}{3}\Gamma_{\alpha}{}^{\beta(\ast)}\wedge\Gamma_{\beta}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\alpha}\right)\\ &\equiv-L_{\mathrm{SKY}}-2\,\mathrm{Ric}_{\alpha\beta}\wedge{}^{\ast}\mathrm{Ric}^{\alpha\beta}+\frac{1}{2}\,\mathrm{Ric}_{\alpha}{}^{\alpha}\wedge{}^{\ast}\mathrm{Ric}_{\beta}{}^{\beta} \end{aligned}$	$dC_{\mathrm{RR}(\star)}:=\frac{1}{2}d\left(\Gamma_{\alpha\beta}\wedge R^{\alpha\beta(\ast)}-\frac{1}{3}\Gamma_{\alpha}{}^{\beta(\ast)}\wedge\Gamma_{\beta}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\alpha}\right)\\ \equiv-L_{\mathrm{SKY}}-2\,\mathrm{Ric}_{\alpha\beta}\wedge{}^{\ast}\mathrm{Ric}^{\alpha\beta}+\frac{1}{2}\,\mathrm{Ric}_{\alpha}{}^{\alpha}\wedge{}^{\ast}\mathrm{Ric}_{\beta}{}^{\beta}$	$dC_{\mathrm{RR}(\ast)}:=\frac{1}{2}d\left(\Gamma_{\alpha\beta}\wedge R^{\alpha\beta(\ast)}-\frac{1}{3}\Gamma_{\alpha}{}^{\beta(\ast)}\wedge\Gamma_{\beta}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\alpha}\right)\\ \equiv-L_{\mathrm{SKY}}-2\mathrm{Ric}_{\alpha\beta}\wedge{}^{\ast}\mathrm{Ric}^{\alpha\beta}+\frac{1}{2}\mathrm{Ric}_{\alpha}{}^{\alpha}\wedge{}^{\ast}\mathrm{Ric}_{\beta}{}^{\beta}$
mielke-geometrodynamics_220 EQ0674 (10.3.31)		0.890 C -5	$\check{\mathrm{E}}_{\{\alpha\}}=-\frac{\theta_{\{\mathrm{T}\}}}{\theta_{\{\mathrm{TL}\}}}\frac{\theta_{\{\mathrm{TL}\}}}{2\ell^2}\left(\frac{1}{\theta_{\{\mathrm{TL}\}}}d\vartheta_{\alpha}-\frac{4\ell}{\theta_{\{\mathrm{T}\}}\theta_{\{\mathrm{L}\}}}\check{\mathcal{J}}_{\alpha}^{\star}\right),$	$\check{\mathcal{E}}_{\alpha}=-\frac{\theta_{\mathrm{T}}\theta_{\mathrm{TL}}}{2\ell^2}\left(\frac{1}{\theta_{\mathrm{TL}}}d\vartheta_{\alpha}-\frac{4\ell}{\theta_{\mathrm{T}}\theta_{\mathrm{L}}}\check{\mathcal{J}}_{\alpha}^{\star}\right),$	$\check{\mathcal{E}}_{\alpha}=-\frac{\theta_{\mathrm{T}}\theta_{\mathrm{TL}}}{2\ell^2}\left(\frac{1}{\theta_{\mathrm{TL}}}d\vartheta_{\alpha}-\frac{4\ell}{\theta_{\mathrm{T}}\theta_{\mathrm{L}}}\check{\mathcal{J}}_{\alpha}^{\star}\right),$
mielke-geometrodynamics_169 EQ0531 (7.7.51)		0.890 N +0	$\left.\left.\left.\left.R:=e_{\beta}\right]e_{\alpha}\right]R^{\alpha\beta},\quad{}^{\ast}R^{\alpha\beta}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}\equiv R^{\alpha\beta}\wedge\eta_{\alpha\beta}\equiv-R\eta,\right.\right.\right.$	$R:=e_{\beta}\rfloor e_{\alpha}\rfloor R^{\alpha\beta},\quad{}^{\ast}R^{\alpha\beta}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}\equiv R^{\alpha\beta}\wedge\eta_{\alpha\beta}\equiv-R\eta,$	$R:=e_{\beta}\rfloor e_{\alpha}\rfloor R^{\alpha\beta},\quad{}^{\ast}R^{\alpha\beta}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}\equiv R^{\alpha\beta}\wedge\eta_{\alpha\beta}\equiv-R\eta,$
mielke-geometrodynamics_176 EQ0540 (8.3.1)		0.891 N +2	$\begin{aligned} (d\,\oplus s)\,\gamma+[\widetilde{\Gamma},\gamma]&=\widetilde{\Theta}\\ (d\,\oplus s)\widetilde{\Theta}+[\widetilde{\Gamma},\widetilde{\Theta}]&\equiv\left[\widetilde{\Omega}^g,\gamma\right] \end{aligned}$	$(d\oplus s)\gamma+[\widetilde{\Gamma},\gamma]=\widetilde{\Theta}$ $(d\oplus s)\widetilde{\Theta}+[\widetilde{\Gamma},\widetilde{\Theta}]\equiv\left[\widetilde{\Omega}^g,\gamma\right]$	$(d\oplus s)\gamma+[\widetilde{\Gamma},\gamma]=\widetilde{\Theta},$ $(d\oplus s)\widetilde{\Theta}+[\widetilde{\Gamma},\widetilde{\Theta}]\equiv[\widetilde{\Omega}^g,\gamma],$
mielke-geometrodynamics_303 EQ0925 (14.1.5)		0.892 N +0	$\begin{aligned} &D^{\wedge\ast}\,\Omega=\tau_e, \quad\&D\,\Omega\equiv 0, \\ &\frac{1}{2}\ast\left(\vartheta\wedge{}^{\ast}\Omega^g\right)=\ell^{\ast 2}\left(\Sigma_{\mathrm{YM}}+\Sigma_{\mathrm{mat}}\right), \end{aligned}$	$D^{\ast}\Omega=\tau_e,$ $D\Omega\equiv 0,$ $\frac{1}{2}\ast\left(\vartheta\wedge{}^{\ast}\Omega^g\right)=\ell^{\ast 2}\left(\Sigma_{\mathrm{YM}}+\Sigma_{\mathrm{mat}}\right),$	$D^{\ast}\Omega=\tau_e,$ $D\Omega\equiv 0,$ $\frac{1}{2}\ast(\vartheta\wedge{}^{\ast}\Omega^g)=\ell^{\ast 2}(\Sigma_{\mathrm{YM}}+\Sigma_{\mathrm{mat}}),$
mielke-geometrodynamics_283 EQ0867 (13.2.3)		0.893 N +0	$B\,\cong\,d\,A,\quad d\,B\,\cong\,d\,F\,\equiv\,0,$	$B\cong dA,\quad dB\cong dF\equiv 0,$	$B\cong dA,\quad dB\cong dF\equiv 0,$
mielke-geometrodynamics_201 EQ0609 (9.2.11)		0.894 N +1	$\begin{aligned} &\&\left(1+\zeta\wedge^2\right)\int_{M^4}L_{\mathrm{SKY}}-(-1)^{\wedge s}\zeta(4\pi)^2\cdot\chi\left(M^4\right)\\ &L_{\{\mathrm{SKY}\}}-(-1)^{\wedge s}\zeta(4\pi)^2\cdot\chi\left(M^4\right)\&\&=\frac{1}{4}\int_{M^4}\left(\Omega^g-\zeta^{\ast}\Omega^{g(\ast)}\right)\wedge^{\ast}\left(\Omega^g-\zeta^{\ast}\Omega^{g(\ast)}\right)\geq 0, \end{aligned}$	$(1+\zeta^2)\int_{M^4}L_{\mathrm{SKY}}-(-1)^s\zeta(4\pi)^2\cdot\chi\left(M^4\right)$ $=\frac{1}{4}\int_{M^4}\left(\Omega^g-\zeta^{\ast}\Omega^{g(\ast)}\right)\wedge^{\ast}\left(\Omega^g-\zeta^{\ast}\Omega^{g(\ast)}\right)\geq 0,$	$(1+\zeta^2)\int_{M^4}L_{\mathrm{SKY}}-(-1)^s\zeta(4\pi)^2\cdot\chi\left(M^4\right)$ $=\frac{1}{4}\int_{M^4}(\Omega^g-\zeta^{\ast}\Omega^{g(\ast)})\wedge^{\ast}(\Omega^g-\zeta^{\ast}\Omega^{g(\ast)})\geq 0,$
mielke-geometrodynamics_309 EQ0959 (14.1.41)		0.894 N +0	$\oint d\,\delta=2\pi\,b_1=2\pi\,v$	$\oint d\delta=2\pi b_1=2\pi v$	$\oint d\delta=2\pi b_1=2\pi v$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_090299 (4.5.10)	102	0.896 N +1	$\begin{aligned} & \& \delta g_{\alpha\beta} = 2\varepsilon_{(\alpha}{}^{\gamma} g_{\beta)\gamma}, \quad \delta\vartheta^{\alpha} = -\vartheta^{\beta}\varepsilon_{\beta}{}^{\alpha}, \\ & \delta\Gamma_{\alpha}{}^{\beta} = D\varepsilon_{\alpha}{}^{\beta}, \quad \delta\Psi = -\varepsilon_{\alpha}{}^{\beta}I^{\alpha}{}_{\beta}\Psi. \end{aligned}$	$\delta g_{\alpha\beta} = 2\varepsilon_{(\alpha}{}^{\gamma} g_{\beta)\gamma}, \quad \delta\vartheta^{\alpha} = -\vartheta^{\beta}\varepsilon_{\beta}{}^{\alpha},$ $\delta\Gamma_{\alpha}{}^{\beta} = D\varepsilon_{\alpha}{}^{\beta}, \quad \delta\Psi = -\varepsilon_{\alpha}{}^{\beta}I^{\alpha}{}_{\beta}\Psi.$	$\delta g_{\alpha\beta} = 2\varepsilon_{(\alpha}{}^{\gamma} g_{\beta)\gamma}, \quad \delta\vartheta^{\alpha} = -\vartheta^{\beta}\varepsilon_{\beta}{}^{\alpha},$ $\delta\Gamma_{\alpha}{}^{\beta} = D\varepsilon_{\alpha}{}^{\beta}, \quad \delta\Psi = -\varepsilon_{\alpha}{}^{\beta}I^{\alpha}{}_{\beta}\Psi.$
mielke-geometrodynamics_090236 (4.3.1)	090	0.896 N +0	$\begin{aligned} & (\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \\ & = g_{ij}\Delta x^i\Delta x^j. \end{aligned}$	$(\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$ $= g_{ij}\Delta x^i\Delta x^j.$	$(\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$ $= g_{ij}\Delta x^i\Delta x^j.$
mielke-geometrodynamics_0901005 (14.3.18)	323	0.896 N -1	$\begin{aligned} & \not{D}^2 \& = \operatorname{Tr} \left(\frac{1}{2} \gamma \wedge \gamma \wedge^* \{D, D\} + \frac{1}{2} \gamma \wedge \gamma \wedge^* \tilde{\Omega} - \gamma \wedge \tilde{\Theta} \wedge^* D \right) \\ & = \left(\square^g + \frac{1}{4} R \right) \eta - \vartheta \wedge \Theta \wedge^* D \end{aligned}$	$D^2 = \operatorname{Tr} \left(\frac{1}{2} \gamma \wedge \gamma \wedge^* \{D, D\} + \frac{1}{2} \gamma \wedge \gamma \wedge^* \tilde{\Omega} - \gamma \wedge \tilde{\Theta} \wedge^* D \right)$ $= \left(\square^g + \frac{1}{4} R \right) \eta - \vartheta \wedge \Theta \wedge^* D$	$\not{D}^2 = \operatorname{Tr} \left(\frac{1}{2} \gamma \wedge \gamma \wedge^* \{D, D\} + \frac{1}{2} \gamma \wedge \gamma \wedge^* \tilde{\Omega} - \gamma \wedge \tilde{\Theta} \wedge^* D \right)$ $= (\square^g + \frac{1}{4} R) \eta - \vartheta \wedge \Theta \wedge^* D$
mielke-geometrodynamics_090230 (4.2.23)	088	0.899 N +0	$\stackrel{*}{\Theta} = \frac{1}{2} T_{\alpha\beta}^{\cdot c} P_c \otimes \vartheta^\alpha \wedge \vartheta^\beta = \frac{1}{2} T_{ij}^{\cdot c} P_c \otimes dx^i \wedge dx^j,$	$\stackrel{*}{\Theta} = \frac{1}{2} T_{\alpha\beta}^{\cdot c} P_c \otimes \vartheta^\alpha \wedge \vartheta^\beta = \frac{1}{2} T_{ij}^{\cdot c} P_c \otimes dx^i \wedge dx^j,$	$\stackrel{*}{\Theta} = \frac{1}{2} T_{\alpha\beta}^{\cdot c} P_c \otimes \vartheta^\alpha \wedge \vartheta^\beta = \frac{1}{2} T_{ij}^{\cdot c} P_c \otimes dx^i \wedge dx^j,$
mielke-geometrodynamics_090723 (10.10.3)	230	0.900 C +4	$\begin{aligned} & \delta_{\text{susy}} \vartheta^\alpha = i\bar{\sigma} \Psi \gamma^\alpha, \quad \delta_{\text{susy}} \Gamma_\alpha^* = i\bar{\sigma} \gamma_\alpha^* D\Psi + ic\bar{\sigma} (\gamma_\alpha \Psi + e_\alpha]{}^* \Psi) \\ & \delta_{\text{susy}} \Psi = 2D\sigma + c\gamma\sigma, \end{aligned}$	$\delta_{\text{susy}} \vartheta^\alpha = i\bar{\sigma} \Psi \gamma^\alpha, \quad \delta_{\text{susy}} \Gamma_\alpha^* = i\bar{\sigma} \gamma_\alpha^* D\Psi + ic\bar{\sigma} (\gamma_\alpha \Psi + e_\alpha]{}^* \Psi)$ $\delta_{\text{susy}} \Psi = 2D\sigma + c\gamma\sigma,$	$\delta_{\text{susy}} \vartheta^\alpha = i\bar{\sigma} \Psi \gamma^\alpha, \quad \delta_{\text{susy}} \Gamma_\alpha^* = i\bar{\sigma} \gamma_\alpha^* D\Psi + ic\bar{\sigma} (\gamma_\alpha \Psi + e_\alpha]{}^* \Psi),$ $\delta_{\text{susy}} \Psi = 2D\sigma + c\gamma\sigma, \tag{1}$
mielke-geometrodynamics_090216 (4.2.8)	086	0.902 N +0	$\widetilde{\omega} \rightarrow^{A^{-1}} \widetilde{\omega} = A^{-1} \widetilde{\omega} A - A^{-1} dA, \quad A \in \mathscr{A}_p.$	$\widetilde{\omega} \rightarrow^{A^{-1}} \widetilde{\omega} = A^{-1} \widetilde{\omega} A - A^{-1} dA, \quad A \in \mathscr{A}_p.$	$\widetilde{\omega} \rightarrow^{A^{-1}} \widetilde{\omega} = A^{-1} \widetilde{\omega} A - A^{-1} dA, \quad A \in \mathscr{A}_p.$
mielke-geometrodynamics_0901042 (15.1.19)	340	0.902 K +0	$\Gamma_{\mu\alpha\beta} : (2^+) \oplus 2(1^-) \oplus (0^+) \oplus (2^-) \oplus 2(1^+) \oplus (0^-),$	$\Gamma_{\mu\alpha\beta} : (2^+) \oplus 2(1^-) \oplus (0^+) \oplus (2^-) \oplus 2(1^+) \oplus (0^-),$	$\Gamma_{\mu\alpha\beta} : (2^+) \oplus 2(1^-) \oplus (0^+) \oplus (2^-) \oplus 2(1^+) \oplus (0^-),$
mielke-geometrodynamics_090481 (7.5.2)	157	0.903 S -1	$\square^c g_{ij} = \square^c \left(\Omega^2 o_{ij}^{\text{Mink}} \right) = \Omega^6 \square^c o_{ij}^{\text{Mink}} = 0$	$\square^c g_{ij} = \square^c \left(\Omega^2 o_{ij}^{\text{Mink}} \right) = \Omega^6 \square^c o_{ij}^{\text{Mink}} = 0$	$\square^c g_{ij} = \square^c (\Omega^2 o_{ij}^{\text{Mink}}) = \Omega^6 \square^c o_{ij}^{\text{Mink}} = 0$
mielke-geometrodynamics_090117 (3.3.18)	059	0.905 N -1	$L_{\text{Max}} := -\frac{1}{2} F \wedge^* F = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) \eta,$	$L_{\text{Max}} := -\frac{1}{2} F \wedge^* F = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) \eta,$	$L_{\text{Max}} := -\frac{1}{2} F \wedge^* F = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) \eta,$
mielke-geometrodynamics_0901110 (B.22)	371	0.905 K +0	$\square \alpha = 0,$	$\square \alpha = 0,$	$\square \alpha = 0,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_00831 (12.1.3)	270	■ 0.907 ● N +0	$\tilde{j}_5:=j_5+\left(1\,/\,96\,\pi^2\right)A\,\wedge\,d\,A$	$\tilde{j}_5 := j_5 + \left(1/96\pi^2\right) A \wedge dA$	$\widetilde{j}_5 \, := \, j_5 + (1/96\pi^2)A \wedge dA$
mielke-geometrodynamics_00409 (6.4.29)	136	■ 0.907 ● C +5	$T^{\mathrm{A}}\!=\!\frac{1}{f}\!\left(\!d\,f\,\wedge\,\vartheta\!\wedge\!\vartheta^{\mathrm{A}}\!-\!2\,\eta^{\mathrm{0A}\beta\gamma}\vartheta_{\beta}\wedge\vartheta_{\gamma}\right)\!=\!-\frac{2}{f}\eta^A{}_{\mu\nu}\vartheta^{\mu}\wedge\vartheta^{\nu}\!=\!\pm\,{}^*T^A,\quad T^{\hat{0}}\!=\!0,$	$T^A = \frac{1}{f} \left(df \wedge \vartheta^A - 2\eta^{0A\beta\gamma} \vartheta_\beta \wedge \vartheta_\gamma \right) = -\frac{2}{f} \eta^A{}_{\mu\nu} \vartheta^\mu \wedge \vartheta^\nu = \pm {}^*T^A, \quad T^{\hat{0}} = 0,$	$T^A = \frac{1}{f} \left(df \wedge \vartheta^A - 2\eta^{0A\beta\gamma} \vartheta_\beta \wedge \vartheta_\gamma \right) = -\frac{2}{f} \eta^A{}_{\mu\nu} \vartheta^\mu \wedge \vartheta^\nu = \pm {}^*T^A, \quad T^{\hat{0}} = 0,$
mielke-geometrodynamics_00657 (10.3.14)	217	■ 0.907 ● N +0	$\left.E_{\alpha}\right _{\delta\ell^2}=\left.e_{\alpha}\right \left\lfloor L(\star)-\frac{2}{\delta\ell^2}T_{\alpha},\right.$	$E_{\alpha}=e_{\alpha}\rfloor L(\star)-\frac{2\gamma}{\delta\ell^2}T_{\alpha},$	$E_{\alpha}=e_{\alpha}\rfloor L(\star)-\frac{2\gamma}{\delta\ell^2}T_{\alpha},$
mielke-geometrodynamics_00396 (6.4.11)	132	■ 0.909 ● W +1	$\begin{aligned} \mathscr{H} &\cong n^{\alpha} \frac{\delta L}{\delta \vartheta^{\alpha}} + \Gamma_{\perp}^{\alpha\beta} \frac{\delta L}{\delta \Gamma^{\alpha\beta}} \\ &\quad - \underline{d}[(n\rfloor\Psi) \wedge \frac{\partial L}{\partial D\Psi} + n^{\alpha} \frac{\partial L}{\partial T^{\alpha}} + \Gamma_{\perp}^{\alpha\beta} \frac{\partial L}{\partial R^{\alpha\beta}}] . \end{aligned}$	$\mathcal{H} \cong n^{\alpha} \frac{\delta L}{\delta \vartheta^{\alpha}} + \Gamma_{\perp}^{\alpha\beta} \frac{\delta L}{\delta \Gamma^{\alpha\beta}} - \underline{d}[(n\rfloor\Psi) \wedge \frac{\partial L}{\partial D\Psi} + n^{\alpha} \frac{\partial L}{\partial T^{\alpha}} + \Gamma_{\perp}^{\alpha\beta} \frac{\partial L}{\partial R^{\alpha\beta}}] .$	$\mathcal{H} \cong n^{\alpha} \frac{\delta L}{\delta \vartheta^{\alpha}} + \Gamma_{\perp}^{\alpha\beta} \frac{\delta L}{\delta \Gamma^{\alpha\beta}} - \underline{d}[(n\rfloor\Psi) \wedge \frac{\partial L}{\partial D\Psi} + n^{\alpha} \frac{\partial L}{\partial T^{\alpha}} + \Gamma_{\perp}^{\alpha\beta} \frac{\partial L}{\partial R^{\alpha\beta}}] .$
mielke-geometrodynamics_00641 (10.2.12)	213	■ 0.909 ● N +0	$d\,s^{\mathrm{2}}\!=\!\!-\!\left(1\!-\!\Lambda_{\mathrm{eff}}\right) r^{\mathrm{2}}\!\right) d\,t^{\mathrm{2}}\!+\!\left(1\!-\!\Lambda_{\mathrm{eff}}\right) r^{\mathrm{2}}\!\right)\!d\,r^{\mathrm{2}}\!+\!r^{\mathrm{2}} d\,\phi^{\mathrm{2}}$	$ds^2 = -\left(1-\Lambda_{\mathrm{eff}}\,r^2\right) dt^2 + \left(1-\Lambda_{\mathrm{eff}}\,r^2\right)^{-1} dr^2 + r^2 d\phi^2$	$ds^2 = - (1-\Lambda_{\mathrm{eff}}\,r^2) dt^2 + (1-\Lambda_{\mathrm{eff}}\,r^2)^{-1} dr^2 + r^2 d\phi^2$
mielke-geometrodynamics_00263 (4.3.28)	094	■ 0.912 ● N +0	$\Gamma_{ij}^{\cdot k}=e_b^{\kappa}\Gamma_{i\alpha}^{\cdot b}E_j^{\alpha}+e_{\beta}^{\kappa}\partial_iE_j^{\beta}$	$\Gamma_{ij}^{\cdot k} = e_b^{\kappa} \Gamma_{i\alpha}^{\cdot b} E_j^{\alpha} + e_{\beta}^{\kappa} \partial_i E_j^{\beta}$	$\Gamma_{ij}^{\cdot \kappa} = e_b^{\kappa} \Gamma_{i\alpha}^{\cdot b} E_j^{\alpha} + e_{\beta}^{\kappa} \partial_i E_j^{\beta}$
mielke-geometrodynamics_00364 (6.2.16)	126	■ 0.912 ● N +0	$D\,D\,\lambda_{\alpha\beta}=-2\,R_{[\alpha\,\mid\beta]}{\,\,}\wedge\,\lambda_{\gamma\mid\beta]}=0$	$DD\lambda_{\alpha\beta}=-2R_{[\alpha]{}^\gamma\wedge\lambda_{\gamma \beta]}=0$	$DD\lambda_{\alpha\beta}=-2R_{[\alpha]{}^\gamma\wedge\lambda_{\gamma \beta]}=0$
mielke-geometrodynamics_00840 (12.2.8)	272	■ 0.916 ● N +1	$\begin{aligned} \tau_{\alpha\beta}&:=\frac{\partial L_{\mathrm{D}}}{\partial \Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\psi}\left({}^*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}{}^*\gamma\right)\psi\\ &=\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\psi}\gamma^{\delta}\gamma_5\psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}=\vartheta_{[\alpha}\wedge\mu_{\beta]}\end{aligned}$	$\tau_{\alpha\beta}:=\frac{\partial L_{\mathrm{D}}}{\partial \Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\psi}\left({}^*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}{}^*\gamma\right)\psi\\ =\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\psi}\gamma^{\delta}\gamma_5\psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}=\vartheta_{[\alpha}\wedge\mu_{\beta]}$	$\tau_{\alpha\beta}:=\frac{\partial L_{\mathrm{D}}}{\partial \Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\psi}\left({}^*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}{}^*\gamma\right)\psi\\ =\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\psi}\gamma^{\delta}\gamma_5\psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}=\vartheta_{[\alpha}\wedge\mu_{\beta]}.$
mielke-geometrodynamics_00232 (4.2.25)	088	■ 0.917 ● K +0	$\Theta\rightarrowtail A^{\{-1\}}\Theta=G^{\{-1\}}\Theta,$	$\Theta \rightarrowtail A^{-1}\Theta = G^{-1}\Theta,$	$\Theta \rightarrowtail A^{-1}\Theta = G^{-1}\Theta,$
mielke-geometrodynamics_00577 (8.6.15)	185	■ 0.920 ● U +25	$-E_{\alpha}=\frac{\theta_{\mathrm{T}}^{\ast}}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma}-\theta_{\mathrm{T}}^{\ast}\Lambda_{\mathrm{eff}}\eta_{\alpha}=\ell^2\,\widehat{\Sigma}_{\alpha}.$	<i>(not rendered)</i>	$-E_{\alpha}=\frac{\theta_{\mathrm{T}}^{\ast}}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma}-\theta_{\mathrm{T}}^{\ast}\Lambda_{\mathrm{eff}}\eta_{\alpha}=\ell^2\,\widehat{\Sigma}_{\alpha}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_337 EQ1031 (15.1.8)		■ 0.920 ● W -1	$\begin{aligned} & \frac{1}{2} * \left(\vartheta^f \wedge * \Omega^f \right) - a \ell^2 \vartheta^f \wedge * \left(2 L_{fg} - \frac{1}{2 b \ell^2} (\Omega^g \wedge \vartheta^g \wedge \vartheta^g + 2 \Lambda \eta) \right) \\ & = \ell^2 * \left\{ \Sigma_{\text{m}}(\text{hadrons}) + \frac{a}{b} * (\vartheta^g \wedge * \Sigma_{\text{m}}(\text{leptons})) \wedge \vartheta^f \right\} \end{aligned}$	$\frac{1}{2} * \left(\vartheta^f \wedge * \Omega^f \right) - a \ell^2 \vartheta^f \wedge * \left\{ 2 L_{fg} - \frac{1}{2 b \ell^2} (\Omega^g \wedge \vartheta^g \wedge \vartheta^g + 2 \Lambda \eta) \right\} \\ = \ell^2 * \left\{ \Sigma_{\text{m}}(\text{hadrons}) + \frac{a}{b} * (\vartheta^g \wedge * \Sigma_{\text{m}}(\text{leptons})) \wedge \vartheta^f \right\}$	$\frac{1}{2} * (\vartheta^f \wedge * \Omega^f) - a \ell^2 \vartheta^f \wedge * \left\{ 2 L_{fg} - \frac{1}{2 b \ell^2} (\Omega^g \wedge \vartheta^g \wedge \vartheta^g + 2 \Lambda \eta) \right\} \\ = \ell^2 * \left\{ \Sigma_{\text{m}}(\text{hadrons}) + \frac{a}{b} * (\vartheta^g \wedge * \Sigma_{\text{m}}(\text{leptons})) \wedge \vartheta^f \right\}$
mielke-geometroynamics_156 EQ0473 (7.4.1)		■ 0.921 ● N +1	$\begin{aligned} & \int \mathcal{D} \Gamma \exp \left[\int L_{\text{SKY}}^{(*)} / \hbar \right] \\ & = \int \mathcal{D} \Gamma \exp \left[- \int \left(R_{\alpha\beta} + * R_{\alpha\beta}^{(*)} \right) \wedge * \left(R^{\alpha\beta} + * R^{\alpha\beta(*)} \right) / 4 \hbar \right] \end{aligned}$	$\int \mathcal{D} \Gamma \exp \left[\int L_{\text{SKY}}^{(*)} / \hbar \right] \\ = \int \mathcal{D} \Gamma \exp \left[- \int \left(R_{\alpha\beta} + * R_{\alpha\beta}^{(*)} \right) \wedge * \left(R^{\alpha\beta} + * R^{\alpha\beta(*)} \right) / 4 \hbar \right]$	$\int \mathcal{D} \Gamma \exp \left[\int L_{\text{SKY}}^{(*)} / \hbar \right] \\ = \int \mathcal{D} \Gamma \exp \left[- \int \left(R_{\alpha\beta} + * R_{\alpha\beta}^{(*)} \right) \wedge * \left(R^{\alpha\beta} + * R^{\alpha\beta(*)} \right) / 4 \hbar \right]$
mielke-geometroynamics_115 EQ0343 (5.4.1)		■ 0.922 ● N +0	$k \partial_k \Gamma_k = \frac{1}{2} \text{STr} \left\{ \left[\Gamma_k^{(2)} + R_k \right]^{-1} (k \partial_k R_k) \right\},$	$k \partial_k \Gamma_k = \frac{1}{2} \text{STr} \left\{ \left[\Gamma_k^{(2)} + R_k \right]^{-1} (k \partial_k R_k) \right\},$	$k \partial_k \Gamma_k = \frac{1}{2} \text{STr} \left\{ \left[\Gamma_k^{(2)} + R_k \right]^{-1} (k \partial_k R_k) \right\},$
mielke-geometroynamics_269 EQ0829 (12.1.1)		■ 0.922 ● W +0	$\bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \rightarrow \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \mathbf{P} \exp \left\{ i \int_x^{x+\varepsilon} A \right\}.$	$\bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \rightarrow \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \mathbf{P} \exp \left\{ i \int_x^{x+\varepsilon} A \right\}.$	$\bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \rightarrow \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \mathbf{P} \exp \left\{ i \int_x^{x+\varepsilon} A \right\}.$
mielke-geometroynamics_238 EQ0736 (11.1.3)		■ 0.922 ● N +0	$\gamma^0 := \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}, \quad \sigma^k : \text{Pauli matrices.}$	$\gamma^0 := \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}, \quad \sigma^k : \text{Pauli matrices.}$	$\gamma^0 := \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}, \quad \sigma^k : \text{Pauli matrices.}$
mielke-geometroynamics_217 EQ0661 (10.3.18)		■ 0.923 ● N +0	$\square H - (-1)^s * D^* D H = dt + \frac{\gamma}{\delta \ell^2} d\mathcal{A} + \frac{1}{\delta \ell} dh,$	$\square H - (-1)^s * D^* D H = dt + \frac{\gamma}{\delta \ell^2} d\mathcal{A} + \frac{1}{\delta \ell} dh,$	$\square H - (-1)^s * D^* D H = dt + \frac{\gamma}{\delta \ell^2} d\mathcal{A} + \frac{1}{\delta \ell} dh,$
mielke-geometroynamics_222 EQ0685 (10.3.43)		■ 0.923 ● K +0	$C_\alpha - E_\alpha = \Sigma_\alpha - D^{\{\}} \mu_\alpha,$	$C_\alpha - E_\alpha = \Sigma_\alpha - D^{\{\}} \mu_\alpha,$	$C_\alpha - E_\alpha = \Sigma_\alpha - D^{\{\}} \mu_\alpha,$
mielke-geometroynamics_225 EQ0702 (10.6.1)		■ 0.924 ● N -1	$L_A = \frac{1}{2} (F \wedge * F + m_{\text{photon}} F \wedge A).$	$L_A = \frac{1}{2} (F \wedge * F + m_{\text{photon}} F \wedge A).$	$L_A = \frac{1}{2} (F \wedge * F + m_{\text{photon}} F \wedge A).$
mielke-geometroynamics_087 EQ0221 (4.2.14)		■ 0.924 ● K +0	$\Omega^T \rightarrow A^{-1} \Omega^T = G^{-1} (\Omega^T - D D T) = G^{-1} \Omega^T - G^{-1} \Omega^L \otimes T.$	$\Omega^T \rightarrow A^{-1} \Omega^T = G^{-1} (\Omega^T - D D T) = G^{-1} \Omega^T - G^{-1} \Omega^L \otimes T.$	$\Omega^T \rightarrow A^{-1} \Omega^T = G^{-1} (\Omega^T - D D T) = G^{-1} \Omega^T - G^{-1} \Omega^L \otimes T.$
mielke-geometroynamics_143 EQ0443 (6.8.14)		■ 0.924 ● N -1	$H_\alpha = -\frac{1}{\ell^2} * \left(\sum_{i=1}^3 a_i^{(i)} T_\alpha \right),$	$H_\alpha = -\frac{1}{\ell^2} * \left(\sum_{i=1}^3 a_i^{(i)} T_\alpha \right),$	$H_\alpha = -\frac{1}{\ell^2} * \left(\sum_{i=1}^3 a_i^{(i)} T_\alpha \right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E00135 (3.4.6)	063	0.940 N -1	$\Pi^{\mathrm{i}}=\frac{2}{\alpha_g}\left(\frac{\partial L}{\partial \mathcal{F}}^*\Omega+\frac{\partial L}{\partial^*\mathcal{F}}\Omega\right)$	$\Pi^\Omega=\frac{2}{\alpha_g}\left(\frac{\partial L}{\partial \mathcal{F}}^*\Omega+\frac{\partial L}{\partial^*\mathcal{F}}\Omega\right)$	$\Pi^\Omega=\frac{2}{\alpha_g}\left(\frac{\partial L}{\partial \mathcal{F}}^*\Omega+\frac{\partial L}{\partial^*\mathcal{F}}\Omega\right)$
mielke-geometrodynamics_E00772 (11.2.1)	245	0.943 N +0	$L_{\mathrm{D}}=\frac{\mathrm{i}}{2}\left(\bar{\psi}\gamma\wedge^*D\psi+(\overline{D\psi})\wedge^*\gamma\psi\right)-m\bar{\psi}\psi\eta.$	$L_{\mathrm{D}}=\frac{i}{2}\left(\bar{\psi}\gamma\wedge^*D\psi+(\overline{D\psi})\wedge^*\gamma\psi\right)-m\bar{\psi}\psi\eta.$	$L_{\mathrm{D}}=\frac{i}{2}\left(\bar{\psi}\gamma\wedge^*D\psi+(\overline{D\psi})\wedge^*\gamma\psi\right)-m\bar{\psi}\psi\eta.$
mielke-geometrodynamics_E00551 (8.4.5)	179	0.945 W -2	$\begin{aligned} L_{\mathrm{FP}}=& \operatorname{Tr}\left[\beta\wedge^*\bar{\Omega}^g-\bar{\chi}\wedge^*D\Psi^{(\pm)}-\bar{\chi}\wedge^*\left[C,\bar{\Omega}^{\pm g}\right]\right.\\ & \left.+\bar{\eta}D^*\Psi+\bar{\Phi}D^*D\Phi+\bar{\Phi}\left[\Psi,^*\Psi\right]+\mathbb{B}d^*\Gamma\right.\\ & \left.-\bar{C}d^*\Psi+\bar{C}d^*(DC)+\frac{1}{2}\rho\beta\wedge^*\beta+\frac{1}{2}\mathbb{B}\wedge^*\mathbb{B}\right]. \end{aligned}$	$L_{\mathrm{FP}}=-\operatorname{Tr}\left\{\beta\wedge^*\bar{\Omega}^g-\bar{\chi}\wedge^*D\Psi^{(\pm)}-\bar{\chi}\wedge^*\left[C,\bar{\Omega}^{\pm g}\right]\right.\\ \left.+\bar{\eta}D^*\Psi+\bar{\Phi}D^*D\Phi+\bar{\Phi}\left[\Psi,^*\Psi\right]+\mathbb{B}d^*\Gamma\right.\\ \left.-\bar{C}d^*\Psi+\bar{C}d^*(DC)+\frac{1}{2}\rho\beta\wedge^*\beta+\frac{1}{2}\mathbb{B}\wedge^*\mathbb{B}\right\}.$	$L_{\mathrm{FP}}=-\operatorname{Tr}\left\{\beta\wedge^*\bar{\Omega}^g-\bar{\chi}\wedge^*D\Psi^{(\pm)}-\bar{\chi}\wedge^*\left[C,\bar{\Omega}^{\pm g}\right]\right.\\ \left.+\bar{\eta}D^*\Psi+\bar{\Phi}D^*D\Phi+\bar{\Phi}\left[\Psi,^*\Psi\right]+\mathbb{B}d^*\Gamma\right.\\ \left.-\bar{C}d^*\Psi+\bar{C}d^*(DC)+\frac{1}{2}\rho\beta\wedge^*\beta+\frac{1}{2}\mathbb{B}\wedge^*\mathbb{B}\right\}.$
mielke-geometrodynamics_E01069 (16.2.8)	356	0.945 N +0	$V_{\mathrm{eff}}^{\iota}(\rho^*)=\frac{1\pm\rho^2}{\rho^2}\left[\iota(\iota+1)-\left(2-\frac{1}{\beta}\right)\rho^2\right],\,_{(\kappa<0)}^{\kappa>0}.$	$V_{\mathrm{eff}}^{\iota}(\rho^*)=\frac{1\pm\rho^2}{\rho^2}\left[\iota(\iota+1)-\left(2-\frac{1}{\beta}\right)\rho^2\right],\,_{(\kappa<0)}^{\kappa>0}.$	$V_{\mathrm{eff}}^{\iota}(\rho^*)=\frac{1\pm\rho^2}{\rho^2}\left[\iota(\iota+1)-\left(2-\frac{1}{\beta}\right)\rho^2\right],\,_{(\kappa<0)}^{\kappa>0}.$
mielke-geometrodynamics_E00593 (9.1.11)	195	0.945 N +0	$R=\left\{\right\}^*\left(\Omega^g\wedge\vartheta\wedge\vartheta\right)=-\frac{6\kappa\gamma}{\ell^{*2}}=4\Lambda_{\mathrm{eff}}.$	$R=^*(\Omega^g\wedge\vartheta\wedge\vartheta)=-\frac{6\kappa\gamma}{\ell^{*2}}=4\Lambda_{\mathrm{eff}}.$	$R=^*(\Omega^g\wedge\vartheta\wedge\vartheta)=-\frac{6\kappa\gamma}{\ell^{*2}}=4\Lambda_{\mathrm{eff}}.$
mielke-geometrodynamics_E00815 (12.3.4)	276	0.946 N +1	$\begin{aligned} & \operatorname{Tr}\left(\gamma_5K_2\right)=-d^*\mathcal{A} \\ & \operatorname{Tr}\left(\gamma_5K_4\right)=\frac{1}{6}\left[\operatorname{Tr}\left(R^{\{\}\wedge R^{\{\}\right)}-\frac{1}{4}d\mathcal{A}\wedge d\mathcal{A}+d\mathcal{K}\right], \end{aligned}$	$\begin{aligned} & \operatorname{Tr}\left(\gamma_5K_2\right)=-d^*\mathcal{A} \\ & \operatorname{Tr}\left(\gamma_5K_4\right)=\frac{1}{6}\left[\operatorname{Tr}\left(R^{\{\}\wedge R^{\{\}\right)}-\frac{1}{4}d\mathcal{A}\wedge d\mathcal{A}+d\mathcal{K}\right], \end{aligned}$	$\begin{aligned} Tr(\gamma_5K_2) &= -d^*\mathcal{A}, \\ Tr(\gamma_5K_4) &= \frac{1}{6}\left[Tr\left(R^{\{\}\wedge R^{\{\}\right)}-\frac{1}{4}d\mathcal{A}\wedge d\mathcal{A}+d\mathcal{K}\right], \end{aligned}$
mielke-geometrodynamics_E00815 (11.4.29)	257	0.946 N +2	$\begin{aligned} & H^{\prime}\simeq\frac{1+\omega}{\beta}F, \\ & F^{\prime}+\frac{2}{\rho}F\simeq\frac{1-\omega}{\beta}H. \end{aligned}$	$\begin{aligned} H' &\simeq \frac{1+\omega}{\beta}F \\ F' + \frac{2}{\rho}F &\simeq \frac{1-\omega}{\beta}H \end{aligned}$	$\begin{aligned} H' &\simeq \frac{1+\omega}{\beta}F, \\ F' + \frac{2}{\rho}F &\simeq \frac{1-\omega}{\beta}H. \end{aligned}$
mielke-geometrodynamics_E01021 (14.4.16)	328	0.947 N -1	$S_{\mathrm{A}}:=4\pi r_+^2=16\pi M_{\mathrm{ir}}^2\frac{\hbar^2}{M^{*4}c^2},$	$S_{\mathrm{A}}:=4\pi r_+^2=16\pi M_{\mathrm{ir}}^2\frac{\hbar^2}{M^{*4}c^2},$	$S_{\mathrm{A}}:=4\pi r_+^2=16\pi M_{\mathrm{ir}}^2\frac{\hbar^2}{M^{*4}c^2},$
mielke-geometrodynamics_E00855 (12.4.2)	277	0.947 N +0	$\mathcal{H}_{\Lambda}\Psi_{\left(\underline{\Gamma}\right)}^{(\pm)}=0$	$\mathcal{H}_{\Lambda}\Psi_{\left(\underline{\Gamma}\right)}^{(\pm)}=0$	$\mathcal{H}_{\Lambda}\Psi_{\left(\underline{\Gamma}\right)}^{(\pm)}=0$
Conf. MathPix confidence: 0.940 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics E09727 (10.10.7)	231	■ 0.955 ● W +2	$\begin{aligned} & i \gamma^\alpha \Psi \wedge \frac{\delta L}{\delta \vartheta^\alpha} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} - 2D \frac{\delta L}{\delta \bar{\Psi}} \\ & \cong i \gamma^\alpha \Psi \wedge \left(\frac{\theta_{\text{TL}}}{\ell} R_\alpha^* + \frac{\theta_{\text{T}}}{\ell^2} T_\alpha + \Sigma_\alpha \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - D \left(i D \Psi + \frac{i}{2} m \gamma \wedge \Psi \right) \\ & = i \gamma^\alpha \Psi \wedge \left(\frac{\theta_{\text{TL}}}{\ell} R_\alpha^* + \frac{\theta_{\text{T}}}{\ell^2} T_\alpha \right) + \gamma^\alpha \Psi \wedge \left(\frac{1}{4} m \bar{\Psi} \gamma_\alpha \Psi \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - i R_\alpha^* \gamma^\alpha \Psi - \frac{i}{2} m T_\alpha \gamma^\alpha \Psi \\ & + \frac{i}{2} m \gamma \wedge D \Psi. \end{aligned}$	$\begin{aligned} & i \gamma^\alpha \Psi \wedge \frac{\delta L}{\delta \vartheta^\alpha} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} - 2D \frac{\delta L}{\delta \bar{\Psi}} \\ & \cong i \gamma^\alpha \Psi \wedge \left(\frac{\theta_{\text{TL}}}{\ell} R_\alpha^* + \frac{\theta_{\text{T}}}{\ell^2} T_\alpha + \Sigma_\alpha \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - D \left(i D \Psi + \frac{i}{2} m \gamma \wedge \Psi \right) \\ & = i \gamma^\alpha \Psi \wedge \left(\frac{\theta_{\text{TL}}}{\ell} R_\alpha^* + \frac{\theta_{\text{T}}}{\ell^2} T_\alpha \right) + \gamma^\alpha \Psi \wedge \left(\frac{1}{4} m \bar{\Psi} \gamma_\alpha \Psi \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - i R_\alpha^* \gamma^\alpha \Psi - \frac{i}{2} m T_\alpha \gamma^\alpha \Psi \\ & + \frac{i}{2} m \gamma \wedge D \Psi. \end{aligned}$	$\begin{aligned} & i \gamma^\alpha \Psi \wedge \frac{\delta L}{\delta \vartheta^\alpha} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} - 2D \frac{\delta L}{\delta \bar{\Psi}} \\ & \cong i \gamma^\alpha \Psi \wedge \left(\frac{\theta_{\text{TL}}}{\ell} R_\alpha^* + \frac{\theta_{\text{T}}}{\ell^2} T_\alpha + \Sigma_\alpha \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - D \left(i D \Psi + \frac{i}{2} m \gamma \wedge \Psi \right) \\ & = i \gamma^\alpha \Psi \wedge \left(\frac{\theta_{\text{TL}}}{\ell} R_\alpha^* + \frac{\theta_{\text{T}}}{\ell^2} T_\alpha \right) + \gamma^\alpha \Psi \wedge \left(\frac{1}{4} m \bar{\Psi} \gamma_\alpha \Psi \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - i R_\alpha^* \gamma^\alpha \Psi - \frac{i}{2} m T_\alpha \gamma^\alpha \Psi \\ & + \frac{i}{2} m \gamma \wedge D \Psi. \end{aligned} \tag{10.1}$
mielke-geometrodynamics E09768 (11.1.35)	244	■ 0.956 ● N +0	$\bar{\psi} \longrightarrow{ }^{\wedge S^{-1}} \bar{\psi} = \bar{\psi} S,$	$\bar{\psi} \longrightarrow^{S^{-1}} \bar{\psi} = \bar{\psi} S,$	
mielke-geometrodynamics E09973 (14.2.14)	315	■ 0.956 ● C +4	$\begin{aligned} & \hat{\Theta} := \hat{D} \hat{\vartheta} = \hat{d} \hat{\vartheta} - [\hat{\omega}, \hat{\vartheta}] \\ & = \left[\begin{array}{c c} \Theta - \hat{\ell} \omega \wedge \alpha^{\text{T}} & \hat{\ell} \left(d \omega - \omega^g \wedge \omega - \omega \wedge \mathring{\omega} \right) + \vartheta \wedge \alpha + \alpha \wedge \mathring{\vartheta} \\ \hline -\alpha^{\text{T}} \wedge \vartheta - \mathring{\vartheta} \wedge \alpha^{\text{T}} & \mathring{\Theta} - \hat{\ell} \alpha^{\text{T}} \wedge \omega \end{array} \right]. \end{aligned}$	$\begin{aligned} & \hat{\Theta} := \hat{D} \hat{\vartheta} = \hat{d} \hat{\vartheta} - [\hat{\omega}, \hat{\vartheta}] \\ & = \left[\begin{array}{c c} \Theta - \hat{\ell} \omega \wedge \alpha^{\text{T}} & \hat{\ell} \left(d \omega - \omega^g \wedge \omega - \omega \wedge \mathring{\omega} \right) + \vartheta \wedge \alpha + \alpha \wedge \mathring{\vartheta} \\ \hline -\alpha^{\text{T}} \wedge \vartheta - \mathring{\vartheta} \wedge \alpha^{\text{T}} & \mathring{\Theta} - \hat{\ell} \alpha^{\text{T}} \wedge \omega \end{array} \right]. \end{aligned}$	$\begin{aligned} & \hat{\Theta} := \hat{D} \hat{\vartheta} = \hat{d} \hat{\vartheta} - [\hat{\omega}, \hat{\vartheta}] \\ & = \left[\begin{array}{c c} \Theta - \hat{\ell} \omega \wedge \alpha^{\text{T}} & \hat{\ell} (d \omega - \omega^g \wedge \omega - \omega \wedge \mathring{\omega}) + \vartheta \wedge \alpha + \alpha \wedge \mathring{\vartheta} \\ \hline -\alpha^{\text{T}} \wedge \vartheta - \mathring{\vartheta} \wedge \alpha^{\text{T}} & \mathring{\Theta} - \hat{\ell} \alpha^{\text{T}} \wedge \omega \end{array} \right]. \end{aligned}$
mielke-geometrodynamics E09761 (11.1.28)	243	■ 0.957 ● N +0	$\frac{1}{N} \operatorname{Tr}(\gamma \otimes_s \gamma) = ds^2 = g_{ij}(m) dx^i \otimes_s dx^j.$	$\frac{1}{N} \operatorname{Tr}(\gamma \otimes_s \gamma) = ds^2 = g_{ij}(m) dx^i \otimes_s dx^j.$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_160 EQ0486 (7.7.2)	160	■ 0.957 ● N +0	$\begin{aligned} & \vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} := \vartheta^{\alpha} + \ell^2 e_{\beta} \rfloor^* H^{\alpha\beta}, \\ & \Gamma_{\alpha}{}^{\beta} \rightarrow \widetilde{\Gamma}_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} + \ell^2 e_{\alpha} \rfloor^* H^{\beta}. \end{aligned}$	$\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} := \vartheta^{\alpha} + \ell^2 e_{\beta} \rfloor^* H^{\alpha\beta},$ $\Gamma_{\alpha}{}^{\beta} \rightarrow \widetilde{\Gamma}_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} + \ell^2 e_{\alpha} \rfloor^* H^{\beta}.$	$\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} := \vartheta^{\alpha} + \ell^2 e_{\beta} \rfloor^* H^{\alpha\beta},$ $\Gamma_{\alpha}{}^{\beta} \rightarrow \widetilde{\Gamma}_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} + \ell^2 e_{\alpha} \rfloor^* H^{\beta}.$
mielke-geometrodynamics_370 EQ1104 (B.16)	370	■ 0.957 ● N +0	$\begin{aligned} & \left\langle \alpha^{(p)}, \beta^{(p)} \right\rangle := \alpha^{(p)} \wedge^* \beta^{(p)} = \beta^{(p)} \wedge^* \alpha^{(p)} \\ & = (-1)^{s*} \left(\alpha^{(p)} \wedge^* \beta^{(p)} \right) \eta \\ & = \frac{(-1)^s}{p!} A^{\alpha_1 \cdots \alpha_p} B_{\alpha_1 \cdots \alpha_p} d\mu(M) \in C^{\infty}(\wedge^n T^*(M)) \end{aligned}$	$\left\langle \alpha^{(p)}, \beta^{(p)} \right\rangle := \alpha^{(p)} \wedge^* \beta^{(p)} = \beta^{(p)} \wedge^* \alpha^{(p)}$ $= (-1)^{s*} \left(\alpha^{(p)} \wedge^* \beta^{(p)} \right) \eta$ $= \frac{(-1)^s}{p!} A^{\alpha_1 \cdots \alpha_p} B_{\alpha_1 \cdots \alpha_p} d\mu(M) \in C^{\infty}(\wedge^n T^*(M))$	$\langle \alpha^{(p)}, \beta^{(p)} \rangle := \alpha^{(p)} \wedge^* \beta^{(p)} = \beta^{(p)} \wedge^* \alpha^{(p)}$ $= (-1)^{s*} (\alpha^{(p)} \wedge^* \beta^{(p)}) \eta$ $= \frac{(-1)^s}{p!} A^{\alpha_1 \cdots \alpha_p} B_{\alpha_1 \cdots \alpha_p} d\mu(M) \in C^{\infty}(\wedge^n T^*(M))$
mielke-geometrodynamics_176 EQ0537 (8.2.4)	176	■ 0.958 ● W +1	$\begin{array}{l} s \, \bar{C} = \mathbb{B}, \quad \& s \, \mathbb{B} = 0, \\ s \, \bar{\chi} = \beta, \quad s \, \beta = 0, \\ s \, \bar{\Phi} = \bar{\eta}, \quad s \, \bar{\eta} = 0. \end{array}$	$s \bar{C} = \mathbb{B}, \quad s \mathbb{B} = 0,$ $s \bar{\chi} = \beta, \quad s \beta = 0,$ $s \bar{\Phi} = \bar{\eta}, \quad s \bar{\eta} = 0.$	$s \bar{C} = \mathbb{B}, \quad s \mathbb{B} = 0,$ $s \bar{\chi} = \beta, \quad s \beta = 0,$ $s \bar{\Phi} = \bar{\eta}, \quad s \bar{\eta} = 0.$
mielke-geometrodynamics_095 EQ0269 (4.3.34)	095	■ 0.958 ● N +0	$R_{\{i \ j\}} := R_{\{\alpha \ i \ j\}}^{\cdots \alpha}$	$R_{ij} := R_{\alpha ij}^{\cdots \alpha}$	$R_{ij} := R_{\alpha ij}^{\cdots \alpha}$
mielke-geometrodynamics_132 EQ0394 (6.4.9)	132	■ 0.958 ● W +2	$\left. \mathscr{H} \cong n^{\alpha} \underline{\Sigma}_{\alpha} + \Gamma_{\perp}^{\alpha\beta} \rho(I_{\alpha\beta}) \underline{\Psi} \wedge \frac{\partial L}{\partial D \Psi} - \underline{d}[(n] \Psi) \wedge \frac{\partial L}{\partial D \Psi} \right].$	$\mathcal{H} \cong n^{\alpha} \underline{\Sigma}_{\alpha} + \Gamma_{\perp}^{\alpha\beta} \rho(I_{\alpha\beta}) \underline{\Psi} \wedge \frac{\partial L}{\partial D \Psi} - \underline{d}[(n] \Psi) \wedge \frac{\partial L}{\partial D \Psi} \Big].$	$\mathcal{H} \cong n^{\alpha} \underline{\Sigma}_{\alpha} + \Gamma_{\perp}^{\alpha\beta} \rho(I_{\alpha\beta}) \underline{\Psi} \wedge \frac{\partial L}{\partial D \Psi} - \underline{d}[(n] \Psi) \wedge \frac{\partial L}{\partial D \Psi} \Big].$
mielke-geometrodynamics_135 EQ0406 (6.4.26)	135	■ 0.959 ● N ·1	$\stackrel{\scriptscriptstyle (\pm)}{\underline{D}} \stackrel{\scriptscriptstyle (\pm)}{\underline{T}}^B = 0$	$\stackrel{\scriptscriptstyle (\pm)}{\underline{D}} \stackrel{\scriptscriptstyle (\pm)}{\underline{T}}^B = 0$	$\stackrel{\scriptscriptstyle (\pm)}{\underline{D}} \stackrel{\scriptscriptstyle (\pm)}{\underline{T}}^B = 0$
mielke-geometrodynamics_152 EQ0458 (7.2.10)	152	■ 0.959 ● K +0	$\mathfrak{so}(4) \approx \mathfrak{so}(3) \times \mathfrak{so}(3) \approx \mathfrak{su}(2) \times \mathfrak{su}(2).$	$\mathfrak{so}(4) \approx \mathfrak{so}(3) \times \mathfrak{so}(3) \approx \mathfrak{su}(2) \times \mathfrak{su}(2).$	$\mathfrak{so}(4) \approx \mathfrak{so}(3) \times \mathfrak{so}(3) \approx \mathfrak{su}(2) \times \mathfrak{su}(2).$
mielke-geometrodynamics_308 EQ0954 (14.1.36)	308	■ 0.959 ● W +0	$\begin{aligned} & \& \left. (-1)^z \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge \xi \end{smallmatrix} \}} \wedge d\delta \right. \\ & = \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge D^* \xi + \xi \wedge \vartheta \wedge D \xi \end{smallmatrix} \}} \\ & = -\frac{1}{2} D \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge \xi \end{smallmatrix} \}}. \end{aligned}$	$(-1)^z \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge \xi \end{smallmatrix} \}} \wedge d\delta$ $= \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge D^* \xi + \xi \wedge \vartheta \wedge D \xi \end{smallmatrix} \}}$ $= -\frac{1}{2} D \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge \xi \end{smallmatrix} \}}.$	$(-1)^z \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge \xi \end{smallmatrix} \}} \wedge d\delta$ $= \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge D^* \xi + \xi \wedge \vartheta \wedge D \xi \end{smallmatrix} \}}$ $= -\frac{1}{2} D \mathrm{Tr}_{\{ \begin{smallmatrix} \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge \xi \end{smallmatrix} \}}.$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E08338 (5.3.7)	112	0.960 C +3	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$
mielke-geometrodynamics_E09926 (14.1.7)	303	0.960 N +0	$D\left(\vartheta\wedge^*\Omega^g\right)\equiv\Theta\wedge^*\Omega^g.$	$D\left(\vartheta\wedge^*\Omega^g\right)\equiv\Theta\wedge^*\Omega^g.$	$D(\vartheta\wedge^*\Omega^g)\equiv\Theta\wedge^*\Omega^g.$
mielke-geometrodynamics_E08248 (4.3.13)	092	0.961 N +0	$D\left(\vartheta\wedge^*\Omega^g\right)\equiv\Theta\wedge^*\Omega^g.$	$D\phi^{(\circ)}=\vartheta^{\alpha}\nabla_{\alpha}\phi^{(\circ)}.$	$D\phi^{(\circ)}=\vartheta^{\alpha}\nabla_{\alpha}\phi^{(\circ)}.$
mielke-geometrodynamics_E08282 (4.4.11)	098	0.961 N +1	$\begin{aligned} d\Theta &= dd\vartheta - d[\omega^L, \vartheta] \\ &= -[d\omega^L, \vartheta] + [\omega^L, d\vartheta] \\ &= -[\Omega^L, \vartheta] - [\omega^L \wedge \omega^L, \vartheta] + [\omega^L, \Theta] + [\omega^L, [\omega^L, \vartheta]] \\ &= [\omega^L, \Theta] + [\vartheta, \Omega^L] \end{aligned}$	$\begin{aligned} d\Theta &= dd\vartheta - d[\omega^L, \vartheta] \\ &= -[d\omega^L, \vartheta] + [\omega^L, d\vartheta] \\ &= -[\Omega^L, \vartheta] - [\omega^L \wedge \omega^L, \vartheta] + [\omega^L, \Theta] + [\omega^L, [\omega^L, \vartheta]] \\ &= [\omega^L, \Theta] + [\vartheta, \Omega^L] \end{aligned}$	$\begin{aligned} d\Theta &= dd\vartheta - d[\omega^L, \vartheta] \\ &= -[d\omega^L, \vartheta] + [\omega^L, d\vartheta] \\ &= -[\Omega^L, \vartheta] - [\omega^L \wedge \omega^L, \vartheta] + [\omega^L, \Theta] + [\omega^L, [\omega^L, \vartheta]] \\ &= [\omega^L, \Theta] + [\vartheta, \Omega^L]. \end{aligned}$
mielke-geometrodynamics_E08018 (2.2.5)	034	0.962 N +0	$T_{\{p\}^{\{q\}}(M):=\bigcup_{m\in M}T_m^*(M)\otimes^qT_m(M).$	$T_p^q(M):=\bigcup_{m\in M}\otimes^pT_m^*(M)\otimes^qT_m(M).$	$T_p^q(M):=\bigcup_{m\in M}\otimes^pT_m^*(M)\otimes^qT_m(M).$
mielke-geometrodynamics_E08348 (5.3.17)	115	0.962 C +4	$\begin{aligned} G_{\alpha} &:= \frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma} \\ &= G_{\alpha}^{\{ \}} + \frac{(-1)^s}{12}(e_{\alpha}\rfloor\mathcal{A}\wedge^*\mathcal{A} - \frac{1}{3}\mathcal{A}\wedge e_{\alpha}\rfloor^*\mathcal{A}) + \frac{(-1)^s}{6}\vartheta_{\alpha}\wedge d\mathcal{A} \end{aligned}$	$\begin{aligned} G_{\alpha} &:= \frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma} \\ &= G_{\alpha}^{\{ \}} + \frac{(-1)^s}{12}(e_{\alpha}\rfloor\mathcal{A}\wedge^*\mathcal{A} - \frac{1}{3}\mathcal{A}\wedge e_{\alpha}\rfloor^*\mathcal{A}) + \frac{(-1)^s}{6}\vartheta_{\alpha}\wedge d\mathcal{A} \end{aligned}$	$\begin{aligned} G_{\alpha} &:= \frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma} \\ &= G_{\alpha}^{\{ \}} + \frac{(-1)^s}{12}\left(e_{\alpha}\rfloor\mathcal{A}\wedge^*\mathcal{A} - \frac{1}{3}\mathcal{A}\wedge e_{\alpha}\rfloor^*\mathcal{A}\right) + \frac{(-1)^s}{6}\vartheta_{\alpha}\wedge d\mathcal{A} \end{aligned} \tag{5.3}$
mielke-geometrodynamics_E08424 (6.7.6)	140	0.962 N +0	$\begin{aligned} \frac{1}{2\kappa}T^{\gamma}\wedge\eta_{\alpha\beta\gamma}-\frac{D\theta_{\mathrm{T}}}{2\ell^2}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}+D\theta_{\mathrm{L}}\wedge R_{\alpha\beta} &= \vartheta_{[\alpha}\wedge\mu_{\beta]} \\ &= \frac{1}{4}\vartheta_{\alpha}\wedge\vartheta_{\beta}\wedge^*j_5. \end{aligned}$	$\begin{aligned} \frac{1}{2\kappa}T^{\gamma}\wedge\eta_{\alpha\beta\gamma}-\frac{D\theta_{\mathrm{T}}}{2\ell^2}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}+D\theta_{\mathrm{L}}\wedge R_{\alpha\beta} &= \vartheta_{[\alpha}\wedge\mu_{\beta]} \\ &= \frac{1}{4}\vartheta_{\alpha}\wedge\vartheta_{\beta}\wedge^*j_5. \end{aligned}$	$\begin{aligned} \frac{1}{2\kappa}T^{\gamma}\wedge\eta_{\alpha\beta\gamma}-\frac{D\theta_{\mathrm{T}}}{2\ell^2}\wedge\vartheta_{\alpha}\wedge\vartheta_{\beta}+D\theta_{\mathrm{L}}\wedge R_{\alpha\beta} &= \vartheta_{[\alpha}\wedge\mu_{\beta]} \\ &= \frac{1}{4}\vartheta_{\alpha}\wedge\vartheta_{\beta}\wedge^*j_5. \end{aligned}$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_00809 (11.4.21)	255	0.962 W +0	$\begin{array}{c} i \, \gamma^0 \vec{\gamma} \cdot \vec{\partial} \left[\begin{array}{cc} i \, H(\rho) & \chi_{\kappa}^m \\ -F(\rho) & \chi_{-\kappa}^m \end{array} \right] = \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \left(\partial_r - \frac{\vec{\sigma} \cdot \vec{L}}{r} \right) \left[\begin{array}{cc} i F(\rho) & \chi_{-\kappa}^m \\ H(\rho) & \chi_{\kappa}^m \end{array} \right]. \\ \\ \begin{array}{c} i \, \gamma^0 \vec{\gamma} \cdot \vec{\partial} \left[\begin{array}{cc} i \, H(\rho) & \chi_{\kappa}^m \\ -F(\rho) & \chi_{-\kappa}^m \end{array} \right] = \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \left(\partial_r - \frac{\vec{\sigma} \cdot \vec{L}}{r} \right) \left[\begin{array}{cc} i F(\rho) & \chi_{-\kappa}^m \\ H(\rho) & \chi_{\kappa}^m \end{array} \right]. \end{array} \end{array}$	$i \gamma^0 \vec{\gamma} \cdot \vec{\partial} \left[\begin{array}{cc} i H(\rho) & \chi_{\kappa}^m \\ -F(\rho) & \chi_{-\kappa}^m \end{array} \right] = \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \left(\partial_r - \frac{\vec{\sigma} \cdot \vec{L}}{r} \right) \left[\begin{array}{cc} i F(\rho) & \chi_{-\kappa}^m \\ H(\rho) & \chi_{\kappa}^m \end{array} \right].$	
mielke-geometrodynamics_00475 (7.4.3)	156	0.963 N -1	$\{ \}^{\ast} R_{\alpha \beta}^{\left(\star \right)} = - C_{\alpha \beta} + \frac{1}{2} R i^{\tau}{}_{\alpha \beta} + \frac{R}{12} \vartheta_{\alpha} \wedge \vartheta_{\beta}.$	${}^{\ast} R_{\alpha \beta}^{\left(\star \right)} = - C_{\alpha \beta} + \frac{1}{2} R i^{\tau}{}_{\alpha \beta} + \frac{R}{12} \vartheta_{\alpha} \wedge \vartheta_{\beta}.$	${}^{\ast} R_{\alpha \beta}^{\left(\star \right)} = - C_{\alpha \beta} + \frac{1}{2} R j \vec{\mathcal{L}}_{\alpha \beta} + \frac{R}{12} \vartheta_{\alpha} \wedge \vartheta_{\beta}.$
mielke-geometrodynamics_00485 (7.7.1)	159	0.963 W -1	$g_{ij} \rightarrow \widetilde{g}_{ij} = g_{ij} + a \operatorname{Ric}_{ij} + b g_{ij} \operatorname{Ric}_k^k$	$g_{ij} \rightarrow \widetilde{g}_{ij} = g_{ij} + a \operatorname{Ric}_{ij} + b g_{ij} \operatorname{Ric}_k^k$	$g_{ij} \rightarrow \widetilde{g}_{ij} = g_{ij} + a \operatorname{Ric}_{ij} + b g_{ij} \operatorname{Ric}_k^k$
mielke-geometrodynamics_01143 (C.11)	376	0.963 N +0	$(A, B) := \operatorname{Tr}(\operatorname{Ad} A \operatorname{Ad} B) = c_{\iota k}^i c_{s i}^k A^{\iota} B^s =: \overset{\circ}{g}_{\iota s} A^{\iota} B^s$	$(A, B) := \operatorname{Tr}(\operatorname{Ad} A \operatorname{Ad} B) = c_{\iota k}^i c_{s i}^k A^{\iota} B^s =: \overset{\circ}{g}_{\iota s} A^{\iota} B^s$	$(A, B) := \operatorname{Tr}(\operatorname{Ad} A \operatorname{Ad} B) = c_{\iota k}^i c_{s i}^k A^{\iota} B^s =: \overset{\circ}{g}_{\iota s} A^{\iota} B^s$
mielke-geometrodynamics_01122 (B.34)	372	0.963 K +0	$H^p\left(S^n\right)=0 \quad \text { for } \quad 1 \leq p \leq n-1 \text { .}$	$H^p\left(S^n\right)=0 \quad \text { for } \quad 1 \leq p \leq n-1.$	$H^p\left(S^n\right)=0 \quad \text { for } \quad 1 \leq p \leq n-1.$
mielke-geometrodynamics_00791 (11.4.3)	252	0.964 N +1	$\left\{ i \gamma^k \nabla_k^{(t)} + \frac{3\varepsilon}{8} \ell^2 \left(\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5 \right) - m \right\} \psi = 0$	$\left\{ i \gamma^k \nabla_k^{(t)} + \frac{3\varepsilon}{8} \ell^2 \left(\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5 \right) - m \right\} \psi = 0$	$\left\{ i \gamma^k \nabla_k^{(t)} + \frac{3\varepsilon}{8} \ell^2 (\overline{\psi} \psi - a \overline{\psi} \gamma^5 \psi \gamma^5) - m \right\} \psi = 0$
mielke-geometrodynamics_00575 (8.6.13)	185	0.964 N -1	$\left. \left. \left. R := e_{\beta} \right] e_{\alpha} \right] R^{\alpha \beta} = \frac{6 \left(\theta_{\mathrm{T}}^{\star} - a_0 \right)}{\ell^2 \left(\theta_{\mathrm{L}}^{\star} - b_6 \right)} \right. \right.$	$R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha \beta} = \frac{6 \left(\theta_{\mathrm{T}}^{\star} - a_0 \right)}{\ell^2 \left(\theta_{\mathrm{L}}^{\star} - b_6 \right)}$	$R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha \beta} = \frac{6(\theta_{\mathrm{T}}^{\star} - a_0)}{\ell^2(\theta_{\mathrm{L}}^{\star} - b_6)}$
mielke-geometrodynamics_00996 (14.3.9)	321	0.965 W -2	$i \, \widehat{D} \widehat{\psi} := i \widehat{\gamma} \wedge {}^* \widehat{D} \widehat{\psi} = i \gamma \wedge {}^* D \widehat{\psi} + i \, \not{D}^{(int)} \widehat{\psi} = 0,$	$i \widehat{D} \widehat{\psi} := i \widehat{\gamma} \wedge {}^* \widehat{D} \widehat{\psi} = i \gamma \wedge {}^* D \widehat{\psi} + i \, \not{D}^{(int)} \widehat{\psi} = 0,$	$i \, \widehat{D} \widehat{\psi} := i \widehat{\gamma} \wedge {}^* \widehat{D} \widehat{\psi} = i \gamma \wedge {}^* D \widehat{\psi} + i \, \not{D}^{(int)} \widehat{\psi} = 0,$
mielke-geometrodynamics_00131 (3.4.2)	062	0.966 N -1	$\{ \}^{\ast} \operatorname{Tr} \left\{ \left(\{ \}^{\ast} (\Omega \wedge \Omega) \wedge \Omega \right) \right\} = \frac{1}{4 \alpha_g} F_{\alpha \beta}{}^{j \ast} F^{\alpha \beta}{}_{j \ast}.$	${}^{\ast} \mathcal{F} := \frac{1}{\alpha_g} \operatorname{Tr} \left\{ {}^{\ast} (\Omega \wedge \Omega) \right\} = \frac{1}{4 \alpha_g} F_{\alpha \beta}{}^{j \ast} F^{\alpha \beta}{}_{j \ast}.$	${}^{\ast} \mathcal{F} := \frac{1}{\alpha_g} \operatorname{Tr} \left\{ {}^{\ast} (\Omega \wedge \Omega) \right\} = \frac{1}{4 \alpha_g} F_{\alpha \beta}{}^j {}^{\ast} F^{\alpha \beta}{}_j.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Conf. MathPix confidence:

0.962

0.5–0.9

<0.5

— no value

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_366 EQ1089 (A.16)	366	■0.967 ● N +0	$\begin{aligned} & \left(\widehat{\gamma}^A\right)^{\mathrm{T}} := \left(\widehat{\gamma}^{\mu}, \widehat{\gamma}^{2p+1}, \widehat{\gamma}^{2p+2}\right), \\ & \widehat{\gamma}^{\mu} := \gamma^{\mu} \otimes \left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]} \sigma^1\right], \\ & \widehat{\gamma}^{2p+1} := i\mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^3 \otimes \left(\bigotimes_{j=1}^{l-p+1} \sigma^1\right), \\ & \widehat{\gamma}^{2p+2} := i\mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^2 \otimes \left(\bigotimes_{j=1}^{l-p+2} \sigma^1\right), \end{aligned}$	$\begin{aligned} & \left(\widehat{\gamma}^A\right)^{\mathrm{T}} := \left(\widehat{\gamma}^{\mu}, \widehat{\gamma}^{2p+1}, \widehat{\gamma}^{2p+2}\right), \\ & \widehat{\gamma}^{\mu} := \gamma^{\mu} \otimes \left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]} \sigma^1\right], \\ & \widehat{\gamma}^{2p+1} := i\mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^3 \otimes \left(\bigotimes_{j=1}^{l-p+1} \sigma^1\right), \\ & \widehat{\gamma}^{2p+2} := i\mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^2 \otimes \left(\bigotimes_{j=1}^{l-p+2} \sigma^1\right), \end{aligned}$	$\left(\widehat{\gamma}^A\right)^{\mathrm{T}} := \left(\widehat{\gamma}^{\mu}, \widehat{\gamma}^{2p+1}, \widehat{\gamma}^{2p+2}\right),$ $\widehat{\gamma}^{\mu} := \gamma^{\mu} \otimes \left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]} \sigma^1\right],$ $\widehat{\gamma}^{2p+1} := i\mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^3 \otimes \left(\bigotimes_{j=1}^{l-p+1} \sigma^1\right),$ $\widehat{\gamma}^{2p+2} := i\mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^2 \otimes \left(\bigotimes_{j=1}^{l-p+2} \sigma^1\right),$
mielke-geometrodynamics_367 EQ1090 (A.17)	367	■0.967 ● N +0	$\stackrel{*}{\alpha}\left(e_{\alpha_1},\ldots,e_{\alpha_p}\right):=\alpha\left(\sigma_*e_{\alpha_1},\ldots,\sigma_*e_{\alpha_p}\right).$	$\stackrel{*}{\sigma}\alpha\left(e_{\alpha_1},\ldots,e_{\alpha_p}\right):=\alpha\left(\sigma_*e_{\alpha_1},\ldots,\sigma_*e_{\alpha_p}\right).$	$\stackrel{*}{\sigma}\alpha\left(e_{\alpha_1},\ldots,e_{\alpha_p}\right):=\alpha\left(\sigma_*e_{\alpha_1},\ldots,\sigma_*e_{\alpha_p}\right).$
mielke-geometrodynamics_195 EQ0592 (9.1.10)	195	■0.968 ● N +0	$\stackrel{+}{\Omega}=\frac{\gamma\kappa}{\ell^{*2}}\vartheta\wedge\vartheta.$	$\stackrel{+}{\Omega}=\frac{\gamma\kappa}{\ell^{*2}}\vartheta\wedge\vartheta.$	$\stackrel{+}{\Omega}^g=\frac{\gamma\kappa}{\ell^{*2}}\vartheta\wedge\vartheta.$
mielke-geometrodynamics_202 EQ0611 (9.3.2)	202	■0.968 ● N +1	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$
mielke-geometrodynamics_272 EQ0841 (12.2.9)	272	■0.968 ● K +0	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5.$	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5.$	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5.$
mielke-geometrodynamics_212 EQ0636 (10.2.7)	212	■0.970 ● K +0	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha},$	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha},$	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha},$
mielke-geometrodynamics_376 EQ1137 (C.5)	376	■0.970 ● N +0	$3c_{[i s}{}^pc_{ jk]}{}^s:=c_{is}{}^pc_{jk}{}^s+c_{js}{}^pc_{ki}{}^s+c_{ks}{}^pc_{ij}{}^s\equiv 0.$	$3c_{[i s}{}^pc_{ jk]}{}^s:=c_{is}{}^pc_{jk}{}^s+c_{js}{}^pc_{ki}{}^s+c_{ks}{}^pc_{ij}{}^s\equiv 0.$	$3c_{[i s}{}^pc_{ jk]}{}^s:=c_{is}{}^pc_{jk}{}^s+c_{js}{}^pc_{ki}{}^s+c_{ks}{}^pc_{ij}{}^s\equiv 0.$
mielke-geometrodynamics_137 EQ0415 (6.5.3)	137	■0.970 ● N +0	$\begin{aligned} & \langle 0 W(\gamma) 0\rangle^{(1)}=\int_{\mathcal{C} \mathcal{G}}\mathcal{D}Ae^{iC}W(A,\gamma) \\ & =\frac{1}{4\pi}\oint ds\oint dt\dot{\gamma}^{\mu}(s)\dot{\gamma}^{\nu}(t)\eta_{0\mu\nu\kappa}\frac{\gamma^{\kappa}(s)-\gamma^{\kappa}(t)}{ \gamma(s)-\gamma(t) ^3} \end{aligned}$	$\begin{aligned} & \langle 0 W(\gamma) 0\rangle^{(1)}=\int_{\mathcal{C} \mathcal{G}}\mathcal{D}Ae^{iC}W(A,\gamma) \\ & =\frac{1}{4\pi}\oint ds\oint dt\dot{\gamma}^{\mu}(s)\dot{\gamma}^{\nu}(t)\eta_{0\mu\nu\kappa}\frac{\gamma^{\kappa}(s)-\gamma^{\kappa}(t)}{ \gamma(s)-\gamma(t) ^3} \end{aligned}$	$\begin{aligned} \langle 0 W(\gamma) 0\rangle^{(1)} &= \int_{\mathcal{C} \mathcal{G}}\mathcal{D}Ae^{iC}W(A,\gamma) \\ & =\frac{1}{4\pi}\oint ds\oint dt\dot{\gamma}^{\mu}(s)\dot{\gamma}^{\nu}(t)\eta_{0\mu\nu\kappa}\frac{\gamma^{\kappa}(s)-\gamma^{\kappa}(t)}{ \gamma(s)-\gamma(t) ^3} \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_00495 (7.7.13)	162	■ 0.971 ● N +0	$\{ \}^*\left(\backslash\vartheta^{\alpha}\backslash\wedge\backslash\vartheta^{\beta}\backslash\wedge\backslash\vartheta^{\gamma}\backslash\wedge\backslash\vartheta^{\delta}\right)=\eta^{\alpha\beta\gamma\delta},\text{ where }\eta_{\alpha\beta\gamma\delta}:=+\delta_{\alpha\beta\gamma\delta}^{0123},$	$*\left(\vartheta^{\alpha}\wedge\vartheta^{\beta}\wedge\vartheta^{\gamma}\wedge\vartheta^{\delta}\right)=\eta^{\alpha\beta\gamma\delta},\text{ where }\eta_{\alpha\beta\gamma\delta}:=+\delta_{\alpha\beta\gamma\delta}^{0123},$	$*(\vartheta^{\alpha}\wedge\vartheta^{\beta}\wedge\vartheta^{\gamma}\wedge\vartheta^{\delta})=\eta^{\alpha\beta\gamma\delta},\text{ where }\eta_{\alpha\beta\gamma\delta}:=+\delta_{\alpha\beta\gamma\delta}^{0123},$
mielke-geometrodynamics_001147 (C.16)	377	■ 0.971 ● W +1	$\begin{aligned} &\left[\lambda_i,\lambda_j\right]=2if_{ij}^k\lambda_k\quad\left(c_{ij}^k=2if_{ig}^k\right),\\ &\left\{\lambda_i,\lambda_j\right\}_+=\frac{4}{N}\delta_{ij}+2d_{ij}^k\lambda_k. \end{aligned}$	$\left[\lambda_i,\lambda_j\right]=2if_{ij}^k\lambda_k\quad\left(c_{ij}^k=2if_{ig}^k\right),\\ \left\{\lambda_i,\lambda_j\right\}_+=\frac{4}{N}\delta_{ij}+2d_{ij}^k\lambda_k.$	$\left[\lambda_i,\lambda_j\right]=2if_{ij}^k\lambda_k\quad\left(c_{ij}^k=2if_{ig}^k\right),\\ \left\{\lambda_i,\lambda_j\right\}_+=\frac{4}{N}\delta_{ij}+2d_{ij}^k\lambda_k.$
mielke-geometrodynamics_00129 (3.3.30)	061	■ 0.972 ● C -6	$\begin{aligned} j:=\frac{\delta L_{\mathrm{matter}}}{\delta A}&=\frac{\partial L_{\mathrm{matter}}}{\partial A}+d\frac{\partial L_{\mathrm{matter}}}{\partial dA}\\ &=\Psi\wedge\frac{\partial L_{\mathrm{matter}}}{\partial D\Psi}+D\frac{\partial L_{\mathrm{matter}}}{\partial B}, \end{aligned}$	$j:=\frac{\delta L_{\mathrm{matter}}}{\delta A}=\frac{\partial L_{\mathrm{matter}}}{\partial A}+d\frac{\partial L_{\mathrm{matter}}}{\partial dA}\\ =\Psi\wedge\frac{\partial L_{\mathrm{matter}}}{\partial D\Psi}+D\frac{\partial L_{\mathrm{matter}}}{\partial B},$	$j:=\frac{\delta L_{\mathrm{matter}}}{\delta A}=\frac{\partial L_{\mathrm{matter}}}{\partial A}+d\frac{\partial L_{\mathrm{matter}}}{\partial dA}\\ =\Psi\wedge\frac{\partial L_{\mathrm{matter}}}{\partial D\Psi}+D\frac{\partial L_{\mathrm{matter}}}{\partial B},$
mielke-geometrodynamics_00784 (11.3.3)	248	■ 0.972 ● N +0	$\begin{aligned} L_{\mathrm{NL}}&=\frac{1}{2}\bar{\psi}\left[\widetilde{K},*\gamma\right]\psi=-3i^*[\Theta,\vartheta]\wedge(\bar{\psi}^*\gamma\psi)\\ &=-3\tau_s\wedge[\Theta,\vartheta]. \end{aligned}$	$L_{\mathrm{NL}}=\frac{1}{2}\bar{\psi}\left[\widetilde{K},*\gamma\right]\psi=-3i^*[\Theta,\vartheta]\wedge(\bar{\psi}^*\gamma\psi)\\ =-3\tau_s\wedge[\Theta,\vartheta].$	$L_{\mathrm{NL}}=\frac{1}{2}\bar{\psi}[\widetilde{K},*\gamma]\psi=-3i^*[\Theta,\vartheta]\wedge(\bar{\psi}^*\gamma\psi)\\ =-3\tau_s\wedge[\Theta,\vartheta].$
mielke-geometrodynamics_00666 (10.3.23)	218	■ 0.972 ● N +0	$\square\vartheta_{\alpha}+\frac{\ell}{\gamma}{}^*D^*(\delta\ell E_{\alpha}+S_{\alpha})=0,$	$\square\vartheta_{\alpha}+\frac{\ell}{\gamma}{}^*D^*(\delta\ell E_{\alpha}+S_{\alpha})=0,$	$\square\vartheta_{\alpha}+\frac{\ell}{\gamma}{}^*D^*(\delta\ell E_{\alpha}+S_{\alpha})=0,$
mielke-geometrodynamics_00945 (14.1.26)	306	■ 0.972 ● N +0	$\begin{aligned} &\Sigma_{\mathrm{YM}}(\Omega)\wedge\Sigma_{\mathrm{YM}}(\Omega)=\frac{1}{4\alpha_g}\mathrm{Tr}\left\{-\xi\wedge\xi\wedge(\vartheta\wedge^*\xi)\wedge(\vartheta\wedge^*\xi)\right.\\ &\left.+\vartheta\wedge\vartheta\wedge^*(\xi\wedge^*\xi)^2-\vartheta\wedge^*\xi\wedge^*(\vartheta\wedge^*\xi)\wedge^*(\xi\wedge^*\xi)\right\}\\ &=\vartheta\wedge\vartheta\wedge^*(\Sigma_{\mathrm{YM}}(\xi)\wedge^*\Sigma_{\mathrm{YM}}(\xi)) \end{aligned}$	$\Sigma_{\mathrm{YM}}(\Omega)\wedge\Sigma_{\mathrm{YM}}(\Omega)=\frac{1}{4\alpha_g}\mathrm{Tr}\left\{-\xi\wedge\xi\wedge(\vartheta\wedge^*\xi)\wedge(\vartheta\wedge^*\xi)\right.\\ \left.+\vartheta\wedge\vartheta\wedge^*(\xi\wedge^*\xi)^2-\vartheta\wedge^*\xi\wedge^*(\vartheta\wedge^*\xi)\wedge^*(\xi\wedge^*\xi)\right\}\\ =\vartheta\wedge\vartheta\wedge^*(\Sigma_{\mathrm{YM}}(\xi)\wedge^*\Sigma_{\mathrm{YM}}(\xi))$	$\Sigma_{\mathrm{YM}}(\Omega)\wedge\Sigma_{\mathrm{YM}}(\Omega)=\frac{1}{4\alpha_g}\mathrm{Tr}\left\{-\xi\wedge\xi\wedge(\vartheta\wedge^*\xi)\wedge(\vartheta\wedge^*\xi)\right.\\ \left.+\vartheta\wedge\vartheta\wedge^*(\xi\wedge^*\xi)^2-\vartheta\wedge^*\xi\wedge^*(\vartheta\wedge^*\xi)\wedge^*(\xi\wedge^*\xi)\right\}\\ =\vartheta\wedge\vartheta\wedge^*(\Sigma_{\mathrm{YM}}(\xi)\wedge^*\Sigma_{\mathrm{YM}}(\xi))$
mielke-geometrodynamics_00462 (7.3.1)	153	■ 0.973 ● N +1	$L_{\mathrm{SKY}}=\mathrm{Tr}(\Omega^g\wedge^*\Omega^g)=-\frac{1}{2}R^{\alpha\beta}\wedge^*R_{\alpha\beta},$	$L_{\mathrm{SKY}}=\mathrm{Tr}(\Omega^g\wedge^*\Omega^g)=-\frac{1}{2}R^{\alpha\beta}\wedge^*R_{\alpha\beta},$	$L_{\mathrm{SKY}}=\mathrm{Tr}(\Omega^g\wedge^*\Omega^g)=-\frac{1}{2}R^{\alpha\beta}\wedge^*R_{\alpha\beta},$
mielke-geometrodynamics_00356 (6.2.7)	124	■ 0.974 ● N +0	$D\,T^{\alpha}\equiv R_{\beta}{}^{\alpha}\wedge\vartheta^{\beta}$	$DT^{\alpha}\equiv R_{\beta}{}^{\alpha}\wedge\vartheta^{\beta}$	$DT^{\alpha}\equiv R_{\beta}{}^{\alpha}\wedge\vartheta^{\beta}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_304 EQ0931 (14.1.12)		■ 0.981 ● N +0	$\Omega \rightarrow e^{*\delta} \Omega := \Omega \cos \delta + (-1)^{z*} \Omega \sin \delta.$ $\delta + (-1)^{z*} \Omega \sin \delta.$	$\Omega \rightarrow e^{*\delta} \Omega := \Omega \cos \delta + (-1)^{z*} \Omega \sin \delta.$	
mielke-geometrodynamics_219 EQ0671 (10.3.28)		■ 0.982 ● N +0	$\check{\mathcal{J}}_{\alpha}^{\star} := dH_{\alpha}^{\star} \cong \tau_{\alpha}^{\star} + E_{\alpha}^{\star} - (-1)^s \eta_{\alpha\beta\gamma} \Gamma^{\star\beta} \wedge H^{\star\gamma},$	$\check{\mathcal{J}}_{\alpha}^{\star} := dH_{\alpha}^{\star} \cong \tau_{\alpha}^{\star} + E_{\alpha}^{\star} - (-1)^s \eta_{\alpha\beta\gamma} \Gamma^{\star\beta} \wedge H^{\star\gamma},$	
mielke-geometrodynamics_366 EQ1086 (A.12)		■ 0.982 ● W -1	$\begin{aligned} & \mathrm{C}(1,1)=\mathrm{C}(2,0)=\mathrm{GL}(2,\mathbb{R}), \\ & \mathrm{C}(0,2)=\mathbb{H} \quad \text{(field of quaternions)}, \end{aligned}$	$\begin{aligned} \mathrm{C}(1,1) &= \mathrm{C}(2,0) = \mathrm{GL}(2, \mathbb{R}), \\ \mathrm{C}(0,2) &= \mathbb{H} \quad \text{(field of quaternions)}, \end{aligned}$	$\begin{aligned} \mathrm{C}(1,1) &= \mathrm{C}(2,0) = \mathrm{GL}(2, \mathbb{R}), \\ \mathrm{C}(0,2) &= \mathbb{H} \quad \text{(field of quaternions)}, \end{aligned}$
mielke-geometrodynamics_088 EQ0229 (4.2.22)		■ 0.982 ● K +0	$\Theta := D\vartheta = d\vartheta - [\omega^L, \vartheta]$	$\Theta := D\vartheta = d\vartheta - [\omega^L, \vartheta]$	
mielke-geometrodynamics_271 EQ0836 (12.2.4)		■ 0.983 ● N +0	$\mathscr{A} := \frac{1}{4} \mathrm{Tr}(\gamma \wedge D\gamma) = \ast(\vartheta^\alpha \wedge T_\alpha) = \frac{1}{2} T^{[\alpha\beta\gamma]} \eta_{\alpha\beta\gamma} = \mathscr{A}_i dx^i,$	$\mathscr{A} := \frac{1}{4} \mathrm{Tr}(\gamma \wedge D\gamma) = \ast(\vartheta^\alpha \wedge T_\alpha) = \frac{1}{2} T^{[\alpha\beta\gamma]} \eta_{\alpha\beta\gamma} = \mathscr{A}_i dx^i,$	
mielke-geometrodynamics_165 EQ0511 (7.7.29)		■ 0.983 ● W -1	$\begin{aligned} \Theta &:= D\gamma = d\gamma + [\Gamma, \gamma] = (d\vartheta^\alpha + \Gamma_\beta^\alpha \wedge \vartheta^\beta) \gamma_\alpha \\ &= T^\alpha \gamma_\alpha = \frac{1}{2} T_{ij}^\alpha \gamma_\alpha dx^i \wedge dx^j \end{aligned}$	$\begin{aligned} \Theta &:= D\gamma = d\gamma + [\Gamma, \gamma] = (d\vartheta^\alpha + \Gamma_\beta^\alpha \wedge \vartheta^\beta) \gamma_\alpha \\ &= T^\alpha \gamma_\alpha = \frac{1}{2} T_{ij}^\alpha \gamma_\alpha dx^i \wedge dx^j, \end{aligned}$	
mielke-geometrodynamics_212 EQ0637 (10.2.8)		■ 0.983 ● N +0	$\Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\} \star} - K_{\alpha}^{\star}$	$\Gamma_{\alpha}^{\star} = \Gamma_{\alpha}^{\{\} \star} - K_{\alpha}^{\star}$	
mielke-geometrodynamics_096 EQ0271 (4.3.36)		■ 0.983 ● N +0	$[D_{\alpha}, D_{\beta}] = R_{\alpha\beta}^{..cd} L_{cd} - T_{\alpha\beta}^{..c} D_c,$	$[D_{\alpha}, D_{\beta}] = R_{\alpha\beta}^{..cd} L_{cd} - T_{\alpha\beta}^{..c} D_c,$	
mielke-geometrodynamics_181 EQ0561 (8.5.4)		■ 0.984 ● N +0	$\sigma^{(\star)} = (-1)^{\mathrm{sig}} \sigma,$	$\sigma^{(\star)} = (-1)^{\mathrm{sig}} \sigma,$	
mielke-geometrodynamics_056 EQ0097 (3.2.7)		■ 0.985 ● K +0	$D \rightarrow {}^{G^{-1}} D = G^{-1} D G.$	$D \rightarrow {}^{G^{-1}} D = G^{-1} D G.$	
mielke-geometrodynamics_152 EQ0459 (7.2.11)		■ 0.985 ● N +0	$\Omega_{\mathrm{C}} = \Omega_{\mathrm{C}}^{(+)} + \Omega_{\mathrm{C}}^{(-)}, \quad \Omega_{\mathrm{C}}^{(\pm)} := \frac{1}{2} \left(\Omega_{\mathrm{C}} \pm \Omega_{\mathrm{C}}^{(*)} \right)$	$\Omega_{\mathrm{C}} = \Omega_{\mathrm{C}}^{(+)} + \Omega_{\mathrm{C}}^{(-)}, \quad \Omega_{\mathrm{C}}^{(\pm)} := \frac{1}{2} \left(\Omega_{\mathrm{C}} \pm \Omega_{\mathrm{C}}^{(*)} \right)$	
mielke-geometrodynamics_344 EQ1051 (15.2.8)		■ 0.985 ● N +0	$\begin{aligned} A_i &= \frac{1}{2} \left\{ A_i^{(f)j} + \ast A_i^{(f)j} \gamma^5 \right\} \lambda_j^{(f)} \\ &\oplus \frac{1}{2} \left\{ A_i^{(c)j} + \ast A_i^{(c)j} \gamma^5 \right\} \lambda_j^{(c)}, \end{aligned}$	$\begin{aligned} A_i &= \frac{1}{2} \left\{ A_i^{(f)j} + \ast A_i^{(f)j} \gamma^5 \right\} \lambda_j^{(f)} \\ &\oplus \frac{1}{2} \left\{ A_i^{(c)j} + \ast A_i^{(c)j} \gamma^5 \right\} \lambda_j^{(c)}, \end{aligned}$	
mielke-geometrodynamics_369 EQ1096 (B.6)		■ 0.985 ● N +0	$d\alpha^{(p)} \equiv 0.$	$d\alpha^{(p)} \equiv 0.$	
mielke-geometrodynamics_081 EQ0191 (4.1.1)		■ 0.985 ● N +0	$x \rightarrow x' = gx + \mathbf{a}, \quad g \in GL(n, \mathbb{R}), \mathbf{a} \in \mathbb{R}^n.$	$x \rightarrow x' = gx + \mathbf{a}, \quad g \in GL(n, \mathbb{R}), \mathbf{a} \in \mathbb{R}^n.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_00694 (10.5.5)	224	0.985 W +0	$\begin{aligned} L_{\mathrm{D}} &= \frac{i}{2} \left(\bar{\psi}^* \gamma \wedge D\psi + D\bar{\psi} \wedge \gamma^* \psi \right) - m \bar{\psi} \psi \eta \\ &= \bar{\psi} \left(i^* \gamma \wedge D - m \eta \right) \psi + d \left(\frac{i}{2} \bar{\psi}^* \gamma \psi \right), \end{aligned}$	$L_{\mathrm{D}} = \frac{i}{2} \left(\bar{\psi}^* \gamma \wedge D\psi + D\bar{\psi} \wedge \gamma^* \psi \right) - m \bar{\psi} \psi \eta$ $= \bar{\psi} \left(i^* \gamma \wedge D - m \eta \right) \psi + d \left(\frac{i}{2} \bar{\psi}^* \gamma \psi \right),$	$L_{\mathrm{D}} = \frac{i}{2} \left(\overline{\psi}^* \gamma \wedge D\psi + D\overline{\psi} \wedge {}^* \gamma \psi \right) - m \overline{\psi} \psi \eta$ $= \overline{\psi} \left(i^* \gamma \wedge D - m \eta \right) \psi + d \left(\frac{i}{2} \overline{\psi}^* \gamma \psi \right),$
mielke-geometrodynamics_00554 (8.4.8)	180	0.986 N +0	$D^* \stackrel{\pm}{\Omega} = 0$	$D^* \stackrel{\pm}{\Omega} = 0$	$D^* \stackrel{\pm}{\Omega}^g = 0$
mielke-geometrodynamics_00658 (2.7.8)	045	0.987 K +0	$b \rightarrow^G b := Gb \quad \in C^\infty(V^\rho).$	$b \rightarrow^G b := Gb \quad \in C^\infty(V^\rho).$	$b \rightarrow^G b := Gb \quad \in C^\infty(V^\rho).$
mielke-geometrodynamics_00285 (4.4.14)	099	0.987 N +0	$D \left(\vartheta \wedge^* \Omega^g \right) = \Theta \wedge^* \Omega^g$	$D \left(\vartheta \wedge^* \Omega^g \right) = \Theta \wedge^* \Omega^g$	$D(\vartheta \wedge {}^* \Omega^g) = \Theta \wedge {}^* \Omega^g$
mielke-geometrodynamics_00834 (12.2.2)	271	0.988 N +0	$i^* \gamma \wedge \left(\mathscr{D} + \frac{i}{4} m \gamma - \frac{1}{2} T \right) \psi = 0,$	$i^* \gamma \wedge \left(\mathscr{D} + \frac{i}{4} m \gamma - \frac{1}{2} T \right) \psi = 0,$	$i^* \gamma \wedge \left(\mathscr{D} + \frac{i}{4} m \gamma - \frac{1}{2} T \right) \psi = 0,$
mielke-geometrodynamics_00602 (2.1.1)	031	0.989 N +0	$x_2 \circ \stackrel{-1}{x}_1 \colon \mathbb{R} \rightarrow \mathbb{R}$	$x_2 \circ \stackrel{-1}{x}_1 \colon \mathbb{R} \rightarrow \mathbb{R}$	$x_2 \circ \stackrel{-1}{x}_1 \colon \mathbb{R} \rightarrow \mathbb{R}$
mielke-geometrodynamics_00339 (5.3.16)	114	0.989 N +0	$L_{\mathrm{EC}} := -\frac{1}{2\kappa} R^{\alpha\beta} \wedge \eta_{\alpha\beta} = L_{\mathrm{HE}} + \frac{1}{12\kappa} \mathscr{A}^* \mathscr{A}$	$L_{\mathrm{EC}} := -\frac{1}{2\kappa} R^{\alpha\beta} \wedge \eta_{\alpha\beta} = L_{\mathrm{HE}} + \frac{1}{12\kappa} \mathscr{A}^* \mathscr{A}$	$L_{\mathrm{EC}} := -\frac{1}{2\kappa} R^{\alpha\beta} \wedge \eta_{\alpha\beta} = L_{\mathrm{HE}} + \frac{1}{12\kappa} \mathscr{A} \wedge {}^* \mathscr{A}$
mielke-geometrodynamics_00624 (2.4.6)	039	0.990 N +0	$[L_i, L_j] = c_{ij}{}^k L_k.$	$[L_i, L_j] = c_{ij}{}^k L_k.$	$[L_i, L_j] = c_{ij}{}^k L_k.$
mielke-geometrodynamics_00507 (7.7.25)	164	0.990 N +1	$\sigma_{\pm} := (\sigma \pm i^* \sigma) / 2 = \frac{1}{2} (1 \mp \gamma_5) \sigma, \quad \text{with} \quad i^* \sigma_{\pm} = \pm \sigma_{\pm}$	$\sigma_{\pm} := (\sigma \pm i^* \sigma) / 2 = \frac{1}{2} (1 \mp \gamma_5) \sigma, \quad \text{with} \quad i^* \sigma_{\pm} = \pm \sigma_{\pm}$	$\sigma_{\pm} := (\sigma \pm i^* \sigma) / 2 = \frac{1}{2} (1 \mp \gamma_5) \sigma, \quad \text{with} \quad i^* \sigma_{\pm} = \pm \sigma_{\pm}$
mielke-geometrodynamics_00572 (8.6.10)	184	0.990 W -2	$\begin{aligned} H_{\alpha}(**) &= \frac{\theta_{\mathrm{T}}^*}{2\ell^2} K^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} = \frac{1}{\ell^2} \left[\theta_{\mathrm{T}}^* K_{\alpha}^{(*)} - \theta_{\mathrm{T}} T_{\alpha} \right] \\ &= \frac{\theta_{\mathrm{T}}^*}{\ell^2} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta}] T^{\beta}) - \frac{1}{2} e_{\alpha} \right] (T^{\beta} \wedge \vartheta_{\beta}) \Big] - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} \\ &\simeq -\partial L_{\theta} / \partial T^{\alpha}. \end{aligned}$	$H_{\alpha}(**) = \frac{\theta_{\mathrm{T}}^*}{2\ell^2} K^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} = \frac{1}{\ell^2} \left[\theta_{\mathrm{T}}^* K_{\alpha}^{(*)} - \theta_{\mathrm{T}} T_{\alpha} \right]$ $= \frac{\theta_{\mathrm{T}}^*}{\ell^2} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta}] T^{\beta}) - \frac{1}{2} e_{\alpha} \right] (T^{\beta} \wedge \vartheta_{\beta}) \Big] - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha}$ $\simeq -\partial L_{\theta} / \partial T^{\alpha}.$	$H_{\alpha}(**) = \frac{\theta_{\mathrm{T}}^*}{2\ell^2} K^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} = \frac{1}{\ell^2} \left[\theta_{\mathrm{T}}^* K_{\alpha}^{(*)} - \theta_{\mathrm{T}} T_{\alpha} \right]$ $= \frac{\theta_{\mathrm{T}}^*}{\ell^2} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta}] T^{\beta}) - \frac{1}{2} e_{\alpha} \right] (T^{\beta} \wedge \vartheta_{\beta}) \Big] - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha}$ $\simeq -\partial L_{\theta} / \partial T^{\alpha}.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_058 EQ0110 (3.3.11)		■ 0.990 ● C -3	$\begin{aligned} F := & E \wedge dt + B \quad \wedge \quad \left(E_x dx + E_y dy + E_z dz \right) \wedge dt \\ & + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy. \end{aligned}$	$F := E \wedge dt + B = (E_x dx + E_y dy + E_z dz) \wedge dt + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$	$F := E \wedge dt + B = (E_x dx + E_y dy + E_z dz) \wedge dt + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$
mielke-geometrodynamics_064 EQ0142 (3.4.13)		■ 0.990 ● W -2	$\begin{aligned} D \tau \wedge \Pi^\Omega &= D D \Pi^\Omega = \left[\Omega, \Pi^\Omega \right] \\ &= \frac{2}{\alpha_g} \left\{ [\Omega, * \Omega] \frac{\partial L}{\partial \mathcal{F}} + (-1)^s [\Omega, \Omega] \frac{\partial L}{\partial * \mathcal{F}} \right\} = 0. \end{aligned}$	$D\tau = DD\Pi^\Omega = [\Omega, \Pi^\Omega] = \frac{2}{\alpha_g} \left\{ [\Omega, * \Omega] \frac{\partial L}{\partial \mathcal{F}} + (-1)^s [\Omega, \Omega] \frac{\partial L}{\partial * \mathcal{F}} \right\} = 0.$	$D\tau = DD\Pi^\Omega = [\Omega, \Pi^\Omega] = \frac{2}{\alpha_g} \left\{ [\Omega, * \Omega] \frac{\partial L}{\partial \mathcal{F}} + (-1)^s [\Omega, \Omega] \frac{\partial L}{\partial * \mathcal{F}} \right\} = 0.$
mielke-geometrodynamics_067 EQ0155 (3.5.3)		■ 0.990 ● K +0	$\stackrel{*}{\sigma} \Omega = \frac{1}{4} F_{\alpha\beta\mu\nu} \sigma^{\mu\nu} \otimes \vartheta^\alpha \wedge \vartheta^\beta$	$\stackrel{*}{\sigma} \Omega = \frac{1}{4} F_{\alpha\beta\mu\nu} \sigma^{\mu\nu} \otimes \vartheta^\alpha \wedge \vartheta^\beta$	$\stackrel{*}{\sigma} \Omega = \frac{1}{4} F_{\alpha\beta\mu\nu} \sigma^{\mu\nu} \otimes \vartheta^\alpha \wedge \vartheta^\beta$
mielke-geometrodynamics_221 EQ0682 (10.3.40)		■ 0.990 ● C -3	$\theta_L C_L - \tilde{\theta}_L \tilde{C}_L = L_{MB} + \frac{\Delta\theta_{TL}}{\ell} d(\Gamma^{*\alpha} \wedge \vartheta_\alpha),$	$\theta_L C_L - \tilde{\theta}_L \tilde{C}_L = L_{MB} + \frac{\Delta\theta_{TL}}{\ell} d(\Gamma^{*\alpha} \wedge \vartheta_\alpha),$	$\theta_L C_L - \tilde{\theta}_L \tilde{C}_L = L_{MB} + \frac{\Delta\theta_{TL}}{\ell} d(\Gamma^{*\alpha} \wedge \vartheta_\alpha),$
mielke-geometrodynamics_284 EQ0874 (13.3.2)		■ 0.990 ● S -2	$\left[Z^A{}_B, Z^C{}_D \right] = \delta_D^A Z^C{}_B - \delta_B^C Z^A{}_D$	$[Z^A{}_B, Z^C{}_D] = \delta_D^A Z^C{}_B - \delta_B^C Z^A{}_D$	$[\mathcal{Z}^A{}_B, \mathcal{Z}^C{}_D] = \delta_D^A \mathcal{Z}^C{}_B - \delta_B^C \mathcal{Z}^A{}_D$
mielke-geometrodynamics_042 EQ0042 (2.6.5)		■ 0.991 ● C -3	$F_{\alpha\beta}{}^j = \partial_\alpha \Gamma_\beta{}^j - \partial_\beta \Gamma_\alpha{}^j - c_{kl}{}^j \Gamma_\alpha{}^k \Gamma_\beta{}^l,$	$F_{\alpha\beta}{}^j = \partial_\alpha \Gamma_\beta{}^j - \partial_\beta \Gamma_\alpha{}^j - c_{kl}{}^j \Gamma_\alpha{}^k \Gamma_\beta{}^l,$	$F_{\alpha\beta}{}^j = \partial_\alpha \Gamma_\beta{}^j - \partial_\beta \Gamma_\alpha{}^j - c_{kl}{}^j \Gamma_\alpha{}^k \Gamma_\beta{}^l,$
mielke-geometrodynamics_088 EQ0233 (4.2.26)		■ 0.991 ● K +0	$\Omega^T = f\Theta + df \wedge \vartheta - \Omega^L \otimes \tilde{\varphi}$	$\Omega^T = f\Theta + df \wedge \vartheta - \Omega^L \otimes \tilde{\varphi}$	$\Omega^T = f\Theta + df \wedge \vartheta - \Omega^L \otimes \tilde{\varphi}$
mielke-geometrodynamics_246 EQ0778		■ 0.991 ● N +0	$\langle -\infty D\tau_A \infty \rangle^{(+)} = \frac{1}{(4.)8\pi^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right).$	$\langle -\infty D\tau_A \infty \rangle^{(+)} = \frac{1}{(4.)8\pi^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right).$	$\langle -\infty D\tau_A \infty \rangle^{(+)} = \frac{1}{(4.)8\pi^2} \text{Tr}(\Omega^g \wedge \Omega^{g(*)}).$
mielke-geometrodynamics_214 EQ0644 (10.3.1)		■ 0.991 ● N +0	$\widetilde{\Gamma}_\alpha^* = \Gamma_\alpha^* - (-1)^s \frac{\varepsilon}{2\ell} \vartheta_\alpha,$	$\widetilde{\Gamma}_\alpha^* = \Gamma_\alpha^* - (-1)^s \frac{\varepsilon}{2\ell} \vartheta_\alpha,$	$\widetilde{\Gamma}_\alpha^* = \Gamma_\alpha^* - (-1)^s \frac{\varepsilon}{2\ell} \vartheta_\alpha,$
mielke-geometrodynamics_065 EQ0149 (3.4.20)		■ 0.992 ● C -3	$L(\mathcal{F}, * \mathcal{F}) = -\mathcal{F}.$	$L(\mathcal{F}, * \mathcal{F}) = -\mathcal{F}.$	$L(\mathcal{F}, * \mathcal{F}) = -\mathcal{F}.$
mielke-geometrodynamics_127 EQ0368 (6.2.19)		■ 0.992 ● N +0	$L_{\parallel}^{(\pm)} := L_{\parallel} \pm i \frac{2\ell^2}{\kappa} dC_{\text{TT}} = L_{\text{HE}} - L_{\text{EC}} \pm i \frac{2\ell^2}{\kappa} d\tilde{C}_{\text{TT}}.$	$L_{\parallel}^{(\pm)} := L_{\parallel} \pm i \frac{2\ell^2}{\kappa} dC_{\text{TT}} = L_{\text{HE}} - L_{\text{EC}} \pm i \frac{2\ell^2}{\kappa} d\tilde{C}_{\text{TT}}.$	$L_{\parallel}^{(\pm)} := L_{\parallel} \pm i \frac{2\ell^2}{\kappa} dC_{\text{TT}} = L_{\text{HE}} - L_{\text{EC}} \pm i \frac{2\ell^2}{\kappa} d\tilde{C}_{\text{TT}}.$
mielke-geometrodynamics_151 EQ0449 (7.2.2)		■ 0.992 ● N +0	$*\Omega^{g(*)} := \frac{1}{4} {}^*R_{\alpha\beta}^{(*)cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta,$	$*\Omega^{g(*)} := \frac{1}{4} {}^*R_{\alpha\beta}^{(*)cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta,$	$*\Omega^{g(*)} := \frac{1}{4} {}^*R_{\alpha\beta}^{(*)cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E01050 (15.2.7)	344	■ 0.992 ● N +0	$\backslash\mathrm{left}\backslash\backslash\mathrm{bar}\{L\}_{j}=\backslash\mathrm{sigma}^{\backslash\alpha\backslash\mathrm{beta}}\backslash\mathrm{otimes}\backslash\mathrm{lambda}_{\{i\}},\backslash\mathrm{quad}\backslash\mathrm{lambda}_{\{i\}},\backslash\mathrm{quad}i\backslash\mathrm{gamma}^{\{5\}}\backslash\mathrm{lambda}_{\{i\}}\backslash\mathrm{mid}j=1,\backslash\mathrm{ldots}8\backslash\mathrm{N}^{\{2\}}-1\backslash\mathrm{right}\backslash,$	$\left\{\bar{L}_j=\sigma^{\alpha\beta}\otimes\lambda_i,\quad\lambda_i,\quad i\gamma^5\lambda_i\mid j=1,\dots 8N^2-1\right\},$	$\left\{\bar{L}_j=\sigma^{\alpha\beta}\otimes\lambda_i,\quad\lambda_i,\quad i\gamma^5\lambda_i\mid j=1,\dots 8N^2-1\right\},$
mielke-geometrodynamics_E0964 (14.2.4)	313	■ 0.992 ● N +0	$\backslash\mathrm{stackrel}\{\mathrm{circ}\}\{\omega\}=b\backslash\mathrm{stackrel}\{\mathrm{circ}\}\{\vartheta\}.$	$\overset{\circ}{\omega}=b\overset{\circ}{\vartheta}.$	$\overset{\circ}{\omega}=b\overset{\circ}{\vartheta}.$
mielke-geometrodynamics_E00256 (4.3.21)	093	■ 0.992 ● W +0	$R_{\backslash\alpha\backslash\mathrm{beta}\backslash c}^{\{.\backslash.\backslash.\backslash d\}}=2\backslash\mathrm{left}\backslash\backslash\mathrm{partial}_{\{[\backslash\alpha\backslash\backslash\mathrm{Gamma}_{\backslash\mathrm{beta}}\backslash c]^{\{.\backslash.\backslash.\backslash d\}}+\backslash\mathrm{Gamma}_{\{[\backslash\alpha\backslash\mathrm{mid}\backslash h\}^{\{.\backslash.\backslash.\backslash d\}}\backslash\mathrm{Gamma}_{\{[\mathrm{mid}\backslash\mathrm{beta}\backslash c]^{\{.\backslash.\backslash.\backslash h\}}\backslash\mathrm{right}\backslash\}}$	$R_{\alpha\beta c}^{\dots d}=2\left\{\partial_{[\alpha}\Gamma_{\beta]c}^{\dots d}+\Gamma_{[\alpha h}^{\dots d}\Gamma_{ \beta]c}^{\dots h}\right\}$	$R_{\alpha\beta c}^{\dots d}=2\left\{\partial_{[\alpha}\Gamma_{\beta]c}^{\dots d}+\Gamma_{[\alpha h}^{\dots d}\Gamma_{ \beta]c}^{\dots h}\right\}$
mielke-geometrodynamics_E00245 (4.3.10)	091	■ 0.992 ● N +0	$\backslash\mathrm{vartheta}\wedge^{\{*\}}\backslash\mathrm{vartheta}=\backslash\mathrm{eta}.$	$\vartheta\wedge^*\vartheta=\eta.$	$\vartheta\wedge^*\vartheta=\eta.$
mielke-geometrodynamics_E00633 (10.2.4)	212	■ 0.992 ● N +0	$-\{(-1)^s\backslash\mathrm{theta}_{\backslash\mathrm{TL}}\}\mathrm{T}_{\backslash\alpha}-\frac{1}{2}\backslash\mathrm{ell}\backslash\mathrm{theta}_{\backslash\mathrm{T}}\backslash\mathrm{eta}_{\backslash\alpha}-\backslash\mathrm{theta}_{\backslash\mathrm{L}}\backslash\mathrm{ell}\mathrm{R}_{\alpha}^{\star}=\backslash\mathrm{ell}\mathrm{R}_{\backslash\alpha}^{\star}\wedge\backslash\mathrm{ell}\mathrm{R}_{\backslash\alpha}^{\star\star}=\backslash\mathrm{ell}\backslash\mathrm{tau}_{\backslash\alpha}^{\star\star}\wedge\backslash\mathrm{ell}\backslash\mathrm{tau}_{\backslash\alpha}^{\star\star};$	$-(-1)^s\theta_{\mathrm{TL}}T_{\alpha}-\frac{1}{2\ell}\theta_{\mathrm{T}}\eta_{\alpha}-\theta_{\mathrm{L}}\ell R_{\alpha}^{\star}=\ell\tau_{\alpha}^{\star};$	$-(-1)^s\theta_{\mathrm{TL}}T_{\alpha}-\frac{1}{2\ell}\theta_{\mathrm{T}}\eta_{\alpha}-\theta_{\mathrm{L}}\ell R_{\alpha}^{\star}=\ell\tau_{\alpha}^{\star};$
mielke-geometrodynamics_E00701 (10.5.12)	225	■ 0.992 ● N +0	$\backslash\mathrm{hat}\{D\}^{\{*\}}\backslash\mathrm{hat}\{D\}\backslash\mathrm{psi}-\mathrm{m}^2\backslash\mathrm{eta}\backslash\mathrm{psi}+\mathrm{i}^*\backslash\mathrm{sigma}\backslash\mathrm{wedge}\backslash\mathrm{left}(\backslash\mathrm{hat}\{R\}+\mathrm{T}^{\{*\}}\backslash\alpha\backslash\mathrm{hat}\{D\}_{\backslash\alpha}\backslash\mathrm{right})\backslash\mathrm{psi}=0,$	$\hat{D}^*\hat{D}\psi-m^2\eta\psi+i^*\sigma\wedge\left(\hat{R}+T^{\alpha}\hat{D}_{\alpha}\right)\psi=0,$	$\hat{D}^*\hat{D}\psi-m^2\eta\psi+i^*\sigma\wedge\left(\hat{R}+T^{\alpha}\hat{D}_{\alpha}\right)\psi=0,$
mielke-geometrodynamics_E00847 (12.2.15)	273	■ 0.992 ● N +0	$\backslash\mathrm{mathscr}\{A\}_{\backslash\alpha}\backslash\mathrm{mathscr}\{A\}^{\{*\}}\backslash\mathrm{eta}=\backslash\mathrm{mathscr}\{A\}\backslash\mathrm{wedge}\{ \}^{\{*\}}\backslash\mathrm{mathscr}\{A\}\backslash\mathrm{cong}\backslash\mathrm{left}(\backslash\mathrm{ell}^{\{4\}}/4\backslash\mathrm{right})^{\{*\}}\backslash\mathrm{j}_{\{5\}}\backslash\mathrm{wedge}\backslash\mathrm{j}_{\{5\}}=0.$	$\mathscr{A}_{\alpha}\mathscr{A}^{\alpha}\eta=\mathscr{A}\wedge^*\mathscr{A}\cong\left(\ell^4/4\right)^*j_5\wedge j_5=0.$	$\mathscr{A}_{\alpha}\mathscr{A}^{\alpha}\eta=\mathscr{A}\wedge^*\mathscr{A}\cong\left(\ell^4/4\right)^*j_5\wedge j_5=0.$
mielke-geometrodynamics_E00388 (6.4.3)	130	■ 0.993 ● N +0	$\backslash\mathrm{left}.\backslash\mathrm{underline}\{\backslash\mathrm{Psi}\}:=\mathrm{n}\backslash\mathrm{rfloor}(\mathrm{d}\backslash\mathrm{t}\backslash\mathrm{wedge}\backslash\mathrm{Psi})=\backslash\mathrm{left}(1-\{ \}^{\{*\}}\backslash\mathrm{perp}\backslash\mathrm{right})\backslash\mathrm{Psi},\backslash\mathrm{quad}\mathrm{n}\backslash\mathrm{right}\backslash\mathrm{rfloor}\backslash\mathrm{underline}\{\backslash\mathrm{Psi}\}\backslash\mathrm{equiv}\backslash\mathrm{0}.$	$\underline{\Psi}:=n\rfloor(dt\wedge\Psi)=(1-\perp)\Psi,\quad n\rfloor\underline{\Psi}\equiv 0.$	$\underline{\Psi}:=n\rfloor(dt\wedge\Psi)=(1-\perp)\Psi,\quad n\rfloor\underline{\Psi}\equiv 0.$
mielke-geometrodynamics_E00228 (4.2.21)	088	■ 0.993 ● N +0	$\backslash\mathrm{widetilde}\{\backslash\mathrm{widetilde}\{\backslash\mathrm{vartheta}\}\}=f\backslash\mathrm{vartheta}.$	$\widetilde{\vartheta}=f\vartheta.$	$\widetilde{\vartheta}=f\vartheta.$
mielke-geometrodynamics_E00207 (4.4.16)	099	■ 0.993 ● K +0	$D\backslash\mathrm{D}\backslash\mathrm{Pi}^{\{*\}}\backslash\mathrm{L}\}+D\backslash\mathrm{Pi}_{\{ \} }^{\{*\}}\backslash\mathrm{T}\}=D\backslash\mathrm{tau}_{\{s\}}.$	$DD\Pi^{\mathrm{L}}+D\Pi_{\emptyset}^{\mathrm{T}}=D\tau_s.$	$DD\Pi^{\mathrm{L}}+D\Pi_{\mathrm{[]}}^{\mathrm{T}}=D\tau_s.$
mielke-geometrodynamics_E00320 (5.2.4)	111	■ 0.993 ● N +0	$\backslash\mathrm{mathscr}\{A\}:=\frac{1}{4}\{ \}^{\{*\}}\backslash\mathrm{operatorname}\{\mathrm{Tr}\}(\backslash\mathrm{gamma}\backslash\mathrm{wedge}\backslash\mathrm{D}\backslash\mathrm{gamma})=\{ \}^{\{*\}}\backslash\mathrm{left}(\backslash\mathrm{vartheta}^{\{*\}}\backslash\alpha\backslash\mathrm{wedge}\mathrm{T}_{\backslash\alpha}\backslash\mathrm{right})=\frac{1}{2}\mathrm{T}^{\{[\backslash\alpha\backslash\mathrm{beta}\backslash\mathrm{gamma}]\}}\backslash\mathrm{eta}_{\backslash\alpha\backslash\mathrm{beta}\backslash\mathrm{gamma}}=\backslash\mathrm{mathscr}\{A\}_{\{i\}}\mathrm{d}\mathrm{x}^{\{i\}},$	$\mathscr{A}:=\frac{1}{4}{}^*\mathrm{Tr}(\gamma\wedge D\gamma)={}^*(\vartheta^{\alpha}\wedge T_{\alpha})=\frac{1}{2}T^{[\alpha\beta\gamma]}\eta_{\alpha\beta\gamma}=\mathscr{A}_i dx^i,$	$\mathscr{A}:=\frac{1}{4}{}^*Tr(\gamma\wedge D\gamma)={}^*(\vartheta^{\alpha}\wedge T_{\alpha})=\frac{1}{2}T^{[\alpha\beta\gamma]}\eta_{\alpha\beta\gamma}=\mathscr{A}_i dx^i,$
mielke-geometrodynamics_E00100 (3.3.1)	056	■ 0.994 ● N -2	$F:=\mathrm{d}A=\frac{1}{2}F_{ij}\mathrm{d}x^i\wedge\mathrm{d}x^j.$	$F:=dA=\frac{1}{2}F_{ij}dx^i\wedge dx^j.$	$F:=dA=\frac{1}{2}F_{ij}dx^i\wedge dx^j.$
mielke-geometrodynamics_E01102 (B.14)	370	■ 0.994 ● N +1	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\backslash\mathrm{d}_{\backslash\mu}(M)\backslash\mathrm{~\&}:=\{ \}^{\{*\}}\backslash\mathrm{1}=\sqrt{\backslash\mathrm{left} \backslash\mathrm{operatorname}\{\mathrm{det}\}\backslash\mathrm{g}_{\backslash\alpha\backslash\mathrm{beta}}\backslash\mathrm{right} }\backslash\mathrm{vartheta}^{\{1\}}\backslash\mathrm{wedge}\backslash\mathrm{cdots}\backslash\mathrm{wedge}\backslash\mathrm{vartheta}^{\{n\}}=\backslash\mathrm{eta}\backslash\mathrm{~\&}=\sqrt{\backslash\mathrm{left} \backslash\mathrm{operatorname}\{\mathrm{det}\}\backslash\mathrm{g}_{\{i\}\{j\}}\backslash\mathrm{right} }\mathrm{d}\mathrm{x}^{\{1\}}\backslash\mathrm{wedge}\backslash\mathrm{cdots}\backslash\mathrm{wedge}\mathrm{d}\mathrm{x}^{\{n\}}=\frac{1}{n}\backslash\mathrm{vartheta}\wedge^{\{*\}}\backslash\mathrm{vartheta}.\backslash\mathrm{end}\{\mathrm{aligned}\}$	$d_{\mu}(M):={}^*1=\sqrt{ \mathrm{det}g_{\alpha\beta} }\vartheta^1\wedge\cdots\wedge\vartheta^n=\eta\\=\sqrt{ \mathrm{det}g_{ij} }dx^1\wedge\cdots\wedge dx^n=\frac{1}{n}\vartheta\wedge^*\vartheta.$	$d_{\mu}(M):={}^*1=\sqrt{ \mathrm{det}g_{\alpha\beta} }\vartheta^1\wedge\cdots\wedge\vartheta^n=\eta\\=\sqrt{ \mathrm{det}g_{ij} }dx^1\wedge\cdots\wedge dx^n=\frac{1}{n}\vartheta\wedge^*\vartheta.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_154 EQ0467 (7.3.6)		■ 0.994 ● N +0	$\begin{aligned} L_{\mathrm{SKY}}^{\star} &= -\frac{1}{2} R^{\alpha\beta} \wedge \star R_{\alpha\beta} - \frac{(-1)^s}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{\star} \\ &= -\frac{1}{4} \left(R_{\alpha\beta} + \star R_{\alpha\beta}^{\star} \right) \wedge \star \left(R^{\alpha\beta} + \star R^{\alpha\beta(\star)} \right) \\ &= \frac{1}{2} \mathrm{Tr} \left(\bar{\Omega}^g \wedge \star \bar{\Omega}^g \right). \end{aligned}$	$\begin{aligned} L_{\mathrm{SKY}}^{(\star)} &= -\frac{1}{2} R^{\alpha\beta} \wedge \star R_{\alpha\beta} - \frac{(-1)^s}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(\star)} \\ &= -\frac{1}{4} \left(R_{\alpha\beta} + \star R_{\alpha\beta}^{(\star)} \right) \wedge \star \left(R^{\alpha\beta} + \star R^{\alpha\beta(\star)} \right) \\ &= \frac{1}{2} \mathrm{Tr} \left(\bar{\Omega}^g \wedge \star \bar{\Omega}^g \right). \end{aligned}$	
mielke-geometrodynamics_163 EQ0498 (7.7.16)		■ 0.994 ● N +0	$\{ \}^{\star} \Phi := \frac{1}{(4-p)!} p! \sqrt{\det g_{ij}} \varepsilon^{i_1 \dots i_p}_{j_1 \dots j_{4-p}} \Phi_{i_1 \dots i_p} dx^{j_1} \wedge \dots \wedge dx^{j_{4-p}}$	$\star \Phi := \frac{1}{(4-p)!p!} \sqrt{\det g_{ij}} \varepsilon^{i_1 \dots i_p}_{j_1 \dots j_{4-p}} \Phi_{i_1 \dots i_p} dx^{j_1} \wedge \dots \wedge dx^{j_{4-p}},$	
mielke-geometrodynamics_230 EQ0724 (10.10.4)		■ 0.994 ● N +0	$\delta_{\mathrm{susy}} L = i \bar{\sigma} \Psi \gamma^\alpha \wedge \frac{\delta L}{\delta \vartheta^\alpha} + \delta_{\mathrm{susy}} \Gamma_\alpha^\star \wedge \frac{\delta L}{\delta \Gamma_\alpha^\star} + (2 \overline{D\sigma} + c \bar{\sigma} \gamma) \wedge \frac{\delta L}{\delta \bar{\Psi}},$	$\delta_{\mathrm{susy}} L = i \bar{\sigma} \Psi \gamma^\alpha \wedge \frac{\delta L}{\delta \vartheta^\alpha} + \delta_{\mathrm{susy}} \Gamma_\alpha^\star \wedge \frac{\delta L}{\delta \Gamma_\alpha^\star} + (2 \overline{D\sigma} + c \bar{\sigma} \gamma) \wedge \frac{\delta L}{\delta \bar{\Psi}},$	
mielke-geometrodynamics_053 EQ0081 (3.1.6)		■ 0.994 ● N +0	$\begin{aligned} \delta L &= \delta \varphi \wedge \star L_\varphi + \delta(d\varphi) \wedge \pi \\ &= \delta \varphi \wedge \star L_\varphi + d(\delta \varphi) \wedge \pi \\ &= \delta \varphi \wedge (\star L_\varphi + d\pi) + d(\delta \varphi \wedge \pi), \end{aligned}$	$\begin{aligned} \delta L &= \delta \varphi \wedge \star L_\varphi + \delta(d\varphi) \wedge \pi \\ &= \delta \varphi \wedge \star L_\varphi + d(\delta \varphi) \wedge \pi \\ &= \delta \varphi \wedge (\star L_\varphi + d\pi) + d(\delta \varphi \wedge \pi), \end{aligned}$	
mielke-geometrodynamics_165 EQ0509 (7.7.27)		■ 0.994 ● K +0	$D = d + [\Gamma, \quad]$	$D = d + [\Gamma, \quad]$	
mielke-geometrodynamics_130 EQ0387 (6.4.2)		■ 0.994 ● K +0	$\left. \{ \}^{\perp} \Psi := dt \wedge \Psi_{\perp} ; \quad \Psi_{\perp} := n \right\} \Psi,$	$\left. \Psi := dt \wedge \Psi_{\perp} ; \quad \Psi_{\perp} := n \right\} \Psi,$	
mielke-geometrodynamics_214 EQ0646 (10.3.3)		■ 0.995 ● N +0	$\Gamma_{\alpha}^{\star} = (-1)^s \frac{\varepsilon}{2\ell} \vartheta_{\alpha},$	$\Gamma_{\alpha}^{\star} = (-1)^s \frac{\varepsilon}{2\ell} \vartheta_{\alpha},$	
mielke-geometrodynamics_255 EQ0806 (11.4.18)		■ 0.995 ● W +1	$\overrightarrow{J^2} \chi_{\kappa}^m = j(j+1) \chi_{\kappa}^m$	$\overrightarrow{J^2} \chi_{\kappa}^m = j(j+1) \chi_{\kappa}^m$	
mielke-geometrodynamics_321 EQ0995 (14.3.8)		■ 0.995 ● W -1	$L_{\mathrm{anom.}} = \frac{\widehat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge \star \Omega \psi = \frac{\ell^{\star}}{4 \sqrt{\alpha_g}} \bar{\psi} F_{\mu\nu} \gamma^5 [\gamma^{\mu}, \gamma^{\nu}] \psi \eta.$	$L_{\mathrm{anom.}} = \frac{\widehat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge \star \Omega \psi = \frac{\ell^{\star}}{4 \sqrt{\alpha_g}} \bar{\psi} F_{\mu\nu} \gamma^5 [\gamma^{\mu}, \gamma^{\nu}] \psi \eta.$	
mielke-geometrodynamics_056 EQ0098 (3.2.8)		■ 0.995 ● K +0	$\Omega \rightarrow {}^{G^{-1}}\Omega = G^{-1} \Omega G.$	$\Omega \rightarrow {}^{G^{-1}}\Omega = G^{-1} \Omega G.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_0304 EQ0930 (14.1.11)		0.995 N +0	$\mathrm{R}=\{ \ }^*\backslash\mathrm{left}(\backslash\mathrm{Omega}^{\mathrm{g}} \ \mathrm{wedge} \ \mathrm{vartheta} \ \mathrm{wedge} \ \mathrm{vartheta}\backslash\mathrm{right})=\emptyset$	$\mathrm{R} = {}^*(\Omega^g \wedge \vartheta \wedge \vartheta) = 0$	$\mathrm{R} = {}^*(\Omega^g \wedge \vartheta \wedge \vartheta) = 0$
mielke-geometrodynamics_0338 EQ1033 (15.1.10)		0.995 N -1	$\frac{1}{2}\{ \ }^*\backslash\mathrm{left}(\backslash\mathrm{vartheta}^{\mathrm{f}} \ \mathrm{wedge}^{\mathrm{*}} \ \backslash\mathrm{Omega}^{\mathrm{f}}\backslash\mathrm{right})+\backslash\mathrm{Lambda}_{\mathrm{eff}}^{\mathrm{f}}\{ \ }^*\backslash\mathrm{vartheta}^{\mathrm{f}}=\emptyset$	$\frac{1}{2}{}^*(\vartheta^f \wedge {}^*\Omega^f) + \Lambda_{\mathrm{eff}}^f \vartheta^f = 0$	$\frac{1}{2}{}^*(\vartheta^f \wedge {}^*\Omega^f) + \Lambda_{\mathrm{eff}}^f \vartheta^f = 0$
mielke-geometrodynamics_0083 EQ0207 (4.1.21)		0.995 N +0	$\delta\!:\!\left\{\begin{array}{lll} A(n,\,\mathbb{R}) & \times A(M) & \longrightarrow \\ \downarrow & \downarrow & \\ (g,\,a) & (e(m),\,x) & = (\,\mathbf{g}.\mathbf{e}\,(m),\,g^{-1}(x-a))\,.\end{array}\right.$	$\delta\!:\!\left\{\begin{array}{lll} A(n,\,\mathbb{R}) & \times A(M) & \longrightarrow \\ \downarrow & \downarrow & \\ (g,\,a) & (e(m),\,x) & = (\,\mathbf{g}.\mathbf{e}\,(m),\,g^{-1}(x-a))\,.\end{array}\right.$	$\delta\!:\!\left\{\begin{array}{lll} A(n,\,\mathbb{R}) & \times A(M) & \longrightarrow \\ \downarrow & \downarrow & \\ (g,\,a) & (e(m),\,x) & = (g.e(m),\,g^{-1}(x-a))\,.\end{array}\right.$
mielke-geometrodynamics_0217 EQ0658 (10.3.15)		0.995 N +0	$\{ \ }^*\mathrm{D}\backslash\mathrm{left}(\backslash\mathrm{vartheta}^{\mathrm{a}} \ \mathrm{wedge} \ \mathrm{H}_{\mathrm{a}} \ \mathrm{right})-\{ \ }^*\backslash\mathrm{left}(\mathrm{T}^{\mathrm{a}} \ \mathrm{wedge} \ \mathrm{H}_{\mathrm{a}} \ \mathrm{right})=\frac{\gamma}{\delta}\{\backslash\mathrm{gamma}\}\{\backslash\mathrm{delta} \ \mathrm{ell}^{\mathrm{2}}\}\{ \ }^*\backslash\mathrm{left}(\backslash\mathrm{vartheta}^{\mathrm{a}} \ \mathrm{wedge} \ \mathrm{T}_{\mathrm{a}} \ \mathrm{right})+\frac{1}{\delta}\{\backslash\mathrm{delta} \ \mathrm{ell}\}\{ \ }^*\backslash\mathrm{left}(\mathrm{eta}^{\mathrm{a}} \ \mathrm{wedge} \ \mathrm{H}_{\mathrm{a}} \ \mathrm{right})\,.$	${}^*D(\vartheta^\alpha \wedge H_\alpha) - {}^*(T^\alpha \wedge H_\alpha) = \frac{\gamma}{\delta \ell^2} {}^*(\vartheta^\alpha \wedge T_\alpha) + \frac{1}{\delta \ell} {}^*(\eta^\alpha \wedge H_\alpha).$	${}^*D(\vartheta^\alpha \wedge H_\alpha) - {}^*(T^\alpha \wedge H_\alpha) = \frac{\gamma}{\delta \ell^2} {}^*(\vartheta^\alpha \wedge T_\alpha) + \frac{1}{\delta \ell} {}^*(\eta^\alpha \wedge H_\alpha).$
mielke-geometrodynamics_0066 EQ0154 (3.5.2)		0.995 N +0	$\stackrel{\scriptscriptstyle (-)}{\sigma}{}_{\mu\nu}:=\left\{\begin{array}{l} \sigma^{ij}=\frac{1}{2}\varepsilon^{ijk}\sigma_k \\ \sigma^{i0}=\binom{-}{-}\frac{1}{2}\sigma^i \end{array}\right \,i,j,k=1,2,3$	$\binom{-}{\sigma}{}_{\mu\nu}:=\left\{\begin{array}{l} \sigma^{ij}=\frac{1}{2}\varepsilon^{ijk}\sigma_k \\ \sigma^{i0}=\binom{-}{-}\frac{1}{2}\sigma^i \end{array}\right \,i,j,k=1,2,3$	$\binom{-}{\sigma}{}_{\mu\nu}:=\left\{\begin{array}{l} \sigma^{ij}=\frac{1}{2}\varepsilon^{ijk}\sigma_k \\ \sigma^{i0}=\binom{-}{-}\frac{1}{2}\sigma^i \end{array}\right \,i,j,k=1,2,3$
mielke-geometrodynamics_0304 EQ0928 (14.1.9)		0.995 W -1	$\begin{aligned} \Sigma_{\mu\nu}^{\mathrm{YM}} &= \frac{1}{\alpha_g} \mathrm{Tr} \left(F_{\mu\alpha} F_{\nu}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right) \\ &= \frac{1}{2\alpha_g} \mathrm{Tr} (F_{\mu\alpha} F_{\nu}^{\alpha} - (-1)^s {}^* F_{\mu\alpha} {}^* F_{\nu}^{\alpha}), \end{aligned}$	$\Sigma_{\mu\nu}^{\mathrm{YM}} = \frac{1}{\alpha_g} \mathrm{Tr} \left(F_{\mu\alpha} F_{\nu}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right) \\ = \frac{1}{2\alpha_g} \mathrm{Tr} (F_{\mu\alpha} F_{\nu}^{\alpha} - (-1)^s {}^* F_{\mu\alpha} {}^* F_{\nu}^{\alpha}),$	$\Sigma_{\mu\nu}^{\mathrm{YM}} = \frac{1}{\alpha_g} \mathrm{Tr} (F_{\mu\alpha} F_{\nu}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}) \\ = \frac{1}{2\alpha_g} \mathrm{Tr} (F_{\mu\alpha} F_{\nu}^{\alpha} - (-1)^s {}^* F_{\mu\alpha} {}^* F_{\nu}^{\alpha}),$
mielke-geometrodynamics_0033 EQ0006 (2.2.1)		0.996 K +0	$\left.e_{\beta}\right]\vartheta^{\alpha}=\delta^{\alpha}_{\beta}.$	$e_{\beta}]\vartheta^{\alpha} = \delta^{\alpha}_{\beta}.$	$e_{\beta}]\vartheta^{\alpha} = \delta^{\alpha}_{\beta}.$
mielke-geometrodynamics_0099 EQ0204 (4.4.13)		0.996 W +2	$\begin{aligned} D[\backslash\mathrm{Theta}, \ \mathrm{vartheta}] &= [D \ \backslash\mathrm{Theta}, \ \mathrm{vartheta}] + [\backslash\mathrm{Theta}, \ D \ \mathrm{vartheta}] \\ &= [[\vartheta, \Omega^g], \vartheta] + [\Theta, \Theta] \\ &= \vartheta \wedge \Omega^g \wedge \vartheta - \Omega^g \wedge \vartheta \wedge \vartheta + \vartheta \wedge \vartheta \wedge \Omega^g - \vartheta \wedge \Omega^g \wedge \vartheta \\ &= [\vartheta \wedge \vartheta, \Omega^g]. \end{aligned}$	$D[\Theta, \vartheta] = [D\Theta, \vartheta] + [\Theta, D\vartheta] \\ = [[\vartheta, \Omega^g], \vartheta] + [\Theta, \Theta] \\ = \vartheta \wedge \Omega^g \wedge \vartheta - \Omega^g \wedge \vartheta \wedge \vartheta + \vartheta \wedge \vartheta \wedge \Omega^g - \vartheta \wedge \Omega^g \wedge \vartheta \\ = [\vartheta \wedge \vartheta, \Omega^g].$	$D[\Theta, \vartheta] = [D\Theta, \vartheta] + [\Theta, D\vartheta] \\ = [[\vartheta, \Omega^g], \vartheta] + [\Theta, \Theta] \\ = \vartheta \wedge \Omega^g \wedge \vartheta - \Omega^g \wedge \vartheta \wedge \vartheta + \vartheta \wedge \vartheta \wedge \Omega^g - \vartheta \wedge \Omega^g \wedge \vartheta \\ = [\vartheta \wedge \vartheta, \Omega^g].$
mielke-geometrodynamics_0103 EQ0308 (4.5.18)		0.996 N +0	$C_{\mathrm{AdS}}=\hat{M}^2+\frac{\Lambda}{3}\hat{J}^2$	$C_{\mathrm{AdS}} = \hat{M}^2 + \frac{\Lambda}{3} \hat{J}^2$	$C_{\mathrm{AdS}} = \hat{M}^2 + \frac{\Lambda}{3} \hat{J}^2$

Conf. MathPix confidence:

0.995

0.5-0.9

<0.5

— no value

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

• not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_00623 (10.1.11)	210	■0.996 ●S -2	$\frac{\delta C_{\mathrm{TL}}}{\delta \vartheta^\alpha} = \frac{1}{\ell} R_\alpha^\star \quad \text{and} \quad \frac{\delta C_{\mathrm{TL}}}{\delta \Gamma^{\star\alpha}} = \frac{1}{\ell} T_\alpha,$ $\frac{\delta C_{\mathrm{TL}}}{\delta \vartheta^\alpha} = \frac{1}{\ell} R_\alpha^\star \quad \text{and} \quad \frac{\delta C_{\mathrm{TL}}}{\delta \Gamma^{\star\alpha}} = \frac{1}{\ell} T_\alpha,$		
mielke-geometroynamics_00915 (13.8.1)	295	■0.996 ●K +0	$\hat{C} := -\frac{1}{2} \left(\Gamma_A{}^B \wedge d\Gamma_B{}^A - \frac{2}{3} \Gamma_A{}^B \wedge \Gamma_B{}^C \wedge \Gamma_C{}^A \right)$	$\hat{C} := -\frac{1}{2} \left(\Gamma_A{}^B \wedge d\Gamma_B{}^A - \frac{2}{3} \Gamma_A{}^B \wedge \Gamma_B{}^C \wedge \Gamma_C{}^A \right)$	$\hat{C} := -\frac{1}{2} \left(\Gamma_A{}^B \wedge d\Gamma_B{}^A - \frac{2}{3} \Gamma_A{}^B \wedge \Gamma_B{}^C \wedge \Gamma_C{}^A \right)$
mielke-geometroynamics_01071 (16.2.10)	357	■0.996 ●N +0	$E_{\{n\}} = \beta m c^2 \left(2n + \iota + \frac{3}{2} + \sqrt{\frac{g}{4} + \frac{1}{\beta^2}} \right), n = 0, 1, 2 \ldots$	$E_n = \beta m c^2 \left(2n + \iota + \frac{3}{2} + \sqrt{\frac{g}{4} + \frac{1}{\beta^2}} \right), n = 0, 1, 2 \ldots$	$E_n = \beta m c^2 \left(2n + \iota + \frac{3}{2} + \sqrt{\frac{g}{4} + \frac{1}{\beta^2}} \right), n = 0, 1, 2 \ldots$
mielke-geometroynamics_00490 (7.7.7)	161	■0.996 ●K +0	$\begin{gathered} \left. \left$		

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mielke-geometrodynamics_EQ0371 (6.2.22)	127	0.997 W +0	$\stackrel{+}{\Pi} \wedge \stackrel{-}{\Pi}_\beta = H^\beta \wedge H_\beta + \ell^{-4} T^\beta \wedge T_\beta = 0.$	$\stackrel{+}{\Pi} \wedge \stackrel{-}{\Pi}_\beta = H^\beta \wedge H_\beta + \ell^{-4} T^\beta \wedge T_\beta = 0.$	$\stackrel{+}{\Pi} \wedge \stackrel{-}{\Pi}_\beta = H^\beta \wedge H_\beta + \ell^{-4} T^\beta \wedge T_\beta = 0.$
mielke-geometrodynamics_EQ0602 (9.2.1)	198	0.997 W +0	$\begin{aligned} p_1(M^4) &= \frac{1}{2(2\pi)^2} \int_{M^4} \text{Tr}(\Omega^L \wedge \Omega^L) \\ &= \frac{(-1)^{s/2}}{(4\pi)^2} \int_{M^4} R_{\alpha\beta\gamma\delta}^* R^{\alpha\beta\gamma\delta} \sqrt{ g } d^4x \end{aligned}$	$p_1(M^4) = \frac{1}{2(2\pi)^2} \int_{M^4} \text{Tr}(\Omega^L \wedge \Omega^L) = \frac{(-1)^{s/2}}{(4\pi)^2} \int_{M^4} R_{\alpha\beta\gamma\delta}^* R^{\alpha\beta\gamma\delta} \sqrt{ g } d^4x$	$p_1(M^4) = \frac{1}{2(2\pi)^2} \int_{M^4} \text{Tr}(\Omega^L \wedge \Omega^L) = \frac{(-1)^{s/2}}{(4\pi)^2} \int_{M^4} R_{\alpha\beta\gamma\delta}^* R^{\alpha\beta\gamma\delta} \sqrt{ g } d^4x$
mielke-geometrodynamics_EQ0717 (10.9.3)	229	0.997 N +0	$\Sigma_\alpha := \frac{\delta L_\Psi}{\delta \vartheta^\alpha} = \frac{\partial L_\Psi}{\partial \vartheta_\alpha} + D \frac{\partial L_\Psi}{\partial T^\alpha},$	$\Sigma_\alpha := \frac{\delta L_\Psi}{\delta \vartheta^\alpha} = \frac{\partial L_\Psi}{\partial \vartheta_\alpha} + D \frac{\partial L_\Psi}{\partial T^\alpha},$	$\Sigma_\alpha := \frac{\delta L_\Psi}{\delta \vartheta^\alpha} = \frac{\partial L_\Psi}{\partial \vartheta_\alpha} + D \frac{\partial L_\Psi}{\partial T^\alpha},$
mielke-geometrodynamics_EQ0766 (11.1.33)	243	0.997 N +0	$\stackrel{*}{\sigma} \tilde{\omega} = \frac{1}{2} \omega^{\alpha\beta} \sigma_{\alpha\beta} = \frac{1}{2} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} \otimes dx^i =: \Gamma_i dx^i.$	$\stackrel{*}{\sigma} \tilde{\omega} = \frac{1}{2} \omega^{\alpha\beta} \sigma_{\alpha\beta} = \frac{1}{2} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} \otimes dx^i =: \Gamma_i dx^i.$	$\stackrel{*}{\sigma} \tilde{\omega} = \frac{1}{2} \omega^{\alpha\beta} \sigma_{\alpha\beta} = \frac{1}{2} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} \otimes dx^i =: \Gamma_i dx^i.$
mielke-geometrodynamics_EQ1003 (14.3.16)	323	0.997 K +0	$D D = \frac{1}{2} \{D, D\} + \frac{1}{2} [D, D] = \frac{1}{2} \{D, D\} + \frac{1}{2} \tilde{\Omega}$	$D D = \frac{1}{2} \{D, D\} + \frac{1}{2} [D, D] = \frac{1}{2} \{D, D\} + \frac{1}{2} \tilde{\Omega}$	$D D = \frac{1}{2} \{D, D\} + \frac{1}{2} [D, D] = \frac{1}{2} \{D, D\} + \frac{1}{2} \tilde{\Omega}$
mielke-geometrodynamics_EQ0225 (4.2.18)	087	0.997 N +0	$\omega^T = \tilde{\vartheta} - D \tilde{\varphi}$	$\omega^T = \tilde{\vartheta} - D \tilde{\varphi}$	$\omega^T = \tilde{\vartheta} - D \tilde{\varphi}$
mielke-geometrodynamics_EQ0048 (2.6.11)	043	0.997 N +0	$D D D b \equiv 0.$	$D D D b \equiv 0.$	$D D D b \equiv 0.$
mielke-geometrodynamics_EQ0099 (3.2.9)	056	0.997 K +0	$\Delta \omega := G^{-1} \omega - \omega = G^{-1} [\omega, G] - G^{-1} dG = -G^{-1} DG = -G^{-1} D.$	$\Delta \omega := G^{-1} \omega - \omega = G^{-1} [\omega, G] - G^{-1} dG = -G^{-1} DG = -G^{-1} D.$	$\Delta \omega := G^{-1} \omega - \omega = G^{-1} [\omega, G] - G^{-1} dG = -G^{-1} DG = -G^{-1} D.$
mielke-geometrodynamics_EQ0219 (4.2.12)	086	0.997 N +0	$\tilde{\tilde{\Omega}} \rightarrow^{A^{-1}} \tilde{\tilde{\Omega}} = A^{-1} \tilde{\tilde{\Omega}} A$	$\tilde{\tilde{\Omega}} \rightarrow^{A^{-1}} \tilde{\tilde{\Omega}} = A^{-1} \tilde{\tilde{\Omega}} A$	$\tilde{\tilde{\Omega}} \rightarrow^{A^{-1}} \tilde{\tilde{\Omega}} = A^{-1} \tilde{\tilde{\Omega}} A$
mielke-geometrodynamics_EQ0411 (6.4.31)	136	0.997 W +0	$n_{\mathrm{NY}} := \int_{R^4} dC_{\mathrm{TT}} = \int_{S_\infty^3} \underline{C}_{\mathrm{TT}} = 3 \text{Vol}(S^3) k = 6\pi^2 k.$	$n_{\mathrm{NY}} := \int_{R^4} dC_{\mathrm{TT}} = \int_{S_\infty^3} \underline{C}_{\mathrm{TT}} = 3 \text{Vol}(S^3) k = 6\pi^2 k.$	$n_{\mathrm{NY}} := \int_{R^4} dC_{\mathrm{TT}} = \int_{S_\infty^3} \underline{C}_{\mathrm{TT}} = 3 \text{Vol}(S^3) k = 6\pi^2 k.$
mielke-geometrodynamics_EQ0492 (7.7.10)	161	0.997 N -1	$H_{\alpha\beta} := -\frac{\partial L}{\partial R^{\alpha\beta}} = \tilde{H}_{\alpha\beta} + \frac{1}{2} R^{\mu\nu} \wedge (\partial \tilde{H}_{\mu\nu} / \partial R^{\alpha\beta})$	$H_{\alpha\beta} := -\frac{\partial L}{\partial R^{\alpha\beta}} = \tilde{H}_{\alpha\beta} + \frac{1}{2} R^{\mu\nu} \wedge (\partial \tilde{H}_{\mu\nu} / \partial R^{\alpha\beta})$	$H_{\alpha\beta} := -\frac{\partial L}{\partial R^{\alpha\beta}} = \tilde{H}_{\alpha\beta} + \frac{1}{2} R^{\mu\nu} \wedge (\partial \tilde{H}_{\mu\nu} / \partial R^{\alpha\beta})$
mielke-geometrodynamics_EQ0877 (13.3.6)	285	0.997 N +0	$[P_\alpha, P_*^\beta] = \frac{1}{\ell^2} L^\beta{}_\alpha.$	$[P_\alpha, P_*^\beta] = \frac{1}{\ell^2} L^\beta{}_\alpha.$	$[P_\alpha, P_*^\beta] = \frac{1}{\ell^2} L^\beta{}_\alpha.$

Conf. MathPix confidence:

≥0.9

0.5-0.9

<0.5

 — no value
Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_312 E09962 (14.2.2)		■ 0.997 ● N +0	$\stackrel{\circ}{\vartheta}=\varphi_j d\xi^j,$	$\overset{\circ}{\vartheta}=\varphi_j d\xi^j,$	$\overset{\circ}{\vartheta}=\varphi_j d\xi^j,$
mielke-geometrodynamics_354 E01060 (16.1.1)		■ 0.997 ● N -1	$R_{ij}^f-\frac{1}{2}\mathrm{f}_{ij}\left(R^f-2\Lambda_{\mathrm{eff}}\right)=\ell^2\mathrm{T}_{ij}$	$R_{ij}^f-\frac{1}{2}\mathrm{f}_{ij}\left(R^f-2\Lambda_{\mathrm{eff}}\right)=\ell^2\mathrm{T}_{ij}$	$R_{ij}^f-\frac{1}{2}\mathrm{f}_{ij}(R^f-2\Lambda_{\mathrm{eff}})=\ell^2\mathrm{T}_{ij}$
mielke-geometrodynamics_315 E09976 (14.2.17)		■ 0.997 ● W -2	$\begin{aligned} \hat{L}_{\mathrm{W}.} &= \frac{1}{2} \operatorname{Tr}(\hat{\Omega} \wedge \underbrace{\hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}}_{2+\mathrm{K} \text{ terms}}) \\ &= \hat{L}_{\mathrm{W}.}^g + \hat{L}_{\mathrm{W}.}^G + \ell^{*2} \hat{L}_{\parallel} - \frac{1}{2} d \operatorname{Tr}(\hat{\alpha} \wedge \hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}) \end{aligned}$	$\begin{aligned} \hat{L}_{\mathrm{W}.} &= \frac{1}{2} \operatorname{Tr}(\hat{\Omega} \wedge \underbrace{\hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}}_{2+\mathrm{K} \text{ terms}}) \\ &= \hat{L}_{\mathrm{W}.}^g + \hat{L}_{\mathrm{W}.}^G + \ell^{*2} \hat{L}_{\parallel} - \frac{1}{2} d \operatorname{Tr}(\hat{\alpha} \wedge \hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}) \end{aligned}$	$\begin{aligned} \hat{L}_{\mathrm{W}.} &= \frac{1}{2} \operatorname{Tr}(\hat{\Omega} \wedge \underbrace{\hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}}_{2+\mathrm{K} \text{ terms}}) \\ &= \hat{L}_{\mathrm{W}.}^g + \hat{L}_{\mathrm{W}.}^G + \ell^{*2} \hat{L}_{\parallel} - \frac{1}{2} d \operatorname{Tr}(\hat{\alpha} \wedge \hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}) \end{aligned}$
mielke-geometrodynamics_153 E09466 (7.3.5)		■ 0.997 ● W +0	$G_{\alpha}^{\{\}}:=\frac{1}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma},\quad G_{ij}:=\operatorname{Ric}_{ij}^{\{\}}-\frac{1}{2}g_{ij},$	$G_{\alpha}^{\{\}}:=\frac{1}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma},\quad G_{ij}:=\operatorname{Ric}_{ij}^{\{\}}-\frac{1}{2}g_{ij},$	$G_{\alpha}^{\{\}}:=\frac{1}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma},\quad G_{ij}:=\operatorname{Ric}_{ij}^{\{\}}-\frac{1}{2}g_{ij},$
mielke-geometrodynamics_058 E0105 (3.3.6)		■ 0.998 ● W -1	$H:=-\frac{\partial L}{\partial dA}=-\frac{\partial L}{\partial F}\quad\text{and}\quad j:=\frac{\delta L_{\mathrm{mat}}}{\delta A},$	$H:=-\frac{\partial L}{\partial dA}=-\frac{\partial L}{\partial F}\quad\text{and}\quad j:=\frac{\delta L_{\mathrm{mat}}}{\delta A},$	$H:=-\frac{\partial L}{\partial dA}=-\frac{\partial L}{\partial F}\quad\text{and}\quad j:=\frac{\delta L_{\mathrm{mat}}}{\delta A},$
mielke-geometrodynamics_063 E0136 (3.4.7)		■ 0.998 ● N +2	$\begin{aligned} \delta\Omega &= \delta(d\omega - \omega \wedge \omega) = \delta d\omega - \delta\omega \wedge \omega - \omega \wedge \delta\omega \\ &= d\delta\omega - [\omega, \delta\omega] = D\delta\omega. \end{aligned}$	$\begin{aligned} \delta\Omega &= \delta(d\omega - \omega \wedge \omega) = \delta d\omega - \delta\omega \wedge \omega - \omega \wedge \delta\omega \\ &= d\delta\omega - [\omega, \delta\omega] = D\delta\omega. \end{aligned}$	$\begin{aligned} \delta\Omega &= \delta(d\omega - \omega \wedge \omega) = \delta d\omega - \delta\omega \wedge \omega - \omega \wedge \delta\omega \\ &= d\delta\omega - [\omega, \delta\omega] = D\delta\omega. \end{aligned}$
mielke-geometrodynamics_131 E00390 (6.4.5)		■ 0.998 ● W +1	$\mathscr{H}=n\rfloor L-\ell_n\underline{\Psi}\wedge\left(\frac{\partial L}{\partial d\Psi}\right).$	$\mathscr{H}=n\rfloor L-\ell_n\underline{\Psi}\wedge\left(\frac{\partial L}{\partial d\Psi}\right).$	$\mathscr{H}=n\rfloor L-\ell_n\underline{\Psi}\wedge\left(\frac{\partial L}{\partial d\Psi}\right).$
mielke-geometrodynamics_195 E00594 (9.1.12)		■ 0.998 ● K +0	$\bar{\Omega}^{\{\}}=0.$	$\bar{\Omega}^{\{\}}=0.$	$\bar{\Omega}^{\{\}}=0.$
mielke-geometrodynamics_326 E01012 (14.4.7)		■ 0.998 ● W -1	$\begin{aligned} S_{\mathrm{A}} &:= 4\pi\left(r_+^2 + \frac{J(J+1)\hbar^2}{M^2c^2}\right) \\ &= \frac{8\pi\hbar^2}{M^{*4}c^2}\left(M^2 + \frac{\alpha_e}{2}M^{*2}\operatorname{Tr}\bar{Q}^2 + M\sqrt{M^2 - J(J+1)\frac{M^{*4}}{M^2} - \alpha_eM^{*2}\operatorname{Tr}\bar{Q}^2}\right) \end{aligned}$	$\begin{aligned} S_{\mathrm{A}} &:= 4\pi\left(r_+^2 + \frac{J(J+1)\hbar^2}{M^2c^2}\right) \\ &= \frac{8\pi\hbar^2}{M^{*4}c^2}\left(M^2 + \frac{\alpha_e}{2}M^{*2}\operatorname{Tr}\bar{Q}^2 + M\sqrt{M^2 - J(J+1)\frac{M^{*4}}{M^2} - \alpha_eM^{*2}\operatorname{Tr}\bar{Q}^2}\right) \end{aligned}$	$\begin{aligned} S_{\mathrm{A}} &:= 4\pi\left(r_+^2 + \frac{J(J+1)\hbar^2}{M^2c^2}\right) \\ &= \frac{8\pi\hbar^2}{M^{*4}c^2}\left(M^2 + \frac{\alpha_e}{2}M^{*2}\operatorname{Tr}\bar{Q}^2 + M\sqrt{M^2 - J(J+1)\frac{M^{*4}}{M^2} - \alpha_eM^{*2}\operatorname{Tr}\bar{Q}^2}\right) \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_049 EQ0073 (2.8.16)		■ 0.998 ● N +0	$\begin{aligned} & \& \stackrel{\scriptstyle}{\infty}\{\omega\}-\stackrel{\scriptstyle}{\infty}\{\omega\} \wedge \stackrel{\scriptstyle}{\infty}\{\omega\} \\ & \wedge \stackrel{\scriptstyle}{\infty}\{\omega\} \wedge \stackrel{\scriptstyle}{\infty}\{\omega\} \quad \& = -d\left(G^{-1}\right) \wedge d G-G^{-1} d d G-\left(G^{-1} d G\right) \wedge\left(G^{-1} d G\right) \\ & \wedge d G-\left(G^{-1}\right) \wedge d G+G^{-1} G d\left(G^{-1}\right) \wedge d G=0 \end{aligned}$	$\begin{aligned} & d \tilde{\omega}-\tilde{\omega} \wedge \tilde{\omega} \\ & =-d\left(G^{-1}\right) \wedge d G-G^{-1} d d G-\left(G^{-1} d G\right) \wedge\left(G^{-1} d G\right) \\ & =-d\left(G^{-1}\right) \wedge d G+G^{-1} G d\left(G^{-1}\right) \wedge d G=0 \end{aligned}$	$\begin{aligned} & d \tilde{\omega}-\tilde{\omega} \wedge \tilde{\omega} \\ & =-d\left(G^{-1}\right) \wedge d G-G^{-1} d d G-\left(G^{-1} d G\right) \wedge\left(G^{-1} d G\right) \\ & =-d\left(G^{-1}\right) \wedge d G+G^{-1} G d\left(G^{-1}\right) \wedge d G=0 \end{aligned}$
mielke-geometrodynamics_086 EQ0214 (4.2.6)		■ 0.998 ● N +0	$\mathrm{scr}\{G\}(n,\mathbb{R}):=\mathrm{C}^{\infty}\left(A(M)\times_{\mathrm{Ad}}GL(n,\mathbb{R})\right)$	$\mathcal{G}(n,\mathbb{R}):=C^{\infty}(A(M)\times_{\mathrm{Ad}}GL(n,\mathbb{R}))$	$\mathcal{G}(n,\mathbb{R}):=C^{\infty}(A(M)\times_{\mathrm{Ad}}GL(n,\mathbb{R}))$
mielke-geometrodynamics_216 EQ0656 (10.3.13)		■ 0.998 ● N +0	$\left.\left.E_{\alpha}\right L(\star)=e_{\alpha}\right\rfloor L(\star)+(-1)^s\frac{2\gamma}{\ell}\left[\left(e_{\alpha}\right\rfloor R^{\star\beta}\right)\wedge\vartheta_{\beta}\Big]$	$E_{\alpha}=e_{\alpha}\rfloor L(\star)+(-1)^s\frac{2\gamma}{\ell}\left[\left(e_{\alpha}\right\rfloor R^{\star\beta}\right)\wedge\vartheta_{\beta}\Big]$	$E_{\alpha}=e_{\alpha}\rfloor L(\star)+(-1)^s\frac{2\gamma}{\ell}\left[\left(e_{\alpha}\rfloor R^{\star\beta}\right)\wedge\vartheta_{\beta}\right]$
mielke-geometrodynamics_256 EQ0813 (11.4.26)		■ 0.998 ● K +0	$d\,\rho^{\ast}:=e^{\left(\mu-\nu\right)/2}d\rho\;.$	$d\rho^{\ast}:=e^{(\mu-\nu)/2}d\rho.$	$d\rho^{\ast}:=e^{(\mu-\nu)/2}d\rho.$
mielke-geometrodynamics_046 EQ0062 (2.8.3)		■ 0.998 ● W +1	$\begin{aligned} & \& \ast\left(d\,\gamma_k\right)=d\,\pi^{\ast}\left(\gamma_k\right)=D\pi^{\ast}\left(\gamma_k\right) \\ & \left(d\,\gamma_k\right)=d\,\pi^{\ast}\left(\gamma_k\right)=D\pi^{\ast}\left(\gamma_k\right) \\ & =\frac{\left(-1\right)^k}{\left(2\pi i\right)^k\left(k-1\right)!}\mathrm{Tr}(D\varOmega\wedge\varOmega\wedge\cdots\wedge\varOmega)=0. \end{aligned}$	$\begin{aligned} & \ast\left(d\gamma_k\right)=d\pi^{\ast}\left(\gamma_k\right)=D\pi^{\ast}\left(\gamma_k\right) \\ & =\frac{\left(-1\right)^k}{\left(2\pi i\right)^k\left(k-1\right)!}\mathrm{Tr}(D\varOmega\wedge\varOmega\wedge\cdots\wedge\varOmega)=0. \end{aligned}$	$\begin{aligned} & \pi^{\ast}(d\gamma_k)=d\pi^{\ast}(\gamma_k)=D\pi^{\ast}(\gamma_k) \\ & =\frac{(-1)^k}{(2\pi i)^k(k-1)!}\mathrm{Tr}(D\varOmega\wedge\varOmega\wedge\cdots\wedge\varOmega)=0. \end{aligned}$
mielke-geometrodynamics_141 EQ0426 (6.7.8)		■ 0.998 ● K +0	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge\ast j_5,$	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge\ast j_5,$	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge\ast j_5,$
mielke-geometrodynamics_048 EQ0069 (2.8.12)		■ 0.998 ● N +0	$\begin{aligned} & c_2(M)\& =\frac{1}{8\pi^2}\int_M\mathrm{Tr}\left(\omega\wedge\Omega+\frac{1}{3}\omega\wedge\omega\wedge\omega\right) \\ & =\frac{1}{8\pi^2}\int_M\mathrm{Tr}\left(\omega\wedge d\omega-\frac{2}{3}\omega\wedge\omega\wedge\omega\right). \end{aligned}$	$\begin{aligned} c_2(M) &= \frac{1}{8\pi^2}\int_M d\mathrm{Tr}\left(\omega\wedge\Omega+\frac{1}{3}\omega\wedge\omega\wedge\omega\right) \\ &= \frac{1}{8\pi^2}\int_{\partial M}\mathrm{Tr}\left(\omega\wedge d\omega-\frac{2}{3}\omega\wedge\omega\wedge\omega\right). \end{aligned}$	$\begin{aligned} c_2(M) &= \frac{1}{8\pi^2}\int_M d\mathrm{Tr}\left(\omega\wedge\varOmega+\frac{1}{3}\omega\wedge\omega\wedge\omega\right) \\ &= \frac{1}{8\pi^2}\int_{\partial M}\mathrm{Tr}\left(\omega\wedge d\omega-\frac{2}{3}\omega\wedge\omega\wedge\omega\right). \end{aligned}$
mielke-geometrodynamics_064 EQ0144 (3.4.15)		■ 0.998 ● N +0	$\iota:=\tau+(-1)^{n+s}\ast\left[\omega,\Pi^{\varOmega}\right]$	$\iota:=\tau+(-1)^{n+s}\ast\left[\omega,\Pi^{\varOmega}\right]$	$\iota:=\tau+(-1)^{n+s}\ast[\omega,\varPi^{\varOmega}]$
mielke-geometrodynamics_110 EQ0318 (5.2.2)		■ 0.998 ● N -1	$j:=\overline{\psi}^{\ast}\gamma\psi=j^{\mu}\eta_{\mu}\quad\text{and}\quad j_5:=\overline{\psi}^{\ast}\gamma\gamma_5\psi=j_5^{\mu}\eta_{\mu},$	$j:=\overline{\psi}^{\ast}\gamma\psi=j^{\mu}\eta_{\mu}\quad\text{and}\quad j_5:=\overline{\psi}^{\ast}\gamma\gamma_5\psi=j_5^{\mu}\eta_{\mu},$	$j:=\overline{\psi}^{\ast}\gamma\psi=j^{\mu}\eta_{\mu}\quad\text{and}\quad j_5:=\overline{\psi}^{\ast}\gamma\gamma_5\psi=j_5^{\mu}\eta_{\mu},$
mielke-geometrodynamics_116 EQ0347 (5.4.5)		■ 0.998 ● N +0	$\mu=\frac{4}{3}\kappa\Lambda=\frac{1}{3}(2\kappa)^2\rho_{\Lambda}\leq\frac{4}{3}g_{\mathrm{N}\ast}\lambda_{\ast}\simeq0.2$	$\mu=\frac{4}{3}\kappa\Lambda=\frac{1}{3}(2\kappa)^2\rho_{\Lambda}\leq\frac{4}{3}g_{\mathrm{N}\ast}\lambda_{\ast}\simeq0.2$	$\mu=\frac{4}{3}\kappa\Lambda=\frac{1}{3}(2\kappa)^2\rho_{\Lambda}\leq\frac{4}{3}g_{\mathrm{N}\ast}\lambda_{\ast}\simeq0.2$
mielke-geometrodynamics_163 EQ0500 (7.7.18)		■ 0.998 ● K +0	$\ast\sigma=\frac{1}{2}\sigma_{\alpha\beta}\ast\left(\vartheta^{\alpha}\wedge\vartheta^{\beta}\right)=\frac{1}{2}\sigma_{\alpha\beta}\eta^{\alpha\beta}=: \sigma^{(\star)}=i\gamma_5\sigma.$	$\ast\sigma=\frac{1}{2}\sigma_{\alpha\beta}\ast\left(\vartheta^{\alpha}\wedge\vartheta^{\beta}\right)=\frac{1}{2}\sigma_{\alpha\beta}\eta^{\alpha\beta}=: \sigma^{(\star)}=i\gamma_5\sigma.$	$\ast\sigma=\frac{1}{2}\sigma_{\alpha\beta}\ast(\vartheta^{\alpha}\wedge\vartheta^{\beta})=\frac{1}{2}\sigma_{\alpha\beta}\eta^{\alpha\beta}=: \sigma^{(\star)}=i\gamma_5\sigma.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_00670 (10.3.27)	219	■ 0.998 ● N +0	$\check{\mathrm{H}}_{\alpha}:=\mathrm{H}_{\alpha}\,\mathrm{cong}\,\Sigma_{\alpha}+\mathrm{E}_{\alpha}-(-1)^s\eta_{\alpha\beta\gamma}\Gamma^{\star\beta}\wedge H^{\gamma}$	$\check{\mathcal{E}}_{\alpha}:=dH_{\alpha}\cong\Sigma_{\alpha}+E_{\alpha}-(-1)^s\eta_{\alpha\beta\gamma}\Gamma^{\star\beta}\wedge H^{\gamma}$	$\check{\mathcal{E}}_{\alpha}:=dH_{\alpha}\cong\Sigma_{\alpha}+E_{\alpha}-(-1)^s\eta_{\alpha\beta\gamma}\Gamma^{\star\beta}\wedge H^{\gamma}$
mielke-geometrodynamics_00680 (10.3.37)	221	■ 0.998 ● W +1	$\theta_{\mathrm{L}}C_{\mathrm{L}}-\tilde{\theta}_{\mathrm{L}}\tilde{C}_{\mathrm{L}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}-\frac{\Lambda}{\ell}\eta+\frac{\theta_{\mathrm{T}}}{2\ell}\vartheta^{\alpha}\wedge T_{\alpha}+\Delta\theta_{\mathrm{L}}C_{\mathrm{L}}+\frac{\Delta\theta_{\mathrm{TL}}}{\ell}d(\Gamma^{\star\alpha}\wedge\vartheta_{\alpha})$	$\theta_{\mathrm{L}}C_{\mathrm{L}}-\tilde{\theta}_{\mathrm{L}}\tilde{C}_{\mathrm{L}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}-\frac{\Lambda}{\ell}\eta+\frac{\theta_{\mathrm{T}}}{2\ell}\vartheta^{\alpha}\wedge T_{\alpha}+\Delta\theta_{\mathrm{L}}C_{\mathrm{L}}+\frac{\Delta\theta_{\mathrm{TL}}}{\ell}d(\Gamma^{\star\alpha}\wedge\vartheta_{\alpha})$	$\theta_{\mathrm{L}}C_{\mathrm{L}}-\tilde{\theta}_{\mathrm{L}}\tilde{C}_{\mathrm{L}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}-\frac{\Lambda}{\ell}\eta+\frac{\theta_{\mathrm{T}}}{2\ell}\vartheta^{\alpha}\wedge T_{\alpha}+\Delta\theta_{\mathrm{L}}C_{\mathrm{L}}+\frac{\Delta\theta_{\mathrm{TL}}}{\ell}d(\Gamma^{\star\alpha}\wedge\vartheta_{\alpha})$
mielke-geometrodynamics_00700 (10.5.11)	225	■ 0.998 ● K +0	$[\hat{D},\hat{D}]=\hat{R}+T^{\alpha}\hat{D}_{\alpha}\;.$	$[\hat{D},\hat{D}]=\hat{R}+T^{\alpha}\hat{D}_{\alpha}.$	$[\hat{D},\hat{D}]=\hat{R}+T^{\alpha}\hat{D}_{\alpha}\;.$
mielke-geometrodynamics_00820 (11.4.34)	258	■ 0.998 ● N -1	$f^l(\rho)=\overline{\rho}^l\left(1+\overline{e}^{4l+2}\right)^{-1/2}$	$f^l(\rho)=\overline{\rho}^l\left(1+\overline{e}^{4l+2}\right)^{-1/2}$	$f^l(\rho)=\overline{\rho}^l(1+\overline{e}^{4l+2})^{-1/2}$
mielke-geometrodynamics_01126 (B.38)	373	■ 0.998 ● N +0	$\partial:C_p(M,\mathbb{R})\rightarrow C_{p-1}(M,\mathbb{R}),$	$\partial:C_p(M,\mathbb{R})\rightarrow C_{p-1}(M,\mathbb{R}),$	$\partial:C_p(M,\mathbb{R})\rightarrow C_{p-1}(M,\mathbb{R}),$
mielke-geometrodynamics_00516 (7.7.34)	166	■ 0.998 ● N +0	$\widetilde{T}=T-3d\varphi,\quad\mathscr{A}^{\theta}=\mathscr{A}-id\theta\;.$	$\widetilde{T}=T-3d\varphi,\quad\mathscr{A}^{\theta}=\mathscr{A}-id\theta.$	$\widetilde{T}=T-3d\varphi,\quad\mathscr{A}^{\theta}=\mathscr{A}-id\theta\;.$
mielke-geometrodynamics_00967 (14.2.8)	313	■ 0.998 ● N +1	$d\stackrel{\circ}{s}=\stackrel{\circ}{g}_{ij}\,d\xi^i\otimes_s d\xi^j$	$d\stackrel{\circ}{s}=\stackrel{\circ}{g}_{ij}\,d\xi^i\otimes_s d\xi^j$	$d\stackrel{\circ}{s}^2=\stackrel{\circ}{g}_{ij}d\xi^i\otimes_s d\xi^j,$
mielke-geometrodynamics_00060 (2.8.1)	046	■ 0.999 ● K +0	$\operatorname{det}\left(1+\frac{i}{2\pi}\Omega\right)=\stackrel{*}{\pi}\left(1+\gamma_1+\cdots+\gamma_m\right)$	$\det\left(1+\frac{i}{2\pi}\Omega\right)=\overset{*}{\pi}\left(1+\gamma_1+\cdots+\gamma_m\right)$	$\det\left(1+\frac{i}{2\pi}\Omega\right)=\overset{*}{\pi}(1+\gamma_1+\cdots+\gamma_m)$
mielke-geometrodynamics_00071 (2.8.14)	048	■ 0.999 ● K +0	$\stackrel{\infty}{\omega}:= -G^{-1}dG,$	$\overset{\infty}{\omega}:= -G^{-1}dG,$	$\overset{\infty}{\omega}:= -G^{-1}dG,$
mielke-geometrodynamics_00235 (4.2.28)	089	■ 0.999 ● K +0	$Q_{\kappa\mu\nu}:= -D_{\kappa}\,g_{\mu\nu}$	$Q_{\kappa\mu\nu}:= -D_{\kappa}g_{\mu\nu}$	$Q_{\kappa\mu\nu}:= -D_{\kappa}\,g_{\mu\nu}$
mielke-geometrodynamics_00278 (4.4.6)	097	■ 0.999 ● W +0	$\begin{aligned} &\& D\,\Pi^{\mathrm{T}}-E=\Sigma\\ &\& D\,\Pi^{\mathrm{L}}+\Pi_{\mathrm{I}}^{\mathrm{T}}=\tau_{\mathrm{s}} \end{aligned}$	$D\Pi^{\mathrm{T}}-E=\Sigma$ $D\Pi^{\mathrm{L}}+\Pi_{\mathrm{I}}^{\mathrm{T}}=\tau_{\mathrm{s}}$	$D\Pi^{\mathrm{T}}-E=\Sigma$ $D\Pi^{\mathrm{L}}+\Pi_{\mathrm{I}}^{\mathrm{T}}=\tau_{\mathrm{s}}$
mielke-geometrodynamics_00313 (4.6.5)	105	■ 0.999 ● N +0	$\Phi(A,\gamma)=P\exp\left[(i/\hbar)\oint A\right],$	$\Phi(A,\gamma)=P\exp[(i/\hbar)\oint A],$	$\Phi(A,\gamma)=P\exp\left[(i/\hbar)\oint A\right],$
mielke-geometrodynamics_00358 (6.2.9)	125	■ 0.999 ● N +0	$L_{\parallel}:= -\frac{1}{2\kappa}T^{\alpha}\wedge\left(^{(1)}T_{\alpha}-2^{(2)}T_{\alpha}-\frac{1}{2}^{(3)}T_{\alpha}\right),$	$L_{\parallel}:= -\frac{1}{2\kappa}T^{\alpha}\wedge\left(^{(1)}T_{\alpha}-2^{(2)}T_{\alpha}-\frac{1}{2}^{(3)}T_{\alpha}\right),$	$L_{\parallel}:= -\frac{1}{2\kappa}T^{\alpha}\wedge\left(^{(1)}T_{\alpha}-2^{(2)}T_{\alpha}-\frac{1}{2}^{(3)}T_{\alpha}\right),$
mielke-geometrodynamics_00461 (7.2.13)	152	■ 0.999 ● N +0	$\stackrel{(\mp)}{\Omega}_{\mathrm{C}}=0$	$\stackrel{(\mp)}{\Omega}_{\mathrm{C}}=0$	$\stackrel{(\mp)}{\Omega}_{\mathrm{C}}=0$
mielke-geometrodynamics_00620 (10.1.8)	209	■ 0.999 ● N +0	$\frac{\delta C_{\mathrm{T}}}{\delta\vartheta^{\alpha}}=\frac{1}{\ell^2}T_{\alpha}\quad\text{and}\quad\frac{\delta C_{\mathrm{T}}}{\delta\Gamma^{\star\alpha}}=\frac{(-1)^s}{\ell^2}\eta_{\alpha},$	$\frac{\delta C_{\mathrm{T}}}{\delta\vartheta^{\alpha}}=\frac{1}{\ell^2}T_{\alpha}\quad\text{and}\quad\frac{\delta C_{\mathrm{T}}}{\delta\Gamma^{\star\alpha}}=\frac{(-1)^s}{\ell^2}\eta_{\alpha},$	$\frac{\delta C_{\mathrm{T}}}{\delta\vartheta^{\alpha}}=\frac{1}{\ell^2}T_{\alpha}\quad\text{and}\quad\frac{\delta C_{\mathrm{T}}}{\delta\Gamma^{\star\alpha}}=\frac{(-1)^s}{\ell^2}\eta_{\alpha},$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E00667 (10.3.24)	218	0.999 N +0	$\left[\square+(-1)^s m^2\right] \vartheta^{\alpha} \cong 0,$ $\cong \emptyset,$	$\left[\square+(-1)^s m^2\right] \vartheta^{\alpha} \cong 0,$	$\left[\square+(-1)^s m^2\right] \vartheta^{\alpha} \cong 0,$
mielke-geometrodynamics_E00678 (10.3.35)	220	0.999 N -2	$\check{\mathrm{E}}_{\alpha}^{\alpha} \check{\mathrm{E}}_{\alpha}^{\alpha} = \frac{\theta_{\mathrm{TL}}}{\ell} d\Gamma_{\alpha}^{\star},$ $\mathrm{d} \Gamma_{\alpha}^{\star},$	$\check{\mathcal{E}}_{\alpha}^{\mathrm{MB}} - \check{\mathcal{E}}_{\alpha} = \frac{\theta_{\mathrm{TL}}}{\ell} d\Gamma_{\alpha}^{\star},$	$\check{\mathcal{E}}_{\alpha}^{\mathrm{MB}} - \check{\mathcal{E}}_{\alpha} = \frac{\theta_{\mathrm{TL}}}{\ell} d\Gamma_{\alpha}^{\star},$
mielke-geometrodynamics_E00740 (11.1.7)	238	0.999 N +0	$\mathrm{XX} = -\mathrm{o}_{\alpha\beta} x^{\alpha} x^{\beta} \mathbb{1}$ $\mathrm{X}^{\alpha} \mathrm{X}^{\beta} \mathbb{1}$	$\mathrm{XX} = -\mathrm{o}_{\alpha\beta} x^{\alpha} x^{\beta} \mathbb{1}$	$\mathrm{XX} = -\mathrm{o}_{\alpha\beta} x^{\alpha} x^{\beta} \mathbb{1}$
mielke-geometrodynamics_E00776 (11.2.5)	246	0.999 S +0	$L_{\mathrm{D}} = \frac{i}{2} \left(\bar{\psi} \gamma \wedge^* d\psi + d\bar{\psi} \wedge^* \gamma \psi \right) - m \bar{\psi} \psi \eta - \frac{1}{2} \bar{\psi} [\tilde{\omega}, \gamma] \psi,$ $L_{\mathrm{D}} = \frac{i}{2} \left(\bar{\psi} \gamma \wedge^* d\psi + d\bar{\psi} \wedge^* \gamma \psi \right) - m \bar{\psi} \psi \eta - \frac{1}{2} \bar{\psi} [\tilde{\omega}, \gamma] \psi,$	$L_{\mathrm{D}} = \frac{i}{2} \left(\bar{\psi} \gamma \wedge^* d\psi + d\bar{\psi} \wedge^* \gamma \psi \right) - m \bar{\psi} \psi \eta - \frac{1}{2} \bar{\psi} [\tilde{\omega}, \gamma] \psi,$	$L_{\mathrm{D}} = \frac{i}{2} (\bar{\psi} \gamma \wedge^* d\psi + d\bar{\psi} \wedge^* \gamma \psi) - m \bar{\psi} \psi \eta - \frac{1}{2} \bar{\psi} [\tilde{\omega}, \gamma] \psi,$
mielke-geometrodynamics_E00805 (11.4.17)	255	0.999 N +1	$\kappa = \mp \left(j + \frac{1}{2} \right) \quad \text{for} \quad j = l \pm \frac{1}{2}.$ $\kappa = \mp \left(j + \frac{1}{2} \right) \quad \text{for} \quad j = l \pm \frac{1}{2}.$	$\kappa = \mp \left(j + \frac{1}{2} \right) \quad \text{for} \quad j = l \pm \frac{1}{2}.$	$\kappa = \mp \left(j + \frac{1}{2} \right) \quad \text{for} \quad j = l \pm \frac{1}{2}.$
mielke-geometrodynamics_E00989 (14.3.2)	320	0.999 W +0	$\hat{L}_{\mathrm{D}} = \frac{i}{2} \left(\bar{\hat{\psi}} \hat{\gamma} \wedge^* \hat{D} \hat{\psi} + (\bar{\hat{D}} \hat{\psi}) \wedge^* \hat{\gamma} \hat{\psi} \right) - \hat{m} \bar{\hat{\psi}} \hat{\psi} \eta$ $\hat{L}_{\mathrm{D}} = \frac{i}{2} \left(\bar{\hat{\psi}} \hat{\gamma} \wedge^* \hat{D} \hat{\psi} + (\bar{\hat{D}} \hat{\psi}) \wedge^* \hat{\gamma} \hat{\psi} \right) - \hat{m} \bar{\hat{\psi}} \hat{\psi} \eta$	$\hat{L}_{\mathrm{D}} = \frac{i}{2} \left(\bar{\hat{\psi}} \hat{\gamma} \wedge^* \hat{D} \hat{\psi} + (\bar{\hat{D}} \hat{\psi}) \wedge^* \hat{\gamma} \hat{\psi} \right) - \hat{m} \bar{\hat{\psi}} \hat{\psi} \eta$	$\hat{L}_{\mathrm{D}} = \frac{i}{2} (\bar{\hat{\psi}} \hat{\gamma} \wedge^* \hat{D} \hat{\psi} + (\bar{\hat{D}} \hat{\psi}) \wedge^* \hat{\gamma} \hat{\psi}) - \hat{m} \bar{\hat{\psi}} \hat{\psi} \eta$
mielke-geometrodynamics_E01057 (15.2.14)	346	0.999 N +0	$i \bar{\gamma} \wedge^* \bar{D} \psi - 6 \ell^2 \bar{\gamma}^5 \bar{\gamma} \wedge^* (\bar{\psi} \bar{\gamma}^5 \bar{\gamma} \psi) \psi - m \psi \eta = 0$ $i \bar{\gamma} \wedge^* \bar{D} \psi - 6 \ell^2 \bar{\gamma}^5 \bar{\gamma} \wedge^* (\bar{\psi} \bar{\gamma}^5 \bar{\gamma} \psi) \psi - m \psi \eta = 0$	$i \bar{\gamma} \wedge^* \bar{D} \psi - 6 \ell^2 \bar{\gamma}^5 \bar{\gamma} \wedge^* (\bar{\psi} \bar{\gamma}^5 \bar{\gamma} \psi) \psi - m \psi \eta = 0$	$i \bar{\gamma} \wedge^* \bar{D} \psi - 6 \ell^2 \bar{\gamma}^5 \bar{\gamma} \wedge^* (\bar{\psi} \bar{\gamma}^5 \bar{\gamma} \psi) \psi - m \psi \eta = 0$
mielke-geometrodynamics_E00669 (10.3.26)	219	0.999 C -12	$m = \frac{\theta_{\mathrm{T}} \theta_{\mathrm{TL}}}{2 (\theta_{\mathrm{TL}}^2 + \theta_{\mathrm{T}} \theta_{\mathrm{L}}) \ell}, \quad \Lambda_{\mathrm{ind}} = \frac{\theta_{\mathrm{T}}^2 [-(-1)^s 3 \theta_{\mathrm{TL}}^2 + 8 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]}{4 [(-1)^s \theta_{\mathrm{TL}}^2 - 2 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]^2 \ell^2},$ $m = \frac{\theta_{\mathrm{T}} \theta_{\mathrm{TL}}}{2 (\theta_{\mathrm{TL}}^2 + \theta_{\mathrm{T}} \theta_{\mathrm{L}}) \ell}, \quad \Lambda_{\mathrm{ind}} = \frac{\theta_{\mathrm{T}}^2 [-(-1)^s 3 \theta_{\mathrm{TL}}^2 + 8 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]}{4 [(-1)^s \theta_{\mathrm{TL}}^2 - 2 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]^2 \ell^2},$	$m = \frac{\theta_{\mathrm{T}} \theta_{\mathrm{TL}}}{2 (\theta_{\mathrm{TL}}^2 + \theta_{\mathrm{T}} \theta_{\mathrm{L}}) \ell}, \quad \Lambda_{\mathrm{ind}} = \frac{\theta_{\mathrm{T}}^2 [-(-1)^s 3 \theta_{\mathrm{TL}}^2 + 8 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]}{4 [(-1)^s \theta_{\mathrm{TL}}^2 - 2 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]^2 \ell^2},$	$m = \frac{\theta_{\mathrm{T}} \theta_{\mathrm{TL}}}{2 (\theta_{\mathrm{TL}}^2 + \theta_{\mathrm{T}} \theta_{\mathrm{L}}) \ell}, \quad \Lambda_{\mathrm{ind}} = \frac{\theta_{\mathrm{T}}^2 [-(-1)^s 3 \theta_{\mathrm{TL}}^2 + 8 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]}{4 [(-1)^s \theta_{\mathrm{TL}}^2 - 2 \theta_{\mathrm{T}} \theta_{\mathrm{L}}]^2 \ell^2},$
mielke-geometrodynamics_E00929 (14.1.10)	304	0.999 N +0	$\begin{aligned} & \operatorname{Tr} \left(\Sigma_{\mathrm{YM}} \wedge^* \vartheta \right) = \operatorname{Tr} \left(\vartheta \wedge^* \Sigma_{\mathrm{YM}} \right) \\ & = \frac{1}{2 \alpha_g} \operatorname{Tr} \left\{ \vartheta \wedge \Omega \wedge^* (\vartheta \wedge^* \Omega) + \vartheta \wedge^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\} = 0 \end{aligned}$ $\begin{aligned} & \operatorname{Tr} \left(\Sigma_{\mathrm{YM}} \wedge^* \vartheta \right) = \operatorname{Tr} \left(\vartheta \wedge^* \Sigma_{\mathrm{YM}} \right) \\ & = \frac{1}{2 \alpha_g} \operatorname{Tr} \left\{ \vartheta \wedge \Omega \wedge^* (\vartheta \wedge^* \Omega) + \vartheta \wedge^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\} = 0 \end{aligned}$	$\operatorname{Tr} (\Sigma_{\mathrm{YM}} \wedge^* \vartheta) = \operatorname{Tr} (\vartheta \wedge^* \Sigma_{\mathrm{YM}})$ $= \frac{1}{2 \alpha_g} \operatorname{Tr} \left\{ \vartheta \wedge \Omega \wedge^* (\vartheta \wedge^* \Omega) + \vartheta \wedge^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\} = 0$	$\operatorname{Tr} (\Sigma_{\mathrm{YM}} \wedge^* \vartheta) = \operatorname{Tr} (\vartheta \wedge^* \Sigma_{\mathrm{YM}})$ $= \frac{1}{2 \alpha_g} \operatorname{Tr} \left\{ \vartheta \wedge \Omega \wedge^* (\vartheta \wedge^* \Omega) + \vartheta \wedge^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\} = 0$
mielke-geometrodynamics_E00093 (3.2.3)	055	0.999 K +0	$\omega^{\prime} \rightarrow^{G^{-1}} \omega^{\prime} = G^{-1} \omega^{\prime} G + (dG^{-1}) G$ $\omega^{\prime} \rightarrow^{G^{-1}} \omega^{\prime} = G^{-1} \omega^{\prime} G + (dG^{-1}) G$	$\omega^{\prime} \rightarrow^{G^{-1}} \omega^{\prime} = G^{-1} \omega^{\prime} G + (dG^{-1}) G$	$\omega^{\prime} \rightarrow^{G^{-1}} \omega^{\prime} = G^{-1} \omega^{\prime} G + (dG^{-1}) G$
mielke-geometrodynamics_E00113 (3.3.14)	059	0.999 N +0	$H = {}^* F.$ $H = {}^* F.$	$H = {}^* F.$	$H = {}^* F.$

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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_071 E08173 (3.7.1)		■ 0.999 ● N +0	$\mathrm{L}_{\mathrm{H}}=-\bar{\Theta}^{\left(1\right)}\wedge^{*}D\phi^0+\bar{\Theta}^{\left(1\right)}\wedge^{*}\Theta^{\left(1\right)}+\frac{\lambda}{4}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)^2\eta.$	$L_{\mathrm{H}}=-\bar{\Theta}^{\left(1\right)}\wedge^{*}D\phi^0+\bar{\Theta}^{\left(1\right)}\wedge^{*}\Theta^{\left(1\right)}+\frac{\lambda}{4}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)^2\eta.$	$L_{\mathrm{H}}=-\bar{\Theta}^{\left(1\right)}\wedge^{*}D\phi^0+\bar{\Theta}^{\left(1\right)}\wedge^{*}\Theta^{\left(1\right)}+\frac{\lambda}{4}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)^2\eta.$
mielke-geometrodynamics_083 E08281 (4.1.15)		■ 0.999 ● N +0	$\delta_g^*\vartheta=g^{-1}\vartheta,\quad g\in GL(n,\mathbb{R}),$	$\delta_g^*\vartheta=g^{-1}\vartheta,\quad g\in GL(n,\mathbb{R}),$	$\delta_g^*\vartheta=g^{-1}\vartheta,\quad g\in GL(n,\mathbb{R}),$
mielke-geometrodynamics_097 E08275 (4.4.3)		■ 0.999 ● W +1	$\begin{aligned} & \mathrm{d}(\delta\vartheta)-\left[\omega^{\mathrm{L}},\delta\vartheta\right]-\left[\delta\omega^{\mathrm{L}},\vartheta\right] \\ & =D(\delta\vartheta)-\left[\delta\omega^{\mathrm{L}},\vartheta\right]. \end{aligned}$	$\begin{aligned} & =d(\delta\vartheta)-\left[\omega^{\mathrm{L}},\delta\vartheta\right]-\left[\delta\omega^{\mathrm{L}},\vartheta\right] \\ & =D(\delta\vartheta)-\left[\delta\omega^{\mathrm{L}},\vartheta\right]. \end{aligned}$	$\begin{aligned} & =d(\delta\vartheta)-[\omega^{\mathrm{L}},\delta\vartheta]-[\delta\omega^{\mathrm{L}},\vartheta] \\ & =D(\delta\vartheta)-[\delta\omega^{\mathrm{L}},\vartheta]. \end{aligned}$
mielke-geometrodynamics_115 E08341 (5.3.18)		■ 0.999 ● K +0	$\mathscr{A}=\frac{\kappa}{2}j_5,$	$^*\mathscr{A}=\frac{\kappa}{2}j_5,$	$^*\mathscr{A}=\frac{\kappa}{2}j_5,$
mielke-geometrodynamics_143 E08441 (6.8.12)		■ 0.999 ● N +0	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}\cong0,$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}\cong0,$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}\cong0,$
mielke-geometrodynamics_156 E08476 (7.4.4)		■ 0.999 ● K +0	$C_{\alpha\beta}=0\quad\text{and}\quad R=0,$	$C_{\alpha\beta}=0\quad\text{and}\quad R=0,$	$C_{\alpha\beta}=0\quad\text{and}\quad R=0,$
mielke-geometrodynamics_166 E08519 (7.7.37)		■ 0.999 ● N +0	$D\Theta\equiv[\Omega^g,\gamma],\quad D\Omega^g\equiv0,$	$D\Theta\equiv[\Omega^g,\gamma],\quad D\Omega^g\equiv0,$	$D\Theta\equiv[\Omega^g,\gamma],\quad D\Omega^g\equiv0,$
mielke-geometrodynamics_183 E08567 (8.6.4)		■ 0.999 ● N +1	$\begin{aligned} & H_{\alpha\beta}(**)=\theta_{\mathrm{L}}R_{\alpha\beta}+\theta_{\mathrm{L}}^*R_{\alpha\beta}^{(*)}+\frac{\theta_{\mathrm{T}}^*}{2\ell^2}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta} \\ & \simeq-\partial L_{\theta}/\partial R^{\alpha\beta} \end{aligned}$	$\begin{aligned} H_{\alpha\beta}(**) &= \theta_{\mathrm{L}}R_{\alpha\beta} + \theta_{\mathrm{L}}^*R_{\alpha\beta}^{(*)} + \frac{\theta_{\mathrm{T}}^*}{2\ell^2}\eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta} \\ & \simeq -\partial L_{\theta}/\partial R^{\alpha\beta} \end{aligned}$	$\begin{aligned} H_{\alpha\beta}(**) &= \theta_{\mathrm{L}}R_{\alpha\beta} + \theta_{\mathrm{L}}^*R_{\alpha\beta}^{(*)} + \frac{\theta_{\mathrm{T}}^*}{2\ell^2}\eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta} \\ & \simeq -\partial L_{\theta}/\partial R^{\alpha\beta}, \end{aligned}$
mielke-geometrodynamics_210 E08624 (10.1.12)		■ 0.999 ● K +0	$DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha},$	$DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha},$	$DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha},$
mielke-geometrodynamics_220 E08673 (10.3.30)		■ 0.999 ● C -4	$H_{\alpha}=-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}-\frac{\theta_{\mathrm{TL}}}{\ell}\Gamma_{\alpha}^{\star},\quad H_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}\Gamma_{\alpha}^{\star},$	$H_{\alpha}=-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}-\frac{\theta_{\mathrm{TL}}}{\ell}\Gamma_{\alpha}^{\star},\quad H_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}\Gamma_{\alpha}^{\star},$	$H_{\alpha}=-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}-\frac{\theta_{\mathrm{TL}}}{\ell}\Gamma_{\alpha}^{\star},\quad H_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}\Gamma_{\alpha}^{\star},$
mielke-geometrodynamics_230 E08722 (10.10.2)		■ 0.999 ● N +0	$\delta L=\delta\vartheta^{\alpha}\wedge\frac{\delta L}{\delta\vartheta^{\alpha}}+\delta\Gamma_{\alpha}^{\star}\wedge\frac{\delta L}{\delta\Gamma_{\alpha}^{\star}}+\delta\bar{\Psi}\wedge\frac{\delta L}{\delta\bar{\Psi}},$	$\delta L=\delta\vartheta^{\alpha}\wedge\frac{\delta L}{\delta\vartheta^{\alpha}}+\delta\Gamma_{\alpha}^{\star}\wedge\frac{\delta L}{\delta\Gamma_{\alpha}^{\star}}+\delta\bar{\Psi}\wedge\frac{\delta L}{\delta\bar{\Psi}},$	$\delta L=\delta\vartheta^{\alpha}\wedge\frac{\delta L}{\delta\vartheta^{\alpha}}+\delta\Gamma_{\alpha}^{\star}\wedge\frac{\delta L}{\delta\Gamma_{\alpha}^{\star}}+\delta\bar{\Psi}\wedge\frac{\delta L}{\delta\bar{\Psi}},$
mielke-geometrodynamics_242 E08759 (11.1.26)		■ 0.999 ● N +0	$\gamma_{\{p\}}=\frac{p!}{(n-p)!}\gamma^{\wedge n+1}\underbrace{\gamma\wedge\cdots\wedge\gamma}_{n-p}.$	$^*(\underbrace{\gamma\wedge\cdots\wedge\gamma}_p)=\frac{p!}{(n-p)!}\gamma^{n+1}\underbrace{\gamma\wedge\cdots\wedge\gamma}_{n-p}.$	$^*(\underbrace{\gamma\wedge\cdots\wedge\gamma}_p)=\frac{p!}{(n-p)!}\gamma^{n+1}\underbrace{\gamma\wedge\cdots\wedge\gamma}_{n-p}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E00780 (11.2.8)	247	0.999 K +0	$(\psi,\chi):=\int_{H^{n-1}}\tau_s(\psi,\chi)=i\int_{H^{n-1}}\psi^*\gamma\chi$	$(\psi,\chi):=\int_{H^{n-1}}\tau_s(\psi,\chi)=i\int_{H^{n-1}}\psi^*\gamma\chi$	$(\psi,\chi):=\int_{H^{n-1}}\tau_s(\psi,\chi)=i\int_{H^{n-1}}\psi^*\gamma\chi$
mielke-geometrodynamics_E00789 (11.4.1)	251	0.999 N +0	$E_r=\sqrt{\alpha}r_{\circ}^2\left[1+\left(\frac{r}{r_{\circ}}\right)^4\right]^{-1/2}\sim\sqrt{\alpha}\frac{1}{r^2}$	$E_r=\sqrt{\alpha}r_{\circ}^2\left[1+\left(\frac{r}{r_{\circ}}\right)^4\right]^{-1/2}\sim\sqrt{\alpha}\frac{1}{r^2}$	$E_r=\sqrt{\alpha}r_{\circ}^2\left[1+\left(\frac{r}{r_{\circ}}\right)^4\right]^{-1/2}\sim\sqrt{\alpha}\frac{1}{r^2}$
mielke-geometrodynamics_E00825 (11.4.39)	259	0.999 N +0	$m_{\pi}=\sqrt{1-\omega^2}m$	$m_{\pi}=\sqrt{1-\omega^2}m$	$m_{\pi}=\sqrt{1-\omega^2}m$
mielke-geometrodynamics_E00892 (13.4.11)	287	0.999 N +0	$\vartheta\stackrel{A^{-1}(x)}{\longrightarrow}\vartheta'=\Lambda^{-1}(x)\vartheta,\quad\theta\stackrel{A^{-1}(x)}{\longrightarrow}\theta'=\theta\Lambda^{-1}(x).$	$\vartheta\stackrel{A^{-1}(x)}{\longrightarrow}\vartheta'=\Lambda^{-1}(x)\vartheta,\quad\theta\stackrel{A^{-1}(x)}{\longrightarrow}\theta'=\theta\Lambda^{-1}(x).$	$\vartheta\stackrel{A^{-1}(x)}{\longrightarrow}\vartheta'=\Lambda^{-1}(x)\vartheta,\quad\theta\stackrel{A^{-1}(x)}{\longrightarrow}\theta'=\theta\Lambda^{-1}(x).$
mielke-geometrodynamics_E00932 (14.1.13)	304	0.999 N +0	$e^{\delta}\circ e^{\varepsilon}=e^{\varepsilon}\circ e^{\delta}=e^{*(\delta+\varepsilon)},$	$e^{\delta}\circ e^{\varepsilon}=e^{\varepsilon}\circ e^{\delta}=e^{*(\delta+\varepsilon)},$	$e^{\delta}\circ e^{\varepsilon}=e^{\varepsilon}\circ e^{\delta}=e^{*(\delta+\varepsilon)},$
mielke-geometrodynamics_E00940 (14.1.21)	305	0.999 N +0	$\xi\wedge\xi=\xi\wedge^*\xi=0$	$\xi\wedge\xi=\xi\wedge^*\xi=0$	$\xi\wedge\xi=\xi\wedge^*\xi=0$
mielke-geometrodynamics_E00951 (14.1.33)	307	0.999 W +2	$\begin{aligned} & D^*\xi-(-1)^z\xi\wedge D\delta=\bar\tau_e\\ & D\xi+(-1)^z*\xi\wedge D\delta=\bar\tau_p \end{aligned}$	$\begin{aligned} & D^*\xi-(-1)^z\xi\wedge D\delta=\bar\tau_e\\ & D\xi+(-1)^z*\xi\wedge D\delta=\bar\tau_p \end{aligned}$	$\begin{aligned} & D^*\xi-(-1)^z\xi\wedge D\delta=\bar\tau_e,\\ & D\xi+(-1)^z*\xi\wedge D\delta=\bar\tau_p. \end{aligned}$
mielke-geometrodynamics_E00968 (14.2.9)	314	0.999 N +2	$\hat{\omega}=\left[\begin{array}{c c} \omega^g&-\alpha\\ \hline \alpha^T&\hat{\omega} \end{array}\right]=:\omega^g\oplus\hat{\omega}-\hat{\alpha};\quad\hat{\alpha}=\left[\begin{array}{c c} 0&\alpha\\ \hline -\alpha^T&0 \end{array}\right]$	$\hat{\omega}=\left[\begin{array}{c c} \omega^g&-\alpha\\ \hline \alpha^T&\hat{\omega} \end{array}\right]=:\omega^g\oplus\hat{\omega}-\hat{\alpha};\quad\hat{\alpha}=\left[\begin{array}{c c} 0&\alpha\\ \hline -\alpha^T&0 \end{array}\right]$	$\hat{\omega}=\left[\begin{array}{c c} \omega^g&-\alpha\\ \hline \alpha^T&\hat{\omega} \end{array}\right]=:\omega^g\oplus\hat{\omega}-\hat{\alpha};\quad\hat{\alpha}=\left[\begin{array}{c c} 0&\alpha\\ \hline -\alpha^T&0 \end{array}\right]$
mielke-geometrodynamics_E01039 (15.1.16)	339	0.999 C +9	$\begin{aligned} & ds_{\{f\}}^2=f_{ij}dx^i\otimes_sdx^j\\ & =-\frac{3\gamma}{2}e^{-\lambda^f}dt^2+2\left[\gamma\left\{1-\left(1+\frac{9\gamma}{4}\right)e^{\lambda^f-\lambda^g}+\frac{9\gamma}{4}e^{2(\lambda^f-\lambda^g)}\right\}\right]^{1/2}dtdr\\ & \quad +e^{\lambda^g}\left\{\frac{2}{3}+\frac{3\gamma}{2}\left(1-e^{\lambda^g-\lambda^f}\right)\right\}dr^2+\frac{2}{3}r^2d\Omega^2 \end{aligned}$	$\begin{aligned} & ds_f^2=f_{ij}dx^i\otimes_sdx^j\\ & =-\frac{3\gamma}{2}e^{-\lambda^f}dt^2+2\left[\gamma\left\{1-\left(1+\frac{9\gamma}{4}\right)e^{\lambda^f-\lambda^g}+\frac{9\gamma}{4}e^{2(\lambda^f-\lambda^g)}\right\}\right]^{1/2}dtdr\\ & \quad +e^{\lambda^g}\left\{\frac{2}{3}+\frac{3\gamma}{2}\left(1-e^{\lambda^g-\lambda^f}\right)\right\}dr^2+\frac{2}{3}r^2d\Omega^2 \end{aligned}$	$\begin{aligned} & ds_f^2=f_{ij}dx^i\otimes_sdx^j\\ & =-\frac{3\gamma}{2}e^{-\lambda^f}dt^2+2\left[\gamma\left\{1-\left(1+\frac{9\gamma}{4}\right)e^{\lambda^f-\lambda^g}+\frac{9\gamma}{4}e^{2(\lambda^f-\lambda^g)}\right\}\right]^{1/2}dtdr\\ & \quad +e^{\lambda^g}\left\{\frac{2}{3}+\frac{3\gamma}{2}\left(1-e^{\lambda^g-\lambda^f}\right)\right\}dr^2+\frac{2}{3}r^2d\Omega^2 \end{aligned}\tag{15.1.16}$
mielke-geometrodynamics_E01088 (A.15)	366	0.999 N +0	$C(1,n-1)=\left(\bigotimes_{i=1}^{\lfloor\frac{n-2}{2}\rfloor}C(0,2)\right)\otimes C(1,1).$	$C(1,n-1)=\left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]}C(0,2)\right]\otimes C(1,1).$	$C(1,n-1)=\left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]}C(0,2)\right]\otimes C(1,1).$
Conf. MathPix confidence: 0.9 0.5-0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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mielke-geometrodynamics_00434 (6.8.5)	142	■ 0.999 ● N -1	$H_{\alpha}:= -\frac{\partial L_g}{\partial d\vartheta^{\alpha}}= -\frac{\partial L_g}{\partial T^{\alpha}}, \quad \text{and} \quad H_{\alpha\beta}:= -\frac{\partial L_g}{\partial d\Gamma^{\alpha\beta}}= -\frac{\partial L_g}{\partial R^{\alpha\beta}}.$	$H_{\alpha}:= -\frac{\partial L_g}{\partial d\vartheta^{\alpha}}= -\frac{\partial L_g}{\partial T^{\alpha}}, \quad \text{and} \quad H_{\alpha\beta}:= -\frac{\partial L_g}{\partial d\Gamma^{\alpha\beta}}= -\frac{\partial L_g}{\partial R^{\alpha\beta}}.$	
mielke-geometrodynamics_00566 (8.6.3)	183	■ 0.999 ● N +0	$H_{\alpha\beta}:= -\partial L_{\mathrm{qPG}}/\partial R^{\alpha\beta} = \frac{a_0}{2\ell^2}\eta_{\alpha\beta} + *\left(\sum_{N=1}^6 b_{(N)}{}^{(N)}R_{\alpha\beta}\right)$	$H_{\alpha\beta}:= -\partial L_{\mathrm{qPG}}/\partial R^{\alpha\beta} = \frac{a_0}{2\ell^2}\eta_{\alpha\beta} + *\left(\sum_{N=1}^6 b_{(N)}{}^{(N)}R_{\alpha\beta}\right)$	$H_{\alpha\beta}:= -\partial L_{\mathrm{qPG}}/\partial R^{\alpha\beta} = \frac{a_0}{2\ell^2}\eta_{\alpha\beta} + *\left(\sum_{N=1}^6 b_{(N)}{}^{(N)}R_{\alpha\beta}\right)$
mielke-geometrodynamics_00798 (11.4.10)	254	■ 0.999 ● W +0	$\begin{aligned} & \& \left\{ i\bar{\gamma}^k\left(\partial_k+i\bar{\Gamma}_k+\frac{3}{4}\partial_k\mu\right) \right. \\ & \& \left. \partial_{\mu}\right)\right.\& \left.+\frac{3}{8}\ell^2e^{\mu/2}\left(\bar{\psi}\psi-\mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5\right)-e^{\mu/2}m\right\}\psi=0. \end{aligned}$	$\left\{ i\bar{\gamma}^k\left(\partial_k+i\bar{\Gamma}_k+\frac{3}{4}\partial_k\mu\right) +\frac{3\varepsilon}{8}\ell^2e^{\mu/2}\left(\bar{\psi}\psi-\mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5\right)-e^{\mu/2}m\right\}\psi=0.$	$\left\{ i\bar{\gamma}^k(\partial_k+i\bar{\Gamma}_k+\frac{3}{4}\partial_k\mu) +\frac{3\varepsilon}{8}\ell^2e^{\mu/2}(\bar{\psi}\psi-\mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5)-e^{\mu/2}m\right\}\psi=0.$
mielke-geometrodynamics_00833 (12.2.1)	270	■ 0.999 ● N +0	$L_{\mathrm{D}}=L(\gamma,\psi,\mathscr{D}\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge\mathscr{D}\psi+\overline{\mathscr{D}\psi}\wedge^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$	$L_{\mathrm{D}}=L(\gamma,\psi,\mathscr{D}\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge\mathscr{D}\psi+\overline{\mathscr{D}\psi}\wedge^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$	$L_{\mathrm{D}}=L(\gamma,\psi,\mathscr{D}\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge\mathscr{D}\psi+\overline{\mathscr{D}\psi}\wedge^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$
mielke-geometrodynamics_00920 (2.4.2)	038	■ 1.000 ● N +0	$V^{\rho}:=V^{\rho}\left(M,\mathbb{C}^N,\rho(G)\subseteq GL(N,\mathbb{C}),P\right),$	$V^{\rho}:=V^{\rho}\left(M,\mathbb{C}^N,\rho(G)\subseteq GL(N,\mathbb{C}),P\right),$	$V^{\rho}:=V^{\rho}(M,\mathbb{C}^N,\rho(G)\subseteq GL(N,\mathbb{C}),P),$
mielke-geometrodynamics_00991 (3.2.1)	055	■ 1.000 ● S +0	$\mathscr{O}\rightarrow^{G^{-1}}\mathscr{O}=G^{-1}\mathscr{O}G.$	$\mathscr{O}\rightarrow^{G^{-1}}\mathscr{O}=G^{-1}\mathscr{O}G.$	$\mathscr{O}\rightarrow^{G^{-1}}\mathscr{O}=G^{-1}\mathscr{O}G.$
mielke-geometrodynamics_00101 (3.3.2)	057	■ 1.000 ● N -1	$A\rightarrow A+d\,\theta(x)\quad\Rightarrow\quad F\rightarrow dA+dd\theta=dA=F.$	$A\rightarrow A+d\theta(x)\quad\Rightarrow\quad F\rightarrow dA+dd\theta=dA=F.$	$A\rightarrow A+d\theta(x)\quad\Rightarrow\quad F\rightarrow dA+dd\theta=dA=F.$
mielke-geometrodynamics_00137 (3.4.8)	063	■ 1.000 ● W +0	$\begin{aligned} \delta L_{\omega} &= \delta \Omega \wedge \Pi^{\Omega} = D(\delta \omega) \wedge \Pi^{\Omega} \\ &= -\delta \omega \wedge D \Pi^{\Omega} + d\left(\delta \omega \wedge \Pi^{\Omega}\right). \end{aligned}$	$\delta L_{\omega} = \delta \Omega \wedge \Pi^{\Omega} = D(\delta \omega) \wedge \Pi^{\Omega} = -\delta \omega \wedge D \Pi^{\Omega} + d\left(\delta \omega \wedge \Pi^{\Omega}\right).$	$\delta L_{\omega} = \delta \Omega \wedge \Pi^{\Omega} = D(\delta \omega) \wedge \Pi^{\Omega} = -\delta \omega \wedge D \Pi^{\Omega} + d(\delta \omega \wedge \Pi^{\Omega}).$
mielke-geometrodynamics_00146 (3.4.17)	065	■ 1.000 ● N +0	$Q=Q^{\left(i\right)}I_i:=\int_{H^3}\iota=\int_{\partial H^3}\Pi^{\Omega},$	$Q=Q^{\left(i\right)}I_i:=\int_{H^3}\iota=\int_{\partial H^3}\Pi^{\Omega},$	$Q=Q^{\left(i\right)}I_i:=\int_{H^3}\iota=\int_{\partial H^3}\Pi^{\Omega},$
mielke-geometrodynamics_00175 (3.7.4)	072	■ 1.000 ● N +0	$\tau_{\mathrm{H}}=-\left[\phi^{(0)},*\Theta^{(1)}\right]$	$\tau_{\mathrm{H}}=-\left[\phi^{(0)},*\Theta^{(1)}\right]$	$\tau_{\mathrm{H}}=-[\phi^{(0)},*\Theta^{(1)}]$
mielke-geometrodynamics_00195 (4.1.5)	082	■ 1.000 ● N +0	$\mathfrak{a}(n,\mathbb{R})=\mathbb{R}^n\otimes\mathfrak{gl}(n,\mathbb{R})$	$\mathfrak{a}(n,\mathbb{R})=\mathbb{R}^n\otimes\mathfrak{gl}(n,\mathbb{R})$	$\mathfrak{a}(n,\mathbb{R})=\mathbb{R}^n\otimes\mathfrak{gl}(n,\mathbb{R})$
mielke-geometrodynamics_00208 (4.1.22)	084	■ 1.000 ● N +0	$T_{\mathrm{A}}(M)=\stackrel{\circ}{V}(M,\mathbb{R}^n,A(n,\mathbb{R}),A(M))\subset V^{\mathrm{id}}$	$T_{\mathrm{A}}(M)=\stackrel{\circ}{V}(M,\mathbb{R}^n,A(n,\mathbb{R}),A(M))\subset V^{\mathrm{id}}$	$T_{\mathrm{A}}(M)=\stackrel{\circ}{V}(M,\mathbb{R}^n,A(n,\mathbb{R}),A(M))\subset V^{\mathrm{id}}$
mielke-geometrodynamics_00212 (4.2.4)	085	■ 1.000 ● K +0	$\Omega^{\mathrm{T}}:=D\omega^{\mathrm{T}}=d\omega^{\mathrm{T}}-\left[\omega^{\mathrm{T}},\omega^{\mathrm{L}}\right],$	$\Omega^{\mathrm{T}}:=D\omega^{\mathrm{T}}=d\omega^{\mathrm{T}}-\left[\omega^{\mathrm{T}},\omega^{\mathrm{L}}\right],$	$\Omega^{\mathrm{T}}:=D\omega^{\mathrm{T}}=d\omega^{\mathrm{T}}-[\omega^{\mathrm{T}},\omega^{\mathrm{L}}],$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E00215 (4.2.7)	086	1.000 ● N +0	$\backslash\mathrm{mscr}\{T\}(n,\backslash\mathrm{mathbb}\{R\}):=\mathcal{C}^{\wedge\{\infty\}}\backslash\mathrm{left}(A(M)\backslash\mathrm{times}_{\backslash\mathrm{mathrm}\{Ad\}}\backslash\mathrm{mathbb}\{R\}^{\wedge\{n\}}\backslash\mathrm{right})$	$\mathcal{T}(n,\mathbb{R}):=C^{\infty}\left(A(M)\times_{\mathrm{Ad}}\mathbb{R}^n\right)$	$\mathcal{T}(n,\mathbb{R}):=C^{\infty}(A(M)\times_{\mathrm{Ad}}\mathbb{R}^n)$
mielke-geometrodynamics_E00217 (4.2.9)	086	1.000 ● N +0	$A=\backslash\mathrm{left}[\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{c}\}\{\mathrm{c}\}~G~\&~T~\backslash\backslash~\backslash\mathrm{hline}~0~\&~1~\backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}~\backslash\mathrm{in}~\backslash\mathrm{mathscr}\{A\}_{\{p\}},~\backslash\mathrm{quad}~G~\backslash\mathrm{in}~\backslash\mathrm{mathscr}\{G\}_{\{p\}},~T~\backslash\mathrm{in}~\backslash\mathrm{mathscr}\{T\}_{\{p\}}$	$A=\left[\begin{array}{c c} G & T \\ 0 & 1 \end{array}\right]\in\mathscr{A}_p,\quad G\in\mathscr{G}_p,T\in\mathscr{T}_p$	$A=\left[\begin{array}{c c} G & T \\ 0 & 1 \end{array}\right]\in\mathscr{A}_p,\quad G\in\mathscr{G}_p,T\in\mathscr{T}_p$
mielke-geometrodynamics_E00243 (4.3.8)	091	1.000 ● W +2	$\backslash\cos\phi=\frac{g_{\mu\nu}\mathrm{d}V^{\mu}\mathrm{d}W^{\nu}}{\sqrt{g_{\alpha\beta}\mathrm{d}V^{\alpha}\mathrm{d}V^{\beta}}\sqrt{g_{\alpha\beta}\mathrm{d}W^{\alpha}\mathrm{d}W^{\beta}}}$	$\cos\phi=\frac{g_{\mu\nu}dV^{\mu}dW^{\nu}}{\sqrt{g_{\alpha\beta}dV^{\alpha}dV^{\beta}}\sqrt{g_{\alpha\beta}dW^{\alpha}dW^{\beta}}}$	$\cos\phi=\frac{g_{\mu\nu}dV^{\mu}dW^{\nu}}{\sqrt{g_{\alpha\beta}dV^{\alpha}dV^{\beta}}\sqrt{g_{\alpha\beta}dW^{\alpha}dW^{\beta}}}.$
mielke-geometrodynamics_E00323 (5.2.7)	111	1.000 ● N +0	$\backslash\mathrm{begin}\{\mathrm{aligned}\}~\backslash\Sigma_{\{\alpha\}}~\backslash\mathrm{rightarrow}~\backslash\sigma_{\{\alpha\}}~\&~\backslash\mathrm{left}.:=\backslash\Sigma_{\{\alpha\}}-D~\backslash\mu_{\{\alpha\}}+e_{\{\beta\}}\backslash\mathrm{right}\backslash\mathrm{rfloor}\backslash\mathrm{left}(T^{\wedge\{\beta\}}~\backslash\mathrm{wedge}~\backslash\mu_{\{\alpha\}}~\backslash\mathrm{right}),~\backslash\backslash~\backslash\tau_{\{\alpha\}~\{\beta\}}~\backslash\mathrm{rightarrow}~\backslash\mathrm{hat}\{\backslash\tau\}_{\{\alpha\}~\{\beta\}}~\&~:=\backslash\tau_{\{\alpha\}~\{\beta\}}-\backslash\mathrm{vartheta}_{\{\alpha\}}~\backslash\mathrm{wedge}~\backslash\mu_{\{\beta\}}]=0~\backslash\mathrm{end}\{\mathrm{aligned}\}$	$\begin{array}{l}\Sigma_{\alpha}\rightarrow\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}+e_{\beta}\rfloor\left(T^{\beta}\wedge\mu_{\alpha}\right),\\ \tau_{\alpha\beta}\rightarrow\hat{\tau}_{\alpha\beta}:=\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge\mu_{\beta]}=0\end{array}$	$\begin{array}{l}\Sigma_{\alpha}\rightarrow\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}+e_{\beta}\rfloor(T^{\beta}\wedge\mu_{\alpha}),\\ \tau_{\alpha\beta}\rightarrow\hat{\tau}_{\alpha\beta}:=\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge\mu_{\beta]}=0\end{array}$
mielke-geometrodynamics_E00328 (5.3.5)	112	1.000 ● N +0	$\backslash\Theta:=D~\backslash\gamma~\backslash\mathrm{quad}~\backslash\mathrm{text}\{~\mathrm{and}~\}~\backslash\mathrm{quad}~\backslash\Omega^g:=d~\backslash\Gamma+\backslash\Gamma~\backslash\mathrm{wedge}~\backslash\Gamma~.$	$\Theta:=D\gamma\quad\mathrm{and}\quad\Omega^g:=d\Gamma+\Gamma\wedge\Gamma.$	$\Theta:=D\gamma\quad\mathrm{and}\quad\Omega^g:=d\Gamma+\Gamma\wedge\Gamma.$
mielke-geometrodynamics_E00370 (6.2.21)	127	1.000 ● N +0	$\backslash\mathrm{stackrel}\{(\backslash\mathrm{pm})\}\{\backslash\mathrm{Pi}\}_{\{\alpha\}}=-\backslash\mathrm{partial}~\backslash\mathrm{stackrel}\{(\backslash\mathrm{pm})\}\{L\}~\backslash\mathrm{partial}~T^{\wedge\{\alpha\}}=H_{\{\alpha\}}^{\parallel}\backslash\mathrm{mp}(i/\kappa)~T_{\alpha}$	$\stackrel{(\pm)}{\Pi}_{\alpha}=-\partial~\stackrel{(\pm)}{L}~\backslash\mathrm{partial}T^{\alpha}=H_{\alpha}^{\parallel}\mp(i/\kappa)T_{\alpha}$	$\stackrel{(\pm)}{\Pi}_{\alpha}=-\partial~\stackrel{(\pm)}{L}~\backslash\mathrm{partial}T^{\alpha}=H_{\alpha}^{\parallel}\mp(i/\kappa)T_{\alpha}$
mielke-geometrodynamics_E00380 (6.2.31)	129	1.000 ● N +0	$\backslash\mathrm{stackrel}\{(\backslash\mathrm{pm})~(\backslash\mathrm{pm})\}\{D\}~\backslash\mathrm{Pi}_{\{\alpha\}}=0,$	$\stackrel{(\pm)}{D}~\stackrel{(\pm)}{\Pi}_{\alpha}=0,$	$\stackrel{(\pm)}{D}~\stackrel{(\pm)}{\Pi}_{\alpha}=0,$
mielke-geometrodynamics_E00428 (6.7.10)	141	1.000 ● N +0	$\backslash\kappa~D~\backslash\theta_{\{\mathrm{mathrm}\{L\}\}}~\backslash\mathrm{wedge}~D~T_{\{\beta\}}=-T~\backslash\mathrm{wedge}~\backslash\eta_{\{\beta\}}=D~\backslash\eta_{\{\beta\}},$	$\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=-T\wedge\eta_{\beta}=D\eta_{\beta},$	$\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=-T\wedge\eta_{\beta}=D\eta_{\beta},$
mielke-geometrodynamics_E00522 (7.7.41)	167	1.000 ● N +1	$\backslash\mathrm{begin}\{\mathrm{gathered}\}~K^{\wedge\{\star\}}:=\backslash\eta_{\{\alpha\}~\{\beta\}~\{\gamma\}}~\backslash\mathrm{wedge}~K^{\wedge\{\beta\}~\{\gamma\}}~\backslash\gamma^{\alpha},~\backslash\backslash~\backslash\Omega^{\wedge\{g\}}(\backslash\mathrm{star}):=\frac{i}{8}~R_{\{\alpha\}~\{\beta\}}~\backslash\eta_{\{\alpha\}~\{\beta\}~\{\gamma\}}~\backslash\mathrm{wedge}~\backslash\sigma_{\{\alpha\}~\{\beta\}}=\frac{1}{4}~R^{\{\alpha\}~\{\beta\}}~\backslash\gamma_5~\backslash\sigma_{\alpha\beta}=i~\gamma_5~\backslash\Omega^g.~\backslash\mathrm{end}\{\mathrm{gathered}\}$	$\begin{array}{l}K^{(\star)}:=\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}\gamma^{\alpha},\\ \Omega^{g(\star)}:=\frac{i}{8}R_{\alpha\beta}\eta^{\alpha\beta\gamma\delta}\sigma_{\gamma\delta}=-\frac{1}{4}R^{\alpha\beta}\gamma_5\sigma_{\alpha\beta}=i\gamma_5\Omega^g.\end{array}$	$\begin{array}{l}K^{(\star)}:=\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}\gamma^{\alpha},\\ \Omega^{g(\star)}:=\frac{i}{8}R_{\alpha\beta}\eta^{\alpha\beta\gamma\delta}\sigma_{\gamma\delta}=-\frac{1}{4}R^{\alpha\beta}\gamma_5\sigma_{\alpha\beta}=i\gamma_5\Omega^g.\end{array}$
mielke-geometrodynamics_E00526 (7.7.46)	168	1.000 ● N -1	$\backslash\mathrm{L}_{\{\mathrm{mathrm}\{NY\}\}}:=d~\backslash\mathrm{C}_{\{\mathrm{mathrm}\{TT\}\}}=\frac{1}{2}~\backslash\mathrm{ell}^{\wedge\{2\}}~\backslash\mathrm{left}(T^{\wedge\{\alpha\}}~\backslash\mathrm{wedge}~T_{\{\alpha\}}+R_{\{\alpha\}~\{\beta\}}~\backslash\mathrm{wedge}~\backslash\mathrm{vartheta}^{\wedge\{\alpha\}}~\backslash\mathrm{wedge}~\backslash\mathrm{vartheta}^{\wedge\{\beta\}}~\backslash\mathrm{right}),$	$L_{\mathrm{NY}}:=dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$	$L_{\mathrm{NY}}:=dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$
mielke-geometrodynamics_E00535 (8.2.2)	175	1.000 ● N +0	$\backslash\mathrm{widetilde}\{\Gamma\}:=\backslash\Gamma~\backslash\mathrm{oplus}~C,~\backslash\mathrm{quad}~\backslash\mathrm{widetilde}\{\Omega\}^g:=\backslash\Omega^g~\backslash\mathrm{oplus}~\backslash\Psi~\backslash\mathrm{oplus}~\backslash\Phi~.$	$\widetilde{\Gamma}:=\Gamma\oplus C,\quad\widetilde{\Omega}^g:=\Omega^g\oplus\Psi\oplus\Phi.$	$\widetilde{\Gamma}:=\Gamma\oplus C,\quad\widetilde{\Omega}^g:=\Omega^g\oplus\Psi\oplus\Phi.$
mielke-geometrodynamics_E00538 (8.2.5)	176	1.000 ● K +0	$\backslash\mathrm{s}_{\{\mathrm{mathrm}\{\mathrm{loc}\}\}}(\backslash\alpha+\backslash\lambda)=\mathrm{d}~\backslash\lambda,\quad\backslash\mathrm{quad}~\backslash\mathrm{s}_{\{\mathrm{mathrm}\{\mathrm{loc}\}\}}~\backslash\lambda=0$	$s_{\mathrm{loc}}(\alpha+\lambda)=-d\lambda,\quad s_{\mathrm{loc}}\lambda=0$	$s_{\mathrm{loc}}(\alpha+\lambda)=-d\lambda,\quad s_{\mathrm{loc}}\lambda=0$
mielke-geometrodynamics_E00539 (8.2.6)	176	1.000 ● K +0	$\backslash\mathrm{left}(\mathrm{d}~\backslash\mathrm{oplus}~\backslash\mathrm{s}_{\{\mathrm{mathrm}\{\mathrm{loc}\}\}}\backslash\mathrm{right})(\backslash\alpha+\backslash\lambda)=\mathrm{d}~\backslash\alpha~.$	$(d\oplus s_{\mathrm{loc}})(\alpha+\lambda)=d\alpha.$	$(d\oplus s_{\mathrm{loc}})(\alpha+\lambda)=d\alpha.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_185 EQ0573 (8.6.11)	185	■ 1.000 ● W +2	$\begin{aligned} L_{\mathrm{qPG}}(* *) &= \frac{\Lambda}{\ell^2} \eta - \frac{a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2} T^\alpha \wedge H_\alpha(* *) - \frac{1}{2} R^{\alpha\beta} \wedge H_{\alpha\beta}(* *) \\ &= \frac{\Lambda}{\ell^2} \eta - \frac{\theta_{\mathrm{T}}^* + a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2\ell^2} T^\alpha \wedge \left[\theta_{\mathrm{T}}^* K_\alpha^{(*)} - \theta_{\mathrm{T}} T_\alpha \right] \\ &\quad - \frac{1}{2} R^{\alpha\beta} \wedge \left[\theta_{\mathrm{L}} R_{\alpha\beta} + \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} - \frac{\theta_{\mathrm{T}}}{2\ell^2} \vartheta_\alpha \wedge \vartheta_\beta \right]. \end{aligned}$	$L_{\mathrm{qPG}}(* *) = \frac{\Lambda}{\ell^2} \eta - \frac{a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2} T^\alpha \wedge H_\alpha(* *) - \frac{1}{2} R^{\alpha\beta} \wedge H_{\alpha\beta}(* *)$ $= \frac{\Lambda}{\ell^2} \eta - \frac{\theta_{\mathrm{T}}^* + a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2\ell^2} T^\alpha \wedge \left[\theta_{\mathrm{T}}^* K_\alpha^{(*)} - \theta_{\mathrm{T}} T_\alpha \right]$ $- \frac{1}{2} R^{\alpha\beta} \wedge \left[\theta_{\mathrm{L}} R_{\alpha\beta} + \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} - \frac{\theta_{\mathrm{T}}}{2\ell^2} \vartheta_\alpha \wedge \vartheta_\beta \right].$	$L_{\mathrm{qPG}}(* *) = \frac{\Lambda}{\ell^2} \eta - \frac{a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2} T^\alpha \wedge H_\alpha(* *) - \frac{1}{2} R^{\alpha\beta} \wedge H_{\alpha\beta}(* *)$ $= \frac{\Lambda}{\ell^2} \eta - \frac{\theta_{\mathrm{T}}^* + a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2\ell^2} T^\alpha \wedge \left[\theta_{\mathrm{T}}^* K_\alpha^{(*)} - \theta_{\mathrm{T}} T_\alpha \right]$ $- \frac{1}{2} R^{\alpha\beta} \wedge \left[\theta_{\mathrm{L}} R_{\alpha\beta} + \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} - \frac{\theta_{\mathrm{T}}}{2\ell^2} \vartheta_\alpha \wedge \vartheta_\beta \right]. \quad (8.6)$
mielke-geometrodynamics_186 EQ0580 (8.6.18)	186	■ 1.000 ● U +6	$G_{\{\alpha\}^{\{\backslash\}\}} \cdot \Lambda_{\mathrm{eff}} \eta_{\alpha} = \kappa_{\mathrm{eff}} \overparen{\Sigma}_{\alpha}.$	(not rendered)	$G_{\alpha}^{\{\}} - \Lambda_{\mathrm{eff}} \eta_{\alpha} = \kappa_{\mathrm{eff}} \widehat{\Sigma}_{\alpha}.$
mielke-geometrodynamics_195 EQ0595 (9.1.13)	195	■ 1.000 ● N +0	$R^{\{\backslash\ \backslash\}} = -\frac{6}{\ell^2} \kappa \gamma.$	$R^{\{\}} = -\frac{6\kappa\gamma}{\ell^{*2}}.$	$R^{\{\}} = -\frac{6\kappa\gamma}{\ell^{*2}}.$
mielke-geometrodynamics_214 EQ0645 (10.3.2)	214	■ 1.000 ● K +0	$\widetilde{T}_{\{\alpha\}} = T_{\{\alpha\}} - \frac{\varepsilon}{\ell} \eta_{\alpha}, \quad \widetilde{R}_{\alpha} = R_{\alpha} - (-1)^s \frac{\varepsilon}{2\ell} T_{\alpha} + (-1)^s \frac{\varepsilon^2}{4\ell^2} \eta_{\alpha}$	$\widetilde{T}_{\alpha} = T_{\alpha} - \frac{\varepsilon}{\ell} \eta_{\alpha}, \quad \widetilde{R}_{\alpha} = R_{\alpha} - (-1)^s \frac{\varepsilon}{2\ell} T_{\alpha} + (-1)^s \frac{\varepsilon^2}{4\ell^2} \eta_{\alpha}$	$\widetilde{T}_{\alpha} = T_{\alpha} - \frac{\varepsilon}{\ell} \eta_{\alpha}, \quad \widetilde{R}_{\alpha} = R_{\alpha} - (-1)^s \frac{\varepsilon}{2\ell} T_{\alpha} + (-1)^s \frac{\varepsilon^2}{4\ell^2} \eta_{\alpha}$
mielke-geometrodynamics_220 EQ0676 (10.3.33)	220	■ 1.000 ● W -2	$\check{\mathrm{E}}_{\{\alpha\}^{\{\mathrm{MB}\}}} = -(-1)^s \frac{\theta_{\mathrm{T}} \theta_{\mathrm{L}}}{2\chi \ell} \left(d\Gamma_{\alpha}^* + \frac{2}{\theta_{\mathrm{L}}} \check{\mathcal{J}}_{\alpha}^{\mathrm{MB}*} \right),$	$\check{\mathcal{E}}_{\alpha}^{\mathrm{MB}} = -(-1)^s \frac{\theta_{\mathrm{T}} \theta_{\mathrm{L}}}{2\chi \ell} \left(d\Gamma_{\alpha}^* + \frac{2}{\theta_{\mathrm{L}}} \check{\mathcal{J}}_{\alpha}^{\mathrm{MB}*} \right),$	$\check{\mathcal{E}}_{\alpha}^{\mathrm{MB}} = -(-1)^s \frac{\theta_{\mathrm{T}} \theta_{\mathrm{L}}}{2\chi \ell} \left(d\Gamma_{\alpha}^* + \frac{2}{\theta_{\mathrm{L}}} \check{\mathcal{J}}_{\alpha}^{\mathrm{MB}*} \right),$
mielke-geometrodynamics_222 EQ0688 (10.3.46)	222	■ 1.000 ● N +0	$G_{\{\alpha\}^{\{\backslash\}\}} \cdot \widetilde{\Lambda} \eta_{\alpha} = \widetilde{\ell} \widetilde{\Sigma}_{\alpha},$	$G_{\alpha}^{\{\}} - \tilde{\Lambda} \eta_{\alpha} = \tilde{\ell} \tilde{\Sigma}_{\alpha},$	$G_{\alpha}^{\{\}} - \tilde{\Lambda} \eta_{\alpha} = \ell \tilde{\Sigma}_{\alpha},$
mielke-geometrodynamics_223 EQ0690 (10.5.1)	223	■ 1.000 ● K +0	$\sigma^1 = \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right), \quad \sigma^2 = \left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 = \left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right).$	$\sigma^1 = \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right), \quad \sigma^2 = \left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 = \left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right).$	$\sigma^1 = \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right), \quad \sigma^2 = \left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 = \left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_227 E08710 (10.7.6)		■ 1.000 ● W +2	$\begin{aligned} \vartheta^{\hat{0}} &= dt + \ell^2 \sigma \rho^{*2} [1 - \cos(\rho/\rho^*)] d\phi \\ \vartheta^{\hat{1}} &= d\rho, \quad \vartheta^{\hat{2}} = \rho^* \sin(\rho/\rho^*) d\phi \\ \Gamma^{\hat{1}\hat{2}} &= \cos(\rho/\rho^*) d\phi = -\Gamma^{\hat{2}\hat{1}} \end{aligned}$	$\vartheta^{\hat{0}} = dt + \ell^2 \sigma \rho^{*2} [1 - \cos(\rho/\rho^*)] d\phi$ $\vartheta^{\hat{1}} = d\rho, \quad \vartheta^{\hat{2}} = \rho^* \sin(\rho/\rho^*) d\phi$ $\Gamma^{\hat{1}\hat{2}} = \cos(\rho/\rho^*) d\phi = -\Gamma^{\hat{2}\hat{1}}$	$\vartheta^{\hat{0}} = dt + \ell^2 \sigma \rho^{*2} [1 - \cos(\rho/\rho^*)] d\phi$ $\vartheta^{\hat{1}} = d\rho, \quad \vartheta^{\hat{2}} = \rho^* \sin(\rho/\rho^*) d\phi,$ $\Gamma^{\hat{1}\hat{2}} = \cos(\rho/\rho^*) d\phi = -\Gamma^{\hat{2}\hat{1}}.$
mielke-geometrodynamics_229 E08716 (10.9.2)		■ 1.000 ● K +0	$D \Psi = d \Psi - \frac{1}{2} \gamma_{\alpha} \Gamma^{*\alpha} \wedge \Psi,$	$D \Psi = d \Psi - \frac{1}{2} \gamma_{\alpha} \Gamma^{*\alpha} \wedge \Psi,$	$D \Psi = d \Psi - \frac{1}{2} \gamma_{\alpha} \Gamma^{*\alpha} \wedge \Psi,$
mielke-geometrodynamics_229 E08720 (10.9.6)		■ 1.000 ● N +0	$\tau_{\alpha}^* := \frac{1}{2} \eta_{\alpha\beta\gamma} \tau^{\beta\gamma} = \frac{(-1)^s}{2} \frac{\delta L_{\Psi}}{\delta \Gamma_{\alpha}^*} = \frac{(-1)^s}{2} \frac{i}{4} \bar{\Psi} \wedge \gamma_{\alpha} \Psi.$	$\tau_{\alpha}^* := \frac{1}{2} \eta_{\alpha\beta\gamma} \tau^{\beta\gamma} = \frac{(-1)^s}{2} \frac{\delta L_{\Psi}}{\delta \Gamma_{\alpha}^*} = \frac{(-1)^s}{2} \frac{i}{4} \bar{\Psi} \wedge \gamma_{\alpha} \Psi.$	$\tau_{\alpha}^* := \frac{1}{2} \eta_{\alpha\beta\gamma} \tau^{\beta\gamma} = \frac{(-1)^s}{2} \frac{\delta L_{\Psi}}{\delta \Gamma_{\alpha}^*} = \frac{(-1)^s}{2} \frac{i}{4} \bar{\Psi} \wedge \gamma_{\alpha} \Psi.$
mielke-geometrodynamics_231 E08728 (10.10.8)		■ 1.000 ● K +0	$\gamma^{\alpha} \Psi \wedge \bar{\Psi} \gamma_{\alpha} \Psi = 0,$	$\gamma^{\alpha} \Psi \wedge \bar{\Psi} \gamma_{\alpha} \Psi = 0,$	$\gamma^{\alpha} \Psi \wedge \bar{\Psi} \gamma_{\alpha} \Psi = 0,$
mielke-geometrodynamics_244 E08770 (11.1.37)		■ 1.000 ● N +0	$D \longrightarrow S^{-1} D = S^{-1} D S.$	$D \longrightarrow S^{-1} D = S^{-1} D S.$	$D \longrightarrow S^{-1} D = S^{-1} D S.$
mielke-geometrodynamics_272 E08839 (12.2.7)		■ 1.000 ● N +0	$j_{\pm} := \frac{1}{2} \bar{\psi} (1 \pm \gamma_5)^* \gamma \psi = \bar{\psi}_{\text{L,R}}^* \gamma \psi_{\text{L,R}}.$	$j_{\pm} := \frac{1}{2} \bar{\psi} (1 \pm \gamma_5)^* \gamma \psi = \bar{\psi}_{\text{L,R}}^* \gamma \psi_{\text{L,R}}.$	$j_{\pm} := \frac{1}{2} \bar{\psi} (1 \pm \gamma_5)^* \gamma \psi = \bar{\psi}_{\text{L,R}}^* \gamma \psi_{\text{L,R}}.$
mielke-geometrodynamics_284 E08873 (13.3.1)		■ 1.000 ● N +0	$\nvdash^{\{A\}\ \ \}_{B}} := L^{\{A\}\ \ \}_{B}} \cdot \frac{1}{n} \delta_B^A L^C_C.$	$\not\vdash^A{}_B := L^A{}_B - \frac{1}{n} \delta_B^A L^C{}_C.$	$\not\vdash^A{}_B := L^A{}_B - \frac{1}{n} \delta_B^A L^C{}_C.$
mielke-geometrodynamics_287 E08893 (13.4.12)		■ 1.000 ● N +0	$\mathrm{A}_{*}(n,\mathbb{R})/\mathrm{GL}(n,\mathbb{R})\approx \mathbb{R}^n\otimes \mathbb{R}^n_{*},$	$A_{*}(n,\mathbb{R})/\mathrm{GL}(n,\mathbb{R})\approx \mathbb{R}^n\otimes \mathbb{R}^n_{*},$	$A_{*}(n,\mathbb{R})/\mathrm{GL}(n,\mathbb{R})\approx \mathbb{R}^n\otimes \mathbb{R}^n_{*},$
mielke-geometrodynamics_290 E09097 (13.5.12)		■ 1.000 ● N +0	$\tilde{L}_{\text{SSB}} + d\hat{C} \cong \frac{1}{2\mu} \eta_{\alpha\beta\gamma\delta} \left[R^{\alpha\beta} - \frac{\Lambda}{3} \vartheta^{\alpha} \wedge \vartheta^{\beta} \right] \wedge \left[R^{\gamma\delta} - \frac{\Lambda}{3} \vartheta^{\gamma} \wedge \vartheta^{\delta} \right].$	$\tilde{L}_{\text{SSB}} + d\hat{C} \cong \frac{1}{2\mu} \eta_{\alpha\beta\gamma\delta} \left[R^{\alpha\beta} - \frac{\Lambda}{3} \vartheta^{\alpha} \wedge \vartheta^{\beta} \right] \wedge \left[R^{\gamma\delta} - \frac{\Lambda}{3} \vartheta^{\gamma} \wedge \vartheta^{\delta} \right].$	$\tilde{L}_{\text{SSB}} + d\hat{C} \cong \frac{1}{2\mu} \eta_{\alpha\beta\gamma\delta} \left[R^{\alpha\beta} - \frac{\Lambda}{3} \vartheta^{\alpha} \wedge \vartheta^{\beta} \right] \wedge \left[R^{\gamma\delta} - \frac{\Lambda}{3} \vartheta^{\gamma} \wedge \vartheta^{\delta} \right].$
mielke-geometrodynamics_307 E09947 (14.1.28)		■ 1.000 ● N +0	$R_{i\alpha}^{\{\}} R_{\cdot j}^{\{\}\alpha} = \frac{1}{4} g_{ij} R_{\alpha\beta}^{\{\}} R^{\{\}\alpha\beta}$	$R_{i\alpha}^{\{\}} R_{\cdot j}^{\{\}\alpha} = \frac{1}{4} g_{ij} R_{\alpha\beta}^{\{\}} R^{\{\}\alpha\beta}$	$R_{i\alpha}^{\{\}} R_{\cdot j}^{\{\}\alpha} = \frac{1}{4} g_{ij} R_{\alpha\beta}^{\{\}} R^{\{\}\alpha\beta}$
mielke-geometrodynamics_323 E01004 (14.3.17)		■ 1.000 ● W +0	$\widetilde{\Theta} := D\gamma = \frac{1}{2} T_{\alpha\beta}^{\cdot\cdot c} \gamma_c \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$	$\widetilde{\Theta} := D\gamma = \frac{1}{2} T_{\alpha\beta}^{\cdot\cdot c} \gamma_c \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$	$\widetilde{\Theta} := D\gamma = \frac{1}{2} T_{\alpha\beta}^{\cdot\cdot c} \gamma_c \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$
mielke-geometrodynamics_326 E01013 (14.4.8)		■ 1.000 ● N +0	$M_{\mathrm{ir}}^2 := \frac{S_{\mathrm{A}}}{16\pi} \frac{M^{*4} c^2}{\hbar^2}$	$M_{\mathrm{ir}}^2 := \frac{S_{\mathrm{A}}}{16\pi} \frac{M^{*4} c^2}{\hbar^2}$	$M_{\mathrm{ir}}^2 := \frac{S_{\mathrm{A}}}{16\pi} \frac{M^{*4} c^2}{\hbar^2}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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mielke-geometrodynamics_E00027 (2.4.9)	039	■ 1.000 ● N +0	$\left[\phi^{\{2\,k\}},\phi^{\{2\,k\}}\right]=-\left[\phi^{\{2\,k\}},\phi^{\{2\,k\}}\right]=0\;.$	$\left[\phi^{(2k)},\phi^{(2k)}\right]=-\left[\phi^{(2k)},\phi^{(2k)}\right]=0.$	$\left[\phi^{(2k)},\phi^{(2k)}\right]=-\left[\phi^{(2k)},\phi^{(2k)}\right]=0.$
mielke-geometrodynamics_E00029 (2.5.2)	040	■ 1.000 ● N +0	$\omega(e\,g)=g^{-1}\,\omega(e)\,g,\quad e\in T(P)\;.$	$\omega(eg)=g^{-1}\omega(e)g,\quad e\in T(P).$	$\omega(eg)=g^{-1}\omega(e)g,\quad e\in T(P).$
mielke-geometrodynamics_E00030 (2.5.3)	040	■ 1.000 ● N -1	$\omega(p\,d\,g)=g^{-1}\,d\,g\quad\text{in}\quad T_e(G)\approx\mathfrak{g}$	$\omega(pdg)=g^{-1}dg\quad\in\quad T_e(G)\approx\mathfrak{g}$	$\omega(p\,dg)=g^{-1}dg\quad\in\quad T_e(G)\approx\mathfrak{g}$
mielke-geometrodynamics_E00031 (2.5.4)	040	■ 1.000 ● N +0	$D:C^{\infty}\left(V^{\rho}\right)\longrightarrow C^{\infty}\left(T_{\mathbb{C}}^{\ast}(M)\otimes V^{\rho}\right)$	$D:C^{\infty}\left(V^{\rho}\right)\longrightarrow C^{\infty}\left(T_{\mathbb{C}}^{\ast}(M)\otimes V^{\rho}\right)$	$D:C^{\infty}\left(V^{\rho}\right)\longrightarrow C^{\infty}\left(T_{\mathbb{C}}^{\ast}(M)\otimes V^{\rho}\right)$
mielke-geometrodynamics_E00032 (2.5.6)	041	■ 1.000 ● N +0	$D\,b=\omega\,\otimes\,b,\quad b\in C^{\infty}\left(V^{\rho}\right)\;.$	$Db=\omega\otimes b,\quad b\in C^{\infty}\left(V^{\rho}\right).$	$Db=\omega\otimes b,\quad b\in C^{\infty}\left(V^{\rho}\right).$
mielke-geometrodynamics_E00033 (2.5.7)	041	■ 1.000 ● C -3	$D\left(\varphi^{\left(p\right)}\otimes b\right)=d\varphi^{\left(p\right)}\otimes b+\left(-1\right)^p\varphi^{\left(p\right)}\otimes Db.$	$D\left(\varphi^{\left(p\right)}\otimes b\right)=d\varphi^{\left(p\right)}\otimes b+\left(-1\right)^p\varphi^{\left(p\right)}\otimes Db.$	$D(\varphi^{(p)}\otimes b)=d\varphi^{(p)}\otimes b+(-1)^p\varphi^{(p)}\otimes Db.$
mielke-geometrodynamics_E00034 (2.5.8)	041	■ 1.000 ● C -3	$D\left(\varphi^{\left(p\right)}\otimes b\right)=\left\{d\varphi^p+\left(-1\right)^p\left[\varphi^{\left(p\right)},\omega\right]\right\}\otimes b.$	$D\left(\varphi^{\left(p\right)}\otimes b\right)=\left\{d\varphi^p+\left(-1\right)^p\left[\varphi^{\left(p\right)},\omega\right]\right\}\otimes b.$	$D(\varphi^{(p)}\otimes b)=\left\{d\varphi^p+(-1)^p[\varphi^{(p)},\omega]\right\}\otimes b.$
mielke-geometrodynamics_E00035 (2.5.9)	041	■ 1.000 ● K +0	$D\,b_{\{A\}}=\omega_{\{A\}}\{\,\}^{\{B\}}\,b_{\{B\}},\,A,B=1,\,\ldots,\,N\;.$	$Db_A=\omega_A{}^Bb_B,A,B=1,\ldots,N.$	$Db_A=\omega_A{}^Bb_B,A,B=1,\ldots,N.$
mielke-geometrodynamics_E00036 (2.5.10)	041	■ 1.000 ● N +0	$\Gamma:=\stackrel{\ast}{\sigma}\,\omega=\Gamma_{\alpha}{}^jL_j\otimes\vartheta^{\alpha},\quad j=1,\ldots,\dim G,$	$\Gamma:=\overset{\ast}{\sigma}\,\omega=\Gamma_{\alpha}{}^jL_j\otimes\vartheta^{\alpha},\quad j=1,\ldots,\dim G,$	$\Gamma:=\overset{\ast}{\sigma}\omega=\Gamma_{\alpha}{}^jL_j\otimes\vartheta^{\alpha},\quad j=1,\ldots,\dim G,$
mielke-geometrodynamics_E00037 (2.5.11)	041	■ 1.000 ● N -1	$A_{\alpha}{}^j{}_{\{m\}}:=\Gamma_{\alpha}{}^j{}_{\{m\}}_{\mid G=U(f)}$	$A_{\alpha}{}^j(m):=\Gamma_{\alpha}{}^j(m)_{ G=U(f)}$	$A_{\alpha}{}^j(m):=\Gamma_{\alpha}{}^j(m)_{ G=U(f)}$
mielke-geometrodynamics_E00038 (2.6.1)	042	■ 1.000 ● N +0	$D\,D\,b=: \,\Omega\,\otimes\,b\;.$	$DDb=: \Omega\otimes b.$	$DDb=: \,\Omega\,\otimes\,b.$
mielke-geometrodynamics_E00039 (2.6.2)	042	■ 1.000 ● N +1	$\begin{aligned} D\,D\,b=&D(\omega\otimes b)\,\&=d\,\omega\otimes b-\omega\wedge D\,b\\ &=(d\,\omega-\omega\wedge\omega)\otimes b\end{aligned}$	$\begin{aligned} DDb=D(\omega\otimes b)&=d\omega\otimes b-\omega\wedge Db\\ &=(d\omega-\omega\wedge\omega)\otimes b\end{aligned}$	$\begin{aligned} DDb=D(\omega\otimes b)&=d\omega\otimes b-\omega\wedge Db\\ &=(d\omega-\omega\wedge\omega)\otimes b\end{aligned}$
mielke-geometrodynamics_E00040 (2.6.3)	042	■ 1.000 ● C +3	$\Omega=d\,\omega-\omega\wedge\omega\wedge\omega=d\,\omega-\frac{1}{2}[\omega,\omega]$	$\Omega=d\omega-\omega\wedge\omega=d\omega-\frac{1}{2}[\omega,\omega]$	$\Omega=d\omega-\omega\wedge\omega=d\omega-\frac{1}{2}[\omega,\omega]\Big .$
mielke-geometrodynamics_E00043 (2.6.6)	042	■ 1.000 ● N +0	$F_{\{i\,j\}}=\partial_{\{i\}}A_{\{j\}}-\partial_{\{j\}}A_{\{i\}}$	$F_{ij}=\partial_iA_j-\partial_jA_i$	$F_{ij}=\partial_iA_j-\partial_jA_i$
mielke-geometrodynamics_E00044 (2.6.7)	042	■ 1.000 ● N +1	$\Delta b=b-b(\circ)=\oint (db)_{\mathrm{hor}}$	$\Delta b=b-b(\circ)=\oint (db)_{\mathrm{hor}}$	$\Delta b=b-b(\circ)=\oint (db)_{\mathrm{hor}},$
mielke-geometrodynamics_E00047 (2.6.10)	043	■ 1.000 ● N +0	$K=\lim_{U\rightarrow 0}\left(\int_U\Omega\otimes b\right)/\left(\int_U1\otimes b\right)$	$K=\lim_{U\rightarrow 0}\left(\int_U\Omega\otimes b\right)/\left(\int_U1\otimes b\right)$	$K=\lim_{U\rightarrow 0}\left(\int_U\Omega\otimes b\right)/\left(\int_U1\otimes b\right)$
mielke-geometrodynamics_E00049 (2.6.12)	043	■ 1.000 ● N +0	$D\,\Omega\equiv 0$	$D\Omega\equiv 0$	$D\Omega\equiv 0$
mielke-geometrodynamics_E00052 (2.7.2)	044	■ 1.000 ● N -1	$G(g\,p)=g\,G(p),\quad g\in G,\quad p\in P\;;$	$G(gp)=gG(p),\quad g\in G,\quad p\in P;$	$G(gp)=g\,G(p),\quad g\in G,\quad p\in P;$
mielke-geometrodynamics_E00053 (2.7.3)	044	■ 1.000 ● K +0	$\pi\circ G=\pi\;.$	$\pi\circ G=\pi.$	$\pi\circ G=\pi.$
mielke-geometrodynamics_E00054 (2.7.4)	045	■ 1.000 ● N +0	$\mathscr{G}_{\{p\}}\approx C^{\infty}\left(P\times_{\mathrm{Ad}}G\right)\;.$	$\mathscr{G}_p\approx C^{\infty}\left(P\times_{\mathrm{Ad}}G\right).$	$\mathscr{G}_p\approx C^{\infty}(P\times_{\mathrm{Ad}}G).$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_045 EQ0055 (2.7.5)	045	■1.000 ● N -1	$G(p)=\exp i\,\theta^{\{k\}}(m)\,L_{\{k\}}\in\mathscr{G}_{\{p\}}\;.$	$G(p)=\exp i\theta^k(m)L_k\in\mathscr{G}_p.$	$G(p)=\exp i\,\theta^k(m)L_k\in\mathscr{G}_p.$
mielke-geometrodynamics_045 EQ0057 (2.7.7)	045	■1.000 ● K +0	$\varphi\rightarrowtail^{G^{-1}}\varphi:=G^{-1}\varphi\in C^\infty(M,\mathbb{C}),G\in\mathscr{G}_V,$ $\varphi\rightarrowtail\varphi\in C^{\{\infty\}}(M,\mathbb{C}),G\in\mathscr{G}_{\{V\}},$	$\varphi\rightarrowtail^{G^{-1}}\varphi:=G^{-1}\varphi\in C^\infty(M,\mathbb{C}),G\in\mathscr{G}_V,$	$\varphi\rightarrowtail^{G^{-1}}\varphi:=G^{-1}\varphi\in C^\infty(M,\mathbb{C}),G\in\mathscr{G}_V,$
mielke-geometrodynamics_046 EQ0063 (2.8.4)	046	■1.000 ● N +0	$p_{\{k\}}\!\left(V^{\{\mathbb{R}\}}\!\right)=(-1)^{\{k\}}\,c_{\{2\,k\}}(V)\;.$	$p_k\left(V^{\mathbb{R}}\right)=(-1)^k c_{2k}(V).$	$p_k(V^{\mathbb{R}})=(-1)^k c_{2k}(V).$
mielke-geometrodynamics_047 EQ0066 (2.8.9)	047	■1.000 ● N +0	$\operatorname{Tr}(\Omega\wedge\Omega)=d\operatorname{Tr}\!\left(\Omega\wedge\Omega+\frac{1}{3}\Omega\wedge\Omega\wedge\Omega\right)$	$\operatorname{Tr}(\Omega\wedge\Omega)=d\operatorname{Tr}\!\left(\omega\wedge\Omega+\frac{1}{3}\omega\wedge\omega\wedge\omega\right)$	$\operatorname{Tr}(\Omega\wedge\Omega)=d\operatorname{Tr}\!\left(\omega\wedge\Omega+\frac{1}{3}\omega\wedge\omega\wedge\omega\right)$
mielke-geometrodynamics_048 EQ0067 (2.8.10)	048	■1.000 ● C +7	$\begin{aligned}&\operatorname{Tr}\!\left\{d\!\left(\Omega\wedge d\Omega-\frac{2}{3}\Omega\wedge\Omega\wedge\Omega\right)\right\}=\operatorname{Tr}\!\left\{d\Omega\wedge d\Omega-\Omega\wedge dd\Omega\right.\\&\quad\left.-\frac{2}{3}d\Omega\wedge\Omega\wedge\Omega+\frac{2}{3}\Omega\wedge d\Omega\wedge\Omega-\frac{2}{3}\Omega\wedge\Omega\wedge d\Omega\right\}\\&=\operatorname{Tr}\!\left\{d\Omega\wedge d\Omega-d\Omega\wedge\Omega\wedge\Omega-\Omega\wedge\Omega\wedge d\Omega\right\}\\&=\operatorname{Tr}\!\left\{(d\Omega-\Omega\wedge\Omega)\wedge(d\Omega-\Omega\wedge\Omega)\right\}=\operatorname{Tr}(\Omega\wedge\Omega)\end{aligned}$	$\operatorname{Tr}\!\left\{d\left(\omega\wedge d\omega-\frac{2}{3}\omega\wedge\omega\wedge\omega\right)\right\}=\operatorname{Tr}\!\left\{d\omega\wedge d\omega-\omega\wedge dd\omega\right.\\&\quad\left.-\frac{2}{3}d\omega\wedge\omega\wedge\omega+\frac{2}{3}\omega\wedge d\omega\wedge\omega-\frac{2}{3}\omega\wedge\omega\wedge d\omega\right\}\\&=\operatorname{Tr}\!\left\{d\omega\wedge d\omega-d\omega\wedge\omega\wedge\omega-\omega\wedge\omega\wedge d\omega\right\}\\&=\operatorname{Tr}\!\left\{(d\omega-\omega\wedge\omega)\wedge(d\omega-\omega\wedge\omega)\right\}=\operatorname{Tr}(\Omega\wedge\Omega)$	$\operatorname{Tr}\!\left\{d\left(\omega\wedge d\omega-\frac{2}{3}\omega\wedge\omega\wedge\omega\right)\right\}=\operatorname{Tr}\!\left\{d\omega\wedge d\omega-\omega\wedge dd\omega\right.\\&\quad\left.-\frac{2}{3}d\omega\wedge\omega\wedge\omega+\frac{2}{3}\omega\wedge d\omega\wedge\omega-\frac{2}{3}\omega\wedge\omega\wedge d\omega\right\}\\&=\operatorname{Tr}\!\left\{d\omega\wedge d\omega-d\omega\wedge\omega\wedge\omega-\omega\wedge\omega\wedge d\omega\right\}\\&=\operatorname{Tr}\!\left\{(d\omega-\omega\wedge\omega)\wedge(d\omega-\omega\wedge\omega)\right\}=\operatorname{Tr}(\Omega\wedge\Omega).$
mielke-geometrodynamics_048 EQ0068 (2.8.11)	048	■1.000 ● K +0	$\operatorname{Tr}\!\left\{\left(\Omega\wedge\Omega\right)\wedge\left(\Omega\wedge\Omega\right)\right\}=-\operatorname{Tr}\!\left\{\left(\Omega\wedge\Omega\right)\wedge\left(\Omega\wedge\Omega\right)\right\}=0\;.$	$\operatorname{Tr}\!\left\{\omega\wedge\left(\omega\wedge\omega\wedge\omega\right)\right\}=-\operatorname{Tr}\!\left\{\left(\omega\wedge\omega\wedge\omega\right)\wedge\omega\right\}=0.$	$\operatorname{Tr}\!\left\{\omega\wedge\left(\omega\wedge\omega\wedge\omega\right)\right\}=-\operatorname{Tr}\!\left\{\left(\omega\wedge\omega\wedge\omega\right)\wedge\omega\right\}=0.$
mielke-geometrodynamics_048 EQ0070 (2.8.13)	048	■1.000 ● N -1	$S_{\mathrm{YM}}=\int_{\mathcal{M}}L_{\mathrm{YM}}$	$S_{\mathrm{YM}}=\int_M L_{\mathrm{YM}}$	$S_{\mathrm{YM}}=\int_M L_{\mathrm{YM}}$
mielke-geometrodynamics_048 EQ0072 (2.8.15)	048	■1.000 ● K +0	$\Omega\sim\stackrel{\infty}{\Omega}$	$\omega\sim\stackrel{\infty}{\omega}$	$\omega\sim\stackrel{\infty}{\omega}$
mielke-geometrodynamics_049 EQ0074 (2.8.17)	049	■1.000 ● K +0	$\Omega(\stackrel{\infty}{\Omega})=0\;.$	$\Omega(\stackrel{\infty}{\omega})=0.$	$\Omega(\stackrel{\infty}{\omega})=0.$
mielke-geometrodynamics_049 EQ0075 (2.8.18)	049	■1.000 ● N +0	$c_{\{2\}}(M)=\frac{1}{24\pi^2}\int_{\partial\mathcal{M}}\operatorname{Tr}\!\left(\stackrel{\infty}{\Omega}\wedge\stackrel{\infty}{\Omega}\wedge\stackrel{\infty}{\Omega}\right),$	$c_2(M)=\frac{1}{24\pi^2}\int_{\partial M}\operatorname{Tr}(\stackrel{\infty}{\omega}\wedge\stackrel{\infty}{\omega}\wedge\stackrel{\infty}{\omega}),$	$c_2(M)=\frac{1}{24\pi^2}\int_{\partial M}\operatorname{Tr}(\stackrel{\infty}{\omega}\wedge\stackrel{\infty}{\omega}\wedge\stackrel{\infty}{\omega}),$
mielke-geometrodynamics_052 EQ0076 (3.1.1)	052	■1.000 ● K +0	$L=L(\varphi,d\varphi)\eta$	$L=L(\varphi,d\varphi)\eta$	$L=L(\varphi,d\varphi)\eta$
mielke-geometrodynamics_052 EQ0077 (3.1.2)	052	■1.000 ● N +0	$\delta\int_{\mathcal{M}}L=0$	$\delta\int_M L=0$	$\delta\int_M L=0$
mielke-geometrodynamics_052 EQ0078 (3.1.3)	052	■1.000 ● N -1	$\delta\varphi=0\quad\text{at the boundary }\partial\mathcal{M}\text{ of }\mathcal{M}$	$\delta\varphi=0\quad\text{at the boundary }\partial M\text{ of }M$	$\delta\varphi=0\quad\text{at the boundary }\partial M\text{ of }M$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_052 EQ0079 (3.1.4)		■ 1.000 ● K +0	$\delta(d\varphi):=d(\varphi+\delta\varphi)-d\varphi=d\delta\varphi$ $\delta\varphi$	$\delta(d\varphi) := d(\varphi + \delta\varphi) - d\varphi = d\delta\varphi$	$\delta(d\varphi) := d(\varphi + \delta\varphi) - d\varphi = d\delta\varphi$
mielke-geometrodynamics_052 EQ0080 (3.1.5)		■ 1.000 ● K +0	$\pi:=\frac{\partial L}{\partial(\partial_\alpha\varphi)}\vartheta^\alpha$ ϑ^α	$\pi := \frac{\partial L}{\partial(\partial_\alpha\varphi)}\vartheta^\alpha$	$\pi := \frac{\partial L}{\partial(\partial_\alpha\varphi)}\vartheta^\alpha$
mielke-geometrodynamics_053 EQ0082 (3.1.7)		■ 1.000 ● N +1	$\int_M d(\delta\varphi\wedge\pi)=\int_{\partial M}\delta\varphi\wedge\pi$ $\delta\varphi\wedge\pi$	$\int_M d(\delta\varphi \wedge \pi) = \int_{\partial M} \delta\varphi \wedge \pi$	$\int_M d(\delta\varphi \wedge \pi) = \int_{\partial M} \delta\varphi \wedge \pi.$
mielke-geometrodynamics_053 EQ0083 (3.1.9)		■ 1.000 ● N +0	$\lambda^\varphi:=(-1)^sL_\varphi+{}^*d\pi=0,$	$\lambda^\varphi := (-1)^s L_\varphi + {}^*d\pi = 0,$	$\lambda^\varphi := (-1)^s L_\varphi + {}^*d\pi = 0,$
mielke-geometrodynamics_053 EQ0084 (3.1.10)		■ 1.000 ● N +0	$\frac{\partial L}{\partial\varphi}-\partial_\alpha\frac{\partial L}{\partial(\partial_\alpha\varphi)}=0.$ $\frac{\partial L}{\partial\varphi}-\partial_\alpha\frac{\partial L}{\partial(\partial_\alpha\varphi)}=0.$	$\frac{\partial L}{\partial\varphi} - \partial_\alpha \frac{\partial L}{\partial(\partial_\alpha\varphi)} = 0.$	$\frac{\partial L}{\partial\varphi} - \partial_\alpha \frac{\partial L}{\partial(\partial_\alpha\varphi)} = 0.$
mielke-geometrodynamics_053 EQ0085 (3.1.11)		■ 1.000 ● K +0	$\{ \}^{\ast}H:=L-d\varphi\wedge\pi$	${}^*H := L - d\varphi \wedge \pi$	${}^*H := L - d\varphi \wedge \pi$
mielke-geometrodynamics_053 EQ0086 (3.1.12)		■ 1.000 ● N +0	$d^{\ast}H=dL+d\varphi\wedge{}^*L_\varphi+d\varphi\wedge d\pi-dd\varphi\wedge\pi$	$d^*H = dL + d\varphi \wedge {}^*L_\varphi + d\varphi \wedge d\pi - dd\varphi \wedge \pi$	$d^*H = dL + d\varphi \wedge {}^*L_\varphi + d\varphi \wedge d\pi - d\,d\varphi \wedge \pi$
mielke-geometrodynamics_053 EQ0087 (3.1.13)		■ 1.000 ● K +0	$d^{\ast}H\simeq 0$	$d^*H \simeq 0$	$d^*H \simeq 0$
mielke-geometrodynamics_054 EQ0088 (3.1.14)		■ 1.000 ● N +0	$\beta\wedge{}^*D\psi=m\psi\eta.$	$\beta \wedge {}^*D\psi = m\psi\eta.$	$\beta \wedge {}^*D\psi = m\psi\eta.$
mielke-geometrodynamics_054 EQ0089 (3.1.15)		■ 1.000 ● N +0	$\square\varphi+m^2\varphi=0$	$\square\varphi + m^2\varphi = 0$	$\square\varphi + m^2\varphi = 0$
mielke-geometrodynamics_054 EQ0090 (3.1.16)		■ 1.000 ● S -1	$\overline{L}_{\mathrm{mat}}=\overline{L}_{\mathrm{mat}}(\overline{\psi}\psi,\overline{\psi}\beta\cdot d\psi)\eta=\overline{\psi}\beta\wedge{}^*d\psi-m\overline{\psi}\psi\eta.$	$\overline{L}_{\mathrm{mat}} = \overline{L}_{\mathrm{mat}}(\overline{\psi}\psi, \overline{\psi}\beta \cdot d\psi)\eta = \overline{\psi}\beta \wedge {}^*d\psi - m\overline{\psi}\psi\eta.$	$\overline{L}_{\mathrm{mat}} = \overline{L}_{\mathrm{mat}}(\overline{\psi}\psi, \overline{\psi}\beta \cdot d\psi)\eta = \overline{\psi}\beta \wedge {}^*d\psi - m\overline{\psi}\psi\eta.$
mielke-geometrodynamics_055 EQ0092 (3.2.2)		■ 1.000 ● N -1	$L_{\mathrm{mat}}=L(\overline{\varphi}\varphi,\overline{\varphi}\beta\cdot(d+\omega')\varphi)\eta=L(\overline{\varphi}\varphi,\overline{\varphi}\beta\cdot D'\varphi)\eta$	$L_{\mathrm{mat}} = L(\overline{\varphi}\varphi, \overline{\varphi}\beta \cdot (d + \omega')\varphi)\eta = L(\overline{\varphi}\varphi, \overline{\varphi}\beta \cdot D'\varphi)\eta$	$L_{\mathrm{mat}} = L(\overline{\varphi}\varphi, \overline{\varphi}\beta \cdot (d + \omega')\varphi)\eta = L(\overline{\varphi}\varphi, \overline{\varphi}\beta \cdot D'\varphi)\eta$
mielke-geometrodynamics_055 EQ0094 (3.2.4)		■ 1.000 ● K +0	$D\left(\{ \}^G b\right)=\{ \}^G\omega\otimes\{ \}^G b.$	$D\left({}^G b\right) = {}^G\omega \otimes {}^G b.$	$D({}^G b) = {}^G\omega \otimes {}^G b.$
mielke-geometrodynamics_055 EQ0095 (3.2.5)		■ 1.000 ● S +0	$D\left({}^G b\right)=dG\otimes b+G\omega\otimes b=\left\{dGG^{-1}+G\omega G^{-1}\right\}\otimes{}^G b.$	$D\left({}^G b\right) = dG \otimes b + G\omega \otimes b = \left\{dGG^{-1} + G\omega G^{-1}\right\} \otimes {}^G b.$	$D({}^G b) = dG \otimes b + G\omega \otimes b = \left\{dGG^{-1} + G\omega G^{-1}\right\} \otimes {}^G b.$
mielke-geometrodynamics_055 EQ0096 (3.2.6)		■ 1.000 ● N +0	$\omega\rightarrow{}^G\omega=G\omega G^{-1}+dGG^{-1}.$	$\omega \rightarrow {}^G\omega = G\omega G^{-1} + dGG^{-1}.$	$\omega \rightarrow {}^G\omega = G\omega G^{-1} + dG\,G^{-1}.$
mielke-geometrodynamics_057 EQ0102 (3.3.3)		■ 1.000 ● N +1	$\theta=-\frac{e}{\hbar}\int\mathbf{A}\cdot d\mathbf{l}$	$\theta = -\frac{e}{\hbar}\int\mathbf{A}\cdot d\mathbf{l}$	$\theta = -\frac{e}{\hbar}\int\mathbf{A}\cdot d\mathbf{l}.$
mielke-geometrodynamics_057 EQ0103 (3.3.4)		■ 1.000 ● N -2	$L=L(A,dA)+L_{\mathrm{mat}}(A,dA,\Psi,d\Psi).$	$L = L(A, dA) + L_{\mathrm{mat}}(A, dA, \Psi, d\Psi).$	$L = L(A, dA) + L_{\mathrm{mat}}(A, dA, \Psi, d\Psi).$
mielke-geometrodynamics_057 EQ0104 (3.3.5)		■ 1.000 ● N -1	$\frac{\delta L}{\delta A}:=\frac{\partial L}{\partial A}+d\frac{\partial L}{\partial dA}=0,$	$\frac{\delta L}{\delta A} := \frac{\partial L}{\partial A} + d\frac{\partial L}{\partial dA} = 0,$	$\frac{\delta L}{\delta A} := \frac{\partial L}{\partial A} + d\frac{\partial L}{\partial dA} = 0,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_058 EQ0106 (3.3.7)		■ 1.000 ● N +1	$d\,F\equiv 0,\quad d\,H=j$	$dF\equiv 0,\quad dH=j$	$dF\equiv 0,\quad dH=j.$
mielke-geometrodynamics_058 EQ0107 (3.3.8)		■ 1.000 ● K +0	$d\,j\,\simeq 0.$	$dj\simeq 0.$	$dj\simeq 0.$
mielke-geometrodynamics_058 EQ0111 (3.3.12)		■ 1.000 ● N -1	$H\!:=\!\frac{H}{d\,x^i\wedge d\,x^j}\wedge d\,t+\frac{D}{d\,x^i\wedge d\,x^j}H_{ij}$	$H:=-\mathfrak{H}\wedge dt+\mathfrak{D}=\frac{1}{2}H_{ij}dx^i\wedge dx^j$	$H:=-\mathfrak{H}\wedge dt+\mathfrak{D}=\frac{1}{2}H_{ij}dx^i\wedge dx^j,$
mielke-geometrodynamics_059 EQ0114 (3.3.15)		■ 1.000 ● W +1	$H_{mn}=\frac{\sqrt{ g }}{2}\,\varepsilon_{mnkl}\,g^{ki}\,g^{lj}\,F_{ij}$	$H_{mn}=\frac{\sqrt{ g }}{2}\varepsilon_{mnkl}g^{ki}g^{lj}F_{ij}.$	$H_{mn}=\frac{\sqrt{ g }}{2}\,\varepsilon_{mnkl}\,g^{ki}\,g^{lj}\,F_{ij}.$
mielke-geometrodynamics_060 EQ0121 (3.3.22)		■ 1.000 ● K +0	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$
mielke-geometrodynamics_060 EQ0122 (3.3.23)		■ 1.000 ● N -1	$\begin{aligned} \tilde{L}_{\mathrm{BF}}&:=-B\wedge dA+\frac{1}{2}B\wedge B\\ &\cong -B\wedge dA+dC \end{aligned}$	$\begin{aligned} \tilde{L}_{\mathrm{BF}}&:=-B\wedge dA+\frac{1}{2}B\wedge B\\ &\cong -B\wedge dA+dC \end{aligned}$	$\begin{aligned} \tilde{L}_{\mathrm{BF}}&:=-B\wedge dA+\frac{1}{2}B\wedge B\\ &\cong -B\wedge dA+dC. \end{aligned}$
mielke-geometrodynamics_060 EQ0123 (3.3.24)		■ 1.000 ● N +0	$B\,\cong dA,\quad dB\,\cong dF\equiv 0$	$B\cong dA,\quad dB\cong dF\equiv 0$	$B\cong dA,\quad dB\cong dF\equiv 0,$
mielke-geometrodynamics_060 EQ0124 (3.3.25)		■ 1.000 ● K +0	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$
mielke-geometrodynamics_061 EQ0125 (3.3.26)		■ 1.000 ● K +0	$A\rightarrow A' = A+d\,\theta(x).$	$A\rightarrow A'=A+d\theta(x).$	$A\rightarrow A'=A+d\theta(x).$
mielke-geometrodynamics_061 EQ0126 (3.3.27)		■ 1.000 ● N +0	$A\rightarrow \widetilde{A}=A+\psi,\quad B\rightarrow \widetilde{B}=B+d\,\psi,$	$A\rightarrow \widetilde{A}=A+\psi,\quad B\rightarrow \widetilde{B}=B+d\psi,$	$A\rightarrow \widetilde{A}=A+\psi,\quad B\rightarrow \widetilde{B}=B+d\psi,$
mielke-geometrodynamics_061 EQ0127 (3.3.28)		■ 1.000 ● K +0	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge {}^*B+L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge {}^*B+L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge {}^*B+L_{\mathrm{matter}},$
mielke-geometrodynamics_061 EQ0128 (3.3.29)		■ 1.000 ● N +0	$-d\,B=d^*F\cong j,$	$-dB=d^*F\cong j,$	$-dB=d^*F\cong j,$
mielke-geometrodynamics_063 EQ0132 (3.4.3)		■ 1.000 ● N +0	$\alpha_g:=g^2/\hbar\,c$	$\alpha_g:=g^2/\hbar c$	$\alpha_g:=g^2/\hbar c$
mielke-geometrodynamics_063 EQ0134 (3.4.5)		■ 1.000 ● K +0	$\delta L_{\omega}:=\delta\Omega\wedge\Pi^{\Omega}.$	$\delta L_{\omega}:=\delta\Omega\wedge\Pi^{\Omega}.$	$\delta L_{\omega}:=\delta\Omega\wedge\Pi^{\Omega}.$
mielke-geometrodynamics_063 EQ0138 (3.4.9)		■ 1.000 ● N -1	$L=L_{\mathrm{mat}}+L_{\omega},$	$L=L_{\mathrm{mat}}+L_{\omega},$	$L=L_{\mathrm{mat}}+L_{\omega},$
mielke-geometrodynamics_063 EQ0139 (3.4.10)		■ 1.000 ● K +0	$\delta L_{\mathrm{mat}}:=\delta\omega\wedge\tau.$	$\delta L_{\mathrm{mat}}:=\delta\omega\wedge\tau.$	$\delta L_{\mathrm{mat}}:=\delta\omega\wedge\tau.$
mielke-geometrodynamics_064 EQ0140 (3.4.11)		■ 1.000 ● N +2	$D\,\Pi^{\Omega}=\tau$	$D\Pi^{\Omega}=\tau$	$\underline{D\Pi^{\Omega}=\tau}.$
mielke-geometrodynamics_064 EQ0141 (3.4.12)		■ 1.000 ● N +0	$D\,\Omega\equiv 0.$	$D\Omega\equiv 0.$	$D\Omega\equiv 0.$
mielke-geometrodynamics_064 EQ0145 (3.4.16)		■ 1.000 ● N +0	$d\,\Pi^{\Omega}+[\omega,\Pi^{\Omega}]=\tau.$	$d\Pi^{\Omega}+[\omega,\Pi^{\Omega}]=\tau.$	$d\,\Pi^{\Omega}+[\omega,\Pi^{\Omega}]=\tau.$
mielke-geometrodynamics_065 EQ0147 (3.4.18)		■ 1.000 ● N +0	$Q_e=\frac{1}{2}Y+I_3.$	$Q_e=\frac{1}{2}Y+I_3.$	$Q_e=\frac{1}{2}Y+I_3.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_065 EQ0148 (3.4.19)		■ 1.000 ● N +0	$L_{\mathrm{YM}}=-\frac{1}{\alpha_g}\operatorname{Tr}\left(\Omega\wedge^*\Omega\right)$.	$L_{\mathrm{YM}}=-\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge^*\Omega)$.	$L_{\mathrm{YM}}=-\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge^*\Omega)$.
mielke-geometrodynamics_065 EQ0150 (3.4.21)		■ 1.000 ● N +1	$\Pi^{\Omega}=-\frac{2}{\alpha_g}\wedge^*\Omega$	$\Pi^{\Omega}=-\frac{2}{\alpha_g}\wedge^*\Omega$	$\Pi^{\Omega}=-\frac{2}{\alpha_g}\wedge^*\Omega$.
mielke-geometrodynamics_066 EQ0153 (3.5.1)		■ 1.000 ● N +1	$\Pi^{\Omega}=\zeta\Omega$.	$\Pi^{\Omega}=\zeta\Omega$.	$\Pi^{\Omega}=\zeta\Omega$.
mielke-geometrodynamics_067 EQ0156 (3.5.4)		■ 1.000 ● N +0	$\wedge^*\sigma^{\mu\nu}=\sigma^{\mu\nu},\quad\wedge^*\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$ $\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$	$\sigma^{\mu\nu}=\sigma^{\mu\nu},\quad\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$	$\sigma^{\mu\nu}=\sigma^{\mu\nu},\quad\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$
mielke-geometrodynamics_067 EQ0158 (3.5.6)		■ 1.000 ● N +0	$A_{\mu}=\mathrm{i}\sigma_{\mu i}\partial^i\ln h$.	$A_{\mu}=\mathrm{i}\sigma_{\mu i}\partial^i\ln h$.	$A_{\mu}=\mathrm{i}\sigma_{\mu i}\partial^i\ln h$.
mielke-geometrodynamics_067 EQ0159 (3.5.7)		■ 1.000 ● N +0	$\square h-h^{-1}\partial_i h\partial^i h=h^3$	$\square h-h^{-1}\partial_i h\partial^i h=h^3$	$\square h-h^{-1}\partial_i h\partial^i h=h^3$
mielke-geometrodynamics_067 EQ0160 (3.5.8)		■ 1.000 ● W +0	$h=\sum_{i=1}^{k+1}\frac{l_{(i)}^2}{(x-x_{(i)})^2},$	$h=\sum_{i=1}^{k+1}\frac{l_{(i)}^2}{(x-x_{(i)})^2},$	$h=\sum_{i=1}^{k+1}\frac{l_{(i)}^2}{(x-x_{(i)})^2},$
mielke-geometrodynamics_067 EQ0161 (3.5.9)		■ 1.000 ● K +0	$p_1(M)=k$.	$p_1(M)=k$.	$p_1(M)=k$.
mielke-geometrodynamics_068 EQ0162 (3.5.10)		■ 1.000 ● N +2	$\langle\infty -\infty\rangle=Z\int_{\mathcal{M}_{\omega}}d\omega\mu[\omega]\exp i\int_M(L_{\omega}+L_{\infty})$	$\langle\infty -\infty\rangle=Z\int_{\mathcal{M}_{\omega}}d\omega\mu[\omega]\exp i\int_M(L_{\omega}+L_{\infty})$	$\langle\infty -\infty\rangle=Z\int_{\mathcal{M}_{\omega}}d\omega\mu[\omega]\exp i\int_M(L_{\omega}+L_{\infty})$.
mielke-geometrodynamics_068 EQ0163 (3.5.11)		■ 1.000 ● N +0	$L_{\infty}=(\stackrel{+}{-})\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge\Omega)$	$L_{\infty}=(\stackrel{+}{-})\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge\Omega)$	$L_{\infty}=(\stackrel{+}{-})\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge\Omega)$
mielke-geometrodynamics_070 EQ0167 (3.6.2)		■ 1.000 ● N +0	$\mathscr{P}_{\pm}:=\bar{\psi}\sigma_{\pm}\psi/2m,$	$\mathscr{P}_{\pm}:=\bar{\psi}\sigma_{\pm}\psi/2m,$	$\mathscr{P}_{\pm}:=\bar{\psi}\sigma_{\pm}\psi/2m,$
mielke-geometrodynamics_070 EQ0168 (3.6.3)		■ 1.000 ● N +0	$D_{\pm}\wedge^*\psi=\left(F^{\pm}-\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\right)\wedge\sigma_{\pm}\psi=0$	$D_{\pm}\wedge^*\psi=\left(F^{\pm}-\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\right)\wedge\sigma_{\pm}\psi=0$	$D_{\pm}\wedge^*\psi\mp i\left(F^{\pm}-\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\right)\wedge\sigma_{\pm}\psi=0$
mielke-geometrodynamics_070 EQ0170 (3.6.5)		■ 1.000 ● N +0	$D^*\psi=\left(\nabla^*\nabla+\frac{1}{4}R+\frac{1}{2}F^+\right)\psi$.	$D^*D\psi=\left(\nabla^*\nabla+\frac{1}{4}R+\frac{1}{2}F^+\right)\psi$.	$D^*D\psi=\left(\nabla^*\nabla+\frac{1}{4}R+\frac{1}{2}F^+\right)\psi$.
mielke-geometrodynamics_070 EQ0172 (3.6.7)		■ 1.000 ● N +0	$\Omega_{\pm}^g=\frac{1}{2\ell^2}\sigma_{\pm},$	$\Omega_{\pm}^g=\frac{1}{2\ell^2}\sigma_{\pm},$	$\Omega_{\pm}^g=\frac{1}{2\ell^2}\sigma_{\pm},$
mielke-geometrodynamics_072 EQ0176 (3.7.5)		■ 1.000 ● N +0	$\begin{aligned} & D^*\Theta^{(1)}-\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta=0, \\ & D\Pi^{\Omega}+\left[\phi^{(0)},*\Theta^{(1)}\right]=0. \end{aligned}$	$D^*\Theta^{(1)}-\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta=0,$ $D\Pi^{\Omega}+\left[\phi^{(0)},*\Theta^{(1)}\right]=0.$	$D^*\Theta^{(1)}-\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta=0,$ $D\Pi^{\Omega}+\left[\phi^{(0)},*\Theta^{(1)}\right]=0.$
mielke-geometrodynamics_072 EQ0177 (3.7.7)		■ 1.000 ● N +0	$\bar{\varphi}\cdot\varphi:=\varphi^{*j}\varphi_j=\frac{\mu^2}{\lambda}.$	$\bar{\varphi}\cdot\varphi:=\varphi^{*j}\varphi_j=\frac{\mu^2}{\lambda}.$	$\bar{\varphi}\cdot\varphi:=\varphi^{*j}\varphi_j=\frac{\mu^2}{\lambda}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_072 EQ0179 (3.7.9)		■ 1.000 ● N -1	$\varphi_{\min} : P \rightarrow C^\infty \left(\mathring{V}(M, \rho(G/H), \rho(G), P) \subset V^\rho \right).$		
mielke-geometrodynamics_073 EQ0180 (3.7.10)		■ 1.000 ● N +0	$H := \left\{ h \in G \mid \rho(h) \mathrm{v}_0 = \mathrm{v}_0 \right\},$	$H := \{ h \in G \mid \rho(h) \mathbf{v}_0 = \mathbf{v}_0 \},$	$H := \{ h \in G \mid \rho(h) \mathbf{v}_0 = \mathbf{v}_0 \},$
mielke-geometrodynamics_073 EQ0181 (3.7.11)		■ 1.000 ● N -2	$Q = \left\{ p \in P \mid \varphi(p) = \sigma(\mathrm{v}_0) \right\}$	$Q = \{ p \in P \mid \varphi(p) = \sigma(\mathbf{v}_0) \}$	$Q = \{ p \in P \mid \varphi(p) = \sigma(\mathbf{v}_0) \}$
mielke-geometrodynamics_073 EQ0182 (3.7.12)		■ 1.000 ● N +0	$D \, \phi^{\{0\}} = 0 \, .$	$D\phi^{(0)} = 0.$	$D\phi^{(0)} = 0.$
mielke-geometrodynamics_073 EQ0183 (3.7.13)		■ 1.000 ● K +0	$\begin{aligned} \Theta^\perp &:= \Omega_0 = d\omega_0 - (\omega \wedge \omega)_0 = d\omega_0 - \omega_0 \wedge \omega - \omega \wedge \omega_0 \\ &= d\omega_0 - [\omega_0, \omega] = D\omega_0 \end{aligned}$	$\Theta^\perp := \Omega_0 = d\omega_0 - (\omega \wedge \omega)_0 = d\omega_0 - \omega_0 \wedge \omega - \omega \wedge \omega_0$ $= d\omega_0 - [\omega_0, \omega] = D\omega_0$	$\Theta^\perp := \Omega_0 = d\omega_0 - (\omega \wedge \omega)_0 = d\omega_0 - \omega_0 \wedge \omega - \omega \wedge \omega_0$ $= d\omega_0 - [\omega_0, \omega] = D\omega_0$
mielke-geometrodynamics_074 EQ0185 (3.7.15)		■ 1.000 ● K +0	$d_{\{0\}} \, \, \, \wedge^{\{*\}} \, \Omega_{\{0\}} = 0 \, .$	$d_0^* \Omega_0 = 0.$	$d_0^* \Omega_0 = 0.$
mielke-geometrodynamics_074 EQ0186 (3.7.16)		■ 1.000 ● N +0	$\left(D^\wedge \{*\} \, \Omega_{\{0\}} \right) = \mathrm{vec} \{ D^\wedge \{*\} \, \Omega_{\{0\}} = \mathrm{vec} \{ D^\wedge \{*\} \, \Theta^\perp \} = 0$	$(D^* \Omega)_0 = \vec{D}^* \Omega_0 = \vec{D}^* \Theta^\perp = 0$	$(D^* \Omega)_0 = \vec{D}^* \Omega_0 = \vec{D}^* \Theta^\perp = 0$
mielke-geometrodynamics_074 EQ0188 (3.7.18)		■ 1.000 ● N -1	$\omega_{\{0\}} = \varphi^j L_j \, \vartheta_0 \, .$	$\omega_0 = \varphi^j L_j \otimes \vartheta_0.$	$\omega_0 = \varphi^j L_j \otimes \vartheta_0.$
mielke-geometrodynamics_080 EQ0190		■ 1.000 ● U +23	$\begin{aligned} \left[T_G \right] : \quad & \quad T_g \psi(m) = D \left(g_m^{-1} g g_{g^{-1}m}, m_0 \right) \psi(g^{-1}m). \\ & \quad g_{g^{-1}m}, m_0 \right) \psi \left(g^{-1}m \right) \\ & \quad . \end{aligned}$	<i>(not rendered)</i>	$T_G : \quad T_g \psi(m) = D \left(g_m^{-1} g g_{g^{-1}m}, m_0 \right) \psi(g^{-1}m).$
mielke-geometrodynamics_081 EQ0193 (4.1.3)		■ 1.000 ● N +0	$A(n, \mathbb{R}) = \mathbb{R}^n \otimes GL(n, \mathbb{R})$	$A(n, \mathbb{R}) = \mathbb{R}^n \otimes GL(n, \mathbb{R})$	$A(n, \mathbb{R}) = \mathbb{R}^n \otimes GL(n, \mathbb{R})$
mielke-geometrodynamics_081 EQ0194 (4.1.4)		■ 1.000 ● N +0	$P := \mathbb{R}^n \otimes 0(1, n-1) \subset A(n, \mathbb{R}),$	$P := \mathbb{R}^n \otimes 0(1, n-1) \subset A(n, \mathbb{R}),$	$P := \mathbb{R}^n \otimes 0(1, n-1) \subset A(n, \mathbb{R}),$
mielke-geometrodynamics_082 EQ0197 (4.1.10)		■ 1.000 ● N +0	$e_{\{\alpha\}}(m) = e_{\alpha}^i \partial_i$	$e_{\alpha}(m) = e_{\alpha}^i \partial_i$	$e_{\alpha}(m) = e_{\alpha}^i \partial_i$
mielke-geometrodynamics_082 EQ0199 (4.1.13)		■ 1.000 ● N +0	$L(M) = \bigcup_{m \in M} L_m(M)$	$L(M) = \bigcup_{m \in M} L_m(M)$	$L(M) = \bigcup_{m \in M} L_m(M)$
mielke-geometrodynamics_083 EQ0200 (4.1.14)		■ 1.000 ● K +0	$L(M) = P(M, GL(n, \mathbb{R}), \pi, \delta)$	$L(M) = P(M, GL(n, \mathbb{R}), \pi, \delta)$	$L(M) = P(M, GL(n, \mathbb{R}), \pi, \delta)$
mielke-geometrodynamics_083 EQ0202 (4.1.16)		■ 1.000 ● K +0	$\vartheta(e) = 0 \Leftrightarrow \pi(e) = 0, e \in T_p(P),$	$\vartheta(e) = 0 \Leftrightarrow \pi(e) = 0, e \in T_p(P),$	$\vartheta(e) = 0 \Leftrightarrow \pi(e) = 0, e \in T_p(P),$
mielke-geometrodynamics_083 EQ0204 (4.1.18)		■ 1.000 ● N +0	$T(M) = V(M, \mathbb{R}^n, GL(n, \mathbb{R}), L(M))$	$T(M) = V(M, \mathbb{R}^n, GL(n, \mathbb{R}), L(M))$	$T(M) = V(M, \mathbb{R}^n, GL(n, \mathbb{R}), L(M))$
mielke-geometrodynamics_083 EQ0205 (4.1.19)		■ 1.000 ● N +0	$A(M) := P(M, A(n, \mathbb{R}), \pi, \delta) = L(M) \times \mathbb{R}^n.$	$A(M) := P(M, A(n, \mathbb{R}), \pi, \delta) = L(M) \times \mathbb{R}^n.$	$A(M) := P(M, A(n, \mathbb{R}), \pi, \delta) = L(M) \times \mathbb{R}^n.$
mielke-geometrodynamics_083 EQ0206 (4.1.20)		■ 1.000 ● N +0	$\beta : A(n, \mathbb{R}) \rightarrow GL(n, \mathbb{R}) = A(n, \mathbb{R})/\mathbb{R}^n.$	$\beta : A(n, \mathbb{R}) \rightarrow GL(n, \mathbb{R}) = A(n, \mathbb{R})/\mathbb{R}^n.$	$\beta : A(n, \mathbb{R}) \rightarrow GL(n, \mathbb{R}) = A(n, \mathbb{R})/\mathbb{R}^n.$
mielke-geometrodynamics_086 EQ0213 (4.2.5)		■ 1.000 ● N +0	$\mathscr{A}(n, \mathbb{R}) := C^\infty(A(M) \times_{\mathrm{Ad}} A(n, \mathbb{R})).$	$\mathscr{A}(n, \mathbb{R}) := C^\infty(A(M) \times_{\mathrm{Ad}} A(n, \mathbb{R})).$	$\mathscr{A}(n, \mathbb{R}) := C^\infty(A(M) \times_{\mathrm{Ad}} A(n, \mathbb{R})).$
mielke-geometrodynamics_087 EQ0223 (4.2.16)		■ 1.000 ● N -2	$Q_{\{A\}} = \left\{ p \in A(M) \mid \widetilde{\varphi}(p) = \sigma(0_m) \right\} = L(M)$	$Q_A = \left\{ p \in A(M) \mid \widetilde{\varphi}(p) = \sigma(0_m) \right\} = L(M)$	$Q_A = \{ p \in A(M) \mid \widetilde{\varphi}(p) = \sigma(0_m) \} = L(M)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_090 EQ0237 (4.3.2)		■ 1.000 ● S +0	$\ell(\mathscr{C}):=\int_{t_0}^{t_1}ds(t)$	$\ell(\mathcal{C}):=\int_{t_0}^{t_1}ds(t)$	$\ell(\mathcal{C}):=\int_{t_0}^{t_1}ds(t)$
mielke-geometrodynamics_090 EQ0238 (4.3.3)		■ 1.000 ● K +0	$\ell\left(\mathscr{C}_{\scriptscriptstyle{1}}+\mathscr{C}_{\scriptscriptstyle{2}}\right)=\ell\left(\mathscr{C}_{\scriptscriptstyle{1}}\right)+\ell\left(\mathscr{C}_{\scriptscriptstyle{2}}\right)$	$\ell\left(\mathcal{C}_1+\mathcal{C}_2\right)=\ell\left(\mathcal{C}_1\right)+\ell\left(\mathcal{C}_2\right)$	$\ell(\mathcal{C}_1+\mathcal{C}_2)=\ell(\mathcal{C}_1)+\ell(\mathcal{C}_2)$
mielke-geometrodynamics_090 EQ0239 (4.3.4)		■ 1.000 ● K +0	$ds=F(m,\vartheta)dt,$	$ds=F(m,\vartheta)dt,$	$ds=F(m,\vartheta)dt,$
mielke-geometrodynamics_091 EQ0242 (4.3.7)		■ 1.000 ● C -3	$ds^2=(F(m,\vartheta)dt)^2=g_{\alpha\beta}\vartheta^\alpha\vartheta^\beta=g_{ij}(m)dx^i\otimes_sdx^j;$	$ds^2=(F(m,\vartheta)dt)^2=g_{\alpha\beta}\vartheta^\alpha\vartheta^\beta=g_{ij}(m)dx^i\otimes_sdx^j;$	$ds^2=(F(m,\vartheta)dt)^2=g_{\alpha\beta}\vartheta^\alpha\vartheta^\beta=g_{ij}(m)dx^i\otimes_sdx^j;$
mielke-geometrodynamics_092 EQ0246 (4.3.11)		■ 1.000 ● N +0	$L^g(M):=P(M,0(s,n-s),\pi,\delta)$.	$L^g(M):=P(M,O(s,n-s),\pi,\delta).$	$L^g(M):=P(M,O(s,n-s),\pi,\delta).$
mielke-geometrodynamics_092 EQ0247 (4.3.12)		■ 1.000 ● N +0	$D\,e_{\alpha}^{\cdot i}=0$	$D\,e_{\alpha}^{\cdot i}=0$	$D\,e_{\alpha}^{\cdot i}=0$
mielke-geometrodynamics_092 EQ0251 (4.3.16)		■ 1.000 ● N +0	$\Gamma_i^{\cdot\alpha\beta}=-\Gamma_i^{\cdot\beta\alpha}$	$\Gamma_i^{\cdot\alpha\beta}=-\Gamma_i^{\cdot\beta\alpha}$	$\Gamma_i^{\cdot\alpha\beta}=-\Gamma_i^{\cdot\beta\alpha}$
mielke-geometrodynamics_093 EQ0252 (4.3.17)		■ 1.000 ● W +0	$g_{\scriptscriptstyle{i\,j}}\stackrel{(*)}{=}\,o_{\scriptscriptstyle{i\,j}}$	$g_{ij}\stackrel{(*)}{=}\,o_{ij}$	$g_{ij}\stackrel{(*)}{=}\,o_{ij}$
mielke-geometrodynamics_093 EQ0253 (4.3.18)		■ 1.000 ● N +0	$\partial_k g_{\scriptscriptstyle{i\,j}}\stackrel{(*)}{=}\,0$.	$\partial_k g_{ij}\stackrel{(*)}{=}\,0.$	$\partial_k g_{ij}\stackrel{(*)}{=}\,0.$
mielke-geometrodynamics_093 EQ0254 (4.3.19)		■ 1.000 ● N +1	$\Omega^g=d\omega^g-\omega^g\wedge\omega^g$	$\Omega^g=d\omega^g-\omega^g\wedge\omega^g$	$\Omega^g=d\omega^g-\omega^g\wedge\omega^g$
mielke-geometrodynamics_094 EQ0257 (4.3.22)		■ 1.000 ● K +0	$\omega^g=\omega^{\{\}}-K$.	$\omega^g=\omega^{\{\}}-K.$	$\omega^g=\omega^{\{\}}-K.$
mielke-geometrodynamics_094 EQ0258 (4.3.23)		■ 1.000 ● K +0	$d\vartheta-\left[\omega^{\{\}},\vartheta\right]=0$	$d\vartheta-\left[\omega^{\{\}},\vartheta\right]=0$	$d\vartheta-\left[\omega^{\{\}},\vartheta\right]=0$
mielke-geometrodynamics_094 EQ0259 (4.3.24)		■ 1.000 ● N +0	$\Theta=[K,\vartheta]$	$\Theta=[K,\vartheta]$	$\Theta=[K,\vartheta]$
mielke-geometrodynamics_095 EQ0265 (4.3.30)		■ 1.000 ● W +0	$\left\{\begin{array}{c}\kappa\\ij\end{array}\right\}=\frac{1}{2}g^{\kappa l}\left(\partial_i g_{jl}+\partial_j g_{il}-\partial_l g_{ij}\right)$	$\left\{\begin{array}{c}\kappa\\ij\end{array}\right\}=\frac{1}{2}g^{\kappa l}\left(\partial_i g_{jl}+\partial_j g_{il}-\partial_l g_{ij}\right)$	$\left\{\begin{array}{c}\kappa\\ij\end{array}\right\}=\frac{1}{2}g^{\kappa l}\left(\partial_i g_{jl}+\partial_j g_{il}-\partial_l g_{ij}\right)$
mielke-geometrodynamics_096 EQ0272 (4.4.1)		■ 1.000 ● N -1	$L_g=L(g_{\alpha\beta};\vartheta,(\omega^g),\Theta,\Omega^g).$	$L_g=L(g_{\alpha\beta};\vartheta,(\omega^g),\Theta,\Omega^g).$	$L_g=L(g_{\alpha\beta};\vartheta,(\omega^g),\Theta,\Omega^g).$
mielke-geometrodynamics_097 EQ0273 (4.4.2)		■ 1.000 ● N +0	$\delta L_g=:\delta\vartheta\wedge E+\delta\Theta\wedge\Pi^{\mathrm{T}}+\delta\Omega^g\wedge\Pi^{\mathrm{L}},$	$\delta L_g=:\delta\vartheta\wedge E+\delta\Theta\wedge\Pi^{\mathrm{T}}+\delta\Omega^g\wedge\Pi^{\mathrm{L}},$	$\delta L_g=:\delta\vartheta\wedge E+\delta\Theta\wedge\Pi^{\mathrm{T}}+\delta\Omega^g\wedge\Pi^{\mathrm{L}},$
mielke-geometrodynamics_097 EQ0274 (4.4.7)		■ 1.000 ● K +0	$\delta\Theta=\delta\left(d\vartheta-\left[\omega^{\mathrm{L}},\vartheta\right]\right)$	$\delta\Theta=\delta\left(d\vartheta-\left[\omega^{\mathrm{L}},\vartheta\right]\right)$	$\delta\Theta=\delta(d\vartheta-\left[\omega^{\mathrm{L}},\vartheta\right])$
mielke-geometrodynamics_097 EQ0277 (4.4.5)		■ 1.000 ● N +0	$S=\int_M(L_g+L_{\mathrm{mat}})$	$S=\int_M(L_g+L_{\mathrm{mat}})$	$S=\int_M(L_g+L_{\mathrm{mat}})$
mielke-geometrodynamics_098 EQ0279 (4.4.8)		■ 1.000 ● N +0	$\delta L_{\text{matter}}=:\delta\vartheta\wedge\Sigma+\delta\omega^g\wedge\tau_s+\delta_{\varphi}\wedge\lambda^{\varphi}$	$\delta L_{\text{matter}}=:\delta\vartheta\wedge\Sigma+\delta\omega^g\wedge\tau_s+\delta_{\varphi}\wedge\lambda^{\varphi}$	$\delta L_{\text{matter}}=:\delta\vartheta\wedge\Sigma+\delta\omega^g\wedge\tau_s+\delta_{\varphi}\wedge\lambda^{\varphi}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_098 EQ0280 (4.4.9)	098	1.000 ● N +0	$D\,\Theta\equiv\left[\vartheta,\Omega^L\right]$	$D\Theta\equiv\left[\vartheta,\Omega^L\right]$	$D\Theta\equiv\left[\vartheta,\Omega^L\right]$
mielke-geometrodynamics_098 EQ0281 (4.4.10)	098	1.000 ● N +0	$D\,\Omega^g\equiv 0,$	$D\Omega^g\equiv 0,$	$D\Omega^g\equiv 0,$
mielke-geometrodynamics_099 EQ0283 (4.4.12)	099	1.000 ● N +0	$D[\,\Theta,\vartheta]\equiv\left[\vartheta\wedge\vartheta,\Omega^g\right]$	$D[\Theta,\vartheta]\equiv\left[\vartheta\wedge\vartheta,\Omega^g\right]$	$D[\Theta,\vartheta]\equiv\left[\vartheta\wedge\vartheta,\Omega^g\right]$
mielke-geometrodynamics_100 EQ0289 (4.4.18)	100	1.000 ● N +0	$\left[\Omega^g,\Pi^L\right]+\mathrm{E}_{\square}=0$	$\left[\Omega^g,\Pi^L\right]+\mathrm{E}_{\square}=0$	$\left[\Omega^g,\Pi^L\right]+\mathrm{E}_{\square}=0$
mielke-geometrodynamics_100 EQ0290 (4.5.1)	100	1.000 ● N +0	$L_{\mathrm{\mathtt{mat}}}=L\left(g_{\alpha\beta},\vartheta^\alpha,\Psi,D\Psi\right).$	$L_{\mathrm{mat}}=L\left(g_{\alpha\beta},\vartheta^\alpha,\Psi,D\Psi\right).$	$L_{\mathrm{mat}}=L(g_{\alpha\beta},\vartheta^\alpha,\Psi,D\Psi).$
mielke-geometrodynamics_100 EQ0291 (4.5.2)	100	1.000 ● W -1	$\Sigma_{\alpha}:=\frac{\partial L}{\partial\vartheta^\alpha},\quad\sigma^{\alpha\beta}:=2\frac{\partial L}{\partial g_{\alpha\beta}},\quad\text{and}\quad\tau_{\alpha\beta}:=\frac{\partial L}{\partial\Gamma_{\alpha\beta}}=I_{\alpha\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial\vartheta^\alpha},\quad\sigma^{\alpha\beta}:=2\frac{\partial L}{\partial g_{\alpha\beta}},\quad\text{and}\quad\tau_{\alpha\beta}:=\frac{\partial L}{\partial\Gamma_{\alpha\beta}}=I_{\alpha\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial\vartheta^\alpha},\quad\sigma^{\alpha\beta}:=2\frac{\partial L}{\partial g_{\alpha\beta}},\quad\text{and}\quad\tau_{\alpha\beta}:=\frac{\partial L}{\partial\Gamma_{\alpha\beta}}=I_{\alpha\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}$
mielke-geometrodynamics_101 EQ0297 (8.10)	101	1.000 ● N +0	$\delta L=\delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^\alpha\wedge\frac{\partial L}{\partial\vartheta^\alpha}+\delta\Psi\wedge\frac{\partial L}{\partial\Psi}+\delta(D\Psi)\wedge\frac{\partial L}{\partial D\Psi}.$	$\delta L=\delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^\alpha\wedge\frac{\partial L}{\partial\vartheta^\alpha}+\delta\Psi\wedge\frac{\partial L}{\partial\Psi}+\delta(D\Psi)\wedge\frac{\partial L}{\partial D\Psi}.$	$\delta L=\delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^\alpha\wedge\frac{\partial L}{\partial\vartheta^\alpha}+\delta\Psi\wedge\frac{\partial L}{\partial\Psi}+\delta(D\Psi)\wedge\frac{\partial L}{\partial D\Psi}.$
mielke-geometrodynamics_102 EQ0300 (4.5.11)	102	1.000 ● N +0	$\delta L=-\varepsilon_{\alpha}{}^{\beta}\left[-\sigma^{\alpha\gamma}g_{\gamma\beta}+\vartheta^\alpha\wedge\frac{\partial L}{\partial\vartheta^\beta}+D\frac{\partial L}{\partial\Gamma_{\alpha}{}^{\beta}}+I^{\alpha}{}_{\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\right].$	$\delta L=-\varepsilon_{\alpha}{}^{\beta}\left[-\sigma^{\alpha\gamma}g_{\gamma\beta}+\vartheta^\alpha\wedge\frac{\partial L}{\partial\vartheta^\beta}+D\frac{\partial L}{\partial\Gamma_{\alpha}{}^{\beta}}+I^{\alpha}{}_{\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\right].$	$\delta L=-\varepsilon_{\alpha}{}^{\beta}\left[-\sigma^{\alpha\gamma}g_{\gamma\beta}+\vartheta^\alpha\wedge\frac{\partial L}{\partial\vartheta^\beta}+D\frac{\partial L}{\partial\Gamma_{\alpha}{}^{\beta}}+I^{\alpha}{}_{\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\right].$
mielke-geometrodynamics_102 EQ0301 (4.5.12)	102	1.000 ● N +1	$D\,\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}=-I_{\alpha\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\simeq 0.$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}=-I_{\alpha\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\simeq 0.$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}=-I_{\alpha\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\simeq 0.$
mielke-geometrodynamics_102 EQ0303 (4.5.13)	102	1.000 ● W -1	$\begin{aligned} & R_{\alpha}{}^{\beta}=d\Gamma_{\alpha}{}^{\beta} \\ & -\Gamma_{\alpha}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\beta} \end{aligned}$	$\begin{aligned} R_{\alpha}{}^{\beta} &= d\Gamma_{\alpha}{}^{\beta} \\ & -\Gamma_{\alpha}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\beta} \end{aligned}$	$\begin{aligned} R_{\alpha}{}^{\beta} &= d\Gamma_{\alpha}{}^{\beta} \\ & -\Gamma_{\alpha}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\beta} \end{aligned}$
mielke-geometrodynamics_103 EQ0307 (4.5.17)	103	1.000 ● N +0	$\hat{M}:=M^3/\left[M^2+\frac{\Lambda}{3}J^2\right],\quad\hat{J}:=JM^2/\left[M^2+\frac{\Lambda}{3}J^2\right],$	$\hat{M}:=M^3/\left[M^2+\frac{\Lambda}{3}J^2\right],\quad\hat{J}:=JM^2/\left[M^2+\frac{\Lambda}{3}J^2\right],$	$\hat{M}:=M^3/[M^2+\frac{\Lambda}{3}J^2],\quad\hat{J}:=JM^2/[M^2+\frac{\Lambda}{3}J^2],$
mielke-geometrodynamics_110 EQ0317 (5.2.1)	110	1.000 ● W +0	$L_{\mathrm{\mathtt{D}}}=L(\gamma,\psi,D\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge D\psi+\overline{D\psi}\wedge{}^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$	$L_{\mathrm{D}}=L(\gamma,\psi,D\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge D\psi+\overline{D\psi}\wedge{}^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$	$L_{\mathrm{D}}=L(\gamma,\psi,D\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge D\psi+\overline{D\psi}\wedge{}^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$
mielke-geometrodynamics_111 EQ0322 (5.2.6)	111	1.000 ● K +0	$\left.d\,U=T^{\alpha}\wedge\mu_{\alpha}-\vartheta^{\alpha}\wedge D\mu_{\alpha}=\vartheta^{\alpha}\wedge\left[e_{\beta}\right]\left(T^{\beta}\wedge\mu_{\alpha}\right)-D\mu_{\alpha}\right]$	$dU=T^{\alpha}\wedge\mu_{\alpha}-\vartheta^{\alpha}\wedge D\mu_{\alpha}=\vartheta^{\alpha}\wedge\left[e_{\beta}\right]\left(T^{\beta}\wedge\mu_{\alpha}\right)-D\mu_{\alpha}\right]$	$dU=T^{\alpha}\wedge\mu_{\alpha}-\vartheta^{\alpha}\wedge D\mu_{\alpha}=\vartheta^{\alpha}\wedge\left[e_{\beta}\right]\left(T^{\beta}\wedge\mu_{\alpha}\right)-D\mu_{\alpha}\right]$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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mielke-geometrodynamics_111 EQ0324 (5.3.1)	111	■ 1.000 ● K +0	$L_{\mathrm{EC}}:=-\frac{1}{2}\kappa R^{\alpha\beta}\wedge\eta_{\alpha\beta},$	$L_{\mathrm{EC}}:=-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta},$	$L_{\mathrm{EC}}:=-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta},$
mielke-geometrodynamics_112 EQ0325 (5.3.2)	112	■ 1.000 ● K +0	$G_{\alpha}:=\frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma}\wedge\eta_{\alpha\beta},$	$G_{\alpha}:=\frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma}=\kappa\Sigma_{\alpha},$	$G_{\alpha}:=\frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma}=\kappa\Sigma_{\alpha},$
mielke-geometrodynamics_112 EQ0327 (5.3.4)	112	■ 1.000 ● N +0	$D^{\{\}\{\}}G_{\alpha}^{\{\}\{\}}\equiv 0$	$D^{\{\}}G_{\alpha}^{\{\}}\equiv 0$	$D^{\{\}}G_{\alpha}^{\{\}}\equiv 0$
mielke-geometrodynamics_112 EQ0331 (5.3.8)	112	■ 1.000 ● N +0	$D[\gamma,\Theta]\equiv 2\,i\left[\sigma,\Omega^g\right],\quad D\left[\gamma,\Omega^g\right]\equiv D[\Theta,\Omega^g],$	$D[\gamma,\Theta]\equiv 2i\left[\sigma,\Omega^g\right],\quad D\left[\gamma,\Omega^g\right]\equiv [\Theta,\Omega^g],$	$D[\gamma,\Theta]\equiv 2i[\sigma,\Omega^g],\quad D[\gamma,\Omega^g]\equiv [\Theta,\Omega^g],$
mielke-geometrodynamics_113 EQ0332 (5.3.9)	113	■ 1.000 ● N +0	$D\,G\equiv i\left[\Theta,\gamma_5\Omega^g\right]$	$DG\equiv i\left[\Theta,\gamma_5\Omega^g\right]$	$DG\equiv i[\Theta,\gamma_5\Omega^g]$
mielke-geometrodynamics_113 EQ0333 (5.3.10)	113	■ 1.000 ● K +0	$\eta_{\alpha\beta\gamma}\wedge T^{\gamma}=\kappa\tau_{\alpha\beta}$	$\eta_{\alpha\beta\gamma}\wedge T^{\gamma}=2\kappa\tau_{\alpha\beta}$	$\eta_{\alpha\beta\gamma}\wedge T^{\gamma}=2\kappa\tau_{\alpha\beta}$
mielke-geometrodynamics_113 EQ0334 (5.3.11)	113	■ 1.000 ● N +0	$\mathrm{T}_{\mu\nu}:=\frac{2}{\sqrt{g}}\frac{\delta L}{\delta g_{\mu\nu}}$	$T_{\mu\nu}:=\frac{2}{\sqrt{g}}\frac{\delta L}{\delta g_{\mu\nu}}$	$T_{\mu\nu}:=\frac{2}{\sqrt{g}}\frac{\delta L}{\delta g_{\mu\nu}}$
mielke-geometrodynamics_114 EQ0338 (5.3.15)	114	■ 1.000 ● N +0	$\{ \}^{\{3\}}T_{\alpha}=\frac{(-1)^s}{3}\ast\left(\vartheta_{\alpha}\wedge\mathscr{A}\right)$	$^{(3)}T_{\alpha}=\frac{(-1)^s}{3}\ast(\vartheta_{\alpha}\wedge\mathscr{A})$	$^{(3)}T_{\alpha}=\frac{(-1)^s}{3}\ast(\vartheta_{\alpha}\wedge\mathscr{A})$
mielke-geometrodynamics_115 EQ0342 (5.3.19)	115	■ 1.000 ● N -1	$\Delta\widetilde{L}:=\kappa\left(L_{\mathrm{EC}}-L_{\mathrm{HE}}\right)\simeq\frac{\kappa^2}{48}j_5\wedge\ast j_5=4f^2j_5\wedge\ast j_5,$	$\Delta\widetilde{L}:=\kappa(L_{\mathrm{EC}}-L_{\mathrm{HE}})\simeq\frac{\kappa^2}{48}j_5\wedge\ast j_5=4f^2j_5\wedge\ast j_5,$	$\Delta\widetilde{L}:=\kappa(L_{\mathrm{EC}}-L_{\mathrm{HE}})\simeq\frac{\kappa^2}{48}j_5\wedge\ast j_5=4f^2j_5\wedge\ast j_5,$
mielke-geometrodynamics_116 EQ0344 (5.4.2)	116	■ 1.000 ● N -1	$L_{\mathrm{HE}}:=L_{\mathrm{HE}}+\frac{\Lambda}{\kappa}\eta$	$L_{\mathrm{HE}}:=L_{\mathrm{HE}}+\frac{\Lambda}{\kappa}\eta$	$L_{\mathrm{HE}}:=L_{\mathrm{HE}}+\frac{\Lambda}{\kappa}\eta$
mielke-geometrodynamics_116 EQ0345 (5.4.3)	116	■ 1.000 ● N +0	$g_{\mathrm{N}}:=\kappa k^2,\quad\lambda:=\Lambda/k^2,$	$g_{\mathrm{N}}:=\kappa k^2,\quad\lambda:=\Lambda/k^2,$	$g_{\mathrm{N}}:=\kappa k^2,\quad\lambda:=\Lambda/k^2,$
mielke-geometrodynamics_116 EQ0346 (5.4.4)	116	■ 1.000 ● N +0	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}},\quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}},\quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}},\quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$
mielke-geometrodynamics_117 EQ0348 (5.4.6)	117	■ 1.000 ● N +0	$f^2=\frac{1}{192}\kappa^2=3\times 2^{-10}\left(\frac{\mu}{\Lambda}\right)^2=2^{-8}\frac{\mu}{\rho_{\Lambda}}$	$f^2=\frac{1}{192}\kappa^2=3\times 2^{-10}\left(\frac{\mu}{\Lambda}\right)^2=2^{-8}\frac{\mu}{\rho_{\Lambda}}$	$f^2=\frac{1}{192}\kappa^2=3\times 2^{-10}\left(\frac{\mu}{\Lambda}\right)^2=2^{-8}\frac{\mu}{\rho_{\Lambda}}$
mielke-geometrodynamics_123 EQ0350 (6.2.1)	123	■ 1.000 ● N +0	$C_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)=-\frac{(-1)^{\mathrm{sig}}}{2\ell^2}\ast\mathscr{A},$	$C_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)=-\frac{(-1)^{\mathrm{sig}}}{2\ell^2}\ast\mathscr{A},$	$C_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)=-\frac{(-1)^{\mathrm{sig}}}{2\ell^2}\ast\mathscr{A},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_124 EQ0351 (6.2.2)		■ 1.000 ● W -1	$\begin{aligned} C_{\mathrm{RR}} &:= -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right) \\ &= -\frac{1}{2} \left(\Gamma_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} + \frac{1}{3} \Gamma_{\alpha}{}^{\beta} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right). \end{aligned}$	$C_{\mathrm{RR}} := -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right) \\ = -\frac{1}{2} \left(\Gamma_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} + \frac{1}{3} \Gamma_{\alpha}{}^{\beta} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right).$	$C_{\mathrm{RR}} := -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right) \\ = -\frac{1}{2} \left(\Gamma_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} + \frac{1}{3} \Gamma_{\alpha}{}^{\beta} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right).$
mielke-geometrodynamics_124 EQ0352 (6.2.3)		■ 1.000 ● N -1	$\hat{C}=C_{\mathrm{RR}}-2\,C_{\mathrm{TT}}$	$\hat{C}=C_{\mathrm{RR}}-2C_{\mathrm{TT}}$	$\hat{C}=C_{\mathrm{RR}}-2C_{\mathrm{TT}}$
mielke-geometrodynamics_124 EQ0353 (6.2.4)		■ 1.000 ● N -1	$d\,C_{\mathrm{TT}}\equiv\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$	$dC_{\mathrm{TT}}\equiv\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$	$dC_{\mathrm{TT}}\equiv\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$
mielke-geometrodynamics_124 EQ0354 (6.2.5)		■ 1.000 ● N -1	$d\,C_{\mathrm{RR}}=-\operatorname{Tr}(\Omega\wedge\Omega)=\frac{1}{2}R^{\alpha\beta}\wedge R_{\alpha\beta}$	$dC_{\mathrm{RR}}=-\operatorname{Tr}(\Omega\wedge\Omega)=\frac{1}{2}R^{\alpha\beta}\wedge R_{\alpha\beta}$	$dC_{\mathrm{RR}}=-\operatorname{Tr}(\Omega\wedge\Omega)=\frac{1}{2}R^{\alpha\beta}\wedge R_{\alpha\beta}$
mielke-geometrodynamics_124 EQ0355 (6.2.6)		■ 1.000 ● N +0	$D\,R_{\alpha\beta}\equiv0,$	$DR_{\alpha\beta}\equiv0,$	$DR_{\alpha\beta}\equiv0,$
mielke-geometrodynamics_125 EQ0359 (6.2.10)		■ 1.000 ● N +0	$H_{\alpha}^{\parallel}:=\partial L_{\parallel}/\partial T^{\alpha}=(1/\kappa)\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}$	$H_{\alpha}^{\parallel}:=\partial L_{\parallel}/\partial T^{\alpha}=(1/\kappa)\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}$	$H_{\alpha}^{\parallel}:=\partial L_{\parallel}/\partial T^{\alpha}=(1/\kappa)\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}$
mielke-geometrodynamics_126 EQ0361 (6.2.12)		■ 1.000 ● N +0	$\tilde{L}_{\parallel}=L_{\parallel}-R^{\alpha\beta}\wedge\lambda_{\alpha\beta}.$	$\tilde{L}_{\parallel}=L_{\parallel}-R^{\alpha\beta}\wedge\lambda_{\alpha\beta}.$	$\tilde{L}_{\parallel}=L_{\parallel}-R^{\alpha\beta}\wedge\lambda_{\alpha\beta}.$
mielke-geometrodynamics_126 EQ0362 (6.2.14)		■ 1.000 ● W +0	$\begin{gathered} D\,H_{\alpha}^{\parallel}-E_{\alpha}^{\parallel}=\Sigma_{\alpha}, \\ D\lambda_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]}^{\parallel}=\tau_{\alpha\beta}, \end{gathered}$	$\begin{gathered} DH_{\alpha}^{\parallel}-E_{\alpha}^{\parallel}=\Sigma_{\alpha}, \\ D\lambda_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]}^{\parallel}=\tau_{\alpha\beta}, \end{gathered}$	$\begin{gathered} DH_{\alpha}^{\parallel}-E_{\alpha}^{\parallel}=\Sigma_{\alpha}, \\ D\lambda_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]}^{\parallel}=\tau_{\alpha\beta}, \end{gathered}$
mielke-geometrodynamics_126 EQ0363 (6.2.15)		■ 1.000 ● N +1	$R^{\alpha\beta}=0.$	$R^{\alpha\beta}=0.$	$R^{\alpha\beta}=0.$
mielke-geometrodynamics_126 EQ0365 (6.2.17)		■ 1.000 ● N +0	$D\left(\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge H_{\beta]}^{\parallel}\right)=0$	$D\left(\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge H_{\beta]}^{\parallel}\right)=0$	$D(\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge H_{\beta]}^{\parallel})=0$
mielke-geometrodynamics_127 EQ0369 (6.2.20)		■ 1.000 ● N +1	$\stackrel{(\pm)}{C}_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge\stackrel{(\pm)}{T}_{\alpha}\right),$	$\stackrel{(\pm)}{C}_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge\stackrel{(\pm)}{T}_{\alpha}\right),$	$\stackrel{(\pm)}{C}_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge\stackrel{(\pm)}{T}_{\alpha}\right),$
mielke-geometrodynamics_127 EQ0373 (6.2.24)		■ 1.000 ● N +0	$\stackrel{(\pm)(\pm)}{D}\Pi_{\alpha}=\Sigma_{\alpha}.$	$\stackrel{(\pm)(\pm)}{D}\Pi_{\alpha}=\Sigma_{\alpha}.$	$\stackrel{(\pm)(\pm)}{D}\Pi_{\alpha}=\Sigma_{\alpha}.$
mielke-geometrodynamics_128 EQ0379 (6.2.30)		■ 1.000 ● K +0	$\stackrel{(\pm)}{L}_{\mathrm{YM}}=\mp(1/4)\operatorname{Tr}\left(\stackrel{(\pm)}{F}\wedge\stackrel{(\pm)}{F}\right)$	$\stackrel{(\pm)}{L}_{\mathrm{YM}}=\mp(1/4)\operatorname{Tr}\left(\stackrel{(\pm)}{F}\wedge\stackrel{(\pm)}{F}\right)$	$\stackrel{(\pm)}{L}_{\mathrm{YM}}=\mp(1/4)\operatorname{Tr}\left(\stackrel{(\pm)}{F}\wedge\stackrel{(\pm)}{F}\right)$
mielke-geometrodynamics_129 EQ0383 (6.3.1)		■ 1.000 ● N +0	$\check{E}_{\alpha}:=E_{\alpha}+\Gamma_{\alpha}{}^{\beta}\wedge H_{\beta}+\Sigma_{\alpha}\cong dH_{\alpha},$	$\check{E}_{\alpha}:=E_{\alpha}+\Gamma_{\alpha}{}^{\beta}\wedge H_{\beta}+\Sigma_{\alpha}\cong dH_{\alpha},$	$\check{E}_{\alpha}:=E_{\alpha}+\Gamma_{\alpha}{}^{\beta}\wedge H_{\beta}+\Sigma_{\alpha}\cong dH_{\alpha},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_130 EQ0386 (6.4.1)	130	■ 1.000 ● N +0	$n:=\partial_t-N^A\partial/\partial x^A,\quad A=1,2,3,$	$n:=\partial_t-N^A\partial/\partial x^A,\quad A=1,2,3,$	$n:=\partial_t-N^A\partial/\partial x^A,\quad A=1,2,3,$
mielke-geometrodynamics_132 EQ0395 (6.4.10)	132	■ 1.000 ● W +0	$\Sigma_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}},\quad \tau_{\alpha\beta}:=\frac{\partial L}{\partial \Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}},\quad \tau_{\alpha\beta}:=\frac{\partial L}{\partial \Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}},\quad \tau_{\alpha\beta}:=\frac{\partial L}{\partial \Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial D\Psi}$
mielke-geometrodynamics_136 EQ0408 (6.4.28)	136	■ 1.000 ● N +0	$d\,s^2=h^2\,dr^2+f^2\left[d\,\psi^2+\sin^2\psi\left(d\theta^2+\sin^2\theta d\phi^2\right)\right],$	$ds^2=h^2dr^2+f^2\left[d\psi^2+\sin^2\psi\left(d\theta^2+\sin^2\theta d\phi^2\right)\right],$	$ds^2=h^2dr^2+f^2\left[d\psi^2+\sin^2\psi\left(d\theta^2+\sin^2\theta d\phi^2\right)\right],$
mielke-geometrodynamics_136 EQ0410 (6.4.30)	136	■ 1.000 ● N -1	$\ell^2\underline{C}_{\mathrm{TT}}=\frac{1}{2}\underline{\vartheta}^{\alpha}\wedge d\underline{\vartheta}_{\alpha}=\frac{1}{2}\underline{\vartheta}^A\wedge d\underline{\vartheta}_A=3\underline{\vartheta}^{\hat{1}}\wedge\underline{\vartheta}^{\hat{2}}\wedge\underline{\vartheta}^{\hat{3}}$	$\ell^2\underline{C}_{\mathrm{TT}}=\frac{1}{2}\underline{\vartheta}^{\alpha}\wedge d\underline{\vartheta}_{\alpha}=\frac{1}{2}\underline{\vartheta}^A\wedge d\underline{\vartheta}_A=3\underline{\vartheta}^{\hat{1}}\wedge\underline{\vartheta}^{\hat{2}}\wedge\underline{\vartheta}^{\hat{3}}$	$\ell^2\underline{C}_{\mathrm{TT}}=\frac{1}{2}\underline{\vartheta}^{\alpha}\wedge d\underline{\vartheta}_{\alpha}=\frac{1}{2}\underline{\vartheta}^A\wedge d\underline{\vartheta}_A=3\underline{\vartheta}^{\hat{1}}\wedge\underline{\vartheta}^{\hat{2}}\wedge\underline{\vartheta}^{\hat{3}}$
mielke-geometrodynamics_136 EQ0412 (6.4.32)	136	■ 1.000 ● N +1	$f=\frac{a\,r^2}{r^2+c^2}$	$f=\frac{ar^2}{r^2+c^2}$	$f=\frac{ar^2}{r^2+c^2}$
mielke-geometrodynamics_137 EQ0414 (6.5.2)	137	■ 1.000 ● N +0	$\Psi(\gamma)=\int_{\mathscr{C}/\mathscr{G}}\mathscr{D}A\,W(A,\gamma)\Psi(A).$	$\Psi(\gamma)=\int_{\mathscr{C}/\mathscr{G}}\mathscr{D}A\,W(A,\gamma)\Psi(A).$	$\Psi(\gamma)=\int_{\mathscr{C}/\mathscr{G}}\mathscr{D}A\,W(A,\gamma)\Psi(A).$
mielke-geometrodynamics_137 EQ0416 (6.5.4)	137	■ 1.000 ● K +0	$\langle 0 W(\gamma) 0\rangle e^{\Lambda G(\gamma)}V(\gamma),$	$\langle 0 W(\gamma) 0\rangle e^{\Lambda G(\gamma)}V(\gamma),$	$\langle 0 W(\gamma) 0\rangle e^{\Lambda G(\gamma)}V(\gamma),$
mielke-geometrodynamics_138 EQ0417 (6.6.1)	138	■ 1.000 ● N +1	$\int T=\int d\,\sigma=2\pi\,N$	$\int T=\int d\sigma=2\pi N$	$\int T=\int d\sigma=2\pi N,$
mielke-geometrodynamics_139 EQ0418 (6.6.2)	139	■ 1.000 ● K +0	$W^{\{N\}}:=S^{\{3\}}\#^N(S^1\times S^2)\,.$	$W^N:=S^3\#^N(S^1\times S^2)\,.$	$W^N:=S^3\#^N(S^1\times S^2).$
mielke-geometrodynamics_140 EQ0419 (6.7.1)	140	■ 1.000 ● K +0	$L:=L_{\mathrm{EC}}+L_{\theta}+L_{\mathrm{D}},$	$L:=L_{\mathrm{EC}}+L_{\theta}+L_{\mathrm{D}},$	$L:=L_{\mathrm{EC}}+L_{\theta}+L_{\mathrm{D}},$
mielke-geometrodynamics_140 EQ0420 (6.7.2)	140	■ 1.000 ● N +1	$L_{\theta}=\theta_{\mathrm{T}}dC_{\mathrm{TT}}+\theta_{\mathrm{L}}dC_{\mathrm{RR}}$	$L_{\theta}=\theta_{\mathrm{T}}dC_{\mathrm{TT}}+\theta_{\mathrm{L}}dC_{\mathrm{RR}}$	$L_{\theta}=\theta_{\mathrm{T}}dC_{\mathrm{TT}}+\theta_{\mathrm{L}}dC_{\mathrm{RR}}$
mielke-geometrodynamics_140 EQ0421 (6.7.3)	140	■ 1.000 ● K +0	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}}=-\frac{\theta_{\mathrm{T}}}{\ell^2}T_{\alpha},$	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}}=-\frac{\theta_{\mathrm{T}}}{\ell^2}T_{\alpha},$	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}}=-\frac{\theta_{\mathrm{T}}}{\ell^2}T_{\alpha},$
mielke-geometrodynamics_140 EQ0422 (6.7.4)	140	■ 1.000 ● N +0	$H_{\alpha\beta}:= -\frac{\partial L}{\partial R^{\alpha\beta}}=\frac{1}{2\kappa}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta}-\theta_{\mathrm{L}}R_{\alpha\beta}$	$H_{\alpha\beta}:= -\frac{\partial L}{\partial R^{\alpha\beta}}=\frac{1}{2\kappa}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta}-\theta_{\mathrm{L}}R_{\alpha\beta}$	$H_{\alpha\beta}:= -\frac{\partial L}{\partial R^{\alpha\beta}}=\frac{1}{2\kappa}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta}-\theta_{\mathrm{L}}R_{\alpha\beta}$
mielke-geometrodynamics_140 EQ0423 (6.7.5)	140	■ 1.000 ● N +0	$D\theta_{\mathrm{T}}\wedge T_{\alpha}+\theta_{\mathrm{T}}DT_{\alpha}+\ell^2E_{\alpha}=-\ell^2\Sigma_{\alpha}.$	$D\theta_{\mathrm{T}}\wedge T_{\alpha}+\theta_{\mathrm{T}}DT_{\alpha}+\ell^2E_{\alpha}=-\ell^2\Sigma_{\alpha}.$	$D\theta_{\mathrm{T}}\wedge T_{\alpha}+\theta_{\mathrm{T}}DT_{\alpha}+\ell^2E_{\alpha}=-\ell^2\Sigma_{\alpha}.$
mielke-geometrodynamics_140 EQ0425 (6.7.7)	140	■ 1.000 ● N +0	$C_{\mathrm{TT}}\cong\frac{\kappa}{4\ell^2}j_5$	$C_{\mathrm{TT}}\cong\frac{\kappa}{4\ell^2}j_5$	$C_{\mathrm{TT}}\cong\frac{\kappa}{4\ell^2}j_5$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_141 EQ0427 (6.7.9)	141	1.000 N +0	$T^{\gamma}\wedge\eta_{\gamma\beta}+\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=0$.	$T^{\gamma}\wedge\eta_{\gamma\beta}+\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=0$.	$T^{\gamma}\wedge\eta_{\gamma\beta}+\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=0$.
mielke-geometrodynamics_141 EQ0429 (6.7.11)	141	1.000 N +0	$\kappa d\theta_{\mathrm{L}}\wedge T_{\beta}=-\eta_{\beta}$	$\kappa d\theta_{\mathrm{L}}\wedge T_{\beta}=-\eta_{\beta}$	$\kappa d\theta_{\mathrm{L}}\wedge T_{\beta}=-\eta_{\beta}$
mielke-geometrodynamics_141 EQ0430 (6.7.12)	141	1.000 N -1	$\kappa\ell^2 d\theta_{\mathrm{L}}\wedge C_{\mathrm{TT}}=2\eta$.	$\kappa\ell^2 d\theta_{\mathrm{L}}\wedge C_{\mathrm{TT}}=2\eta$.	$\kappa\ell^2 d\theta_{\mathrm{L}}\wedge C_{\mathrm{TT}}=2\eta$.
mielke-geometrodynamics_141 EQ0431 (6.7.13)	141	1.000 N +0	$D\,T^{\gamma}\wedge\eta_{\alpha\beta\gamma}+\kappa D\theta_{\mathrm{T}}\wedge T_{[\alpha}\wedge\vartheta_{\beta]}/\ell^2=2\kappa\left(T_{[\alpha}\wedge\mu_{\beta]}-\vartheta_{[\alpha}\wedge D\mu_{\beta]}\right)$.	$DT^{\gamma}\wedge\eta_{\alpha\beta\gamma}+\kappa D\theta_{\mathrm{T}}\wedge T_{[\alpha}\wedge\vartheta_{\beta]}/\ell^2=2\kappa\left(T_{[\alpha}\wedge\mu_{\beta]}-\vartheta_{[\alpha}\wedge D\mu_{\beta]}\right)$.	$DT^{\gamma}\wedge\eta_{\alpha\beta\gamma}+\kappa D\theta_{\mathrm{T}}\wedge T_{[\alpha}\wedge\vartheta_{\beta]}/\ell^2=2\kappa\left(T_{[\alpha}\wedge\mu_{\beta]}-\vartheta_{[\alpha}\wedge D\mu_{\beta]}\right)$.
mielke-geometrodynamics_142 EQ0432 (6.8.1)	142	1.000 N +0	$W=\int\left[L\left(\vartheta^{\alpha},T^{\alpha},R^{\alpha\beta},\Psi,D\Psi\right)+L_g\left(\vartheta^{\alpha},T^{\alpha},R^{\alpha\beta}\right)\right]$.	$W=\int\left[L\left(\vartheta^{\alpha},T^{\alpha},R^{\alpha\beta},\Psi,D\Psi\right)+L_g\left(\vartheta^{\alpha},T^{\alpha},R^{\alpha\beta}\right)\right]$.	$W=\int\left[L\left(\vartheta^{\alpha},T^{\alpha},R^{\alpha\beta},\Psi,D\Psi\right)+L_g\left(\vartheta^{\alpha},T^{\alpha},R^{\alpha\beta}\right)\right]$.
mielke-geometrodynamics_142 EQ0433 (6.8.3)	142	1.000 C -3	$\begin{gathered} \frac{\delta L}{\delta\Psi}=\frac{\partial L}{\partial\Psi}-(-1)^pD\frac{\partial L}{\partial D\Psi}=0, \quad (\text{MATTER}) \\ DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha}, \quad (\text{FIRST}) \\ DH_{\alpha\beta}-E_{\alpha\beta}=\tau_{\alpha\beta}. \quad (\text{SECOND}) \end{gathered}$	$\frac{\delta L}{\delta\Psi}=\frac{\partial L}{\partial\Psi}-(-1)^pD\frac{\partial L}{\partial D\Psi}=0, \quad (\text{MATTER}) \\ DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha}, \quad (\text{FIRST}) \\ DH_{\alpha\beta}-E_{\alpha\beta}=\tau_{\alpha\beta}. \quad (\text{SECOND})$	$\frac{\delta L}{\delta\Psi}=\frac{\partial L}{\partial\Psi}-(-1)^pD\frac{\partial L}{\partial D\Psi}=0, \quad (\text{MATTER}) \\ DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha}, \quad (\text{FIRST}) \\ DH_{\alpha\beta}-E_{\alpha\beta}=\tau_{\alpha\beta}. \quad (\text{SECOND})$
mielke-geometrodynamics_142 EQ0435 (6.8.6)	142	1.000 N +1	$\Sigma_{\alpha}:=\frac{\delta L}{\delta\vartheta^{\alpha}}=\frac{\partial L}{\partial\vartheta^{\alpha}}+D\frac{\partial L}{\partial T^{\alpha}}$	$\Sigma_{\alpha}:=\frac{\delta L}{\delta\vartheta^{\alpha}}=\frac{\partial L}{\partial\vartheta^{\alpha}}+D\frac{\partial L}{\partial T^{\alpha}}$	$\Sigma_{\alpha}:=\frac{\delta L}{\delta\vartheta^{\alpha}}=\frac{\partial L}{\partial\vartheta^{\alpha}}+D\frac{\partial L}{\partial T^{\alpha}}$
mielke-geometrodynamics_142 EQ0436 (6.8.7)	142	1.000 W +0	$\tau_{\alpha\beta}:=\frac{\delta L}{\delta\Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial(D\Psi)}+\vartheta_{[\alpha}\wedge\frac{\partial L}{\partial T^{\beta]}}+D\frac{\partial L}{\partial R^{\alpha\beta}},$	$\tau_{\alpha\beta}:=\frac{\delta L}{\delta\Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial(D\Psi)}+\vartheta_{[\alpha}\wedge\frac{\partial L}{\partial T^{\beta]}}+D\frac{\partial L}{\partial R^{\alpha\beta}},$	$\tau_{\alpha\beta}:=\frac{\delta L}{\delta\Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial(D\Psi)}+\vartheta_{[\alpha}\wedge\frac{\partial L}{\partial T^{\beta]}}+D\frac{\partial L}{\partial R^{\alpha\beta}},$
mielke-geometrodynamics_142 EQ0438 (6.8.9)	142	1.000 N +0	$E_{\alpha\beta}:=-\vartheta_{[\alpha}\wedge H_{\beta]}$	$E_{\alpha\beta}:=-\vartheta_{[\alpha}\wedge H_{\beta]}$	$E_{\alpha\beta}:=-\vartheta_{[\alpha}\wedge H_{\beta]}$
mielke-geometrodynamics_143 EQ0439 (6.8.10)	143	1.000 N +0	$\vartheta^{\alpha}\wedge E_{\alpha}=4L_g+2T^{\alpha}\wedge H_{\alpha}+2R^{\alpha\beta}\wedge H_{\alpha\beta}$.	$\vartheta^{\alpha}\wedge E_{\alpha}=4L_g+2T^{\alpha}\wedge H_{\alpha}+2R^{\alpha\beta}\wedge H_{\alpha\beta}$.	$\vartheta^{\alpha}\wedge E_{\alpha}=4L_g+2T^{\alpha}\wedge H_{\alpha}+2R^{\alpha\beta}\wedge H_{\alpha\beta}$.
mielke-geometrodynamics_143 EQ0442 (6.8.13)	143	1.000 N +0	$L_{\mathrm{qPG}}=-\frac{1}{2}T^{\alpha}\wedge H_{\alpha}-\frac{1}{2}R^{\alpha\beta}\wedge\left(H_{\alpha\beta}-\frac{a_0}{2\ell^2}\eta_{\alpha\beta}\right),$	$L_{\mathrm{qPG}}=-\frac{1}{2}T^{\alpha}\wedge H_{\alpha}-\frac{1}{2}R^{\alpha\beta}\wedge\left(H_{\alpha\beta}-\frac{a_0}{2\ell^2}\eta_{\alpha\beta}\right),$	$L_{\mathrm{qPG}}=-\frac{1}{2}T^{\alpha}\wedge H_{\alpha}-\frac{1}{2}R^{\alpha\beta}\wedge\left(H_{\alpha\beta}-\frac{a_0}{2\ell^2}\eta_{\alpha\beta}\right),$
mielke-geometrodynamics_143 EQ0444 (6.8.15)	143	1.000 N +0	$H_{\alpha\beta}=-\frac{a_0}{2\ell^2}\eta_{\alpha\beta}-\frac{1}{\kappa}\ast\left(\sum_{j=1}^6b_j^{(j)}R_{\alpha\beta}\right).$	$H_{\alpha\beta}=-\frac{a_0}{2\ell^2}\eta_{\alpha\beta}-\frac{1}{\kappa}\ast\left(\sum_{j=1}^6b_j^{(j)}R_{\alpha\beta}\right).$	$H_{\alpha\beta}=-\frac{a_0}{2\ell^2}\eta_{\alpha\beta}-\frac{1}{\kappa}\ast\left(\sum_{j=1}^6b_j^{(j)}R_{\alpha\beta}\right).$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

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mielke-geometrodynamics_161 EQ0491 (7.7.9)		■ 1.000 ● K +0	$\widetilde{L} \rightarrow L = -\frac{1}{2} \left(R^{\alpha\beta} \wedge \widetilde{H}_{\alpha\beta} - \widetilde{L} \right)$	$\widetilde{L} \rightarrow L = -\frac{1}{2} \left(R^{\alpha\beta} \wedge \widetilde{H}_{\alpha\beta} - \widetilde{L} \right)$	$\widetilde{L} \rightarrow L = -\frac{1}{2} \left(R^{\alpha\beta} \wedge \widetilde{H}_{\alpha\beta} - \widetilde{L} \right)$
mielke-geometrodynamics_161 EQ0493 (7.7.11)		■ 1.000 ● N +0	$\widetilde{H}_{\alpha\beta\mu\nu} := \frac{\partial^2 \widetilde{L}}{\partial R^{\mu\nu} \partial R^{\alpha\beta}} = -\frac{\partial \widetilde{H}_{\mu\nu}}{\partial R^{\alpha\beta}}$	$\widetilde{H}_{\alpha\beta\mu\nu} := \frac{\partial^2 \widetilde{L}}{\partial R^{\mu\nu} \partial R^{\alpha\beta}} = -\frac{\partial \widetilde{H}_{\mu\nu}}{\partial R^{\alpha\beta}}$	$\widetilde{H}_{\alpha\beta\mu\nu} := \frac{\partial^2 \widetilde{L}}{\partial R^{\mu\nu} \partial R^{\alpha\beta}} = -\frac{\partial \widetilde{H}_{\mu\nu}}{\partial R^{\alpha\beta}}$
mielke-geometrodynamics_161 EQ0494 (7.7.12)		■ 1.000 ● N -1	$\widetilde{\eta}_{\alpha\beta} := \frac{\theta_{\mathrm{T}}}{2} \eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2} \vartheta_{\alpha} \wedge \vartheta_{\beta} - \ell^2 \widetilde{H}_{\alpha\beta} = 0$	$\widetilde{\eta}_{\alpha\beta} := \frac{\theta_{\mathrm{T}}^*}{2} \eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2} \vartheta_{\alpha} \wedge \vartheta_{\beta} - \ell^2 \widetilde{H}_{\alpha\beta} = 0$	$\widetilde{\eta}_{\alpha\beta} := \frac{\theta_{\mathrm{T}}^*}{2} \eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2} \vartheta_{\alpha} \wedge \vartheta_{\beta} - \ell^2 \widetilde{H}_{\alpha\beta} = 0$
mielke-geometrodynamics_162 EQ0496 (7.7.14)		■ 1.000 ● K +0	$\eta := \eta_{\alpha\beta\gamma\delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta} / 4!$	$\eta := \eta_{\alpha\beta\gamma\delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta} / 4!$	$\eta := \eta_{\alpha\beta\gamma\delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta} / 4!$
mielke-geometrodynamics_163 EQ0499 (7.7.17)		■ 1.000 ● K +0	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta} \mathbf{1}_4.$	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta} \mathbf{1}_4.$	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta} \mathbf{1}_4.$
mielke-geometrodynamics_163 EQ0501 (7.7.19)		■ 1.000 ● K +0	$\gamma := \gamma^{\alpha} \vartheta_{\alpha}$	$\gamma := \gamma^{\alpha} \vartheta_{\alpha}$	$\gamma := \gamma^{\alpha} \vartheta_{\alpha}$
mielke-geometrodynamics_163 EQ0502 (7.7.20)		■ 1.000 ● N +0	${}^* \gamma = \gamma^{\alpha} \eta_{\alpha} = \frac{i}{6} \gamma_5 \gamma \wedge \gamma \wedge \gamma,$	${}^* \gamma = \gamma^{\alpha} \eta_{\alpha} = \frac{i}{6} \gamma_5 \gamma \wedge \gamma \wedge \gamma,$	${}^* \gamma = \gamma^{\alpha} \eta_{\alpha} = \frac{i}{6} \gamma_5 \gamma \wedge \gamma \wedge \gamma,$
mielke-geometrodynamics_164 EQ0503 (7.7.21)		■ 1.000 ● N +1	$\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta} = \frac{i}{2} \gamma \wedge \gamma, \quad {}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma.$	$\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta} = \frac{i}{2} \gamma \wedge \gamma, \quad {}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma.$	$\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta} = \frac{i}{2} \gamma \wedge \gamma, \quad {}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma.$
mielke-geometrodynamics_164 EQ0504 (7.7.22)		■ 1.000 ● N +0	$\sigma^{(*)} := i \gamma_5 \sigma, \quad \sigma^{(**)} = i^2 \gamma_5^2 \sigma = -\sigma.$	$\sigma^{(*)} := i \gamma_5 \sigma, \quad \sigma^{(**)} = i^2 \gamma_5^2 \sigma = -\sigma.$	$\sigma^{(*)} := i \gamma_5 \sigma, \quad \sigma^{(**)} = i^2 \gamma_5^2 \sigma = -\sigma.$
mielke-geometrodynamics_164 EQ0505 (7.7.23)		■ 1.000 ● N +0	$[\gamma, \sigma^{(*)}] = \gamma \wedge i \gamma_5 \sigma - i \gamma_5 \sigma \wedge \gamma = -i \gamma_5 (\gamma \wedge \sigma + \sigma \wedge \gamma) = 2 \gamma_5 \gamma \wedge \gamma \wedge \gamma$	$[\gamma, \sigma^{(*)}] = \gamma \wedge i \gamma_5 \sigma - i \gamma_5 \sigma \wedge \gamma = -i \gamma_5 (\gamma \wedge \sigma + \sigma \wedge \gamma) = 2 \gamma_5 \gamma \wedge \gamma \wedge \gamma$	$[\gamma, \sigma^{(*)}] = \gamma \wedge i \gamma_5 \sigma - i \gamma_5 \sigma \wedge \gamma = -i \gamma_5 (\gamma \wedge \sigma + \sigma \wedge \gamma) = 2 \gamma_5 \gamma \wedge \gamma \wedge \gamma$
mielke-geometrodynamics_165 EQ0508 (7.7.26)		■ 1.000 ● N +0	$\gamma \rightarrow \gamma^{\theta} = e^{i \gamma^5 \theta} \gamma e^{-i \gamma^5 \theta}$	$\gamma \rightarrow \gamma^{\theta} = e^{i \gamma^5 \theta} \gamma e^{-i \gamma^5 \theta}$	$\gamma \rightarrow \gamma^{\theta} = e^{i \gamma^5 \theta} \gamma e^{-i \gamma^5 \theta}$
mielke-geometrodynamics_165 EQ0510 (7.7.28)		■ 1.000 ● K +0	$\gamma := \vartheta^{\alpha} \gamma_{\alpha}, \quad \Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta} = \Gamma^{\{\}} - K,$	$\gamma := \vartheta^{\alpha} \gamma_{\alpha}, \quad \Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta} = \Gamma^{\{\}} - K,$	$\gamma := \vartheta^{\alpha} \gamma_{\alpha}, \quad \Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta} = \Gamma^{\{\}} - K,$
mielke-geometrodynamics_166 EQ0513 (7.7.31)		■ 1.000 ● N +0	$\Omega^{(*)} = \frac{1}{4} R_{\alpha\beta}^{(*)} \sigma^{\alpha\beta}, \quad R_{\alpha\beta}^{(*)} := \frac{1}{2} \eta_{\alpha\beta\gamma\delta} R^{\gamma\delta}.$	$\Omega^{(*)} = \frac{1}{4} R_{\alpha\beta}^{(*)} \sigma^{\alpha\beta}, \quad R_{\alpha\beta}^{(*)} := \frac{1}{2} \eta_{\alpha\beta\gamma\delta} R^{\gamma\delta}.$	$\Omega^{(*)} = \frac{1}{4} R_{\alpha\beta}^{(*)} \sigma^{\alpha\beta}, \quad R_{\alpha\beta}^{(*)} := \frac{1}{2} \eta_{\alpha\beta\gamma\delta} R^{\gamma\delta}.$
mielke-geometrodynamics_166 EQ0514 (7.7.32)		■ 1.000 ● K +0	$D \sigma = \frac{i}{2} [\Theta, \gamma]$	$D \sigma = \frac{i}{2} [\Theta, \gamma]$	$D \sigma = \frac{i}{2} [\Theta, \gamma]$
mielke-geometrodynamics_166 EQ0517 (7.7.35)		■ 1.000 ● N +0	$d^* \gamma = 0, \quad d^* \Gamma = 0$	$d^* \gamma = 0, \quad d^* \Gamma = 0$	$d^* \gamma = 0, \quad d^* \Gamma = 0$

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mielke-geometrodynamics_166 EQ0518 (7.7.36)	166	■ 1.000 ● N +0	$D \ D \ \Phi=\left[\Omega^g, \ \Phi\right],$	$DD\Phi = [\Omega^g, \Phi],$	$DD\Phi = [\Omega^g, \Phi],$
mielke-geometrodynamics_166 EQ0520 (7.7.38)	166	■ 1.000 ● C +3	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$	$G := G_{\alpha}\gamma^{\alpha} := \frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda} = -i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g) = i[\gamma,\gamma_5\Omega^g].$	$G := G_{\alpha}\gamma^{\alpha} := \frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda} = -i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g) = i[\gamma,\gamma_5\Omega^g].$
mielke-geometrodynamics_167 EQ0521 (7.7.39)	167	■ 1.000 ● W +0	$D[\gamma, \Theta] \equiv 2i\left[\sigma, \Omega^g\right], \quad D\left[\gamma, \Omega^g\right] \equiv [\Theta, \Omega^g], \quad DG \equiv i\left[\Theta, \gamma_5\Omega^g\right],$	$D[\gamma, \Theta] \equiv 2i\left[\sigma, \Omega^g\right], \quad D\left[\gamma, \Omega^g\right] \equiv [\Theta, \Omega^g], \quad DG \equiv i\left[\Theta, \gamma_5\Omega^g\right],$	$D[\gamma, \Theta] \equiv 2i[\sigma, \Omega^g], \quad D[\gamma, \Omega^g] \equiv [\Theta, \Omega^g], \quad DG \equiv i[\Theta, \gamma_5\Omega^g],$
mielke-geometrodynamics_167 EQ0523 (7.7.42)	167	■ 1.000 ● N +0	$D \ R_{\alpha \beta}^{\left(\star\right)} \equiv 0,$	$DR_{\alpha\beta}^{(\star)} \equiv 0,$	$DR_{\alpha\beta}^{(\star)} \equiv 0,$
mielke-geometrodynamics_169 EQ0532 (7.7.52)	169	■ 1.000 ● K +0	$\left\{ \right\} ^{\left(6\right)}R^{\alpha \beta}=-\frac{1}{12}R\vartheta^{\alpha}\wedge\vartheta^{\beta}$	$^{(6)}R^{\alpha\beta} = -\frac{1}{12}R\vartheta^{\alpha}\wedge\vartheta^{\beta}$	$^{(6)}R^{\alpha\beta} = -\frac{1}{12}R\vartheta^{\alpha}\wedge\vartheta^{\beta}$
mielke-geometrodynamics_177 EQ0541 (8.3.2)	177	■ 1.000 ● N +0	$s \ \gamma=\psi-[C, \ \gamma]$	$s\gamma = \psi - [C, \gamma]$	$s\gamma = \psi - [C, \gamma]$
mielke-geometrodynamics_177 EQ0542 (8.3.3)	177	■ 1.000 ● K +0	$s \ \sigma:=\frac{i}{2} s(\gamma\wedge\gamma)=\frac{i}{2}([\psi,\gamma]-[[C,\gamma],\gamma])=\frac{i}{2}[\psi,\gamma]-[C,\sigma].$	$s\sigma := \frac{i}{2}s(\gamma\wedge\gamma) = \frac{i}{2}([\psi,\gamma]-[[C,\gamma],\gamma]) = \frac{i}{2}[\psi,\gamma]-[C,\sigma].$	$s\sigma:=\frac{i}{2}s(\gamma\wedge\gamma)=\frac{i}{2}([\psi,\gamma]-[[C,\gamma],\gamma])=\frac{i}{2}[\psi,\gamma]-[C,\sigma].$
mielke-geometrodynamics_177 EQ0543 (8.3.4)	177	■ 1.000 ● N +0	$s \ \bar{c}=b, \quad \quad s \ b=0 \ .$	$s\bar{c} = b, \quad sb = 0.$	$s\bar{c} = b, \quad sb = 0.$
mielke-geometrodynamics_178 EQ0546 (8.3.7)	178	■ 1.000 ● N +0	$d \ s^2=\frac{1}{4} \operatorname{Tr}\{\gamma\otimes\gamma\}^*o_{\alpha\beta}D\xi^{\alpha}\otimes D\xi^{\beta}$	$ds^2 = \frac{1}{4}\operatorname{Tr}\{\gamma\otimes\gamma\}^*o_{\alpha\beta}D\xi^{\alpha}\otimes D\xi^{\beta}$	$ds^2 = \frac{1}{4}\operatorname{Tr}\{\gamma\otimes\gamma\}^*o_{\alpha\beta}D\xi^{\alpha}\otimes D\xi^{\beta}$
mielke-geometrodynamics_179 EQ0550 (8.4.4)	179	■ 1.000 ● N ·1	$L_{\mathrm{TG}}=dC_{\mathrm{RR}}+L_{\mathrm{FP}}=dC_{\mathrm{RR}}+s\{\cdots\}$	$L_{\mathrm{TG}} = dC_{\mathrm{RR}} + L_{\mathrm{FP}} = dC_{\mathrm{RR}} + s\{\cdots\}$	$L_{\mathrm{TG}} = dC_{\mathrm{RR}} + L_{\mathrm{FP}} = dC_{\mathrm{RR}} + s\{\cdots\}$
mielke-geometrodynamics_181 EQ0559 (8.5.2)	181	■ 1.000 ● N +0	$\sigma:=\frac{i}{2}\gamma\wedge\gamma.$	$\sigma := \frac{i}{2}\gamma\wedge\gamma.$	$\sigma:=\frac{i}{2}\gamma\wedge\gamma.$
mielke-geometrodynamics_182 EQ0563 (8.5.6)	182	■ 1.000 ● N +0	$\Omega^{g(\theta)}:=\sqrt{2}\left(\Omega^g\sin\theta+{}^*\Omega^{g(\star)}\cos\theta\right)$	$\Omega^{g(\theta)} := \sqrt{2}\left(\Omega^g\sin\theta+{}^*\Omega^{g(\star)}\cos\theta\right)$	$\Omega^{g(\theta)}:=\sqrt{2}\left(\Omega^g\sin\theta+{}^*\Omega^{g(\star)}\cos\theta\right)$
mielke-geometrodynamics_183 EQ0564 (8.6.1)	183	■ 1.000 ● K +0	$L_{\mathrm{qPG}}=\frac{\Lambda}{\ell^2}\eta-\frac{a_0}{4\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta}-\frac{1}{2}T^{\alpha}\wedge H_{\alpha}-\frac{1}{2}R^{\alpha\beta}\wedge H_{\alpha\beta}.$	$L_{\mathrm{qPG}} = \frac{\Lambda}{\ell^2}\eta - \frac{a_0}{4\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta} - \frac{1}{2}T^{\alpha}\wedge H_{\alpha} - \frac{1}{2}R^{\alpha\beta}\wedge H_{\alpha\beta}.$	$L_{\mathrm{qPG}} = \frac{\Lambda}{\ell^2}\eta - \frac{a_0}{4\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta} - \frac{1}{2}T^{\alpha}\wedge H_{\alpha} - \frac{1}{2}R^{\alpha\beta}\wedge H_{\alpha\beta}.$
mielke-geometrodynamics_183 EQ0565 (8.6.2)	183	■ 1.000 ● N +1	$H_{\alpha}:= -\partial L_{\mathrm{qPG}}/\partial T^{\alpha} = \frac{1}{\ell^2}*\left(\sum_{M=1}^3a_{(M)}{}^{(M)}T_{\alpha}\right),$	$H_{\alpha} := -\partial L_{\mathrm{qPG}}/\partial T^{\alpha} = \frac{1}{\ell^2}*\left(\sum_{M=1}^3a_{(M)}{}^{(M)}T_{\alpha}\right),$	$H_{\alpha}:= -\partial L_{\mathrm{qPG}}/\partial T^{\alpha} = \frac{1}{\ell^2}*\left(\sum_{M=1}^3a_{(M)}{}^{(M)}T_{\alpha}\right),$
mielke-geometrodynamics_184 EQ0569 (8.6.7)	184	■ 1.000 ● N +0	$\begin{aligned} D\,H_{\alpha}-E_{\alpha} \, &\&= \Sigma_{\alpha} \\ D\,H_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]} \, &= \tau_{\alpha\beta}, \end{aligned}$	$\begin{aligned} DH_{\alpha}-E_{\alpha} &= \Sigma_{\alpha}, \\ DH_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]} &= \tau_{\alpha\beta}, \end{aligned}$	$\begin{aligned} DH_{\alpha}-E_{\alpha} &= \Sigma_{\alpha}, \\ DH_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]} &= \tau_{\alpha\beta}, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_184 EQ9571 (8.6.9)	184	■ 1.000 ● N -1	$\frac{\theta_{\mathrm{T}}^{\ast}}{2\ell^2}\eta_{\alpha\beta\gamma}T^{\gamma}+\frac{\theta_{\mathrm{T}}}{\ell^2}\vartheta_{[\alpha}\wedge T_{\beta]}+\vartheta_{[\alpha}\wedge H_{\beta]}(**)=\tau_{\alpha\beta}.$	$\frac{\theta_{\mathrm{T}}^{\ast}}{2\ell^2}\eta_{\alpha\beta\gamma}T^{\gamma}+\frac{\theta_{\mathrm{T}}}{\ell^2}\vartheta_{[\alpha}\wedge T_{\beta]}+\vartheta_{[\alpha}\wedge H_{\beta]}(**)=\tau_{\alpha\beta}.$	$\frac{\theta_{\mathrm{T}}^{\ast}}{2\ell^2}\eta_{\alpha\beta\gamma}T^{\gamma}+\frac{\theta_{\mathrm{T}}}{\ell^2}\vartheta_{[\alpha}\wedge T_{\beta]}+\vartheta_{[\alpha}\wedge H_{\beta]}(**)=\tau_{\alpha\beta}.$
mielke-geometrodynamics_185 EQ9576 (8.6.14)	185	■ 1.000 ● N -1	$\Lambda_{\mathrm{eff}}=\frac{1}{\theta_{\mathrm{T}}^{\ast}}\left[\Lambda-\frac{1}{4}(\theta_{\mathrm{T}}^{\ast}-a_0)R\right]=\frac{\Lambda}{\theta_{\mathrm{T}}^{\ast}}-\Lambda_{\theta}$	$\Lambda_{\mathrm{eff}}=\frac{1}{\theta_{\mathrm{T}}^{\ast}}\left[\Lambda-\frac{1}{4}(\theta_{\mathrm{T}}^{\ast}-a_0)R\right]=\frac{\Lambda}{\theta_{\mathrm{T}}^{\ast}}-\Lambda_{\theta}$	$\Lambda_{\mathrm{eff}}=\frac{1}{\theta_{\mathrm{T}}^{\ast}}\left[\Lambda-\frac{1}{4}(\theta_{\mathrm{T}}^{\ast}-a_0)R\right]=\frac{\Lambda}{\theta_{\mathrm{T}}^{\ast}}-\Lambda_{\theta}$
mielke-geometrodynamics_186 EQ9578 (8.6.16)	186	■ 1.000 ● K +0	$\widetilde{L}_{\mathrm{qPG}}=L_{\mathrm{qPG}}+\lambda_{\alpha}\wedge T^{\alpha}.$	$\widetilde{L}_{\mathrm{qPG}}=L_{\mathrm{qPG}}+\lambda_{\alpha}\wedge T^{\alpha}.$	$\widetilde{L}_{\mathrm{qPG}}=L_{\mathrm{qPG}}+\lambda_{\alpha}\wedge T^{\alpha}.$
mielke-geometrodynamics_188 EQ9582 (8.6.20)	188	■ 1.000 ● N +0	$\Lambda_{\theta}=\frac{3\left(\theta_{\mathrm{T}}^{\ast}-a_0\right)^2}{2\ell^2\left(\theta_{\mathrm{L}}^{\star}-b_6\right)\theta_{\mathrm{T}}^{\ast}}$	$\Lambda_{\theta}=\frac{3\left(\theta_{\mathrm{T}}^{\ast}-a_0\right)^2}{2\ell^2\left(\theta_{\mathrm{L}}^{\star}-b_6\right)\theta_{\mathrm{T}}^{\ast}}$	$\Lambda_{\theta}=\frac{3\left(\theta_{\mathrm{T}}^{\ast}-a_0\right)^2}{2\ell^2\left(\theta_{\mathrm{L}}^{\star}-b_6\right)\theta_{\mathrm{T}}^{\ast}}$
mielke-geometrodynamics_193 EQ9586 (9.1.4)	193	■ 1.000 ● N +0	$ds^2=\left(\frac{N(r\pm t)}{r}\right)^2(-dt^2+dr^2+r^2d\Omega^2),$	$ds^2=\left(\frac{N(r\pm t)}{r}\right)^2(-dt^2+dr^2+r^2d\Omega^2),$	$ds^2=\left(\frac{N(r\pm t)}{r}\right)^2(-dt^2+dr^2+r^2d\Omega^2),$
mielke-geometrodynamics_194 EQ9588 (9.1.6)	194	■ 1.000 ● N +0	$u^2-v^2=e^{r^{\ast}/4\mu}$	$u^2-v^2=e^{r^{\ast}/4\mu}$	$u^2-v^2=e^{r^{\ast}/4\mu}$
mielke-geometrodynamics_194 EQ9589 (9.1.7)	194	■ 1.000 ● N +0	$(u+v)/(u-v)=e^{t/2\mu},$	$(u+v)/(u-v)=e^{t/2\mu},$	$(u+v)/(u-v)=e^{t/2\mu},$
mielke-geometrodynamics_194 EQ9590 (9.1.8)	194	■ 1.000 ● N +0	$ds^2=16\mu^2e^{\nu-r^{\ast}/2\mu}(-dv^2+du^2)+r^2d\Omega^2,$	$ds^2=16\mu^2e^{\nu-r^{\ast}/2\mu}(-dv^2+du^2)+r^2d\Omega^2,$	$ds^2=16\mu^2e^{\nu-r^{\ast}/2\mu}(-dv^2+du^2)+r^2d\Omega^2,$
mielke-geometrodynamics_195 EQ9591 (9.1.9)	195	■ 1.000 ● N +0	$\Omega^{\{\}}=\Omega_{\mathrm{C}}^{\{\}}+\bar{\Omega}^{\{\}}-\frac{1}{6}R^{\{\}}\vartheta\wedge\vartheta,$	$\Omega^{\{\}}=\Omega_{\mathrm{C}}^{\{\}}+\bar{\Omega}^{\{\}}-\frac{1}{6}R^{\{\}}\vartheta\wedge\vartheta,$	$\Omega^{\{\}}=\Omega_{\mathrm{C}}^{\{\}}+\bar{\Omega}^{\{\}}-\frac{1}{6}R^{\{\}}\vartheta\wedge\vartheta,$
mielke-geometrodynamics_197 EQ9600 (9.1.18)	197	■ 1.000 ● W +1	$\begin{aligned} ds_{\mathrm{Sds}}^2&=\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)d\tau^2\\&+\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)^{-1}dr^2+r^2d\Omega^2, \end{aligned}$	$\begin{aligned} ds_{\mathrm{Sds}}^2&=\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)d\tau^2\\&+\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)^{-1}dr^2+r^2d\Omega^2 \end{aligned}$	$ds_{\mathrm{Sds}}^2=\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)d\tau^2+\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)^{-1}dr^2+r^2d\Omega^2,$
mielke-geometrodynamics_197 EQ9601 (9.1.19)	197	■ 1.000 ● N +0	$\begin{aligned} f&=\frac{2\mu}{r^2}\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)^{-1/2}\\g&=h=k=0 \end{aligned}$	$\begin{aligned} f&=\frac{2\mu}{r^2}\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)^{-1/2}\\g&=h=k=0 \end{aligned}$	$\begin{aligned} f&=\frac{2\mu}{r^2}\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{\ast 2}}\right)^{-1/2}\\g&=h=k=0 \end{aligned}$
mielke-geometrodynamics_198 EQ9603 (9.2.2)	198	■ 1.000 ● K +0	$p_1(M^4)=-3\tau$	$p_1(M^4)=-3\tau$	$p_1(M^4)=-3\tau$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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mielke-geometrodynamics_EQ0607 (9.2.8)	200	■1.000 ●W +0	$\begin{aligned} \chi(M \# N) &= \chi(M) + \chi(N) - \chi(S^n) \\ \chi(M \times N) &= \chi(M) \cdot \chi(N), \\ \chi(M^2) &= 2(g-1), \\ \chi(\mathbb{C}P^2) &= 3 \end{aligned}$	$\begin{aligned} \chi(M \# N) &= \chi(M) + \chi(N) - \chi(S^n), \\ \chi(M \times N) &= \chi(M) \cdot \chi(N), \\ \chi(M^2) &= 2(g-1), \\ \chi(\mathbb{C}P^2) &= 3 \end{aligned}$	$\begin{aligned} \chi(M \# N) &= \chi(M) + \chi(N) - \chi(S^n), \\ \chi(M \times N) &= \chi(M) \cdot \chi(N), \\ \chi(M^2) &= 2(g-1), \\ \chi(\mathbb{C}P^2) &= 3 \end{aligned}$
mielke-geometrodynamics_EQ0608 (9.2.10)	201	■1.000 ●K +0	$\chi\left(M^{2k+1}\right)=0$	$\chi\left(M^{2k+1}\right)=0$	$\chi\left(M^{2k+1}\right)=0$
mielke-geometrodynamics_EQ0612 (10.0.1)	208	■1.000 ●N +0	$L_{\mathrm{EC}}=-\frac{\chi}{\ell}\vartheta^\alpha\wedge R_\alpha^\star\wedge R_{\alpha}^\star$	$L_{\mathrm{EC}}=-\frac{\chi}{\ell}\vartheta^\alpha\wedge R_\alpha^\star$	$L_{\mathrm{EC}}=-\frac{\chi}{\ell}\vartheta^\alpha\wedge R_\alpha^\star$
mielke-geometrodynamics_EQ0614 (10.1.2)	209	■1.000 ●N +0	$T^\alpha:=d\vartheta^\alpha-(-1)^s\eta^{\alpha\beta}\wedge\Gamma_\beta^\star\wedge\Gamma_\beta^\star$	$T^\alpha:=d\vartheta^\alpha-(-1)^s\eta^{\alpha\beta}\wedge\Gamma_\beta^\star$	$T^\alpha:=d\vartheta^\alpha-(-1)^s\eta^{\alpha\beta}\wedge\Gamma_\beta^\star$
mielke-geometrodynamics_EQ0615 (10.1.3)	209	■1.000 ●W +0	$R_\alpha^\star=\frac{1}{2}\eta_{\alpha\beta\gamma}R^{\beta\gamma}:=d\Gamma_\alpha^\star+\frac{(-1)^s}{2}\eta_\alpha^{\beta\gamma}\Gamma_\beta^\star\wedge\Gamma_\gamma^\star,\frac{(-1)^s}{2}\eta_{\alpha\beta\gamma}R^{\beta\gamma}:=d\Gamma_\alpha^\star+\frac{(-1)^s}{2}\eta_\alpha^{\beta\gamma}\Gamma_\beta^\star\wedge\Gamma_\gamma^\star,$	$R_\alpha^\star=\frac{1}{2}\eta_{\alpha\beta\gamma}R^{\beta\gamma}:=d\Gamma_\alpha^\star+\frac{(-1)^s}{2}\eta_\alpha^{\beta\gamma}\Gamma_\beta^\star\wedge\Gamma_\gamma^\star,$	$R_\alpha^\star=\frac{1}{2}\eta_{\alpha\beta\gamma}R^{\beta\gamma}:=d\Gamma_\alpha^\star+\frac{(-1)^s}{2}\eta_\alpha^{\beta\gamma}\Gamma_\beta^\star\wedge\Gamma_\gamma^\star,$
mielke-geometrodynamics_EQ0616 (10.1.4)	209	■1.000 ●N +0	$DT^\alpha\equiv(-1)^s\eta^{\alpha\beta}\wedge R_\beta^\star\wedge R_{\beta}^\star$	$DT^\alpha\equiv(-1)^s\eta^{\alpha\beta}\wedge R_\beta^\star$	$DT^\alpha\equiv(-1)^s\eta^{\alpha\beta}\wedge R_\beta^\star$
mielke-geometrodynamics_EQ0617 (10.1.5)	209	■1.000 ●N +0	$DR_\alpha^\star\equiv 0.$	$DR_\alpha^\star\equiv 0.$	$DR_\alpha^\star\equiv 0.$
mielke-geometrodynamics_EQ0618 (10.1.6)	209	■1.000 ●N +0	$C_{\mathrm{T}}:=\frac{1}{2\ell^2}\vartheta^\alpha\wedge T_\alpha=-\frac{(-1)^s}{\ell^2}\eta^\alpha\wedge K_\alpha^\star,$	$C_{\mathrm{T}}:=\frac{1}{2\ell^2}\vartheta^\alpha\wedge T_\alpha=-\frac{(-1)^s}{\ell^2}\eta^\alpha\wedge K_\alpha^\star,$	$C_{\mathrm{T}}:=\frac{1}{2\ell^2}\vartheta^\alpha\wedge T_\alpha=-\frac{(-1)^s}{\ell^2}\eta^\alpha\wedge K_\alpha^\star,$
mielke-geometrodynamics_EQ0619 (10.1.7)	209	■1.000 ●K +0	$C_{\mathrm{L}}:=(-1)^s\Gamma^{\star\alpha}\wedge R_\alpha^\star-\frac{1}{3!}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\Gamma^{\star\gamma}.$	$C_{\mathrm{L}}:=(-1)^s\Gamma^{\star\alpha}\wedge R_\alpha^\star-\frac{1}{3!}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\Gamma^{\star\gamma}.$	$C_{\mathrm{L}}:=(-1)^s\Gamma^{\star\alpha}\wedge R_\alpha^\star-\frac{1}{3!}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\Gamma^{\star\gamma}.$
mielke-geometrodynamics_EQ0621 (10.1.9)	209	■1.000 ●N -2	$\frac{\delta C_{\mathrm{L}}}{\delta \vartheta^\alpha}=0\quad\text{and}\quad\frac{\delta C_{\mathrm{L}}}{\delta \Gamma^{\star\alpha}}=(-1)^s2R_\alpha^\star,$	$\frac{\delta C_{\mathrm{L}}}{\delta \vartheta^\alpha}=0\quad\text{and}\quad\frac{\delta C_{\mathrm{L}}}{\delta \Gamma^{\star\alpha}}=(-1)^s2R_\alpha^\star,$	$\frac{\delta C_{\mathrm{L}}}{\delta \vartheta^\alpha}=0\quad\text{and}\quad\frac{\delta C_{\mathrm{L}}}{\delta \Gamma^{\star\alpha}}=(-1)^s2R_\alpha^\star,$
mielke-geometrodynamics_EQ0622 (10.1.10)	210	■1.000 ●N -1	$C_{\mathrm{TL}}:=\frac{1}{\ell}\left(\Gamma^{\star\alpha}\wedge T_\alpha-\frac{(-1)^s}{2}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\vartheta^\gamma\right).$	$C_{\mathrm{TL}}:=\frac{1}{\ell}\left(\Gamma^{\star\alpha}\wedge T_\alpha-\frac{(-1)^s}{2}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\vartheta^\gamma\right).$	$C_{\mathrm{TL}}:=\frac{1}{\ell}\left(\Gamma^{\star\alpha}\wedge T_\alpha-\frac{(-1)^s}{2}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\vartheta^\gamma\right).$
mielke-geometrodynamics_EQ0625 (10.1.13)	210	■1.000 ●N +0	$DH_\alpha^\star-S_\alpha=\tau_\alpha^\star,$	$DH_\alpha^\star-S_\alpha=\tau_\alpha^\star,$	$DH_\alpha^\star-S_\alpha=\tau_\alpha^\star,$
mielke-geometrodynamics_EQ0626 (10.1.14)	211	■1.000 ●N +0	$H_\alpha:=-\frac{\partial L}{\partial T^\alpha},\quad H_\alpha^\star:=-\frac{(-1)^s}{2}\frac{\partial L}{\partial R^{\star\alpha}},$	$H_\alpha:=-\frac{\partial L}{\partial T^\alpha},\quad H_\alpha^\star:=-\frac{(-1)^s}{2}\frac{\partial L}{\partial R^{\star\alpha}},$	$H_\alpha:=-\frac{\partial L}{\partial T^\alpha},\quad H_\alpha^\star:=-\frac{(-1)^s}{2}\frac{\partial L}{\partial R^{\star\alpha}},$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					

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mielke-geometrodynamics_211 EQ0628 (10.1.16)		■ 1.000 ● N +0	$S_{\alpha}:=\frac{(-1)^s}{2}\frac{\partial L}{\partial \Gamma^{\star\alpha}}=\frac{1}{2}\eta_{\alpha\beta}\wedge H^\beta$	$S_\alpha := \frac{(-1)^s}{2} \frac{\partial L}{\partial \Gamma^{\star\alpha}} = \frac{1}{2} \eta_{\alpha\beta} \wedge H^\beta.$	$S_\alpha := \frac{(-1)^s}{2} \frac{\partial L}{\partial \Gamma^{\star\alpha}} = \frac{1}{2} \eta_{\alpha\beta} \wedge H^\beta.$
mielke-geometrodynamics_211 EQ0629 (10.1.17)		■ 1.000 ● N +0	$\Sigma_{\alpha}:=\frac{\delta L_{\mathrm{MAT}}}{\delta \vartheta^{\alpha}},\quad \tau_{\alpha}^{\star}:=\frac{(-1)^s}{2}\frac{\delta L_{\mathrm{MAT}}}{\delta \Gamma^{\star\alpha}},$	$\Sigma_\alpha := \frac{\delta L_{\mathrm{MAT}}}{\delta \vartheta^\alpha}, \quad \tau_\alpha^\star := \frac{(-1)^s}{2} \frac{\delta L_{\mathrm{MAT}}}{\delta \Gamma^{\star\alpha}},$	$\Sigma_\alpha := \frac{\delta L_{\mathrm{MAT}}}{\delta \vartheta^\alpha}, \quad \tau_\alpha^\star := \frac{(-1)^s}{2} \frac{\delta L_{\mathrm{MAT}}}{\delta \Gamma^{\star\alpha}},$
mielke-geometrodynamics_211 EQ0630 (10.2.1)		■ 1.000 ● N -1	$L_{\mathrm{EC}}:=-\frac{\chi}{\ell}\vartheta^\alpha\wedge R_\alpha^\star\equiv-\chi\,C_{\mathrm{TL}}-\frac{\chi}{\ell}d\left(\Gamma_\alpha^\star\wedge\vartheta^\alpha\right)$	$L_{\mathrm{EC}} := -\frac{\chi}{\ell} \vartheta^\alpha \wedge R_\alpha^\star \equiv -\chi C_{\mathrm{TL}} - \frac{\chi}{\ell} d(\Gamma_\alpha^\star \wedge \vartheta^\alpha)$	$L_{\mathrm{EC}} := -\frac{\chi}{\ell} \vartheta^\alpha \wedge R_\alpha^\star \equiv -\chi C_{\mathrm{TL}} - \frac{\chi}{\ell} d(\Gamma_\alpha^\star \wedge \vartheta^\alpha)$
mielke-geometrodynamics_211 EQ0631 (10.2.2)		■ 1.000 ● C -5	$L_{\mathrm{MB}}(\vartheta^\alpha,\Gamma_\alpha^\star)=\theta_{\mathrm{T}}C_{\mathrm{T}}+\theta_{\mathrm{L}}C_{\mathrm{L}}+\theta_{\mathrm{TL}}C_{\mathrm{TL}},$	$L_{\mathrm{MB}}(\vartheta^\alpha,\Gamma_\alpha^\star)=\theta_{\mathrm{T}}C_{\mathrm{T}}+\theta_{\mathrm{L}}C_{\mathrm{L}}+\theta_{\mathrm{TL}}C_{\mathrm{TL}},$	$L_{\mathrm{MB}}(\vartheta^\alpha,\Gamma_\alpha^\star)=\theta_{\mathrm{T}}C_{\mathrm{T}}+\theta_{\mathrm{L}}C_{\mathrm{L}}+\theta_{\mathrm{TL}}C_{\mathrm{TL}},$
mielke-geometrodynamics_212 EQ0632 (10.2.3)		■ 1.000 ● C -3	$-\theta_{\mathrm{TL}}R_\alpha^\star-\frac{1}{\ell}\theta_{\mathrm{T}}T_\alpha=\ell\Sigma_\alpha$	$-\theta_{\mathrm{TL}}R_\alpha^\star-\frac{1}{\ell}\theta_{\mathrm{T}}T_\alpha=\ell\Sigma_\alpha$	$-\theta_{\mathrm{TL}}R_\alpha^\star-\frac{1}{\ell}\theta_{\mathrm{T}}T_\alpha=\ell\Sigma_\alpha$
mielke-geometrodynamics_212 EQ0634 (10.2.5)		■ 1.000 ● N -2	$T_\alpha-\frac{\varepsilon}{\ell}\eta_\alpha=\frac{2}{A}\ell\left(\theta_{\mathrm{TL}}\tau_\alpha^\star-\theta_{\mathrm{L}}\ell\Sigma_\alpha\right)$	$T_\alpha-\frac{\varepsilon}{\ell}\eta_\alpha=\frac{2}{A}\ell\left(\theta_{\mathrm{TL}}\tau_\alpha^\star-\theta_{\mathrm{L}}\ell\Sigma_\alpha\right)$	$T_\alpha-\frac{\varepsilon}{\ell}\eta_\alpha=\frac{2}{A}\ell\left(\theta_{\mathrm{TL}}\tau_\alpha^\star-\theta_{\mathrm{L}}\ell\Sigma_\alpha\right)$
mielke-geometrodynamics_212 EQ0635 (10.2.6)		■ 1.000 ● C -3	$R_\alpha^\star-\frac{\rho}{\ell^2}\eta_\alpha=\frac{2}{A}\left(\theta_{\mathrm{TL}}\ell\Sigma_\alpha-\theta_{\mathrm{T}}\tau_\alpha^\star\right)$	$R_\alpha^\star-\frac{\rho}{\ell^2}\eta_\alpha=\frac{2}{A}\left(\theta_{\mathrm{TL}}\ell\Sigma_\alpha-\theta_{\mathrm{T}}\tau_\alpha^\star\right)$	$R_\alpha^\star-\frac{\rho}{\ell^2}\eta_\alpha=\frac{2}{A}\left(\theta_{\mathrm{TL}}\ell\Sigma_\alpha-\theta_{\mathrm{T}}\tau_\alpha^\star\right)$
mielke-geometrodynamics_213 EQ0638 (10.2.9)		■ 1.000 ● N +0	$R_\alpha^\star\equiv R_\alpha^{\{\}\star}-DK_\alpha^\star+\frac{1}{2}\eta_{\alpha\beta\gamma}K^{\star\beta}\wedge K^{\star\gamma}$	$R_\alpha^\star \equiv R_\alpha^{\{\}\star} - DK_\alpha^\star + \frac{1}{2} \eta_{\alpha\beta\gamma} K^{\star\beta} \wedge K^{\star\gamma}$	$R_\alpha^\star \equiv R_\alpha^{\{\}\star} - DK_\alpha^\star + \frac{1}{2} \eta_{\alpha\beta\gamma} K^{\star\beta} \wedge K^{\star\gamma}$
mielke-geometrodynamics_213 EQ0640 (10.2.11)		■ 1.000 ● C -4	$\Lambda_{\mathrm{eff}}=-\left[4\rho-\left(1+2(-1)^s\right)\varepsilon^2\right]/4\ell^2=\theta_{\mathrm{T}}^2\left[\left(9+2(-1)^s\right)\theta_{\mathrm{TL}}^2+8\theta_{\mathrm{T}}\theta_{\mathrm{L}}\right]/(2A\ell)^2$	$\Lambda_{\mathrm{eff}} = -\left[4\rho - (1 + 2(-1)^s)\varepsilon^2\right]/4\ell^2 = \theta_{\mathrm{T}}^2\left[(9 + 2(-1)^s)\theta_{\mathrm{TL}}^2 + 8\theta_{\mathrm{T}}\theta_{\mathrm{L}}\right]/(2A\ell)^2$	$\Lambda_{\mathrm{eff}} = -[4\rho - (1 + 2(-1)^s)\varepsilon^2]/4\ell^2 = \theta_{\mathrm{T}}^2[(9 + 2(-1)^s)\theta_{\mathrm{TL}}^2 + 8\theta_{\mathrm{T}}\theta_{\mathrm{L}}]/(2A\ell)^2$
mielke-geometrodynamics_213 EQ0642 (10.2.13)		■ 1.000 ● N +0	$ds^2=-N^2(r)dt^2+N^{-2}(r)dr^2+r^2\left[d\phi+N^\phi(r)dt\right]^2$	$ds^2 = -N^2(r)dt^2 + N^{-2}(r)dr^2 + r^2\left[d\phi + N^\phi(r)dt\right]^2$	$ds^2 = -N^2(r)dt^2 + N^{-2}(r)dr^2 + r^2\left[d\phi + N^\phi(r)dt\right]^2$
mielke-geometrodynamics_213 EQ0643 (10.2.14)		■ 1.000 ● N -1	$N^{\{2\}}(r)=-M-\Lambda_{\mathrm{eff}}r^2+\frac{J^2}{4r^2},\quad N^{\phi}(r)=-\frac{J}{2r^2},$	$N^2(r) = -M - \Lambda_{\mathrm{eff}} r^2 + \frac{J^2}{4r^2}, \quad N^\phi(r) = -\frac{J}{2r^2},$	$N^2(r) = -M - \Lambda_{\mathrm{eff}} r^2 + \frac{J^2}{4r^2}, \quad N^\phi(r) = -\frac{J}{2r^2},$
mielke-geometrodynamics_214 EQ0647 (10.3.4)		■ 1.000 ● K +0	$S_{\alpha}=\frac{\ell}{\varepsilon}E_{\alpha}$	$S_\alpha = \frac{\ell}{\varepsilon} E_\alpha$	$S_\alpha = \frac{\ell}{\varepsilon} E_\alpha$
mielke-geometrodynamics_215 EQ0648 (10.3.5)		■ 1.000 ● N +0	$H_{\alpha}^{\star}=\frac{\ell}{\varepsilon}H_{\alpha}$	$H_\alpha^\star = \frac{\ell}{\varepsilon} H_\alpha$	$H_\alpha^\star = \frac{\ell}{\varepsilon} H_\alpha$
mielke-geometrodynamics_215 EQ0649 (10.3.6)		■ 1.000 ● N +1	$T_\alpha=\frac{\varepsilon}{\ell}\eta_\alpha,\quad R_\alpha^\star=\frac{\rho}{\ell^2}\eta_\alpha$	$T_\alpha = \frac{\varepsilon}{\ell} \eta_\alpha, \quad R_\alpha^\star = \frac{\rho}{\ell^2} \eta_\alpha$	$T_\alpha = \frac{\varepsilon}{\ell} \eta_\alpha, \quad R_\alpha^\star = \frac{\rho}{\ell^2} \eta_\alpha,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E00650 (10.3.7)	215	■ 1.000 ● N +0	$H_{\alpha}^{\star}=\delta\ell\ H_{\alpha}+\frac{\gamma}{\ell}\vartheta_{\alpha}$	$H_{\alpha}^{\star}=\delta\ell H_{\alpha}+\frac{\gamma}{\ell}\vartheta_{\alpha}$	$H_{\alpha}^{\star}=\delta\ell H_{\alpha}+\frac{\gamma}{\ell}\vartheta_{\alpha}$
mielke-geometrodynamics_E00651 (10.3.8)	215	■ 1.000 ● N +1	$\delta\ell\ D\ H_{\alpha}+\frac{\gamma}{\ell}\{T_{\alpha}-\frac{1}{2}\eta_{\alpha\beta}\wedge H^{\beta}=\tau_{\alpha}^{\star}$	$\delta\ell DH_{\alpha}+\frac{\gamma}{\ell}T_{\alpha}-\frac{1}{2}\eta_{\alpha\beta}\wedge H^{\beta}=\tau_{\alpha}^{\star}$	$\delta\ell DH_{\alpha}+\frac{\gamma}{\ell}T_{\alpha}-\frac{1}{2}\eta_{\alpha\beta}\wedge H^{\beta}=\tau_{\alpha}^{\star}$
mielke-geometrodynamics_E00653 (10.3.10)	216	■ 1.000 ● K +0	$H_{\alpha}^{\star}=\frac{\gamma}{\ell(1+\delta\varepsilon)}\vartheta_{\alpha}=-\frac{\ell}{\varepsilon}H_{\alpha},$	$H_{\alpha}^{\star}=\frac{\gamma}{\ell(1+\delta\varepsilon)}\vartheta_{\alpha}=-\frac{\ell}{\varepsilon}H_{\alpha},$	$H_{\alpha}^{\star}=\frac{\gamma}{\ell(1+\delta\varepsilon)}\vartheta_{\alpha}=-\frac{\ell}{\varepsilon}H_{\alpha},$
mielke-geometrodynamics_E00659 (10.3.16)	217	■ 1.000 ● K +0	$H=\left(\vartheta^{\alpha}\wedge H_{\alpha}\right),$	$H=\left(\vartheta^{\alpha}\wedge H_{\alpha}\right),$	$H=\left(\vartheta^{\alpha}\wedge H_{\alpha}\right),$
mielke-geometrodynamics_E00660 (10.3.17)	217	■ 1.000 ● N +0	$t=\left(T^{\alpha}\wedge H_{\alpha}\right),\quad \mathscr{A}:=\left(\vartheta^{\alpha}\wedge T_{\alpha}\right),\quad h=\left(\eta^{\alpha}\wedge H_{\alpha}\right),$	$t=\left(T^{\alpha}\wedge H_{\alpha}\right),\quad \mathscr{A}:=\left(\vartheta^{\alpha}\wedge T_{\alpha}\right),\quad h=\left(\eta^{\alpha}\wedge H_{\alpha}\right),$	$t=\left(T^{\alpha}\wedge H_{\alpha}\right),\quad \mathscr{A}:=\left(\vartheta^{\alpha}\wedge T_{\alpha}\right),\quad h=\left(\eta^{\alpha}\wedge H_{\alpha}\right),$
mielke-geometrodynamics_E00663 (10.3.20)	217	■ 1.000 ● N +2	$ds^2=(-1)^s\,dt^2+d\mathscr{A}^2+dh^2$	$ds^2=(-1)^s\,dt^2+d\mathscr{A}^2+dh^2$	$ds^2=(-1)^s\,dt^2+d\mathscr{A}^2+dh^2$
mielke-geometrodynamics_E00665 (10.3.22)	218	■ 1.000 ● N +0	$D^{\star}\vartheta_{\alpha}:=\frac{\ell}{\gamma}\left[D^{\star}H_{\alpha}^{\star}-\delta\ell D^{\star}H_{\alpha}\right]$	$D^{\star}\vartheta_{\alpha}:=\frac{\ell}{\gamma}\left[D^{\star}H_{\alpha}^{\star}-\delta\ell D^{\star}H_{\alpha}\right]$	$D^{\star}\vartheta_{\alpha}:=\frac{\ell}{\gamma}\left[D^{\star}H_{\alpha}^{\star}-\delta\ell D^{\star}H_{\alpha}\right]$
mielke-geometrodynamics_E00668 (10.3.25)	219	■ 1.000 ● N +0	$D^{\star}\vartheta^{\alpha}\cong 0,$	$D^{\star}\vartheta^{\alpha}\cong 0,$	$D^{\star}\vartheta^{\alpha}\cong 0,$
mielke-geometrodynamics_E00672 (10.3.29)	220	■ 1.000 ● N +0	$P_{\alpha}=\int\check{\mathcal{E}}_{\alpha},\quad J_{\alpha}=\int\check{\mathcal{J}}_{\alpha}^{\star}.$	$P_{\alpha}=\int\check{\mathcal{E}}_{\alpha},\quad J_{\alpha}=\int\check{\mathcal{J}}_{\alpha}^{\star}.$	$P_{\alpha}=\int\check{\mathcal{E}}_{\alpha},\quad J_{\alpha}=\int\check{\mathcal{J}}_{\alpha}^{\star}.$
mielke-geometrodynamics_E00675 (10.3.32)	220	■ 1.000 ● N -1	$\check{\mathcal{J}}_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$	$\check{\mathcal{J}}_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$	$\check{\mathcal{J}}_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$
mielke-geometrodynamics_E00677 (10.3.34)	220	■ 1.000 ● N -1	$\check{\mathcal{J}}_{\alpha}^{\mathrm{MB}\star}=(-1)^s\frac{\chi}{2\ell}d\vartheta_{\alpha}-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$	$\check{\mathcal{J}}_{\alpha}^{\mathrm{MB}\star}=(-1)^s\frac{\chi}{2\ell}d\vartheta_{\alpha}-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$	$\check{\mathcal{J}}_{\alpha}^{\mathrm{MB}\star}=(-1)^s\frac{\chi}{2\ell}d\vartheta_{\alpha}-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$
mielke-geometrodynamics_E00679 (10.3.36)	221	■ 1.000 ● K +0	$\widetilde{C}_{\mathrm{L}}=(-1)^s\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{R}_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{\Gamma}^{\star\beta}\wedge\widetilde{\Gamma}^{\star\gamma},$	$\widetilde{C}_{\mathrm{L}}=(-1)^s\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{R}_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{\Gamma}^{\star\beta}\wedge\widetilde{\Gamma}^{\star\gamma},$	$\widetilde{C}_{\mathrm{L}}=(-1)^s\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{R}_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{\Gamma}^{\star\beta}\wedge\widetilde{\Gamma}^{\star\gamma},$
mielke-geometrodynamics_E00683 (10.3.41)	221	■ 1.000 ● N -1	$c=12\cdot 4\pi\theta_{\mathrm{L}},\quad \tilde{c}=12\cdot 4\pi\tilde{\theta}_{\mathrm{L}},$	$c=12\cdot 4\pi\theta_{\mathrm{L}},\quad \tilde{c}=12\cdot 4\pi\tilde{\theta}_{\mathrm{L}},$	$c=12\cdot 4\pi\theta_{\mathrm{L}},\quad \tilde{c}=12\cdot 4\pi\tilde{\theta}_{\mathrm{L}},$
mielke-geometrodynamics_E00684 (10.3.42)	222	■ 1.000 ● K +0	$\widetilde{L}=L+\lambda_{\alpha}\wedge T^{\alpha}.$	$\widetilde{L}=L+\lambda_{\alpha}\wedge T^{\alpha}.$	$\widetilde{L}=L+\lambda_{\alpha}\wedge T^{\alpha}.$
mielke-geometrodynamics_E00691 (10.5.2)	223	■ 1.000 ● K +0	$\left[\sigma^{\alpha},\sigma^{\beta}\right]=2i\eta^{\alpha\beta\gamma}\sigma_{\gamma}.$	$\left[\sigma^{\alpha},\sigma^{\beta}\right]=2i\eta^{\alpha\beta\gamma}\sigma_{\gamma}.$	$\left[\sigma^{\alpha},\sigma^{\beta}\right]=2i\eta^{\alpha\beta\gamma}\sigma_{\gamma}.$
mielke-geometrodynamics_E00692 (10.5.3)	224	■ 1.000 ● K +0	$\gamma_0=i\sigma^2,\quad \gamma_1=\sigma^1,\quad \gamma_2=\sigma^3.$	$\gamma_0=i\sigma^2,\quad \gamma_1=\sigma^1,\quad \gamma_2=\sigma^3.$	$\gamma_0=i\sigma^2,\quad \gamma_1=\sigma^1,\quad \gamma_2=\sigma^3.$
mielke-geometrodynamics_E00693 (10.5.4)	224	■ 1.000 ● N +0	$\gamma_{\alpha}\gamma_{\beta}=g_{\alpha\beta}\mathbf{1}+\eta_{\alpha\beta\nu}\gamma^{\nu},$	$\gamma_{\alpha}\gamma_{\beta}=g_{\alpha\beta}\mathbf{1}+\eta_{\alpha\beta\nu}\gamma^{\nu},$	$\gamma_{\alpha}\gamma_{\beta}=g_{\alpha\beta}\mathbf{1}+\eta_{\alpha\beta\nu}\gamma^{\nu},$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_224 EQ0695 (10.5.6)		■ 1.000 ● N +0	$i^{\{*\}} \wedge \gamma \wedge D \psi - m \psi \eta - \frac{i}{2} (D^* \gamma) \psi = 0$	$i^* \gamma \wedge D \psi - m \psi \eta - \frac{i}{2} (D^* \gamma) \psi = 0$	$i^* \gamma \wedge D \psi - m \psi \eta - \frac{i}{2} (D^* \gamma) \psi = 0$
mielke-geometroynamics_224 EQ0697 (10.5.8)		■ 1.000 ● K +0	$\hat{D} := D - \frac{1}{2} T,$	$\hat{D} := D - \frac{1}{2} T,$	$\hat{D} := D - \frac{1}{2} T,$
mielke-geometroynamics_224 EQ0698 (10.5.9)		■ 1.000 ● N +0	$i^{\{*\}} \wedge \gamma \wedge \hat{D} \psi - m \eta \psi = 0,$	$i^* \gamma \wedge \hat{D} \psi - m \eta \psi = 0,$	$i^* \gamma \wedge \hat{D} \psi - m \eta \psi = 0,$
mielke-geometroynamics_224 EQ0699 (10.5.10)		■ 1.000 ● K +0	$\hat{R} = R - \frac{1}{2} dT = \frac{i}{4} R^{\alpha \beta} \sigma_{\alpha \beta} - \frac{1}{2} dT$	$\hat{R} = R - \frac{1}{2} dT = \frac{i}{4} R^{\alpha \beta} \sigma_{\alpha \beta} - \frac{1}{2} dT$	$\hat{R} = R - \frac{1}{2} dT = \frac{i}{4} R^{\alpha \beta} \sigma_{\alpha \beta} - \frac{1}{2} dT$
mielke-geometroynamics_225 EQ0703 (10.6.2)		■ 1.000 ● N +0	$d^{\{*\}} F - m_{\text{photon}} F = 0$	$d^* F - m_{\text{photon}} F = 0$	$d^* F - m_{\text{photon}} F = 0$
mielke-geometroynamics_226 EQ0705 (10.7.1)		■ 1.000 ● N +0	$X := x_{\alpha} \vartheta^{\alpha}, \quad Y := y_{\alpha} \vartheta^{\alpha}.$	$X := x_{\alpha} \vartheta^{\alpha}, \quad Y := y_{\alpha} \vartheta^{\alpha}.$	$X := x_{\alpha} \vartheta^{\alpha}, \quad Y := y_{\alpha} \vartheta^{\alpha}.$
mielke-geometroynamics_226 EQ0706 (10.7.2)		■ 1.000 ● N +0	$\Sigma_{\alpha} = \varepsilon x_{\alpha} X \wedge Y, \quad \tau_{\alpha}^* = \sigma y_{\alpha} X \wedge Y,$	$\Sigma_{\alpha} = \varepsilon x_{\alpha} X \wedge Y, \quad \tau_{\alpha}^* = \sigma y_{\alpha} X \wedge Y,$	$\Sigma_{\alpha} = \varepsilon x_{\alpha} X \wedge Y, \quad \tau_{\alpha}^* = \sigma y_{\alpha} X \wedge Y,$
mielke-geometroynamics_226 EQ0707 (10.7.3)		■ 1.000 ● N +0	$\vartheta^{\alpha} \wedge \Sigma_{\alpha} = 0, \quad \vartheta^{\alpha} \wedge \tau_{\alpha}^* = 0$	$\vartheta^{\alpha} \wedge \Sigma_{\alpha} = 0, \quad \vartheta^{\alpha} \wedge \tau_{\alpha}^* = 0$	$\vartheta^{\alpha} \wedge \Sigma_{\alpha} = 0, \quad \vartheta^{\alpha} \wedge \tau_{\alpha}^* = 0$
mielke-geometroynamics_227 EQ0708 (10.7.4)		■ 1.000 ● N +0	$R^{\alpha \beta} = \varepsilon \ell^2 x^{[\alpha} y^{\beta]} X \wedge Y,$	$R^{\alpha \beta} = \varepsilon \ell^2 x^{[\alpha} y^{\beta]} X \wedge Y,$	$R^{\alpha \beta} = \varepsilon \ell^2 x^{[\alpha} y^{\beta]} X \wedge Y,$
mielke-geometroynamics_228 EQ0711 (10.7.7)		■ 1.000 ● N +0	$\mathscr{A} := * (\vartheta^{\alpha} \wedge T_{\alpha}) = -(-1)^s \frac{\varepsilon}{\ell^2}$	$\mathscr{A} := * (\vartheta^{\alpha} \wedge T_{\alpha}) = -(-1)^s \frac{\varepsilon}{\ell^2}$	$\mathscr{A} := * (\vartheta^{\alpha} \wedge T_{\alpha}) = -(-1)^s \frac{\varepsilon}{\ell^2}$
mielke-geometroynamics_228 EQ0713 (10.8.2)		■ 1.000 ● K +0	$\gamma = \gamma_{\alpha} \vartheta^{\alpha}.$	$\gamma = \gamma_{\alpha} \vartheta^{\alpha}.$	$\gamma = \gamma_{\alpha} \vartheta^{\alpha}.$
mielke-geometroynamics_228 EQ0714		■ 1.000 ● K +0	$\bar{\Psi} \wedge \Psi = 0, \quad \bar{\Psi} \wedge \gamma_5 \gamma^{\alpha} \Psi = 0, \quad \bar{\Psi} \wedge \gamma_5 \Psi = 0.$	$\bar{\Psi} \wedge \Psi = 0, \quad \bar{\Psi} \wedge \gamma_5 \gamma^{\alpha} \Psi = 0, \quad \bar{\Psi} \wedge \gamma_5 \Psi = 0.$	$\bar{\Psi} \wedge \Psi = 0, \quad \bar{\Psi} \wedge \gamma_5 \gamma^{\alpha} \Psi = 0, \quad \bar{\Psi} \wedge \gamma_5 \Psi = 0.$
mielke-geometroynamics_229 EQ0715 (10.9.1)		■ 1.000 ● N -1	$L_{\text{RS}} = \frac{i}{4} (\bar{\Psi} \wedge D \Psi - \Psi \wedge \overline{D \Psi}) + \frac{i}{4} m \bar{\Psi} \wedge \gamma \wedge \Psi,$	$L_{\text{RS}} = \frac{i}{4} (\bar{\Psi} \wedge D \Psi - \Psi \wedge \overline{D \Psi}) + \frac{i}{4} m \bar{\Psi} \wedge \gamma \wedge \Psi,$	$L_{\text{RS}} = \frac{i}{4} (\bar{\Psi} \wedge D \Psi - \Psi \wedge \overline{D \Psi}) + \frac{i}{4} m \bar{\Psi} \wedge \gamma \wedge \Psi,$
mielke-geometroynamics_229 EQ0719 (10.9.5)		■ 1.000 ● N +1	$\Sigma_{\alpha} = -\frac{i}{4} m \bar{\Psi} \wedge \gamma_{\alpha} \Psi = -(-1)^s 2 m \tau_{\alpha}^*,$	$\Sigma_{\alpha} = -\frac{i}{4} m \bar{\Psi} \wedge \gamma_{\alpha} \Psi = -(-1)^s 2 m \tau_{\alpha}^*,$	$\Sigma_{\alpha} = -\frac{i}{4} m \bar{\Psi} \wedge \gamma_{\alpha} \Psi = -(-1)^s 2 m \tau_{\alpha}^*,$
mielke-geometroynamics_230 EQ0721 (10.10.1)		■ 1.000 ● N +0	$L = L_{\infty} (\vartheta^{\alpha}, \Gamma_{\alpha}^*, \Psi) = L_{\text{MB}} + L_{\Psi}$	$L = L_{\infty} (\vartheta^{\alpha}, \Gamma_{\alpha}^*, \Psi) = L_{\text{MB}} + L_{\Psi}$	$L = L_{\infty} (\vartheta^{\alpha}, \Gamma_{\alpha}^*, \Psi) = L_{\text{MB}} + L_{\Psi}$
mielke-geometroynamics_230 EQ0726 (10.10.6)		■ 1.000 ● N +0	$\frac{2}{i} \frac{\delta L}{\delta \Psi} = D \Psi + \frac{1}{2} m \gamma \wedge \Psi \cong 0$	$\frac{2}{i} \frac{\delta L}{\delta \Psi} = D \Psi + \frac{1}{2} m \gamma \wedge \Psi \cong 0$	$\frac{2}{i} \frac{\delta L}{\delta \Psi} = D \Psi + \frac{1}{2} m \gamma \wedge \Psi \cong 0$
mielke-geometroynamics_231 EQ0729 (10.10.9)		■ 1.000 ● N +0	$\gamma \wedge \gamma = -2 \gamma^{\alpha} \eta_{\alpha}$	$\gamma \wedge \gamma = -2 \gamma^{\alpha} \eta_{\alpha}$	$\gamma \wedge \gamma = -2 \gamma^{\alpha} \eta_{\alpha}$
mielke-geometroynamics_231 EQ0730 (10.10.10)		■ 1.000 ● W -2	$i \left[\left(\frac{\theta_{\text{TL}}}{\ell} - 1 \right) R_{\alpha}^* + \left(\frac{\theta_{\text{T}}}{\ell^2} - \frac{m}{2} \right) T_{\alpha} + \frac{m^2}{2} \eta_{\alpha} \right] \wedge \gamma^{\alpha} \Psi = 0,$	$i \left[\left(\frac{\theta_{\text{TL}}}{\ell} - 1 \right) R_{\alpha}^* + \left(\frac{\theta_{\text{T}}}{\ell^2} - \frac{m}{2} \right) T_{\alpha} + \frac{m^2}{2} \eta_{\alpha} \right] \wedge \gamma^{\alpha} \Psi = 0,$	$i \left[\left(\frac{\theta_{\text{TL}}}{\ell} - 1 \right) R_{\alpha}^* + \left(\frac{\theta_{\text{T}}}{\ell^2} - \frac{m}{2} \right) T_{\alpha} + \frac{m^2}{2} \eta_{\alpha} \right] \wedge \gamma^{\alpha} \Psi = 0,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics EQ0731 (10.10.11)	231	■ 1.000 ● W +0	$\begin{aligned} & i\left[\left(\theta_{\mathrm{L}}+\frac{\theta_{\mathrm{TL}}}{\ell}-1\right)R_{\alpha}^{\star}+\left((-1)^s\frac{\theta_{\mathrm{TL}}}{\ell}+\frac{\theta_{\mathrm{T}}}{\ell^2}-\frac{m}{2}\right)T_{\alpha}+\frac{1}{2}\left(\frac{\theta_{\mathrm{T}}}{\ell^2}+m^2\right)\eta_{\alpha}\right. \\ & \left.+\tau_{\alpha}^{\star}\right]\wedge\gamma^{\alpha}\Psi\simeq 0, \end{aligned}$	$i\left[\left(\theta_{\mathrm{L}}+\frac{\theta_{\mathrm{TL}}}{\ell}-1\right)R_{\alpha}^{\star}+\left((-1)^s\frac{\theta_{\mathrm{TL}}}{\ell}+\frac{\theta_{\mathrm{T}}}{\ell^2}-\frac{m}{2}\right)T_{\alpha}+\frac{1}{2}\left(\frac{\theta_{\mathrm{T}}}{\ell^2}+m^2\right)\eta_{\alpha}\right. \\ \left.+\tau_{\alpha}^{\star}\right]\wedge\gamma^{\alpha}\Psi\simeq 0$	$i\left[\left(\theta_{\mathrm{L}}+\frac{\theta_{\mathrm{TL}}}{\ell}-1\right)R_{\alpha}^{\star}+\left((-1)^s\frac{\theta_{\mathrm{TL}}}{\ell}+\frac{\theta_{\mathrm{T}}}{\ell^2}-\frac{m}{2}\right)T_{\alpha}+\frac{1}{2}\left(\frac{\theta_{\mathrm{T}}}{\ell^2}+m^2\right)\eta_{\alpha}\right. \\ \left.+\tau_{\alpha}^{\star}\right]\wedge\gamma^{\alpha}\Psi\simeq 0, \quad (10.10)$
mielke-geometrodynamics EQ0732 (10.10.12)	232	■ 1.000 ● W +2	$\theta_{\mathrm{T}}\simeq -m^2\ell^2, \theta_{\mathrm{TL}}\simeq \frac{(-1)^s}{2}m(2m+1)\ell, \theta_{\mathrm{L}}\simeq 1-\frac{\theta_{\mathrm{TL}}}{\ell}=1-\frac{(-1)^s}{2}m(2m+1)$	$\theta_{\mathrm{T}}\simeq -m^2\ell^2, \theta_{\mathrm{TL}}\simeq \frac{(-1)^s}{2}m(2m+1)\ell, \theta_{\mathrm{L}}\simeq 1-\frac{\theta_{\mathrm{TL}}}{\ell}=1-\frac{(-1)^s}{2}m(2m+1)$	$\theta_{\mathrm{T}}\simeq -m^2\ell^2, \theta_{\mathrm{TL}}\simeq \frac{(-1)^s}{2}m(2m+1)\ell, \theta_{\mathrm{L}}\simeq 1-\frac{\theta_{\mathrm{TL}}}{\ell}=1-\frac{(-1)^s}{2}m(2m+1)$
mielke-geometrodynamics EQ0733 (10.10.13)	232	■ 1.000 ● N +0	$\vartheta_{\alpha}=(-1)^s\ell\Gamma_{\alpha}^{\star}+\bar{\sigma}\gamma_{\alpha}\Psi.$	$\vartheta_{\alpha}=(-1)^s\ell\Gamma_{\alpha}^{\star}+\bar{\sigma}\gamma_{\alpha}\Psi.$	$\vartheta_{\alpha}=(-1)^s\ell\Gamma_{\alpha}^{\star}+\bar{\sigma}\gamma_{\alpha}\Psi.$
mielke-geometrodynamics EQ0735 (11.1.2)	238	■ 1.000 ● N +0	$\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathbf{o}_{\alpha\beta}\mathbb{1}$	$\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathbf{o}_{\alpha\beta}\mathbb{1}$	$\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathbf{o}_{\alpha\beta}\mathbb{1}$
mielke-geometrodynamics EQ0738 (11.1.5)	238	■ 1.000 ● N +0	$\mathrm{X}:=\imath x^{\alpha}\gamma_{\alpha}\longleftrightarrow x^{\alpha}=-\frac{l}{N}\mathrm{Tr}(\gamma^{\alpha}X).$	$\mathrm{X}:=\imath x^{\alpha}\gamma_{\alpha}\longleftrightarrow x^{\alpha}=-\frac{l}{N}\mathrm{Tr}(\gamma^{\alpha}X).$	$\mathrm{X}:=\imath x^{\alpha}\gamma_{\alpha}\longleftrightarrow x^{\alpha}=-\frac{l}{N}\mathrm{Tr}(\gamma^{\alpha}X).$
mielke-geometrodynamics EQ0739 (11.1.6)	238	■ 1.000 ● K +0	$\det\mathrm{X}=-\mathbf{o}_{\alpha\beta}x^{\alpha}x^{\beta},$	$\det\mathrm{X}=-\mathbf{o}_{\alpha\beta}x^{\alpha}x^{\beta},$	$\det\mathrm{X}=-\mathbf{o}_{\alpha\beta}x^{\alpha}x^{\beta},$
mielke-geometrodynamics EQ0741 (11.1.8)	238	■ 1.000 ● N +0	$\mathrm{X}-\mathrm{X}'=SXS^{-1},\quad S\in\widetilde{SO}_{\circ}(s,n-s).$	$\mathrm{X}-\mathrm{X}'=SXS^{-1},\quad S\in\widetilde{SO}_{\circ}(s,n-s).$	$\mathrm{X}-\mathrm{X}'=SXS^{-1},\quad S\in\widetilde{SO}_{\circ}(s,n-s).$
mielke-geometrodynamics EQ0743 (11.1.10)	239	■ 1.000 ● N +0	$\sigma^{\alpha\beta}:=\frac{i}{4}\left[\gamma^{\alpha},\gamma^{\beta}\right],\quad\Lambda\left(\sigma^{\alpha\beta}\right)=L^{\alpha\beta},$	$\sigma^{\alpha\beta}:=\frac{i}{4}\left[\gamma^{\alpha},\gamma^{\beta}\right],\quad\Lambda\left(\sigma^{\alpha\beta}\right)=L^{\alpha\beta},$	$\sigma^{\alpha\beta}:=\frac{i}{4}\left[\gamma^{\alpha},\gamma^{\beta}\right],\quad\Lambda\left(\sigma^{\alpha\beta}\right)=L^{\alpha\beta},$
mielke-geometrodynamics EQ0744 (11.1.11)	239	■ 1.000 ● N +0	$L_{\circ}^g(M):=P\left(M,SO_{\circ}(s,n-s),\pi,\delta\right)$	$L_{\circ}^g(M):=P\left(M,SO_{\circ}(s,n-s),\pi,\delta\right)$	$L_{\circ}^g(M):=P\left(M,SO_{\circ}(s,n-s),\pi,\delta\right)$
mielke-geometrodynamics EQ0745 (11.1.12)	239	■ 1.000 ● N +0	$\widetilde{L}(M):=P\left(M,\widetilde{SO}_{\circ}(s,n-2),\tilde{\pi},\tilde{\delta}\right)\ni\tilde{p}$	$\widetilde{L}(M):=P\left(M,\widetilde{SO}_{\circ}(s,n-2),\tilde{\pi},\tilde{\delta}\right)\ni\tilde{p}$	$\widetilde{L}(M):=P\left(M,\widetilde{SO}_{\circ}(s,n-2),\tilde{\pi},\tilde{\delta}\right)\ni\tilde{p}$
mielke-geometrodynamics EQ0746 (11.1.13)	239	■ 1.000 ● N +0	$f(\tilde{p}S)=f(\tilde{p})\Lambda(S),S\in\widetilde{SO}_{\circ}(s,n-2).$	$f(\tilde{p}S)=f(\tilde{p})\Lambda(S),S\in\widetilde{SO}_{\circ}(s,n-2).$	$f(\tilde{p}S)=f(\tilde{p})\Lambda(S),S\in\widetilde{SO}_{\circ}(s,n-2).$
mielke-geometrodynamics EQ0748 (11.1.15)	240	■ 1.000 ● N +0	$\tilde{L}\left(M^4\right)=M^4\times SL(2,\mathbb{C})$	$\tilde{L}\left(M^4\right)=M^4\times SL(2,\mathbb{C})$	$\tilde{L}\left(M^4\right)=M^4\times SL(2,\mathbb{C})$
mielke-geometrodynamics EQ0749 (11.1.16)	241	■ 1.000 ● N +0	$V^{\bar{s}}:=V\left(M,\mathbb{C}^N,\rho^{\bar{s}}\left(\widetilde{SO}_{\circ}(s,n-s)\right)\subset GL(N,\mathbb{C}),\tilde{L}(M)\right)$	$V^{\bar{s}}:=V\left(M,\mathbb{C}^N,\rho^{\bar{s}}\left(\widetilde{SO}_{\circ}(s,n-s)\right)\subset GL(N,\mathbb{C}),\tilde{L}(M)\right)$	$V^{\bar{s}}:=V\left(M,\mathbb{C}^N,\rho^{\bar{s}}\left(\widetilde{SO}_{\circ}(s,n-s)\right)\subset GL(N,\mathbb{C}),\tilde{L}(M)\right)$
mielke-geometrodynamics EQ0751 (11.1.18)	241	■ 1.000 ● K +0	$D^{\frac{1}{2}}:=D^{(-\frac{1}{2},\frac{3}{2})}\oplus D^{(\frac{1}{2},\frac{3}{2})}\left(\widetilde{SO}_{\circ}(1,3)\right)$	$D^{\frac{1}{2}}:=D^{(-\frac{1}{2},\frac{3}{2})}\oplus D^{(\frac{1}{2},\frac{3}{2})}\left(\widetilde{SO}_{\circ}(1,3)\right)$	$D^{\frac{1}{2}}:=D^{(-\frac{1}{2},\frac{3}{2})}\oplus D^{(\frac{1}{2},\frac{3}{2})}\left(\widetilde{SO}_{\circ}(1,3)\right)$
mielke-geometrodynamics EQ0754 (11.1.21)	241	■ 1.000 ● N +0	$\psi\rightarrow^{S^{-1}}\psi:=S^{-1}\psi$	$\psi\rightarrow^{S^{-1}}\psi:=S^{-1}\psi$	$\psi\rightarrow^{S^{-1}}\psi:=S^{-1}\psi$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00755 (11.1.22)	242	■ 1.000 ● W +0	$\begin{gathered} S(p) \in \mathscr{G}_{\{S\}} \approx C^{\infty} \left(\widetilde{L}(M) \times_{\mathrm{Ad}} \widetilde{SO}_{\circ}(s, n-s) \right) \\ \Lambda \left(S_{(p)} \right) = G(p) \in \mathscr{G}_p \end{gathered}$	$S(p) \in \mathscr{G}_S \approx C^{\infty} \left(\tilde{L}(M) \times_{\mathrm{Ad}} \widetilde{SO}_{\circ}(s, n-s) \right)$ $\Lambda \left(S_{(p)} \right) = G(p) \in \mathscr{G}_p$	$S(p) \in \mathscr{G}_S \approx C^{\infty}(\tilde{L}(M) \times_{\mathrm{Ad}} \widetilde{SO}_{\circ}(s, n-s))$ $\Lambda(S_{(p)}) = G(p) \in \mathscr{G}_p$
mielke-geometrodynamics_E00757 (11.1.24)	242	■ 1.000 ● N +0	$S^{-1}(p) \, \gamma_{\alpha} \, S(p) = \gamma_{\beta} G_{\alpha}{}^{\beta}(p), \quad [G_{\alpha}{}^{\beta}(p)] \in \mathscr{G}_p.$	$S^{-1}(p) \gamma_{\alpha} S(p) = \gamma_{\beta} G_{\alpha}{}^{\beta}(p), \quad [G_{\alpha}{}^{\beta}(p)] \in \mathscr{G}_p.$	$S^{-1}(p) \gamma_{\alpha} S(p) = \gamma_{\beta} G_{\alpha}{}^{\beta}(p), \quad [G_{\alpha}{}^{\beta}(p)] \in \mathscr{G}_p.$
mielke-geometrodynamics_E00762 (11.1.29)	243	■ 1.000 ● N +0	$\bar{\psi} := \psi^+ \gamma^0,$	$\bar{\psi} := \psi^+ \gamma^0,$	$\bar{\psi} := \psi^+ \gamma^0,$
mielke-geometrodynamics_E00763 (11.1.30)	243	■ 1.000 ● W -2	$\bar{\sigma}^{\alpha\beta} := \gamma^0 \sigma^{\alpha\beta} \gamma^0 = \sigma^{\alpha\beta}, \quad \bar{\gamma}^{\alpha} := \gamma^0 \gamma^{\alpha} \gamma^0 = \gamma^{\alpha}.$	$\bar{\sigma}^{\alpha\beta} := \gamma^0 \sigma^{\alpha\beta} \gamma^0 = \sigma^{\alpha\beta}, \quad \bar{\gamma}^{\alpha} := \gamma^0 \gamma^{\alpha} \gamma^0 = \gamma^{\alpha}.$	$\bar{\sigma}^{\alpha\beta} := \gamma^0 \sigma^{\alpha\beta} \gamma^0 = \sigma^{\alpha\beta}, \quad \bar{\gamma}^{\alpha} := \gamma^0 \gamma^{\alpha} \gamma^0 = \gamma^{\alpha}.$
mielke-geometrodynamics_E00764 (11.1.31)	243	■ 1.000 ● N +0	$D \, \psi = d\psi + i \widetilde{\omega} \psi,$	$D\psi = d\psi + i \widetilde{\omega} \psi,$	$D\psi = d\psi + i \widetilde{\omega} \psi,$
mielke-geometrodynamics_E00765 (11.1.32)	243	■ 1.000 ● N +0	$\Lambda(i \widetilde{\omega}) = \omega^g$	$\Lambda(i \widetilde{\omega}) = \omega^g$	$\Lambda(i \widetilde{\omega}) = \omega^g$
mielke-geometrodynamics_E00771 (11.1.38)	244	■ 1.000 ● K +0	$\gamma \longrightarrow {}^{S^{-1}} \gamma = S^{-1} \gamma S$	$\gamma \longrightarrow {}^{S^{-1}} \gamma = S^{-1} \gamma S$	$\gamma \longrightarrow {}^{S^{-1}} \gamma = S^{-1} \gamma S$
mielke-geometrodynamics_E00775 (11.2.4)	246	■ 1.000 ● K +0	$\widetilde{\omega} = \gamma^0 \widetilde{\omega}^+ \gamma^0 = \widetilde{\omega}$	$\widetilde{\omega} = \gamma^0 \widetilde{\omega}^+ \gamma^0 = \widetilde{\omega}$	$\widetilde{\omega} = \gamma^0 \widetilde{\omega}^+ \gamma^0 = \widetilde{\omega}$
mielke-geometrodynamics_E00777 (11.2.6)	246	■ 1.000 ● N +0	$\tau_s = i \left(\bar{\psi}^* \gamma \psi \right).$	$\tau_s = i \left(\bar{\psi}^* \gamma \psi \right).$	$\tau_s = i (\bar{\psi}^* \gamma \psi).$
mielke-geometrodynamics_E00779 (11.2.7)	247	■ 1.000 ● K +0	$d\tau_s = 0$	$d\tau_s = 0$	$d\tau_s = 0$
mielke-geometrodynamics_E00781 (11.2.9)	247	■ 1.000 ● N +1	$(\psi, \chi) = i \int_{H^{n-1}} \psi^+ \chi d^{n-1} x$	$(\psi, \chi) = i \int_{H^{n-1}} \psi^+ \chi d^{n-1} x$	$(\psi, \chi) = i \int_{H^{n-1}} \psi^+ \chi d^{n-1} x.$
mielke-geometrodynamics_E00782 (11.3.1)	247	■ 1.000 ● N +0	$\widetilde{\omega} = \widetilde{\omega}^{\{\}} - \widetilde{K}, \quad \Lambda(i \widetilde{K}) = K$	$\widetilde{\omega} = \widetilde{\omega}^{\{\}} - \widetilde{K}, \quad \Lambda(i \widetilde{K}) = K$	$\widetilde{\omega} = \widetilde{\omega}^{\{\}} - \widetilde{K}, \quad \Lambda(i \widetilde{K}) = K$
mielke-geometrodynamics_E00785 (11.3.4)	248	■ 1.000 ● N +0	$[\Theta, \vartheta] = \ell^{*2} \tau_s.$	$[\Theta, \vartheta] = \ell^{*2} \tau_s.$	$[\Theta, \vartheta] = \ell^{*2} \tau_s.$
mielke-geometrodynamics_E00790 (11.4.2)	252	■ 1.000 ● S +0	$\begin{aligned} & \left(\bar{\psi} i \gamma^5 \gamma_{\mu} \psi \right) \left(\bar{\psi} i \gamma^5 \gamma^{\mu} \psi \right) = \left(\bar{\psi} \gamma_{\mu} \psi \right) \left(\bar{\psi} \gamma^{\mu} \psi \right) \\ & \hspace{10em} = \left(\bar{\psi} \psi \right)^2 - \left(\bar{\psi} \gamma^5 \psi \right)^2. \end{aligned}$	$\begin{aligned} & (\bar{\psi} i \gamma^5 \gamma_{\mu} \psi) (\bar{\psi} i \gamma^5 \gamma^{\mu} \psi) = (\bar{\psi} \gamma_{\mu} \psi) (\bar{\psi} \gamma^{\mu} \psi) \\ & \hspace{10em} = (\bar{\psi} \psi)^2 - (\bar{\psi} \gamma^5 \psi)^2. \end{aligned}$	$(\bar{\psi} i \gamma^5 \gamma_{\mu} \psi) (\bar{\psi} i \gamma^5 \gamma^{\mu} \psi) = (\bar{\psi} \gamma_{\mu} \psi) (\bar{\psi} \gamma^{\mu} \psi)$ $= (\bar{\psi} \psi)^2 - (\bar{\psi} \gamma^5 \psi)^2.$
mielke-geometrodynamics_E00793 (11.4.5)	253	■ 1.000 ● N +0	$\rho := \frac{\sqrt{2\pi}}{\ell^*} r,$	$\rho := \frac{\sqrt{2\pi}}{\ell^*} r,$	$\rho := \frac{\sqrt{2\pi}}{\ell^*} r,$
mielke-geometrodynamics_E00794 (11.4.6)	253	■ 1.000 ● N +0	$\bar{g}_{kl} \longleftarrow g_{kl} = e^{\mu} \bar{g}_{kl}.$	$\bar{g}_{kl} \longleftarrow g_{kl} = e^{\mu} \bar{g}_{kl}.$	$\bar{g}_{kl} \longleftarrow g_{kl} = e^{\mu} \bar{g}_{kl}.$
mielke-geometrodynamics_E00795 (11.4.7)	253	■ 1.000 ● N -2	$\bar{E}_{\cdot k}^{\alpha} \longleftarrow E_{\cdot k}^{\alpha} = e^{\mu/2} \bar{E}_{\cdot k}^{\alpha}$	$\bar{E}_{\cdot k}^{\alpha} \longleftarrow E_{\cdot k}^{\alpha} = e^{\mu/2} \bar{E}_{\cdot k}^{\alpha}$	$\bar{E}_{\cdot k}^{\alpha} \longleftarrow E_{\cdot k}^{\alpha} = e^{\mu/2} \bar{E}_{\cdot k}^{\alpha}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_253 EQ0796 (11.4.8)	253	■ 1.000 ● N +0	$\left.\begin{array}{c} \alpha \\\ k l \end{array}\right\}=\overline{\left.\begin{array}{c} \alpha \\\ k l \end{array}\right\}}+\frac{1}{2}\left(\delta_k^{\alpha} \partial_l+\delta_l^{\alpha} \partial_k-g_{k l} \partial^{\alpha}\right) \mu$	$\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}=\overline{\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}}+\frac{1}{2}\left(\delta_k^{\alpha} \partial_l+\delta_l^{\alpha} \partial_k-g_{k l} \partial^{\alpha}\right) \mu$	$\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}=\overline{\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}}+\frac{1}{2}\left(\delta_k^{\alpha} \partial_l+\delta_l^{\alpha} \partial_k-g_{k l} \partial^{\alpha}\right) \mu$
mielke-geometrodynamics_253 EQ0797 (11.4.9)	253	■ 1.000 ● K +0	$\Gamma_{k}=\bar{\Gamma}_{k}-\frac{1}{4} \delta_k^{[\alpha} \partial^{\beta]} \mu \sigma_{\alpha \beta} .$	$\Gamma_k=\bar{\Gamma}_k-\frac{1}{4} \delta_k^{[\alpha} \partial^{\beta]} \mu \sigma_{\alpha \beta} .$	$\Gamma_k=\bar{\Gamma}_k-\frac{1}{4} \delta_k^{[\alpha} \partial^{\beta]} \mu \sigma_{\alpha \beta} .$
mielke-geometrodynamics_254 EQ0801 (11.4.13)	254	■ 1.000 ● N +1	$\begin{aligned} & \& \bar{\Gamma}_0=\frac{i}{4} e^{(\nu-\mu) / 2} \partial_r(\nu-\mu) \gamma_0 \gamma_1 \\ & \& \bar{\Gamma}_1=0, \quad \bar{\Gamma}_2=\frac{i}{2} \gamma_2 \gamma_1 \\ & \& \bar{\Gamma}_3=\frac{i}{2}(\sin \vartheta \gamma_3 \gamma_1+\cos \vartheta \gamma_3 \gamma_2) \end{aligned}$	$\begin{aligned} & \bar{\Gamma}_0=\frac{i}{4} e^{(\nu-\mu) / 2} \partial_r(\nu-\mu) \gamma_0 \gamma_1 \\ & \bar{\Gamma}_1=0, \quad \bar{\Gamma}_2=\frac{i}{2} \gamma_2 \gamma_1 \\ & \bar{\Gamma}_3=\frac{i}{2}(\sin \vartheta \gamma_3 \gamma_1+\cos \vartheta \gamma_3 \gamma_2) \end{aligned}$	$\begin{aligned} \bar{T}_0 &= \frac{i}{4} e^{(v-\mu) / 2} \partial_r(v-\mu) \gamma_0 \gamma_1 \\ \bar{T}_1 &= 0, \quad \bar{T}_2 = \frac{i}{2} \gamma_2 \gamma_1 \\ \bar{T}_3 &= \frac{i}{2}(\sin \vartheta \gamma_3 \gamma_1 + \cos \vartheta \gamma_3 \gamma_2) \end{aligned}$
mielke-geometrodynamics_255 EQ0804 (11.4.16)	255	■ 1.000 ● N +0	$\chi_{\kappa}^m=\sum_{\bar{m}} \bar{\chi}_{\kappa}^m=\pm 1 / 2 \mathrm{C}\left(\frac{1}{2} j ; m-\bar{m}, \bar{m}\right) Y_l^{m-\bar{m}}(\vartheta, \phi) \chi^{\bar{m}} .$	$\chi_{\kappa}^m=\sum_{\bar{m}=\pm 1 / 2} C\left(l \frac{1}{2} j ; m-\bar{m}, \bar{m}\right) Y_l^{m-\bar{m}}(\vartheta, \phi) \chi^{\bar{m}} .$	$\chi_{\kappa}^m=\sum_{\bar{m}=\pm 1 / 2} C\left(l \frac{1}{2} j ; m-\bar{m}, \bar{m}\right) Y_l^{m-\bar{m}}(\vartheta, \phi) \chi^{\bar{m}} .$
mielke-geometrodynamics_255 EQ0808 (11.4.20)	255	■ 1.000 ● W +0	$\frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \chi_{\kappa}^m=-\chi_{-\kappa}^m$	$\frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \chi_{\kappa}^m=-\chi_{-\kappa}^m$	$\frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \chi_{\kappa}^m=-\chi_{-\kappa}^m$
mielke-geometrodynamics_257 EQ0814 (11.4.27)	257	■ 1.000 ● W +1	$\beta=\frac{M^*}{2 m}, \quad M^*:=\frac{\sqrt{8 \pi} \hbar}{c \ell^*} .$	$\beta=\frac{M^*}{2 m}, \quad M^*:=\frac{\sqrt{8 \pi} \hbar}{c \ell^*} .$	$\beta=\frac{M^*}{2 m}, \quad M^*:=\frac{\sqrt{8 \pi} \hbar}{c \ell^*} .$
mielke-geometrodynamics_257 EQ0816 (11.4.30)	257	■ 1.000 ● K +0	$F^{\prime \prime}+\frac{2}{\rho} F^{\prime}-\frac{2}{\rho^2} F \simeq \frac{1-\omega^2}{\beta^2} F$	$F^{\prime \prime}+\frac{2}{\rho} F^{\prime}-\frac{2}{\rho^2} F \simeq \frac{1-\omega^2}{\beta^2} F$	$F^{\prime \prime}+\frac{2}{\rho} F^{\prime}-\frac{2}{\rho^2} F \simeq \frac{1-\omega^2}{\beta^2} F$
mielke-geometrodynamics_257 EQ0817 (11.4.31)	257	■ 1.000 ● N +0	$F \sim C_{\infty}\left(\frac{1}{\bar{\rho}}+\frac{1}{\bar{\rho}^2}\right) e^{-\bar{\rho}}$	$F \sim C_{\infty}\left(\frac{1}{\bar{\rho}}+\frac{1}{\bar{\rho}^2}\right) e^{-\bar{\rho}}$	$F \sim C_{\infty}\left(\frac{1}{\bar{\rho}}+\frac{1}{\bar{\rho}^2}\right) e^{-\bar{\rho}}$
mielke-geometrodynamics_257 EQ0818 (11.4.32)	257	■ 1.000 ● N +0	$H \sim C_{\infty} \sqrt{\frac{1+\omega}{1-\omega}} \frac{1}{\bar{\rho}} e^{-\bar{\rho}}$	$H \sim C_{\infty} \sqrt{\frac{1+\omega}{1-\omega}} \frac{1}{\bar{\rho}} e^{-\bar{\rho}}$	$H \sim C_{\infty} \sqrt{\frac{1+\omega}{1-\omega}} \frac{1}{\bar{\rho}} e^{-\bar{\rho}}$
mielke-geometrodynamics_257 EQ0819 (11.4.33)	257	■ 1.000 ● N +0	$\bar{\rho}:=\frac{1}{\beta} \sqrt{1-\omega^2} \rho$	$\bar{\rho}:=\frac{1}{\beta} \sqrt{1-\omega^2} \rho$	$\bar{\rho}:=\frac{1}{\beta} \sqrt{1-\omega^2} \rho$
mielke-geometrodynamics_258 EQ0821 (11.4.35)	258	■ 1.000 ● N +0	$F(\rho) \simeq C_0 f^1(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$F(\rho) \simeq C_0 f^1(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$F(\rho) \simeq C_0 f^1(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$
mielke-geometrodynamics_258 EQ0822 (11.4.36)	258	■ 1.000 ● N +0	$H(\rho) \simeq C_0 \sqrt{\frac{1+\omega}{1-\omega}} f^0(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$H(\rho) \simeq C_0 \sqrt{\frac{1+\omega}{1-\omega}} f^0(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$H(\rho) \simeq C_0 \sqrt{\frac{1+\omega}{1-\omega}} f^0(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_258 EQ0823 (11.4.37)	258	■ 1.000 ● K +0	$C_{\{0\}}=(1-\omega)\sqrt{\frac{-2}{\varepsilon(\omega+1)}}$	$C_0=(1-\omega)\sqrt{\frac{-2}{\varepsilon(\omega+1)}}$	$C_0=(1-\omega)\sqrt{\frac{-2}{\varepsilon(\omega+1)}}$
mielke-geometrodynamics_259 EQ0826 (11.4.40)	259	■ 1.000 ● N +0	$m_{\{s\}}=\frac{1}{c}\int_{H^3}\Sigma_{\{\mathrm{D}\}}^0$	$m_s=\frac{1}{c}\int_{H^3}\Sigma_{\mathrm{D}}^0$	$m_s=\frac{1}{c}\int_{H^3}\Sigma_{\mathrm{D}}^0$
mielke-geometrodynamics_262 EQ0827 (11.5.1)	262	■ 1.000 ● N +1	$Q:=i\,e\int_{H^3}\bar{\psi}^*\gamma\psi=k\,e,\quad k=0,1,2,\ldots$	$Q:=ie\int_{H^3}\bar{\psi}^*\gamma\psi=ke,\quad k=0,1,2,\ldots$	$Q:=ie\int_{H^3}\bar{\psi}^*\gamma\psi=ke,\quad k=0,1,2,\ldots$
mielke-geometrodynamics_263 EQ0828 (11.5.2)	263	■ 1.000 ● N +0	$i\,\gamma\wedge^*d\psi(x)+e^2\int_{y\in M}\bar{\psi}(y)\gamma\psi(y)D_{\mathrm{F}}(y-x)\wedge^*\gamma\psi(x)=m\psi(x)\eta$	$i\gamma\wedge^*d\psi(x)+e^2\int_{y\in M}\bar{\psi}(y)\gamma\psi(y)D_{\mathrm{F}}(y-x)\wedge^*\gamma\psi(x)=m\psi(x)\eta$	$i\gamma\wedge^*d\psi(x)+e^2\int_{y\in M}\bar{\psi}(y)\gamma\psi(y)D_{\mathrm{F}}(y-x)\wedge^*\gamma\psi(x)=m\psi(x)\eta$
mielke-geometrodynamics_269 EQ0830 (12.1.2)	269	■ 1.000 ● N +0	$\left\langle dj_5(x)\right\rangle =2im\left\langle P\right\rangle -\left(1/96\pi^2\right)F\wedge F$	$\langle dj_5(x)\rangle=2im\langle P\rangle-(1/96\pi^2)F\wedge F$	$\langle dj_5(x)\rangle=2im\langle P\rangle-(1/96\pi^2)F\wedge F$
mielke-geometrodynamics_271 EQ0838 (12.2.6)	271	■ 1.000 ● N +0	$d\,j_5=2\,im\,P=2\,im\,\bar{\psi}\gamma_5\psi\eta$	$dj_5=2imP=2im\bar{\psi}\gamma_5\psi\eta$	$dj_5=2imP=2im\bar{\psi}\gamma_5\psi\eta$
mielke-geometrodynamics_272 EQ0843 (12.2.11)	272	■ 1.000 ● N +0	$L_{a\psi\psi}=\frac{1}{2}d\theta\wedge j_5=\frac{1}{2f_a}da\wedge\bar{\psi}^*\gamma\gamma_5\psi,$	$L_{a\psi\psi}=\frac{1}{2}d\theta\wedge j_5=\frac{1}{2f_a}da\wedge\bar{\psi}^*\gamma\gamma_5\psi,$	$L_{a\psi\psi}=\frac{1}{2}d\theta\wedge j_5=\frac{1}{2f_a}da\wedge\bar{\psi}^*\gamma\gamma_5\psi,$
mielke-geometrodynamics_273 EQ0846 (12.2.14)	273	■ 1.000 ● N -1	$d\,j_5\cong4\,dC_{\mathrm{TT}}=\frac{2}{\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right)$	$dj_5\cong4dC_{\mathrm{TT}}=\frac{2}{\ell^2}(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta)$	$dj_5\cong4dC_{\mathrm{TT}}=\frac{2}{\ell^2}(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta)$
mielke-geometrodynamics_275 EQ0849 (12.3.2)	275	■ 1.000 ● N +0	$K\left(t,\,x,\,\not{D}^2\right)=\left(4\pi\right)^{-n/2}\sum_{k=0}^{\infty}t^{\left(k-n\right)/2}K_k\left(x,\,\not{D}^2\right)$	$K\left(t,x,\not{D}^2\right)=(4\pi)^{-n/2}\sum_{k=0}^{\infty}t^{(k-n)/2}K_k\left(x,\not{D}^2\right)$	$K(t,x,\not{D}^2)=(4\pi)^{-n/2}\sum_{k=0}^{\infty}t^{(k-n)/2}K_k(x,\not{D}^2)$
mielke-geometrodynamics_276 EQ0852 (12.3.5)	276	■ 1.000 ● N -1	$-K_2/t=\left(2\ell^2/t\right)dC_{\mathrm{TT}}\cong\left(\ell^2/2t\right)dj_5=\left(im\ell^2/t\right)\bar{\psi}\gamma_5\psi\sim\ell^2M^2m^4\rightarrow0.$	$-K_2/t=(2\ell^2/t)dC_{\mathrm{TT}}\cong(\ell^2/2t)dj_5=(im\ell^2/t)\bar{\psi}\gamma_5\psi\sim\ell^2M^2m^4\rightarrow0.$	$-K_2/t=(2\ell^2/t)dC_{\mathrm{TT}}\cong(\ell^2/2t)dj_5=(im\ell^2/t)\bar{\psi}\gamma_5\psi\sim\ell^2M^2m^4\rightarrow0.$
mielke-geometrodynamics_277 EQ0856 (12.4.3)	277	■ 1.000 ● N +0	$\mathscr{G}^A\cong0,\quad A=1,2,3$	$\mathscr{G}^A\cong0,\quad A=1,2,3$	$\mathscr{G}^A\cong0,\quad A=1,2,3$
mielke-geometrodynamics_278 EQ0858 (12.4.5)	278	■ 1.000 ● K +0	$\gamma:=\gamma_\alpha\vartheta^\alpha,\quad{}^*\gamma=\gamma^\alpha\eta_\alpha.$	$\gamma:=\gamma_\alpha\vartheta^\alpha,\quad{}^*\gamma=\gamma^\alpha\eta_\alpha.$	$\gamma:=\gamma_\alpha\vartheta^\alpha,\quad{}^*\gamma=\gamma^\alpha\eta_\alpha.$
mielke-geometrodynamics_278 EQ0859 (12.4.6)	278	■ 1.000 ● N +0	$\Theta:=D\gamma=T^\alpha\gamma_\alpha,\quad\Omega:=d\Gamma+\Gamma\wedge\Gamma=\frac{i}{4}R^{\alpha\beta}\sigma_{\alpha\beta}$	$\Theta:=D\gamma=T^\alpha\gamma_\alpha,\quad\Omega:=d\Gamma+\Gamma\wedge\Gamma=\frac{i}{4}R^{\alpha\beta}\sigma_{\alpha\beta}$	$\Theta:=D\gamma=T^\alpha\gamma_\alpha,\quad\Omega:=d\Gamma+\Gamma\wedge\Gamma=\frac{i}{4}R^{\alpha\beta}\sigma_{\alpha\beta}$
mielke-geometrodynamics_278 EQ0860 (12.4.7)	278	■ 1.000 ● N -1	$C_{\mathrm{RR}}:=-\operatorname{Tr}\left(\Gamma\wedge\Omega-\frac{1}{3}\Gamma\wedge\Gamma\wedge\Gamma\right).$	$C_{\mathrm{RR}}:=-\operatorname{Tr}\left(\Gamma\wedge\Omega-\frac{1}{3}\Gamma\wedge\Gamma\wedge\Gamma\right).$	$C_{\mathrm{RR}}:=-\operatorname{Tr}\left(\Gamma\wedge\Omega-\frac{1}{3}\Gamma\wedge\Gamma\wedge\Gamma\right).$
mielke-geometrodynamics_279 EQ0862 (12.4.9)	279	■ 1.000 ● N +1	$C_{\mathrm{TT}}:=\frac{1}{8\ell^2}\operatorname{Tr}(\gamma\wedge\Theta)=\frac{1}{2\ell^2}\vartheta^\alpha\wedge T_\alpha.$	$C_{\mathrm{TT}}:=\frac{1}{8\ell^2}\operatorname{Tr}(\gamma\wedge\Theta)=\frac{1}{2\ell^2}\vartheta^\alpha\wedge T_\alpha.$	$C_{\mathrm{TT}}:=\frac{1}{8\ell^2}\operatorname{Tr}(\gamma\wedge\Theta)=\frac{1}{2\ell^2}\vartheta^\alpha\wedge T_\alpha.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E08863 (12.4.10)	279	■ 1.000 ● C +5	$d\,C_{\mathrm{TT}}=\frac{1}{8}\ell^2\operatorname{Tr}(\Theta\wedge\Theta-4i\Omega\wedge\sigma)=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right).$	$dC_{\mathrm{TT}}=\frac{1}{8\ell^2}\operatorname{Tr}(\Theta\wedge\Theta-4i\Omega\wedge\sigma)=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right).$	$dC_{\mathrm{TT}}=\frac{1}{8\ell^2}\operatorname{Tr}(\Theta\wedge\Theta-4i\Omega\wedge\sigma)=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right).$
mielke-geometrodynamics_E08864 (12.4.11)	279	■ 1.000 ● N -2	$\hat{C}_{\mathrm{C}}_{\mathrm{RR}}=C_{\mathrm{RR}}-2\,C_{\mathrm{TT}}$	$\hat{C}_{\mathrm{RR}}=C_{\mathrm{RR}}-2C_{\mathrm{TT}}$	$\hat{C}_{\mathrm{RR}}=C_{\mathrm{RR}}-2C_{\mathrm{TT}}$
mielke-geometrodynamics_E08865 (13.2.1)	283	■ 1.000 ● N -1	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$
mielke-geometrodynamics_E08866 (13.2.2)	283	■ 1.000 ● N +1	$\tilde{L}_{\mathrm{BF}}:=-B\wedge dA+\frac{1}{2}B\wedge B\cong-B\wedge dA+dC$	$\tilde{L}_{\mathrm{BF}}:=-B\wedge dA+\frac{1}{2}B\wedge B\cong-B\wedge dA+dC$	$\tilde{L}_{\mathrm{BF}}:=-B\wedge dA+\frac{1}{2}B\wedge B\cong-B\wedge dA+dC.$
mielke-geometrodynamics_E08868 (13.2.4)	283	■ 1.000 ● N +0	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$
mielke-geometrodynamics_E08869 (13.2.5)	283	■ 1.000 ● K +0	$A\rightarrow A'=A+d\theta(x),$	$A\rightarrow A'=A+d\theta(x),$	$A\rightarrow A'=A+d\theta(x),$
mielke-geometrodynamics_E08870 (13.2.6)	284	■ 1.000 ● N +0	$A\rightarrow\widetilde{A}=A+\sigma,\quad B\rightarrow\widetilde{B}=B+d\sigma,$	$A\rightarrow\widetilde{A}=A+\sigma,\quad B\rightarrow\widetilde{B}=B+d\sigma,$	$A\rightarrow\widetilde{A}=A+\sigma,\quad B\rightarrow\widetilde{B}=B+d\sigma,$
mielke-geometrodynamics_E08871 (13.2.7)	284	■ 1.000 ● K +0	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$
mielke-geometrodynamics_E08872 (13.2.8)	284	■ 1.000 ● N +1	$-dB=d^*F\cong j,$	$-dB=d^*F\cong j,$	$-dB=d^*F\cong j,$
mielke-geometrodynamics_E08875 (13.3.3)	284	■ 1.000 ● N +0	$\left[L^{\alpha}{}_{\beta},L^{\gamma}{}_{\delta}\right]=\delta^{\alpha}_{\delta}L^{\gamma}{}_{\beta}-\delta^{\gamma}_{\beta}L^{\alpha}{}_{\delta},$	$[L^{\alpha}{}_{\beta},L^{\gamma}{}_{\delta}]=\delta^{\alpha}_{\delta}L^{\gamma}{}_{\beta}-\delta^{\gamma}_{\beta}L^{\alpha}{}_{\delta},$	$[L^{\alpha}{}_{\beta},L^{\gamma}{}_{\delta}]=\delta^{\alpha}_{\delta}L^{\gamma}{}_{\beta}-\delta^{\gamma}_{\beta}L^{\alpha}{}_{\delta},$
mielke-geometrodynamics_E08880 (13.4.1)	285	■ 1.000 ● N +0	$A(x):=\left(\begin{array}{cc}\Lambda(x)&\tau(x)\\ \tau_*(x)&1\end{array}\right),$	$A(x):=\left\{\begin{array}{cc}\Lambda(x)&\tau(x)\\ \tau_*(x)&1\end{array}\right\},$	$A(x):=\left\{\begin{array}{cc}\Lambda(x)&\tau(x)\\ \tau_*(x)&1\end{array}\right\},$
mielke-geometrodynamics_E08881 (13.4.2)	285	■ 1.000 ● N +0	$\hat{\Gamma}=\left(\begin{array}{cc}\Gamma_{\mathrm{*}}^{(L)}&\Gamma_{\mathrm{*}}^{(T)}\\ \Gamma_{\mathrm{*}}^{(T)}&0\end{array}\right)=\left(\begin{array}{cc}\Gamma_{\alpha}^{(L)\beta}L_{\beta}^{\alpha}&\Gamma^{(T)\alpha}\ell P_{\alpha}\\ \Gamma_{\beta}^{*(T)}&\ell P_{\mathrm{*}}^{\beta}\end{array}\right)$	$\hat{\Gamma}=\left(\begin{array}{cc}\Gamma_{\mathrm{*}}^{(L)}&\Gamma_{\mathrm{*}}^{(T)}\\ \Gamma_{\mathrm{*}}^{(T)}&0\end{array}\right)=\left(\begin{array}{cc}\Gamma_{\alpha}^{(L)\beta}L_{\beta}^{\alpha}&\Gamma^{(T)\alpha}\ell P_{\alpha}\\ \Gamma_{\beta}^{*(T)}&\ell P_{\mathrm{*}}^{\beta}\end{array}\right)$	$\hat{I}=\left(\begin{array}{cc}\Gamma_{\mathrm{*}}^{(L)}&\Gamma_{\mathrm{*}}^{(T)}\\ \Gamma_{\mathrm{*}}^{(T)}&0\end{array}\right)=\left(\begin{array}{cc}\Gamma_{\alpha}^{(L)\beta}L_{\beta}^{\alpha}&\Gamma^{(T)\alpha}\ell P_{\alpha}\\ \Gamma_{\beta}^{*(T)}&\ell P_{\mathrm{*}}^{\beta}\end{array}\right)$
mielke-geometrodynamics_E08882 (13.4.3)	286	■ 1.000 ● N +1	$\hat{\Gamma}\xrightarrow{A^{-1}(x)}\hat{\Gamma}'=A^{-1}(x)\hat{\Gamma}A(x)+A^{-1}(x)dA(x).$	$\hat{\Gamma}\xrightarrow{A^{-1}(x)}\hat{\Gamma}'=A^{-1}(x)\hat{\Gamma}A(x)+A^{-1}(x)dA(x).$	$\hat{I}\xrightarrow{A^{-1}(x)}\hat{I}'=A^{-1}(x)\hat{I}A(x)+A^{-1}(x)dA(x).$
mielke-geometrodynamics_E08883	286	■ 1.000 ● N +0	$\tau\tau_{\mathrm{*}}=-\Lambda\tau_{\mathrm{*}}\Lambda^{-1}\tau.$	$\tau\tau_{\mathrm{*}}=-\Lambda\tau_{\mathrm{*}}\Lambda^{-1}\tau.$	$\tau\tau_{\mathrm{*}}=-\Lambda\tau_{\mathrm{*}}\Lambda^{-1}\tau.$
mielke-geometrodynamics_E08884 (13.4.4)	286	■ 1.000 ● S +1	$\hat{R}:=d\hat{\Gamma}+\hat{\Gamma}\wedge\hat{\Gamma}=\left(\begin{array}{cc}R^{(L)}+\Gamma^{(L)}\wedge\Gamma_{\mathrm{*}}^{(L)}&d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}\\ d\Gamma_{\mathrm{*}}^{(T)}+\Gamma_{\mathrm{*}}^{(T)}\wedge\Gamma^{(L)}&0\end{array}\right)$	$\hat{R}:=d\hat{\Gamma}+\hat{\Gamma}\wedge\hat{\Gamma}=\left(\begin{array}{cc}R^{(L)}+\Gamma^{(L)}\wedge\Gamma_{\mathrm{*}}^{(L)}&d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}\\ d\Gamma_{\mathrm{*}}^{(T)}+\Gamma_{\mathrm{*}}^{(T)}\wedge\Gamma^{(L)}&0\end{array}\right),$	$\hat{R}:=d\hat{\Gamma}+\hat{\Gamma}\wedge\hat{\Gamma}=\left(\begin{array}{cc}R^{(L)}+\Gamma^{(L)}\wedge\Gamma_{\mathrm{*}}^{(L)}&d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}\\ d\Gamma_{\mathrm{*}}^{(T)}+\Gamma_{\mathrm{*}}^{(T)}\wedge\Gamma^{(L)}&0\end{array}\right),$
mielke-geometrodynamics_E08885	286	■ 1.000 ● K +0	$R^{(L)}:=d\Gamma^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}$	$R^{(L)}:=d\Gamma^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}$	$R^{(L)}:=d\Gamma^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ0886 (13.4.5)	286	■ 1.000 ● N +0	$R^{\{T\}}:=d\,\Gamma^{\{T\}}+\Gamma^{\{L\}}\wedge\Gamma^{\{T\}}=\left(T^{\beta}-R_{\alpha}^{\beta}\xi^{\alpha}\right)\ell P_{\beta}$ $\left(T^{\beta}-R_{\alpha}^{\beta}\xi^{\alpha}\right)\ell P_{\beta}$	$R^{(T)}:=d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}=\left(T^{\beta}-R_{\alpha}^{\beta}\xi^{\alpha}\right)\ell P_{\beta}$	$R^{(T)}:=d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}=\left(T^{\beta}-R_{\alpha}^{\beta}\xi^{\alpha}\right)\ell P_{\beta}$
mielke-geometrodynamics_EQ0887 (13.4.6)	286	■ 1.000 ● N +1	$\hat{R}\mapsto\hat{R}'=A^{-1}(x)\hat{R}A(x).$	$\hat{R}\xrightarrow{A^{-1}(x)}\hat{R}'=A^{-1}(x)\hat{R}A(x).$	$\hat{R}\xrightarrow{A^{-1}(x)}\hat{R}'=A^{-1}(x)\hat{R}A(x).$
mielke-geometrodynamics_EQ0888 (13.4.7)	286	■ 1.000 ● N +0	$\Gamma^{\{L\}}\mapsto\Gamma^{\{L\}'}=\Lambda^{-1}(x)\Gamma^{\{L\}}\Lambda(x)+\Lambda^{-1}(x)d\Lambda(x)$	$\Gamma^{(L)}\xrightarrow{A^{-1}(x)}\Gamma^{(L)'}=\Lambda^{-1}(x)\Gamma^{(L)}\Lambda(x)+\Lambda^{-1}(x)d\Lambda(x)$	$\Gamma^{(L)}\xrightarrow{A^{-1}(x)}\Gamma^{(L)'}=\Lambda^{-1}(x)\Gamma^{(L)}\Lambda(x)+\Lambda^{-1}(x)d\Lambda(x)$
mielke-geometrodynamics_EQ0891 (13.4.10)	287	■ 1.000 ● N +0	$\vartheta:=\Gamma^{\{T\}}+D\xi:=\vartheta^{\alpha}P_{\alpha},\quad\theta:=\Gamma_{*}^{\{T\}}+D\xi_{*}:=\theta_{\beta}P_{*}^{\beta},$	$\vartheta:=\Gamma^{(T)}+D\xi:=\vartheta^{\alpha}P_{\alpha},\quad\theta:=\Gamma_{*}^{\{T\}}+D\xi_{*}:=\theta_{\beta}P_{*}^{\beta},$	$\vartheta:=\Gamma^{(T)}+D\xi:=\vartheta^{\alpha}P_{\alpha},\quad\theta:=\Gamma_{*}^{\{T\}}+D\xi_{*}:=\theta_{\beta}P_{*}^{\beta},$
mielke-geometrodynamics_EQ0894 (13.4.13)	287	■ 1.000 ● N +0	$D\,\xi\stackrel{*}{=}0,\quad D\,\xi_{*}\stackrel{*}{=}0,$	$D\xi\stackrel{*}{=}0,\quad D\xi_{*}\stackrel{*}{=}0,$	$D\xi\stackrel{*}{=}0,\quad D\xi_{*}\stackrel{*}{=}0,$
mielke-geometrodynamics_EQ0895 (13.4.14)	287	■ 1.000 ● N +1	$\overline{\Gamma}=\left(\begin{array}{cc}\Gamma^{\{L\}}&\vartheta\\ \theta&0\end{array}\right),$	$\overline{\Gamma}=\left(\begin{array}{cc}\Gamma^{\{L\}}&\vartheta\\ \theta&0\end{array}\right),$	$\overline{\Gamma}=\left(\begin{array}{cc}\Gamma^{\{L\}}&\vartheta\\ \theta&0\end{array}\right),$
mielke-geometrodynamics_EQ0896 (13.5.1)	288	■ 1.000 ● N +0	$\hat{\Gamma}=\Gamma_{\alpha}^{\beta}L^{\alpha}_{\beta}+\vartheta^{\alpha}P_{\alpha}+\theta_{\beta}P_{*}^{\beta},$	$\hat{\Gamma}=\Gamma_{\alpha}^{\beta}L^{\alpha}_{\beta}+\vartheta^{\alpha}P_{\alpha}+\theta_{\beta}P_{*}^{\beta},$	$\hat{\Gamma}=\Gamma_{\alpha}^{\beta}L^{\alpha}_{\beta}+\vartheta^{\alpha}P_{\alpha}+\theta_{\beta}P_{*}^{\beta},$
mielke-geometrodynamics_EQ0899 (13.5.4)	288	■ 1.000 ● K +0	$\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}=-\mathrm{Tr}\{\hat{B}\wedge\hat{R}\}-d\hat{C}.$	$\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}=-\mathrm{Tr}\{\hat{B}\wedge\hat{R}\}-d\hat{C}.$	$\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}=-\mathrm{Tr}\{\hat{B}\wedge\hat{R}\}-d\hat{C}.$
mielke-geometrodynamics_EQ0900 (13.5.5)	289	■ 1.000 ● N +0	$\hat{D}B_A{}^B\cong\hat{D}\hat{R}_A{}^B\equiv0.$	$\hat{D}B_A{}^B\cong\hat{D}\hat{R}_A{}^B\equiv0.$	$\hat{D}B_A{}^B\cong\hat{D}\hat{R}_A{}^B\equiv0.$
mielke-geometrodynamics_EQ0901 (13.5.6)	289	■ 1.000 ● N -1	$\tilde{L}_{\mathrm{SSB}}=\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}+\frac{1}{2}\eta_{ABCDE}B^{AB}\wedge B^{CD}\Phi^E.$	$\tilde{L}_{\mathrm{SSB}}=\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}+\frac{1}{2}\eta_{ABCDE}B^{AB}\wedge B^{CD}\Phi^E.$	$\tilde{L}_{\mathrm{SSB}}=\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}+\frac{1}{2}\eta_{ABCDE}B^{AB}\wedge B^{CD}\Phi^E.$
mielke-geometrodynamics_EQ0902 (13.5.7)	289	■ 1.000 ● N +0	$\hat{g}_{AB}\Phi^A\Phi^B=\mu^2,\quad\Phi_E\hat{D}\Phi^E=0$	$\hat{g}_{AB}\Phi^A\Phi^B=\mu^2,\quad\Phi_E\hat{D}\Phi^E=0,$	$\hat{g}_{AB}\Phi^A\Phi^B=\mu^2,\quad\Phi_E\hat{D}\Phi^E=0,$
mielke-geometrodynamics_EQ0903 (13.5.8)	289	■ 1.000 ● K +0	$\left\langle\Phi^E\right\rangle=\Phi_0^E=(0,0,0,0,\mu)^T$	$\langle\Phi^E\rangle=\Phi_0^E=(0,0,0,0,\mu)^T$	$\langle\Phi^E\rangle=\Phi_0^E=(0,0,0,0,\mu)^T$
mielke-geometrodynamics_EQ0904 (13.5.9)	289	■ 1.000 ● K +0	$\tilde{L}_{\mathrm{SSB}}=-\mathrm{Tr}\{\hat{B}\wedge\hat{R}\}+\frac{\mu}{2}\eta_{\alpha\beta\gamma\delta}B^{\alpha\beta}\wedge B^{\gamma\delta}-dC_{\mathrm{RR}}+2dC_{\mathrm{TT}}$	$\tilde{L}_{\mathrm{SSB}}=-\mathrm{Tr}\{\hat{B}\wedge\hat{R}\}+\frac{\mu}{2}\eta_{\alpha\beta\gamma\delta}B^{\alpha\beta}\wedge B^{\gamma\delta}-dC_{\mathrm{RR}}+2dC_{\mathrm{TT}}$	$\tilde{L}_{\mathrm{SSB}}=-\mathrm{Tr}\{\hat{B}\wedge\hat{R}\}+\frac{\mu}{2}\eta_{\alpha\beta\gamma\delta}B^{\alpha\beta}\wedge B^{\gamma\delta}-dC_{\mathrm{RR}}+2dC_{\mathrm{TT}}$
mielke-geometrodynamics_EQ0905 (13.5.10)	290	■ 1.000 ● N +1	$d\,\hat{C}=d\,C_{\mathrm{RR}}-2\,d\,C_{\mathrm{TT}},$	$d\hat{C}=dC_{\mathrm{RR}}-2dC_{\mathrm{TT}},$	$d\hat{C}=dC_{\mathrm{RR}}-2dC_{\mathrm{TT}},$
mielke-geometrodynamics_EQ0906 (13.5.11)	290	■ 1.000 ● N +0	$B_{\alpha\beta}\cong\frac{1}{\mu}\eta_{\alpha\beta\gamma\delta}\left[R^{\gamma\delta}-\frac{\Lambda}{3}\theta^{\gamma}\wedge\vartheta^{\delta}\right]$	$B_{\alpha\beta}\cong\frac{1}{\mu}\eta_{\alpha\beta\gamma\delta}\left[R^{\gamma\delta}-\frac{\Lambda}{3}\theta^{\gamma}\wedge\vartheta^{\delta}\right]$	$B_{\alpha\beta}\cong\frac{1}{\mu}\eta_{\alpha\beta\gamma\delta}\left[R^{\gamma\delta}-\frac{\Lambda}{3}\theta^{\gamma}\wedge\vartheta^{\delta}\right]$
mielke-geometrodynamics_EQ0909 (13.5.14)	291	■ 1.000 ● N -1	$\tilde{L}_{\mathrm{SSB}}\cong-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\frac{\Lambda}{\kappa}\eta$	$\tilde{L}_{\mathrm{SSB}}\cong-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\frac{\Lambda}{\kappa}\eta$	$\tilde{L}_{\mathrm{SSB}}\cong-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\frac{\Lambda}{\kappa}\eta$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_E09910 (13.5.15)	291	■ 1.000 ● N +0	$\mu=\frac{4}{3}\kappa\Lambda=\frac{1}{3}(2\kappa)^2\rho_\Lambda$ ρ_{Λ} .	$\mu = \frac{4}{3}\kappa\Lambda = \frac{1}{3}(2\kappa)^2\rho_\Lambda.$	$\mu = \frac{4}{3}\kappa\Lambda = \frac{1}{3}(2\kappa)^2\rho_\Lambda.$
mielke-geometrodynamics_E09912 (13.7.1)	293	■ 1.000 ● N +0	$g_{\mathrm{N}}:=\kappa\,k^2$, $\quad\quad\quad\lambda:=\Lambda/k^2$,	$g_{\mathrm{N}} := \kappa k^2, \quad \lambda := \Lambda/k^2,$	$g_{\mathrm{N}} := \kappa k^2, \quad \lambda := \Lambda/k^2,$
mielke-geometrodynamics_E09913 (13.7.2)	293	■ 1.000 ● S +0	$k\,\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}}$, $\quad k\,\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda)$,	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}}, \quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}}, \quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$
mielke-geometrodynamics_E09914 (13.7.3)	293	■ 1.000 ● N +0	$\mu_*\leq\frac{4}{3}g_{\mathrm{N}*}\lambda_*\simeq0.2$	$\mu_*\leq\frac{4}{3}g_{\mathrm{N}*}\lambda_*\simeq0.2$	$\mu_*\leq\frac{4}{3}g_{\mathrm{N}*}\lambda_*\simeq0.2$
mielke-geometrodynamics_E09916 (13.8.2)	295	■ 1.000 ● N -1	$d\,C_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right)$	$dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right)$	$dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right)$
mielke-geometrodynamics_E09917 (13.8.3)	295	■ 1.000 ● K +0	$d\,C_{\mathrm{RR}}=-\frac{1}{2}R_\alpha{}^\beta\wedge R_\beta{}^\alpha$	$dC_{\mathrm{RR}}=-\frac{1}{2}R_\alpha{}^\beta\wedge R_\beta{}^\alpha$	$dC_{\mathrm{RR}}=-\frac{1}{2}R_\alpha{}^\beta\wedge R_\beta{}^\alpha$
mielke-geometrodynamics_E09918 (13.8.4)	295	■ 1.000 ● N +0	$D\,R_\alpha{}^\beta\equiv0$,	$DR_\alpha{}^\beta\equiv0,$	$DR_\alpha{}^\beta\equiv0,$
mielke-geometrodynamics_E09921 (13.8.7)	296	■ 1.000 ● N +1	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta}$,	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$
mielke-geometrodynamics_E09922 (14.1.1)	302	■ 1.000 ● N +0	$L=L_g+L_\omega+L_{\mathrm{mat}}$	$L=L_g+L_\omega+L_{\mathrm{mat}}$	$L=L_g+L_\omega+L_{\mathrm{mat}}$
mielke-geometrodynamics_E09923 (14.1.2)	302	■ 1.000 ● N +1	$L_{\omega-g}=\operatorname{Tr}\left\{\left(\vartheta\wedge\Omega^g\right)\wedge{}^*\left(\vartheta\wedge\Omega\right)\right\}$	$L_{\omega-g}=\operatorname{Tr}\left\{\wedge^*(\vartheta\wedge\Omega^g)\wedge^*(\vartheta\wedge\Omega)\right\}$	$L_{\omega-g}=\operatorname{Tr}\left\{\wedge^*(\vartheta\wedge\Omega^g)\wedge^*(\vartheta\wedge\Omega)\right\}$
mielke-geometrodynamics_E09924 (14.1.3)	303	■ 1.000 ● N -1	$L_{\mathrm{GMD.}}=\frac{1}{\ell^*2}L_{\mathrm{W.}}+L_{\mathrm{YM}}+L_{\mathrm{mat}}$	$L_{\mathrm{GMD.}}=\frac{1}{\ell^*2}L_{\mathrm{W.}}+L_{\mathrm{YM}}+L_{\mathrm{mat}}$	$L_{\mathrm{GMD.}}=\frac{1}{\ell^*2}L_{\mathrm{W.}}+L_{\mathrm{YM}}+L_{\mathrm{mat}}$
mielke-geometrodynamics_E09933 (14.1.14)	305	■ 1.000 ● N +0	$e^{*\pi/2}=(-1)^z*$, $\quad z:=\frac{s-1}{2}$.	$e^{*\pi/2}=(-1)^z*, \quad z:=\frac{s-1}{2}.$	$e^{*\pi/2}=(-1)^z*, \quad z:=\frac{s-1}{2}.$
mielke-geometrodynamics_E09934 (14.1.15)	305	■ 1.000 ● K +0	$\xi:=e^{*(-\delta)}\Omega$	$\xi:=e^{*(-\delta)}\Omega$	$\xi:=e^{*(-\delta)}\Omega$
mielke-geometrodynamics_E09935 (14.1.16)	305	■ 1.000 ● N +0	$\Omega\wedge{}^*\Omega\rightarrow\xi\wedge{}^*\xi=\Omega\wedge{}^*\Omega\cos2\delta-(-1)^z\Omega\wedge\Omega\sin2\delta$	$\Omega\wedge{}^*\Omega\rightarrow\xi\wedge{}^*\xi=\Omega\wedge{}^*\Omega\cos2\delta-(-1)^z\Omega\wedge\Omega\sin2\delta$	$\Omega\wedge{}^*\Omega\rightarrow\xi\wedge{}^*\xi=\Omega\wedge{}^*\Omega\cos2\delta-(-1)^z\Omega\wedge\Omega\sin2\delta$
mielke-geometrodynamics_E09936 (14.1.17)	305	■ 1.000 ● N +0	$\Omega\wedge\Omega\rightarrow\xi\wedge\xi=\Omega\wedge\Omega\cos2\delta-(-1)^z\Omega\wedge{}^*\Omega\sin2\delta.$	$\Omega\wedge\Omega\rightarrow\xi\wedge\xi=\Omega\wedge\Omega\cos2\delta-(-1)^z\Omega\wedge{}^*\Omega\sin2\delta.$	$\Omega\wedge\Omega\rightarrow\xi\wedge\xi=\Omega\wedge\Omega\cos2\delta-(-1)^z\Omega\wedge{}^*\Omega\sin2\delta.$
mielke-geometrodynamics_E09937 (14.1.18)	305	■ 1.000 ● N +0	$\vec{E}=\{E_j:=F_{j0}\mid j=1,2,3\}$	$\vec{E}=\{E_j:=F_{j0}\mid j=1,2,3\}$	$\vec{E}=\{E_j:=F_{j0}\mid j=1,2,3\}$
mielke-geometrodynamics_E09938 (14.1.19)	305	■ 1.000 ● N +0	$\vec{B}=\left\{B_j:=\frac{1}{2}\epsilon_{ijk}F^{jk}\right\}\Big _{i,j,k=1,2,3}$	$\vec{B}=\left\{B_j:=\frac{1}{2}\epsilon_{ijk}F^{jk}\Big _{i,j,k=1,2,3}\right\}$	$\vec{B}=\{B_j:=\frac{1}{2}\epsilon_{ijk}\,F^{jk} i,j,k=1,2,3\}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E09943 (14.1.24)	306	■ 1.000 ● N +1	$\Omega = e^{\delta} \xi$.	$\Omega = e^{*\delta} \xi$.	$\Omega = e^{*\delta} \xi$.
mielke-geometrodynamics_E09944 (14.1.25)	306	■ 1.000 ● N -1	$\Sigma_{\mathrm{YM}}(\Omega) = \Sigma_{\mathrm{YM}}(\xi)$	$\Sigma_{\mathrm{YM}}(\Omega) = \Sigma_{\mathrm{YM}}(\xi)$	$\Sigma_{\mathrm{YM}}(\Omega) = \Sigma_{\mathrm{YM}}(\xi)$
mielke-geometrodynamics_E09950 (14.1.31)	307	■ 1.000 ● K +0	$\tau_e = \bar{\tau}_e \cos \delta - (-1)^z \bar{\tau}_p \sin \delta$	$\tau_e = \bar{\tau}_e \cos \delta - (-1)^z \bar{\tau}_p \sin \delta$	$\tau_e = \bar{\tau}_e \cos \delta - (-1)^z \bar{\tau}_p \sin \delta$
mielke-geometrodynamics_E09952 (14.1.34)	307	■ 1.000 ● N +0	$D \delta = d \delta = 0,$	$D \delta = d \delta = 0,$	$D \delta = d \delta = 0,$
mielke-geometrodynamics_E09955 (14.1.37)	308	■ 1.000 ● N +0	$(-1)^z \Sigma_{\mathrm{YM}}(\xi) \wedge d \delta = D \Sigma_{\mathrm{YM}}(\xi).$	$(-1)^z \Sigma_{\mathrm{YM}}(\xi) \wedge d \delta = D \Sigma_{\mathrm{YM}}(\xi).$	$(-1)^z \Sigma_{\mathrm{YM}}(\xi) \wedge d \delta = D \Sigma_{\mathrm{YM}}(\xi).$
mielke-geometrodynamics_E09958 (14.1.40)	309	■ 1.000 ● N +1	$\delta = \int_{c_1} d \delta + \delta_0$	$\delta = \int_{c_1} d \delta + \delta_0$	$\delta = \int_{c_1} d \delta + \delta_0.$
mielke-geometrodynamics_E09969 (14.2.10)	314	■ 1.000 ● N +1	$\hat{\omega}^{AB} = -\hat{\omega}^{BA}, \quad A, B = 0, \dots, 3 + K$	$\hat{\omega}^{AB} = -\hat{\omega}^{BA}, \quad A, B = 0, \dots, 3 + K$	$\hat{\omega}^{AB} = -\hat{\omega}^{BA}, \quad A, B = 0, \dots, 3 + K$
mielke-geometrodynamics_E09975 (14.2.16)	315	■ 1.000 ● N +1	$\Omega^{\prime} := \pi \left(d \omega - \omega \wedge \overset{\circ}{\omega} - \omega^g \wedge \omega \right) = \Omega - \omega^g \wedge \omega.$	$\Omega^{\prime} := \pi \left(d \omega - \omega \wedge \overset{\circ}{\omega} - \omega^g \wedge \omega \right) = \Omega - \omega^g \wedge \omega.$	$\Omega^{\prime} := \pi (d \omega - \omega \wedge \overset{\circ}{\omega} - \omega^g \wedge \omega) = \Omega - \omega^g \wedge \omega.$
mielke-geometrodynamics_E09980 (14.2.21)	316	■ 1.000 ● N +0	$\hat{g}_{AB} \left(x, \xi^j \right) = \sum_{n=-\infty}^{+\infty} g_{AB}^{(n)} (x) Y_n \left(\xi^j \right)$	$\hat{g}_{AB} \left(x, \xi^j \right) = \sum_{n=-\infty}^{+\infty} g_{AB}^{(n)} (x) Y_n \left(\xi^j \right)$	$\hat{g}_{AB} (x, \xi^j) = \sum_{n=-\infty}^{+\infty} g_{AB}^{(n)} (x) Y_n (\xi^j)$
mielke-geometrodynamics_E09981 (14.2.22)	317	■ 1.000 ● N -1	$M^* = \sqrt{\hbar c / G_{\mathrm{N}}} \simeq 10^{19} \mathrm{GeV} / c^2.$	$M^* = \sqrt{\hbar c / G_{\mathrm{N}}} \simeq 10^{19} \mathrm{GeV} / c^2.$	$M^* = \sqrt{\hbar c / G_{\mathrm{N}}} \simeq 10^{19} \mathrm{GeV} / c^2.$
mielke-geometrodynamics_E09983 (14.2.24)	318	■ 1.000 ● K +0	$G = SU(3) \times SU(2) \times U(1)$	$G = SU(3) \times SU(2) \times U(1)$	$G = SU(3) \times SU(2) \times U(1)$
mielke-geometrodynamics_E09984 (14.2.25)	318	■ 1.000 ● N +0	$B = \mathbb{C} P^2 \times S^2 \times S^1,$	$B = \mathbb{C} P^2 \times S^2 \times S^1,$	$B = \mathbb{C} P^2 \times S^2 \times S^1,$
mielke-geometrodynamics_E09985 (14.2.26)	318	■ 1.000 ● N +1	$G/H = \frac{SU(3) \times SU(2) \times U(1)}{SU(2) \times U(1) \times U(1)},$	$G/H = \frac{SU(3) \times SU(2) \times U(1)}{SU(2) \times U(1) \times U(1)},$	$G/H = \frac{SU(3) \times SU(2) \times U(1)}{SU(2) \times U(1) \times U(1)},$
mielke-geometrodynamics_E09986 (14.2.27)	318	■ 1.000 ● K +0	$\widetilde{S}^7 := \frac{SO(5) \times SO(3)}{SO(3) \times SO(3)}.$	$\widetilde{S}^7 := \frac{SO(5) \times SO(3)}{SO(3) \times SO(3)}.$	$\widetilde{S}^7 := \frac{SO(5) \times SO(3)}{SO(3) \times SO(3)}.$
mielke-geometrodynamics_E09988 (14.3.1)	319	■ 1.000 ● K +0	$\varphi \rightarrow \varphi^{\mathrm{C}} := \bar{\varphi}$	$\varphi \rightarrow \varphi^{\mathrm{C}} := \bar{\varphi}$	$\varphi \rightarrow \varphi^{\mathrm{C}} := \bar{\varphi}$
mielke-geometrodynamics_E09991 (14.3.4)	320	■ 1.000 ● N +0	$\hat{\psi} = \sum_{F=1}^{\dim \rho(G)} \left(\psi_-^F(x) \chi_{\rho F}^-(\xi) + \psi_+^F(x) \chi_{\rho F}^+(\xi) \right).$	$\hat{\psi} = \sum_{F=1}^{\dim \rho(G)} \left(\psi_-^F(x) \chi_{\rho F}^-(\xi) + \psi_+^F(x) \chi_{\rho F}^+(\xi) \right).$	$\hat{\psi} = \sum_{F=1}^{\dim \rho(G)} (\psi_-^F(x) \chi_{\rho F}^-(\xi) + \psi_+^F(x) \chi_{\rho F}^+(\xi)).$
mielke-geometrodynamics_E09994 (14.3.7)	321	■ 1.000 ● N +0	$D^{g+\omega} := d + i \widetilde{\omega}^g \otimes i \omega$	$D^{g+\omega} := d + i \widetilde{\omega}^g \otimes i \omega$	$D^{g+\omega} := d + i \widetilde{\omega}^g \otimes i \omega$
mielke-geometrodynamics_E09999 (14.3.12)	322	■ 1.000 ● N +0	$(\widehat{\gamma}^{n+1})^2 = -1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp i \widehat{\psi}.$	$(\widehat{\gamma}^{n+1})^2 = -1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp i \widehat{\psi}.$	$(\widehat{\gamma}^{n+1})^2 = -1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp i \widehat{\psi}.$
mielke-geometrodynamics_E01000 (14.3.13)	322	■ 1.000 ● N +0	$(\widehat{\gamma}^{n+1})^2 = +1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp \widehat{\psi}.$	$(\widehat{\gamma}^{n+1})^2 = +1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp \widehat{\psi}.$	$(\widehat{\gamma}^{n+1})^2 = +1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp \widehat{\psi}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_324 EQ1006 (14.4.1)	324	■ 1.000 ● N +0	$\bar{Q}^{\{i\}}=Q^{\{i\}}\cos\delta+P^{\{i\}}\sin\delta,$ $\quad Q^{\{\circ\}}\equiv Q_e.$	$\bar{Q}^{(i)}=Q^{(i)}\cos\delta+P^{(i)}\sin\delta,$ $\quad Q^{(\circ)}\equiv Q_e.$	$\bar{Q}^{(i)}=Q^{(i)}\cos\delta+P^{(i)}\sin\delta,$ $\quad Q^{(\circ)}\equiv Q_e.$
mielke-geometrodynamics_325 EQ1009 (14.4.4)	325	■ 1.000 ● N +0	$\rho^2:=r^2+\frac{J(J+1)}{M^2}\hbar^2\cos^2\vartheta$	$\rho^2:=r^2+\frac{J(J+1)\hbar^2}{M^2c^2}\cos^2\vartheta$	$\rho^2:=r^2+\frac{J(J+1)\hbar^2}{M^2c^2}\cos^2\vartheta$
mielke-geometrodynamics_325 EQ1010 (14.4.5)	325	■ 1.000 ● W +0	$\Delta:=r^2-\frac{2M\hbar}{M^{*2}c}r+\frac{J(J+1)}{M^2}\hbar^2\cos^2\vartheta$ $\quad +\frac{\alpha_e}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$	$\Delta:=r^2-\frac{2M\hbar}{M^{*2}c}r+\frac{J(J+1)\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$	$\Delta:=r^2-\frac{2M\hbar}{M^{*2}c}r+\frac{J(J+1)\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$
mielke-geometrodynamics_327 EQ1017 (14.4.12)	327	■ 1.000 ● N +0	$d s^2=-\frac{\bar{\Delta}}{\bar{\rho}^2}\left(\frac{4N\hbar}{Mc}\sin^2\left(\frac{\vartheta}{2}\right)d\phi+c dt\right)^2+\frac{\bar{\rho}^2}{\bar{\Delta}}(dr^2+\bar{\Delta}d\Omega^2),$	$ds^2=-\frac{\bar{\Delta}}{\bar{\rho}^2}\left(\frac{4N\hbar}{Mc}\sin^2\left(\frac{\vartheta}{2}\right)d\phi+c dt\right)^2+\frac{\bar{\rho}^2}{\bar{\Delta}}(dr^2+\bar{\Delta}d\Omega^2),$	$ds^2=-\frac{\bar{\Delta}}{\bar{\rho}^2}\left(\frac{4N\hbar}{Mc}\sin^2\left(\frac{\vartheta}{2}\right)d\phi+c dt\right)^2+\frac{\bar{\rho}^2}{\bar{\Delta}}(dr^2+\bar{\Delta}d\Omega^2),$
mielke-geometrodynamics_328 EQ1018 (14.4.13)	328	■ 1.000 ● N +0	$\bar{\Delta}:=r^2-\frac{2M\hbar}{M^{*2}c}r-\frac{N^2\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$	$\bar{\Delta}:=r^2-\frac{2M\hbar}{M^{*2}c}r-\frac{N^2\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$	$\bar{\Delta}:=r^2-\frac{2M\hbar}{M^{*2}c}r-\frac{N^2\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$
mielke-geometrodynamics_328 EQ1019 (14.4.14)	328	■ 1.000 ● N +0	$\bar{\rho}^2:=r^2+\frac{N^2\hbar^2}{M^2c^2}$	$\bar{\rho}^2:=r^2+\frac{N^2\hbar^2}{M^2c^2}$	$\bar{\rho}^2:=r^2+\frac{N^2\hbar^2}{M^2c^2}$
mielke-geometrodynamics_328 EQ1020 (14.4.15)	328	■ 1.000 ● W +1	$\begin{aligned} A = & -\frac{1}{\bar{\rho}^2}(\bar{Q}\cos\delta+N\bar{Q}\sin\delta)c dt \\ & +\frac{1-\cos\vartheta}{\bar{\rho}^2}\bar{Q}\left[-\bar{\rho}^2\sin\delta+2r\frac{N\hbar}{Mc}\cos\delta\right]d\phi. \end{aligned}$	$A = -\frac{1}{\bar{\rho}^2}(\bar{Q}\cos\delta+N\bar{Q}\sin\delta)c dt$ $+\frac{1-\cos\vartheta}{\bar{\rho}^2}\bar{Q}\left[-\bar{\rho}^2\sin\delta+2r\frac{N\hbar}{Mc}\cos\delta\right]d\phi.$	$A = -\frac{1}{\bar{\rho}^2}(\bar{Q}\cos\delta+N\bar{Q}\sin\delta)c dt$ $+\frac{1-\cos\vartheta}{\bar{\rho}^2}\bar{Q}\left[-\bar{\rho}^2\sin\delta+2r\frac{N\hbar}{Mc}\cos\delta\right]d\phi.$
mielke-geometrodynamics_328 EQ1022 (14.4.17)	328	■ 1.000 ● W +0	$M^2/M_{\mathrm{ir}}^2=\frac{\left[1+\frac{1}{4}(\alpha_e^2\mathrm{Tr}\bar{Q}^2-2N^2)(M^*/M_{\mathrm{ir}})^4\right]^2}{1-\frac{1}{4}N^2(M^*/M_{\mathrm{ir}})^4},$	$M^2/M_{\mathrm{ir}}^2=\frac{\left[1+\frac{1}{4}(\alpha_e^2\mathrm{Tr}\bar{Q}^2-2N^2)(M^*/M_{\mathrm{ir}})^4\right]^2}{1-\frac{1}{4}N^2(M^*/M_{\mathrm{ir}})^4},$	$M^2/M_{\mathrm{ir}}^2=\frac{\left[1+\frac{1}{4}(\alpha_e^2\mathrm{Tr}\bar{Q}^2-2N^2)(M^*/M_{\mathrm{ir}})^4\right]^2}{1-\frac{1}{4}N^2(M^*/M_{\mathrm{ir}})^4},$
mielke-geometrodynamics_335 EQ1023 (15.0.1)	335	■ 1.000 ● N +0	$\ell:=2\pi\hbar/(M_Pc)=\ell^*/\sqrt{\kappa}\simeq1.3\times10^{-13}\,\mathrm{cm}$	$\ell:=2\pi\hbar/(M_Pc)=\ell^*/\sqrt{\kappa}\simeq1.3\times10^{-13}\,\mathrm{cm}$	$\ell:=2\pi\hbar/(M_Pc)=\ell^*/\sqrt{\kappa}\simeq1.3\times10^{-13}\,\mathrm{cm}$
mielke-geometrodynamics_335 EQ1024 (15.0.2)	335	■ 1.000 ● N +0	$V_{\mathrm{Yu}}(r)\sim\frac{1}{r}e^{-rm_\pi c/\hbar},$	$V_{\mathrm{Yu}}(r)\sim\frac{1}{r}e^{-rm_\pi c/\hbar},$	$V_{\mathrm{Yu}}(r)\sim\frac{1}{r}e^{-rm_\pi c/\hbar},$
mielke-geometrodynamics_336 EQ1026 (15.1.2)	336	■ 1.000 ● N +0	$L_{fg}=P(\vartheta^f,\vartheta^g)\eta,$	$L_{fg}=P(\vartheta^f,\vartheta^g)\eta,$	$L_{fg}=P(\vartheta^f,\vartheta^g)\eta,$
mielke-geometrodynamics_337 EQ1027 (15.1.3)	337	■ 1.000 ● N +0	$\delta L_{fg}=\delta\vartheta^f\wedge\Sigma^f+\delta\vartheta^g\wedge\Sigma^g.$	$\delta L_{fg}=\delta\vartheta^f\wedge\Sigma^f+\delta\vartheta^g\wedge\Sigma^g.$	$\delta L_{fg}=\delta\vartheta^f\wedge\Sigma^f+\delta\vartheta^g\wedge\Sigma^g.$
mielke-geometrodynamics_337 EQ1028 (15.1.4)	337	■ 1.000 ● N +0	$\frac{1}{a}\vartheta^f\wedge\Sigma^f+\frac{1}{b}\vartheta^g\wedge\Sigma^g=2L_{fg}.$	$\frac{1}{a}\vartheta^f\wedge\Sigma^f+\frac{1}{b}\vartheta^g\wedge\Sigma^g=2L_{fg}.$	$\frac{1}{a}\vartheta^f\wedge\Sigma^f+\frac{1}{b}\vartheta^g\wedge\Sigma^g=2L_{fg}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_337 EQ1029 (15.1.5)		■ 1.000 ● W +0	$L_{\{f\}g}^{\prime}=\frac{m_{\{f\}}^2}{\ell^2} \operatorname{det} g_{ij} ^u \operatorname{det} f_{ij} ^{1/2-u}(\vartheta^f-\vartheta^g)\wedge(\vartheta^f-\vartheta^g)\wedge\vartheta^g\wedge\vartheta^g$	$L'_{fg}=\frac{m_f^2}{\ell^2} \det g_{ij} ^u \det f_{ij} ^{1/2-u}(\vartheta^f-\vartheta^g)\wedge(\vartheta^f-\vartheta^g)\wedge\vartheta^g\wedge\vartheta^g$	$L'_{fg}=\frac{m_f^2}{\ell^2} \det g_{ij} ^u \det f_{ij} ^{1/2-u}(\vartheta^f-\vartheta^g)\wedge(\vartheta^f-\vartheta^g)\wedge\vartheta^g\wedge\vartheta^g$
mielke-geometrodynamics_338 EQ1035 (15.1.12)		■ 1.000 ● N +1	$\vartheta^{\mathrm{m}_{\{f\}}}:=\vartheta^{\mathrm{f}}-\vartheta^{\mathrm{g}}$	$\vartheta^{m_f}:=\vartheta^f-\vartheta^g$	$\vartheta^{m_f}:=\vartheta^f-\vartheta^g$
mielke-geometrodynamics_338 EQ1036 (15.1.13)		■ 1.000 ● W +1	$\begin{aligned}ds^2&\&=\frac{1}{1+\kappa}g^{\alpha\beta}\left(\vartheta_{\alpha}^g\otimes_s\vartheta_{\beta}^g+\kappa\vartheta_{\alpha}^f\otimes\vartheta_{\beta}^f\right)\\&=\frac{1}{1+\kappa}(g_{ij}+\kappa f_{ij})dx^i\otimes_sdx^j\end{aligned}$	$ds^2=\frac{1}{1+\kappa}g^{\alpha\beta}\left(\vartheta_{\alpha}^g\otimes_s\vartheta_{\beta}^g+\kappa\vartheta_{\alpha}^f\otimes\vartheta_{\beta}^f\right)\\=\frac{1}{1+\kappa}(g_{ij}+\kappa f_{ij})dx^i\otimes_sdx^j$	$ds^2=\frac{1}{1+\kappa}g^{\alpha\beta}\left(\vartheta_{\alpha}^g\otimes_s\vartheta_{\beta}^g+\kappa\vartheta_{\alpha}^f\otimes\vartheta_{\beta}^f\right)\\=\frac{1}{1+\kappa}(g_{ij}+\kappa f_{ij})dx^i\otimes_sdx^j.$
mielke-geometrodynamics_338 EQ1037 (15.1.14)		■ 1.000 ● N +0	$\kappa=\ell^{*2}/\ell^2=G_{\mathrm{N}}/G_{\mathrm{S}}=0.38\times10^{-38}$	$\kappa=\ell^{*2}/\ell^2=G_{\mathrm{N}}/G_{\mathrm{S}}=0.38\times10^{-38}$	$\kappa=\ell^{*2}/\ell^2=G_{\mathrm{N}}/G_{\mathrm{S}}=0.38\times10^{-38}$
mielke-geometrodynamics_339 EQ1038 (15.1.15)		■ 1.000 ● W +1	$e^{-\lambda^{\mathrm{g},f}}:=1-\frac{2\mu^{\mathrm{g},f}}{r}-\frac{1}{3}\Lambda^{\mathrm{g},f}r^2,$	$e^{-\lambda^{g,f}}:=1-\frac{2\mu^{g,f}}{r}-\frac{1}{3}\Lambda^{g,f}r^2,$	$e^{-\lambda^{g,f}}:=1-\frac{2\mu^{g,f}}{r}-\frac{1}{3}\Lambda^{g,f}r^2,$
mielke-geometrodynamics_339 EQ1040 (15.1.17)		■ 1.000 ● N +0	$e^{-\lambda^{\mathrm{f}}}\simeq1-\frac{2\mu^{\mathrm{f}}}{r}\exp\left(-rm_fc/\hbar\right).$	$e^{-\lambda^f}\simeq1-\frac{2\mu^f}{r}\exp\left(-rm_fc/\hbar\right).$	$e^{-\lambda^f}\simeq1-\frac{2\mu^f}{r}\exp\left(-rm_fc/\hbar\right).$
mielke-geometrodynamics_340 EQ1041 (15.1.18)		■ 1.000 ● K +0	$E_{\alpha\beta}:(2^+)\oplus2(1^-)\oplus2(0^+)\oplus(1^+)$	$E_{\alpha\beta}:(2^+)\oplus2(1^-)\oplus2(0^+)\oplus(1^+)$	$E_{\alpha\beta}:(2^+)\oplus2(1^-)\oplus2(0^+)\oplus(1^+)$
mielke-geometrodynamics_342 EQ1043 (15.1.20)		■ 1.000 ● N +0	$\frac{M_{\mathrm{U}}}{M_{\mathrm{P}}}=\frac{\ell_{\mathrm{H}}}{\ell^*}\cdot\frac{\ell}{\ell^*}=\frac{1}{\kappa}\frac{\ell_{\mathrm{H}}}{\ell}\simeq10^{80}$	$\frac{M_{\mathrm{U}}}{M_{\mathrm{P}}}=\frac{\ell_{\mathrm{H}}}{\ell^*}\cdot\frac{\ell}{\ell^*}=\frac{1}{\kappa}\frac{\ell_{\mathrm{H}}}{\ell}\simeq10^{80}$	$\frac{M_{\mathrm{U}}}{M_{\mathrm{P}}}=\frac{\ell_{\mathrm{H}}}{\ell^*}\cdot\frac{\ell}{\ell^*}=\frac{1}{\kappa}\frac{\ell_{\mathrm{H}}}{\ell}\simeq10^{80}$
mielke-geometrodynamics_343 EQ1044 (15.2.1)		■ 1.000 ● W +0	$\begin{aligned}\bar{G}&=SL(2f,\mathbb{C})\otimes SL(2c,\mathbb{C})\\&\supset SU(f)_{\mathrm{L}}\otimes SU(f)_{\mathrm{R}}\otimes SU(c)_{\mathrm{L}}\otimes SU(c)_{\mathrm{R}}\end{aligned}$	$\bar{G}=SL(2f,\mathbb{C})\otimes SL(2c,\mathbb{C})\\ \supset SU(f)_{\mathrm{L}}\otimes SU(f)_{\mathrm{R}}\otimes SU(c)_{\mathrm{L}}\otimes SU(c)_{\mathrm{R}}$	$\begin{aligned}\bar{G}&=SL(2f,\mathbb{C})\otimes SL(2c,\mathbb{C})\\&\supset SU(f)_{\mathrm{L}}\otimes SU(f)_{\mathrm{R}}\otimes SU(c)_{\mathrm{L}}\otimes SU(c)_{\mathrm{R}}\end{aligned}$
mielke-geometrodynamics_343 EQ1045 (15.2.2)		■ 1.000 ● N +0	$\bar{L}(M):=M^4\times\bar{G}\supset M^4\times SL(2,\mathbb{C})$	$\bar{L}(M):=M^4\times\bar{G}\supset M^4\times SL(2,\mathbb{C})$	$\bar{L}(M):=M^4\times\bar{G}\supset M^4\times SL(2,\mathbb{C})$
mielke-geometrodynamics_344 EQ1046 (15.2.3)		■ 1.000 ● N -1	$\bar{\gamma}=\bar{E}_j^{\alpha}(m)\gamma_{\alpha}\otimes dx^j=\bar{E}_jdx^j$	$\bar{\gamma}=\bar{E}_j^{\alpha}(m)\gamma_{\alpha}\otimes dx^j=\bar{E}_jdx^j$	$\bar{\gamma}=\bar{E}_j^{\alpha}(m)\gamma_{\alpha}\otimes dx^j=\bar{E}_jdx^j$
mielke-geometrodynamics_344 EQ1048 (15.2.5)		■ 1.000 ● K +0	$\frac{1}{4}\operatorname{Tr}(\bar{\gamma}\otimes_s\bar{\gamma})=f_{ij}dx^i\otimes_sdx^j$	$\frac{1}{4}\operatorname{Tr}(\bar{\gamma}\otimes_s\bar{\gamma})=f_{ij}dx^i\otimes_sdx^j$	$\frac{1}{4}\operatorname{Tr}(\bar{\gamma}\otimes_s\bar{\gamma})=f_{ij}dx^i\otimes_sdx^j$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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mielke-geometrodynamics_345 EQ1052 (15.2.9)		■ 1.000 ● N +0	$\bar{\omega}_{\mathrm{i}}=\frac{1}{2}\,\Gamma_{\mathrm{i}}^{\mathrm{(f)}}\alpha\beta j\,\sigma_{\alpha\beta}\lambda_j^{\mathrm{(f)}}\oplus\frac{1}{2}\Gamma_{\mathrm{i}}^{\mathrm{(c)}}\alpha\beta j\,\sigma_{\alpha\beta}\lambda_j^{\mathrm{(c)}}\oplus A_{\mathrm{i}}\,.$	$\bar{\omega}_i = \frac{1}{2}\Gamma_i^{(f)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(f)} \oplus \frac{1}{2}\Gamma_i^{(c)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(c)} \oplus A_i.$	$\bar{\omega}_i = \frac{1}{2}\Gamma_i^{(f)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(f)} \oplus \frac{1}{2}\Gamma_i^{(c)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(c)} \oplus A_i.$
mielke-geometrodynamics_345 EQ1053 (15.2.10)		■ 1.000 ● N +0	$\bar{\Theta}:=\bar{D}\bar{\gamma}=\bar{d}\bar{\gamma}-[\bar{\omega},\bar{\gamma}]$	$\bar{\Theta} := \bar{D}\bar{\gamma} = d\bar{\gamma} - [\bar{\omega}, \bar{\gamma}]$	$\bar{\Theta} := \bar{D}\bar{\gamma} = d\bar{\gamma} - [\bar{\omega}, \bar{\gamma}]$
mielke-geometrodynamics_345 EQ1054 (15.2.11)		■ 1.000 ● W +1	$\bar{\Omega} = d\bar{\omega} - \bar{\omega} \wedge \bar{\omega}$	$\bar{\Omega} = d\bar{\omega} - \bar{\omega} \wedge \bar{\omega}$	$\bar{\Omega} = d\bar{\omega} - \bar{\omega} \wedge \bar{\omega}$
mielke-geometrodynamics_346 EQ1056 (15.2.13)		■ 1.000 ● N +0	$\psi=\left\{\psi^{\left(q_f,q_c\right)}\mid q_f=1,\ldots,f;q_c=1,\ldots,c\right\}.$	$\psi = \left\{\psi^{(q_f,q_c)} \mid q_f = 1, \dots, f; q_c = 1, \dots, c\right\}.$	$\psi = \{\psi^{(q_f,q_c)} q_f = 1, \dots, f; \, q_c = 1, \dots, c\}.$
mielke-geometrodynamics_354 EQ1061 (16.1.2)		■ 1.000 ● C -3	$B=\frac{2}{\ell^{*2}}\hbar c\Lambda_{\mathrm{eff}}=-\frac{3\kappa}{2}\frac{\hbar c}{\ell^{*4}}$	$B = \frac{2\hbar c}{\ell^{*2}}\Lambda_{\mathrm{eff}} = -\frac{3\kappa}{2}\frac{\hbar c}{\ell^{*4}}$	$B = \frac{2\hbar c}{\ell^{*2}}\Lambda_{\mathrm{eff}} = -\frac{3\kappa}{2}\frac{\hbar c}{\ell^{*4}}$
mielke-geometrodynamics_355 EQ1063 (16.2.2)		■ 1.000 ● N +0	$\square=\frac{1}{\sqrt{ f }}\partial_i\left(f^{ij}\sqrt{ f }\partial_j\right)$	$\square = \frac{1}{\sqrt{ f }}\partial_i\left(f^{ij}\sqrt{ f }\partial_j\right)$	$\square = \frac{1}{\sqrt{ f }}\partial_i(f^{ij}\sqrt{ f }\partial_j)$
mielke-geometrodynamics_355 EQ1064 (16.2.3)		■ 1.000 ● N +0	$\varphi=m\frac{c}{\hbar}\frac{1}{\rho}F_{nm}^{\iota}(\rho^{*})Y_{\iota}^m(\vartheta,\phi)e^{-it\omega mc^2/\hbar}.$	$\varphi = m\frac{c}{\hbar}\frac{1}{\rho}F_{nm}^{\iota}(\rho^{*})Y_{\iota}^m(\vartheta,\phi)e^{-it\omega mc^2/\hbar}.$	$\varphi = m\frac{c}{\hbar}\frac{1}{\rho}F_{nm}^{\iota}(\rho^{*})Y_{\iota}^m(\vartheta,\phi)e^{-it\omega mc^2/\hbar}.$
mielke-geometrodynamics_355 EQ1067 (16.2.6)		■ 1.000 ● K +0	$\beta=\frac{\sqrt{2\pi\kappa}}{\ell^{*}}/\frac{mc}{\hbar}=\frac{\ell_m}{\ell}.$	$\beta = \frac{\sqrt{2\pi\kappa}}{\ell^{*}}/\frac{mc}{\hbar} = \frac{\ell_m}{\ell}.$	$\beta = \frac{\sqrt{2\pi\kappa}}{\ell^{*}}/\frac{mc}{\hbar} = \frac{\ell_m}{\ell}.$
mielke-geometrodynamics_358 EQ1072 (16.3.1)		■ 1.000 ● K +0	$\operatorname{Tr}\bar{Q}^2=-\frac{4b}{\alpha_e}Y+I(I+1)-\frac{1}{4}Y^2.$	$\operatorname{Tr}\bar{Q}^2 = -\frac{4b}{\alpha_e}Y + I(I+1) - \frac{1}{4}Y^2.$	$\operatorname{Tr}\bar{Q}^2 = -\frac{4b}{\alpha_e}Y + I(I+1) - \frac{1}{4}Y^2.$
mielke-geometrodynamics_358 EQ1073 (16.3.2)		■ 1.000 ● N +0	$Q_{\mathrm{e}}=I_3+\frac{1}{2}Y-\frac{2}{3}C\,.$	$Q_e = I_3 + \frac{1}{2}Y - \frac{2}{3}C.$	$Q_e = I_3 + \frac{1}{2}Y - \frac{2}{3}C.$
mielke-geometrodynamics_359 EQ1074 (16.3.3)		■ 1.000 ● N +0	$\frac{M^2}{M^{*2}}=\left\{1-bY+\frac{\alpha_e}{4}\left[I(I+1)-\frac{1}{4}Y^2\right]\right\}^2+\frac{1}{4}J(J+1)$	$\frac{M^2}{M^{*2}} = \left\{1 - bY + \frac{\alpha_e}{4}\left[I(I+1) - \frac{1}{4}Y^2\right]\right\}^2 + \frac{1}{4}J(J+1)$	$\frac{M^2}{M^{*2}} = \left\{1 - bY + \frac{\alpha_e}{4}[I(I+1) - \frac{1}{4}Y^2]\right\}^2 + \frac{1}{4}J(J+1)$
mielke-geometrodynamics_360 EQ1075 (16.3.4)		■ 1.000 ● N +0	$T_{\bullet}=\frac{2\hbar c}{S_Ak_B}\left(r_+-\frac{M}{M^{*2}}\frac{\hbar}{c}\right).$	$T_{\bullet} = \frac{2\hbar c}{S_Ak_B}\left(r_+ - \frac{M}{M^{*2}}\frac{\hbar}{c}\right).$	$T_{\bullet} = \frac{2\hbar c}{S_Ak_B}\left(r_+ - \frac{M}{M^{*2}}\frac{\hbar}{c}\right).$
mielke-geometrodynamics_364 EQ1076 (A.1)		■ 1.000 ● N +0	$x^A y_A := \sum_{A=0}^n \sum_{B=0}^n x^A y^B g_{AB}.$	$x^A y_A := \sum_{A=0}^n \sum_{B=0}^n x^A y^B g_{AB}.$	$x^A y_A := \sum_{A=0}^n \sum_{B=0}^n x^A y^B g_{AB}.$
mielke-geometrodynamics_364 EQ1077 (A.2)		■ 1.000 ● N +0	$c=\hbar=1\,.$	$c = \hbar = 1.$	$c = \hbar = 1.$
mielke-geometrodynamics_365 EQ1080 (A.5)		■ 1.000 ● W +0	$\left\{\sigma^i,\sigma^j\right\}=g^{ij}$	$\left\{\sigma^i,\sigma^j\right\} = g^{ij}$	$\{\sigma^i,\sigma^j\} = g^{ij}$
mielke-geometrodynamics_365 EQ1081 (A.6)		■ 1.000 ● K +0	$\left(\gamma^{\alpha}\right)^{\mathrm{T}}:=\left(\gamma^0,\gamma^k\right),$	$(\gamma^{\alpha})^{\mathrm{T}} := (\gamma^0, \gamma^k),$	$(\gamma^{\alpha})^{\mathrm{T}} := (\gamma^0, \gamma^k),$
mielke-geometrodynamics_365 EQ1083 (A.8)		■ 1.000 ● N +0	$\gamma^{\alpha}\gamma^{\beta}+\gamma^{\beta}\gamma^{\alpha}=2g^{\alpha\beta}$	$\gamma^{\alpha}\gamma^{\beta} + \gamma^{\beta}\gamma^{\alpha} = 2g^{\alpha\beta}$	$\gamma^{\alpha}\gamma^{\beta} + \gamma^{\beta}\gamma^{\alpha} = 2g^{\alpha\beta}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ1084 (A.9)	365	■ 1.000 ● N +0	$\gamma^{\{0\}}=\gamma^{\{0\}}, \quad \gamma^{\{k\}}=-\gamma^{\{k\}},$ $\gamma^{\{5\}}=\gamma^{\{5\}} \quad .$	$\gamma^{0+} = \gamma^0, \quad \gamma^{k+} = -\gamma^k, \quad \gamma^{5+} = \gamma^5.$	$\gamma^{0+} = \gamma^0, \quad \gamma^{k+} = -\gamma^k, \quad \gamma^{5+} = \gamma^5.$
mielke-geometrodynamics_EQ1087 (A.14)	366	■ 1.000 ● K +0	$C(1,3)=C(0,2) \otimes C(1,1) \quad .$	$C(1,3) = C(0,2) \otimes C(1,1).$	$C(1,3) = C(0,2) \otimes C(1,1).$
mielke-geometrodynamics_EQ1091 (B.1)	368	■ 1.000 ● N +0	$\alpha^{\{p\}}=\frac{1}{p!} A_{\alpha_1\alpha_2\cdots\alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}(\wedge^p T^*(M)).$	$\alpha^{(p)} = \frac{1}{p!} A_{\alpha_1 \cdots \alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}(\wedge^p T^*(M)).$	$\alpha^{(p)} = \frac{1}{p!} A_{\alpha_1 \cdots \alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}(\wedge^p T^*(M)).$
mielke-geometrodynamics_EQ1092 (B.2)	368	■ 1.000 ● W +0	$\begin{gathered} \alpha^{\{p\}} \wedge \beta^{\{q\}} := \\ \frac{1}{(p!) \cdot (q!)} A_{\alpha_1 \cdots \alpha_p} \cdot B_{\beta_1 \cdots \beta_q} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \\ \wedge \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_q} \end{gathered}$	$\alpha^{(p)} \wedge \beta^{(q)} := \frac{1}{(p!) \cdot (q!)} A_{\alpha_1 \cdots \alpha_p} \cdot B_{\beta_1 \cdots \beta_q} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \\ \wedge \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_q}$	$\alpha^{(p)} \wedge \beta^{(q)} := \frac{1}{(p!) \cdot (q!)} A_{\alpha_1 \cdots \alpha_p} \cdot B_{\beta_1 \cdots \beta_q} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \\ \wedge \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_q}$
mielke-geometrodynamics_EQ1093 (B.3)	368	■ 1.000 ● N +0	$\alpha^{\{p\}} \wedge \beta^{\{q\}} = (-1)^{pq} \beta^{\{q\}} \wedge \alpha^{\{p\}},$ $\alpha^{\{p\}} \wedge \beta^{\{q\}} = (-1)^{pq} \beta^{\{q\}} \wedge \alpha^{\{p\}},$	$\alpha^{(p)} \wedge \beta^{(q)} = (-1)^{pq} \beta^{(q)} \wedge \alpha^{(p)},$	$\alpha^{(p)} \wedge \beta^{(q)} = (-1)^{pq} \beta^{(q)} \wedge \alpha^{(p)},$
mielke-geometrodynamics_EQ1094 (B.4)	368	■ 1.000 ● N +0	$D(M) := \sum_{p=0}^{\infty} \wedge^p T^*(M)$	$D(M) := \sum_{p=0}^n \wedge^p T^*(M)$	$D(M) := \sum_{p=0}^n \wedge^p T^*(M)$
mielke-geometrodynamics_EQ1095 (B.5)	369	■ 1.000 ● N +1	$d(\alpha^{\{p\}} \wedge \beta^{\{q\}}) = d\alpha^{\{p\}} \wedge \beta^{\{q\}} + (-1)^p \alpha^{\{p\}} \wedge d\beta^{\{q\}}$	$d\left(\alpha^{(p)} \wedge \beta^{(q)}\right) = d\alpha^{(p)} \wedge \beta^{(q)} + (-1)^p \alpha^{(p)} \wedge d\beta^{(q)}$	$d(\alpha^{(p)} \wedge \beta^{(q)}) = d\alpha^{(p)} \wedge \beta^{(q)} + (-1)^p \alpha^{(p)} \wedge d\beta^{(q)}$
mielke-geometrodynamics_EQ1097 (B.7)	369	■ 1.000 ● N +0	$\sigma: M \rightarrow N, \quad \sigma \in \mathcal{D}(M)$	$\sigma: M \rightarrow N, \quad \sigma \in \mathcal{D}(M)$	$\sigma: M \rightarrow N, \quad \sigma \in \mathcal{D}(M)$
mielke-geometrodynamics_EQ1098 (B.8)	369	■ 1.000 ● N +0	$\sigma_*: \wedge^p T^*(N) \rightarrow \wedge^p T^*(M)$	$\sigma_*: \wedge^p T^*(N) \rightarrow \wedge^p T^*(M)$	$\sigma_*: \wedge^p T^*(N) \rightarrow \wedge^p T^*(M)$
mielke-geometrodynamics_EQ1103 (B.15)	370	■ 1.000 ● K +0	$\{ \}^{\{ * \}} \alpha^{\{p\}} = (-1)^{p(n+1)+s} \alpha^{\{p\}} \quad .$	$**\alpha^{(p)} = (-1)^{p(n+1)+s} \alpha^{(p)}.$	$**\alpha^{(p)} = (-1)^{p(n+1)+s} \alpha^{(p)}.$
mielke-geometrodynamics_EQ1105 (B.17)	370	■ 1.000 ● N +0	$(\alpha, \beta) := \int_M \alpha^{\{p\}} \wedge \beta^{\{p\}} = (\beta, \alpha)$	$(\alpha, \beta) := \int_M \alpha^{(p)} \wedge \beta^{(p)} = (\beta, \alpha)$	$(\alpha, \beta) := \int_M \alpha^{(p)} \wedge \beta^{(p)} = (\beta, \alpha)$
mielke-geometrodynamics_EQ1107 (B.19)	370	■ 1.000 ● N +0	$\square \alpha^{\{p\}} := d\delta \alpha^{\{p\}} + \delta d \alpha^{\{p\}} \quad .$	$\square \alpha^{(p)} := d\delta \alpha^{(p)} + \delta d \alpha^{(p)}.$	$\square \alpha^{(p)} := d\delta \alpha^{(p)} + \delta d \alpha^{(p)}.$
mielke-geometrodynamics_EQ1108 (B.20)	371	■ 1.000 ● N +0	$\begin{aligned} (\alpha, \square \alpha) &= (\alpha, d\delta \alpha) + (\alpha, \delta d \alpha) \\ &= (\delta \alpha, \delta \alpha) + (d\alpha, d\alpha). \end{aligned}$	$\begin{aligned} (\alpha, \square \alpha) &= (\alpha, d\delta \alpha) + (\alpha, \delta d \alpha) \\ &= (\delta \alpha, \delta \alpha) + (d\alpha, d\alpha). \end{aligned}$	$\begin{aligned} (\alpha, \square \alpha) &= (\alpha, d\delta \alpha) + (\alpha, \delta d \alpha) \\ &= (\delta \alpha, \delta \alpha) + (d\alpha, d\alpha). \end{aligned}$
mielke-geometrodynamics_EQ1109 (B.21)	371	■ 1.000 ● K +0	$d\alpha = 0 \quad .$	$d\alpha = 0.$	$d\alpha = 0.$
mielke-geometrodynamics_EQ1111 (B.23)	371	■ 1.000 ● N +0	$\beta^{\{p\}} = d\alpha^{\{p-1\}} \in C^{\infty}(B(M))$	$\beta^{(p)} = d\alpha^{(p-1)} \in C^{\infty}(B(M))$	$\beta^{(p)} = d\alpha^{(p-1)} \in C^{\infty}(B(M))$
mielke-geometrodynamics_EQ1112 (B.24)	371	■ 1.000 ● N +0	$H(M) := Z(M) / B(M)$	$H(M) := Z(M) / B(M)$	$H(M) := Z(M) / B(M)$

Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_371 EQ1113 (B.25)		■ 1.000 ● N +0	$H(M)=\sum_{p=0}^n H^p(M):=\sum_{p=0}^n Z^p(M) / B^p(M) \text{ .}$	$H(M)=\sum_{p=0}^n H^p(M):=\sum_{p=0}^n Z^p(M)/B^p(M).$	$H(M)=\sum_{p=0}^n H^p(M):=\sum_{p=0}^n Z^p(M)/B^p(M).$
mielke-geometrodynamics_371 EQ1115 (B.27)		■ 1.000 ● N +0	$H^n(M)=\wedge^n T^*(M) / B^n(M) \text{ .}$	$H^n(M)=\wedge^n T^*(M)/B^n(M).$	$H^n(M)=\wedge^n T^*(M)/B^n(M).$
mielke-geometrodynamics_371 EQ1116 (B.28)		■ 1.000 ● N +0	$H^{\{0\}}(M)=Z^{\{0\}}(M)$	$H^0(M)=Z^0(M)$	$H^0(M)=Z^0(M)$
mielke-geometrodynamics_372 EQ1117 (B.29)		■ 1.000 ● N +0	$H^{\{0\}}(M) \approx \mathbb{R}$	$H^0(M)\approx \mathbb{R}$	$H^0(M)\approx \mathbb{R}$
mielke-geometrodynamics_372 EQ1118 (B.30)		■ 1.000 ● N -1	$b_{\{p\}}:=\operatorname{dim} H^{\{p\}}(M)$	$b_p:=\dim H^p(M)$	$b_p:=\dim H^p(M)$
mielke-geometrodynamics_372 EQ1119 (B.31)		■ 1.000 ● N +1	$\mathrm{f}_{\mathrm{M}}(t)=\sum_{p=0}^n b_{\{p\}} t^{\{p\}}$	$\mathrm{f}_{\mathrm{M}}(t)=\sum_{p=0}^n b_p t^p$	$\mathrm{f}_{\mathrm{M}}(t)=\sum_{p=0}^n b_p t^p,$
mielke-geometrodynamics_372 EQ1120 (B.32)		■ 1.000 ● N +0	$\chi(M):=\mathrm{f}_{\mathrm{M}}(-1)=\sum_{p=0}^n (-1)^{\{p\}} b_{\{p\}},$	$\chi(M):=\mathrm{f}_{\mathrm{M}}(-1)=\sum_{p=0}^n (-1)^p b_p,$	$\chi(M):=\mathrm{f}_{\mathrm{M}}(-1)=\sum_{p=0}^n (-1)^p b_p,$
mielke-geometrodynamics_372 EQ1121 (B.33)		■ 1.000 ● N +0	$\mathrm{H}^0(S^{\{n\}}\!\left(S^{\{n\}}\right)=H^{\{n\}}\!\left(S^{\{n\}}\right) \approx \mathbb{R}$	$H^0(S^n)=H^n(S^n)\approx \mathbb{R}$	$H^0(S^n)=H^n(S^n)\approx \mathbb{R}$
mielke-geometrodynamics_372 EQ1123 (B.35)		■ 1.000 ● N +0	$\mathrm{f}_{S^n}(t)=1+t^{\{n\}},$	$\mathrm{f}_{S^n}(t)=1+t^n,$	$\mathrm{f}_{S^n}(t)=1+t^n,$
mielke-geometrodynamics_372 EQ1124 (B.36)		■ 1.000 ● N +0	$\chi\!\left(S^{\{n\}}\right)=\!\left(\!\begin{array}{l} 0 \text{ for } n=2k+1, \\ 2 \text{ for } n=2k, \end{array}\!\right).$	$\chi(S^n)=\left\{\begin{array}{l} 0 \text{ for } n=2k+1, \\ 2 \text{ for } n=2k, \end{array}\right.$	$\chi(S^n)=\left\{\begin{array}{l} 0 \text{ for } n=2k+1, \\ 2 \text{ for } n=2k, \end{array}\right.$
mielke-geometrodynamics_372 EQ1125 (B.37)		■ 1.000 ● N +0	$\Delta_{\{p\}}:=\!\left\{x\in E^{\{n\}}\mid x=\sum_{i=0}^p \lambda^i \eta_i, \lambda^i\geq 0, \sum_{i=0}^p \lambda^i=1\right\}.$	$\Delta_p:=\left\{x\in E^n\mid x=\sum_{i=0}^p \lambda^i \eta_i, \lambda^i\geq 0, \sum_{i=0}^p \lambda^i=1\right\}.$	$\Delta_p:=\left\{x\in E^n x=\sum_{i=0}^p \lambda^i \eta_i, \lambda^i\geq 0, \sum_{i=0}^p \lambda^i=1\right\}.$
mielke-geometrodynamics_373 EQ1127 (B.39)		■ 1.000 ● N +0	$\partial c=\sum_{i=0}^p (-1)^{\{i\}} c\!\left(\sum_{j=0}^{\{i-1\}} \lambda^j \eta_j+\sum_{j=i}^{\{p-1\}} \lambda^j \eta_j\right).$	$\partial c=\sum_{i=0}^p (-1)^ic\left(\sum_{j=0}^{i-1} \lambda^j \eta_j+\sum_{j=i}^{p-1} \lambda^j \eta_{j+1}\right).$	$\partial c=\sum_{i=0}^p (-1)^ic\left(\sum_{j=0}^{i-1} \lambda^j \eta_j+\sum_{j=i}^{p-1} \lambda^j \eta_{j+1}\right).$
mielke-geometrodynamics_373 EQ1129 (B.41)		■ 1.000 ● N +0	$H_{\{\mathrm{p}\}}(M,\mathbb{R}):=Z_{\{\mathrm{p}\}}(M,\mathbb{R}) / B_{\{\mathrm{p}\}}(M,\mathbb{R})$	$H_{\mathrm{p}}(M,\mathbb{R}):=Z_{\mathrm{p}}(M,\mathbb{R})/B_{\mathrm{p}}(M,\mathbb{R})$	$H_{\mathrm{p}}(M,\mathbb{R}):=Z_{\mathrm{p}}(M,\mathbb{R})/B_{\mathrm{p}}(M,\mathbb{R})$
mielke-geometrodynamics_373 EQ1130 (B.42)		■ 1.000 ● N +1	$\int_{c_{\{p+1\}}} d\alpha^{\{p\}}=\int \partial c_{\{p+1\}} \alpha^{\{p\}}$	$\int_{c_{p+1}} d\alpha^{(p)}=\int_{\partial c_{p+1}} \alpha^{(p)}$	$\int_{c_{p+1}} d\alpha^{(p)}=\int_{\partial c_{p+1}} \alpha^{(p)}.$
mielke-geometrodynamics_373 EQ1132 (B.44)		■ 1.000 ● K +0	$\langle d\alpha,c\rangle_{\mathrm{M}}=\langle \alpha,\partial c\rangle_{\mathrm{M}}$	$\langle d\alpha,c\rangle_{\mathrm{M}}=\langle \alpha,\partial c\rangle_{\mathrm{M}}$	$\langle d\alpha,c\rangle_{\mathrm{M}}=\langle \alpha,\partial c\rangle_{\mathrm{M}}$
mielke-geometrodynamics_375 EQ1133 (C.1)		■ 1.000 ● N +0	$I_{\{j\}}:=\!\left.\frac{\partial g(\xi)}{\partial \xi^{\{j\}}}\right _{g=e}\in T_e(G)\approx \mathfrak{g}$	$I_j:=\left.\frac{\partial g(\xi)}{\partial \xi^j}\right _{g=e}\in T_e(G)\approx \mathfrak{g}$	$I_j:=\frac{\partial g(\xi)}{\partial \xi^j}\big _{g=e}\in T_e(G)\approx \mathfrak{g}$
mielke-geometrodynamics_376 EQ1138 (C.6)		■ 1.000 ● N +0	$[\mathrm{AB}]:=\mathrm{A}^{\{i\}}\mathrm{B}^{\{j\}}\mathrm{c}_{ij}^{\{k\}}\mathrm{I}_k.$	$[\mathrm{AB}]:=\mathrm{A}^i\mathrm{B}^j\mathrm{c}_{ij}^k\mathrm{I}_k.$	$[\mathrm{AB}]:=\mathrm{A}^i\mathrm{B}^j\mathrm{c}_{ij}^k\mathrm{I}_k.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

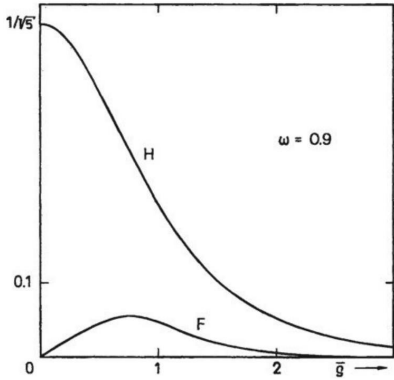
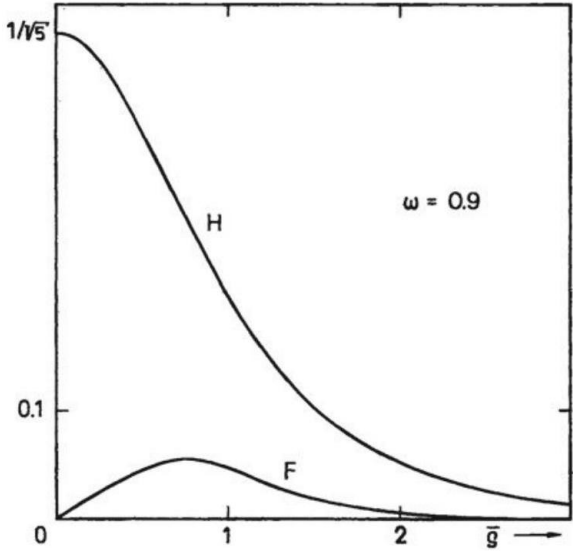
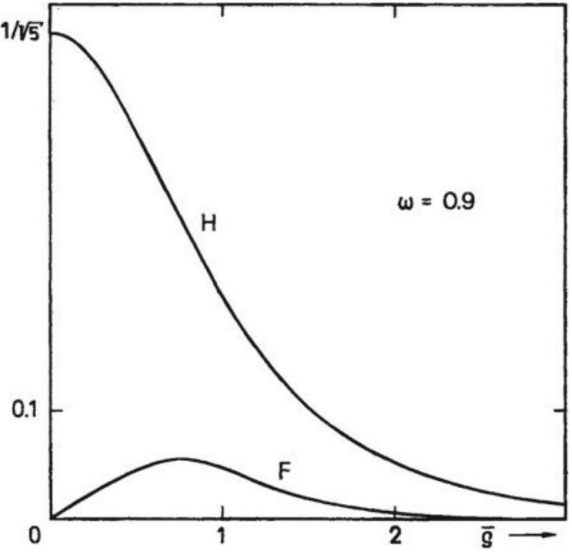
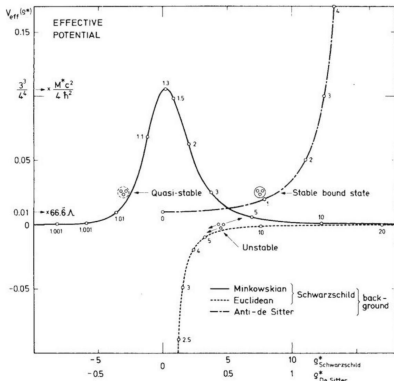
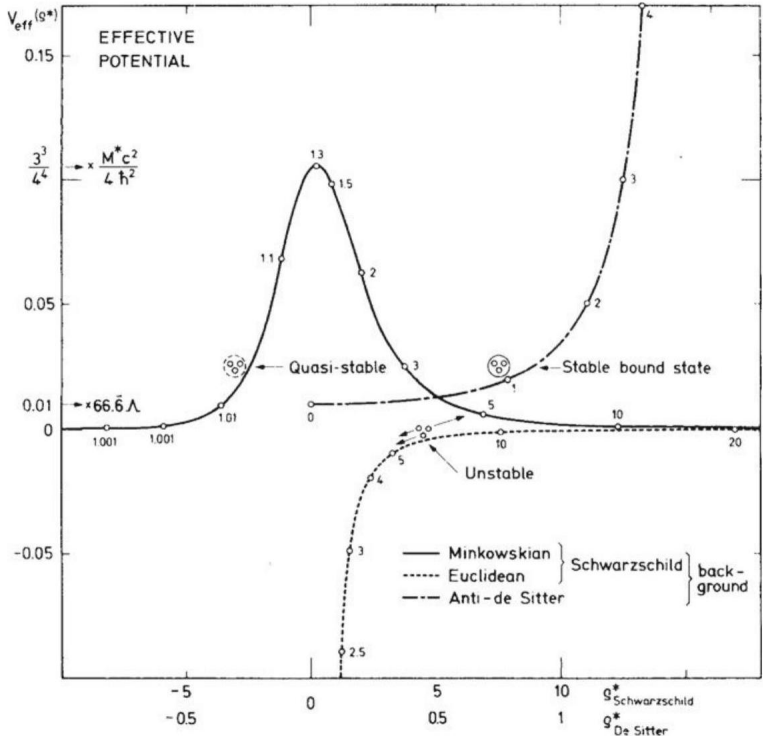
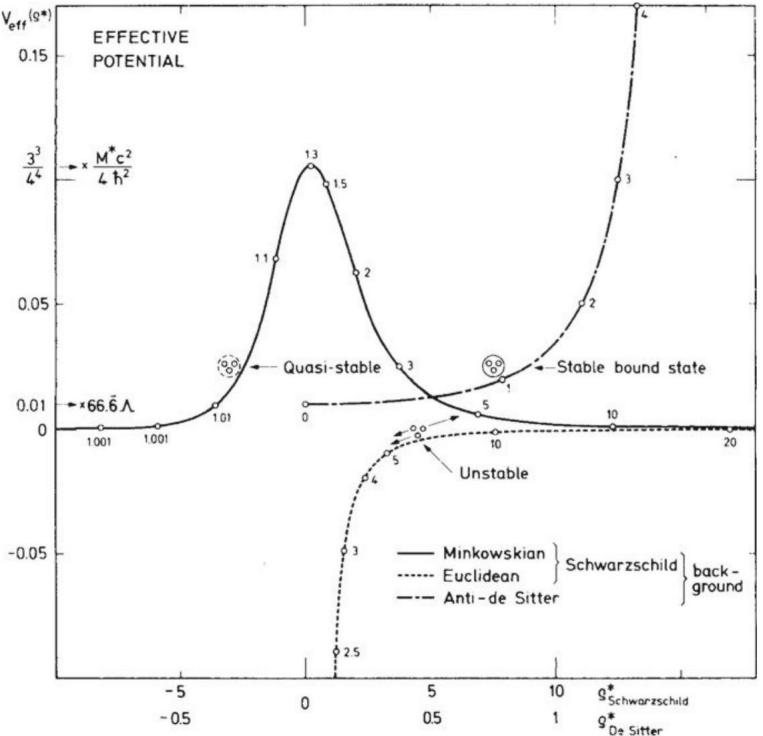
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mielke-geometrodynamics_376 EQ1139 (C.7)	376	■ 1.000 ● W +0	$\rho_{A}^C\left(g_1\right)\rho_C^B\left(g_2\right)=\rho_A^B\left(g_1g_2\right)$	$\rho_A^C\left(g_1\right)\rho_C^B\left(g_2\right)=\rho_A^B\left(g_1g_2\right)$	$\rho_A^C\left(g_1\right)\rho_C^B\left(g_2\right)=\rho_A^B\left(g_1g_2\right)$
mielke-geometrodynamics_376 EQ1140 (C.8)	376	■ 1.000 ● K +0	$\left[A,B\right]\longrightarrow\rho\left(\left[A,B\right]\right)=\left[\rho(A),\rho(B)\right].$	$\left[A,B\right]\longrightarrow\rho\left(\left[A,B\right]\right)=\left[\rho(A),\rho(B)\right].$	$\left[A,B\right]\longrightarrow\rho\left(\left[A,B\right]\right)=\left[\rho(A),\rho(B)\right].$
mielke-geometrodynamics_376 EQ1141 (C.9)	376	■ 1.000 ● K +0	$\operatorname{Ad}A(B):=\left[A,B\right];\quad A,B\in\mathfrak{g},$	$\operatorname{Ad}A(B):=\left[A,B\right];\quad A,B\in\mathfrak{g},$	$\operatorname{Ad}A(B):=\left[A,B\right];\quad A,B\in\mathfrak{g},$
mielke-geometrodynamics_376 EQ1142 (C.10)	376	■ 1.000 ● N +0	$\left(\operatorname{Ad}A\right)_{j\left\{ \right\} ^k}=\mathrm{c}_{ij\left\{ \right\} ^k}A^i.$	$\left(\operatorname{Ad}A\right)_j^k=c_{ij}^kA^i.$	$\left(\operatorname{Ad}A\right)_j^k=c_{ij}^kA^i.$
mielke-geometrodynamics_376 EQ1144 (C.12)	376	■ 1.000 ● K +0	$\left(A,B\right)=2N\operatorname{Tr}\left(AB\right)-2\operatorname{Tr}\left(A\right)\operatorname{Tr}\left(B\right)$	$\left(A,B\right)=2N\operatorname{Tr}\left(AB\right)-2\operatorname{Tr}\left(A\right)\operatorname{Tr}\left(B\right)$	$\left(A,B\right)=2N\operatorname{Tr}\left(AB\right)-2\operatorname{Tr}\left(A\right)\operatorname{Tr}\left(B\right)$
mielke-geometrodynamics_377 EQ1146 (C.14)	377	■ 1.000 ● N +1	$g=\exp^{\left(-\frac{i}{2}\xi^j\lambda_j\right)}\in SU\left(N\right),j=1,\dots,N^2-1.$	$g=\exp^{\left(-\frac{i}{2}\xi^j\lambda_j\right)}\in SU\left(N\right),j=1,\dots,N^2-1.$	$g=\exp^{\left(-\frac{i}{2}\xi^j\lambda_j\right)}\in SU\left(N\right),j=1,\dots,N^2-1.$
mielke-geometrodynamics_377 EQ1148 (C.17)	377	■ 1.000 ● W +0	$\lambda_i\lambda_j=\frac{2}{N}\delta_{ij}+\left(d_{ij}^k+if_{ij}^k\right)\lambda_k.$	$\lambda_i\lambda_j=\frac{2}{N}\delta_{ij}+\left(d_{ij}^k+if_{ij}^k\right)\lambda_k.$	$\lambda_i\lambda_j=\frac{2}{N}\delta_{ij}+\left(d_{ij}^k+if_{ij}^k\right)\lambda_k.$
mielke-geometrodynamics_377 EQ1149 (C.18)	377	■ 1.000 ● N +0	$\operatorname{Tr}\left(\lambda_i\lambda_j\right)=2\delta_{ij}.$	$\operatorname{Tr}\left(\lambda_i\lambda_j\right)=2\delta_{ij}.$	$\operatorname{Tr}\left(\lambda_i\lambda_j\right)=2\delta_{ij}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Tables

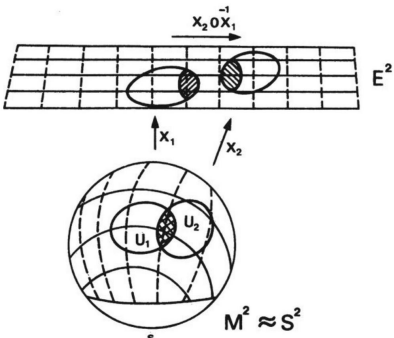
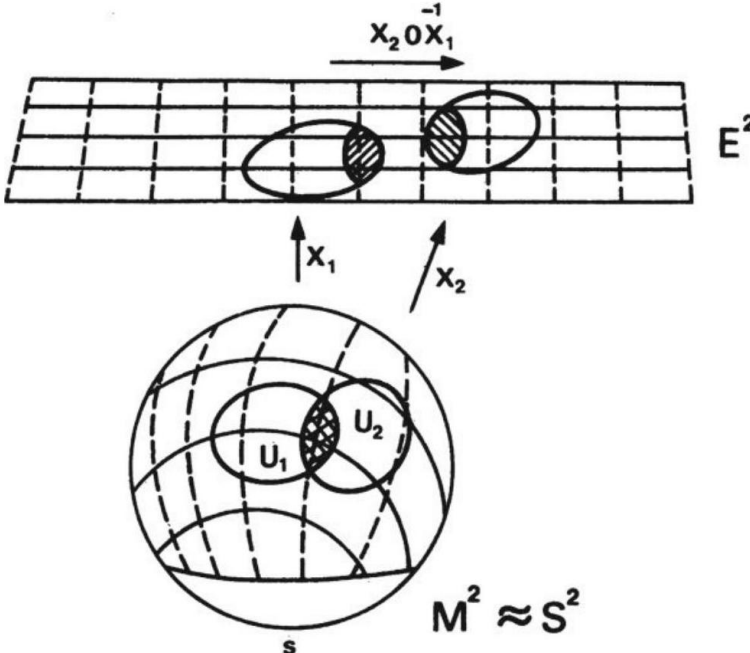
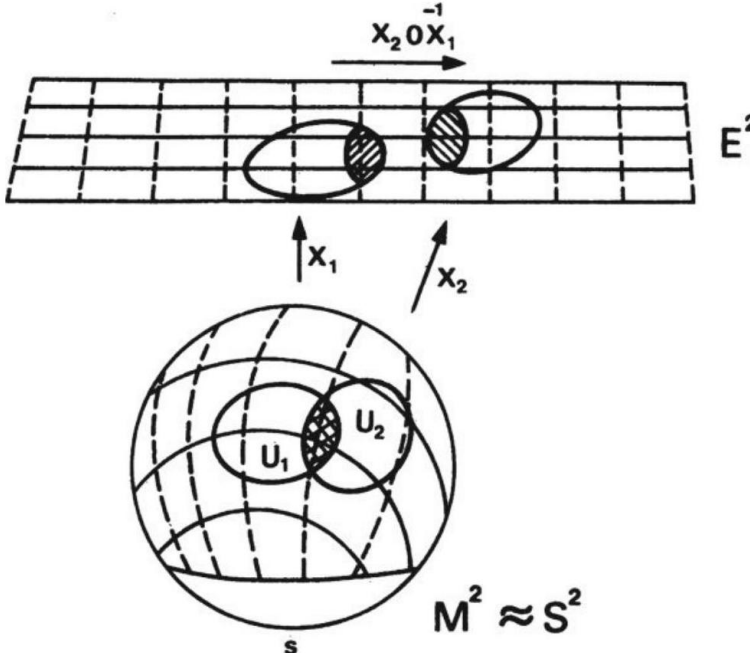
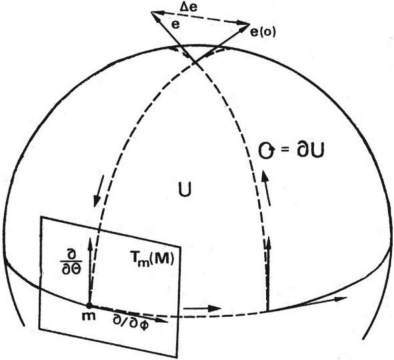
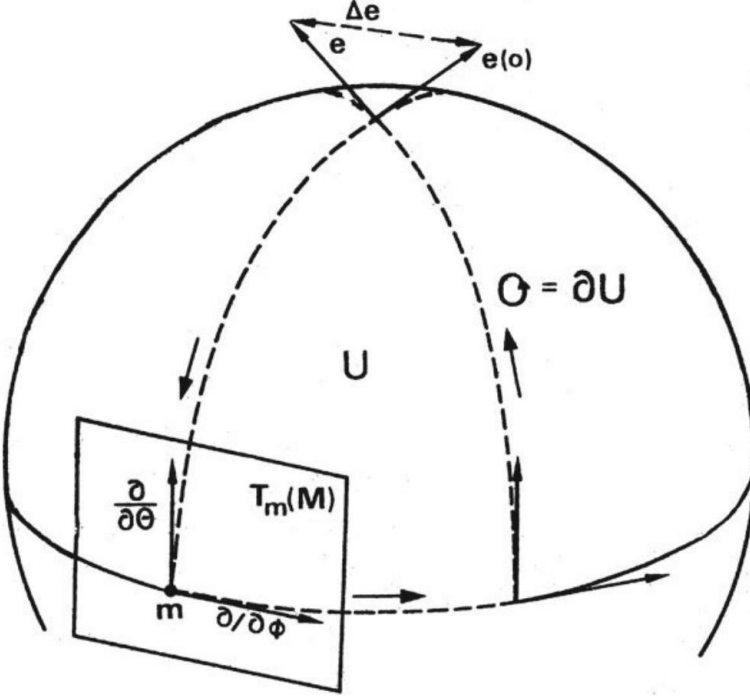
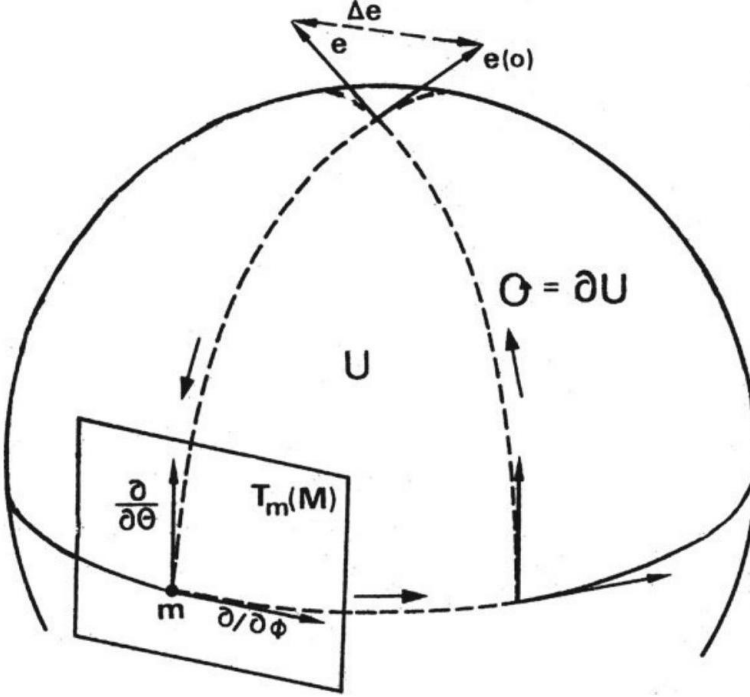
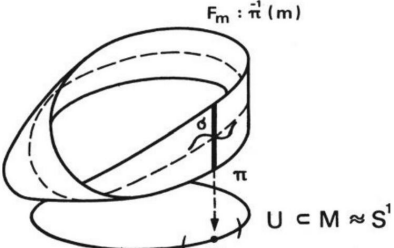
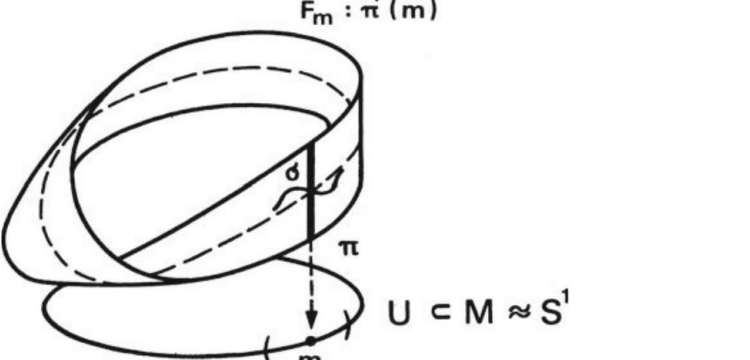
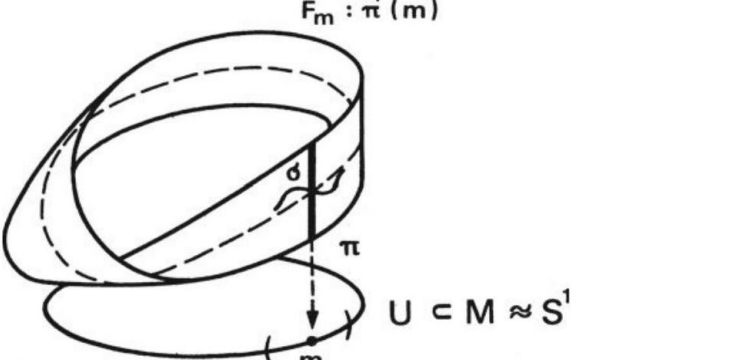
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																								
mielke-geometrodynamics_075	075	—	(no LaTeX source; 1111 × 569 px region)	—	<table><tr><th>Gauge Field Terminology</th><th>Fiber Bundle Terminology</th></tr><tr><td>spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength</td><td>base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form</td></tr><tr><td>electromagnetism electromagnetism with monopoles Dirac’s monopole quantization</td><td>dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$ classification of U(1)-bundles according to the first Chern class</td></tr><tr><td>Yang–Mills theory instanton number</td><td>dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over S^4 according to the second Chern class</td></tr></table>	Gauge Field Terminology	Fiber Bundle Terminology	spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength	base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form	electromagnetism electromagnetism with monopoles Dirac’s monopole quantization	dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$ classification of U(1)-bundles according to the first Chern class	Yang–Mills theory instanton number	dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over S^4 according to the second Chern class																																																
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mielke-geometrodynamics_102	102	—	(no LaTeX source; 902 × 202 px region)	—	<table><tr><th>potential</th><th>field strength</th><th>Bianchi identity</th><th>Noether identity</th></tr><tr><td>ϑ^α $\Gamma_\alpha{}^\beta$</td><td>$T^\alpha = D\vartheta^\alpha$ $R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta$</td><td>$DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$ $DR_\alpha{}^\beta \equiv 0$</td><td>$\widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_{\beta\gamma}) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}$ $D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta$</td></tr></table>	potential	field strength	Bianchi identity	Noether identity	ϑ^α $\Gamma_\alpha{}^\beta$	$T^\alpha = D\vartheta^\alpha$ $R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta$	$DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma$ $DR_\alpha{}^\beta \equiv 0$	$\widehat{D} \Sigma_\alpha \cong (e_\alpha \lrcorner R_{\beta\gamma}) \wedge \Delta^\beta{}_\gamma - \frac{1}{2} (e_\alpha \lrcorner Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}$ $D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta$																																																
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mielke-geometrodynamics_210	210	—	(no LaTeX source; 1159 × 360 px region)	—	<table><tr><th></th><th>Objects</th><th>p-form</th><th>Components</th><th>n = 4</th><th>3</th><th>2</th></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_\alpha^\star$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	p -form	Components	n = 4	3	2	ϑ^α	Vector	1	n^2	16	9	1	$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1	T^α	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_α	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_α	Vector	$n-1$	n^2	16	9	4
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mielke-geometrodynamics_275	275	—	(no LaTeX source; 441 × 116 px region)	—	<table><tr><th>Spin</th><th>Gravitational</th><th>YM anomaly</th></tr><tr><td>1/2</td><td>1</td><td>1</td></tr><tr><td>3/2</td><td>−21</td><td>3</td></tr></table>	Spin	Gravitational	YM anomaly	1/2	1	1	3/2	−21	3																																															
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mielke-geometrodynamics_364	364	—	(no LaTeX source; 1058 × 670 px region)	—	<table><tr><td>:=</td><td>: Equal by definition</td></tr><tr><td>≡</td><td>: Identically equal</td></tr><tr><td>≈</td><td>: Isomorphic</td></tr><tr><td>≅</td><td>: Approximately equal</td></tr><tr><td>∼</td><td>: Asymptotic expansion</td></tr><tr><td>×</td><td>: Topological product</td></tr><tr><td>⊗</td><td>: Direct product</td></tr><tr><td>⊕</td><td>: Direct sum</td></tr><tr><td>⋈</td><td>: Semidirect product</td></tr><tr><td>G_o</td><td>: Connected component of the group</td></tr><tr><td>$\tilde{G}$</td><td>: Simply connected covering group</td></tr><tr><td>[A, B] := AB − BA</td><td>: Commutator</td></tr><tr><td>{A, B} := AB + BA</td><td>: Anticommutator</td></tr><tr><td>A* = C A</td><td>: Complex-conjugate matrix</td></tr><tr><td>A⁺ = A*^T</td><td>: Hermitian adjoint matrix</td></tr><tr><td>$\pi^{-1}(N)$</td><td>: Original domain of N with respect to the mapping $\pi : M \rightarrow N$</td></tr><tr><td>$T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$</td><td>: Symmetrized second-rank tensor</td></tr><tr><td>$T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$</td><td>: Antisymmetrized second-rank tensor</td></tr></table>	:=	: Equal by definition	≡	: Identically equal	≈	: Isomorphic	≅	: Approximately equal	∼	: Asymptotic expansion	×	: Topological product	⊗	: Direct product	⊕	: Direct sum	⋈	: Semidirect product	G _o	: Connected component of the group	$\tilde{G}$	: Simply connected covering group	[A, B] := AB − BA	: Commutator	{A, B} := AB + BA	: Anticommutator	A* = C A	: Complex-conjugate matrix	A ⁺ = A* ^T	: Hermitian adjoint matrix	$\pi^{-1}(N)$	: Original domain of N with respect to the mapping $\pi : M \rightarrow N$	$T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$	: Symmetrized second-rank tensor	$T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$	: Antisymmetrized second-rank tensor																				
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Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured																																																													

Image regions — rendered against scan

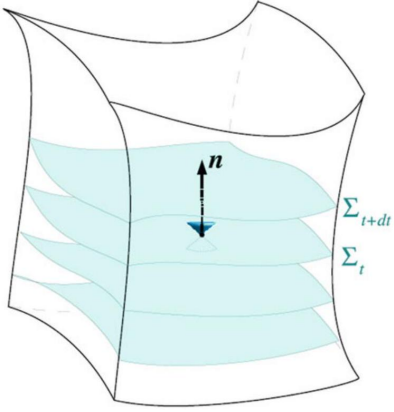
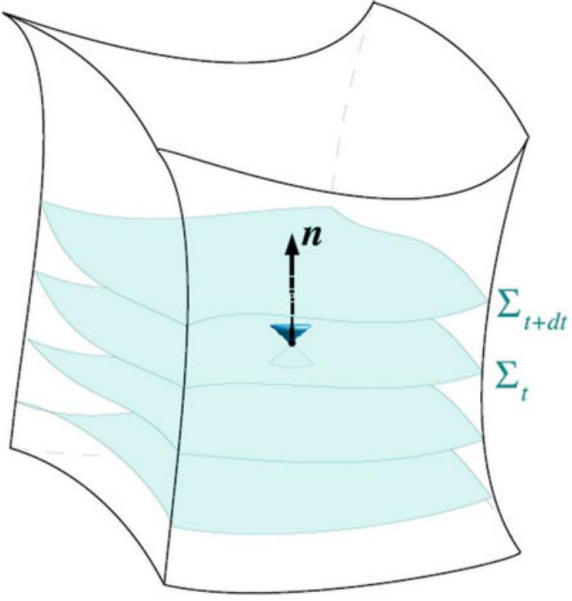
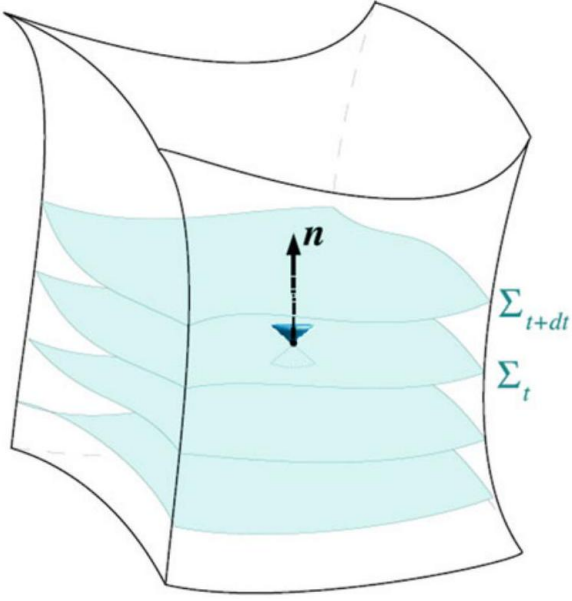
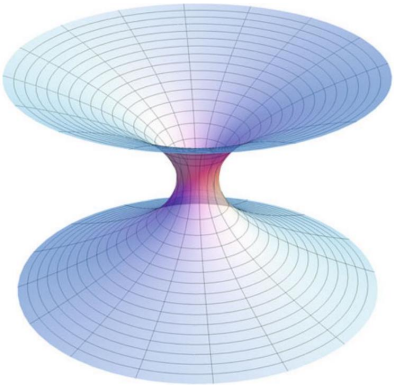
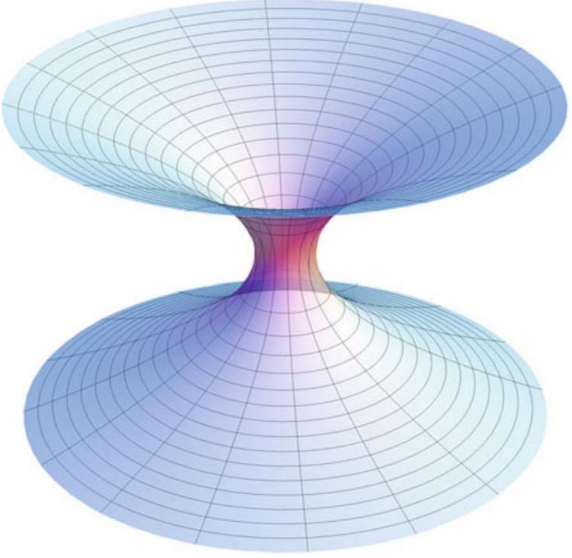
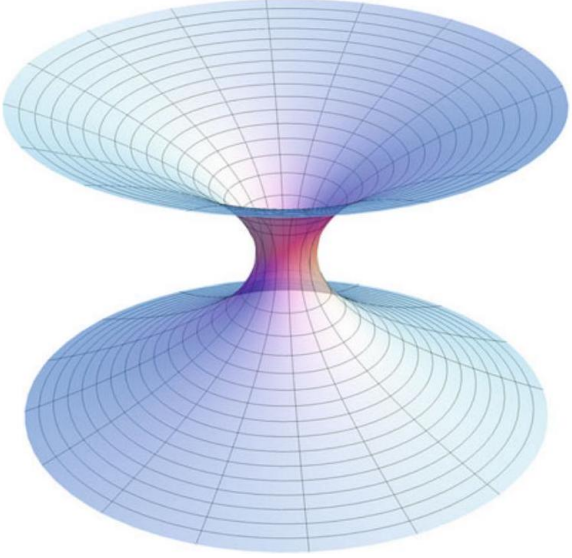
Two comparable cells per row. **Rendered:** the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan:** the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
						
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mielke-geometrodynamics_0020003	359	(dup)	b173365e-9184-4902-bd66-732270e80db9-359_902_600_206_737.jpg			
mielke-geometrodynamics_0020001	025	(dup)	b173365e-9184-4902-bd66-732270e80db9-025_885_748_204_592.jpg			

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